

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ E08584 (11.4) [conf 0.001]	188	■ 0.001	$\left.\left.\left.\left[\begin{array}{cccccccc} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & \ddots & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & -1 & \\ & & & & & & & 1 \\ & & & & & & & & -1 \\ & & & & & & & & & \ddots \end{array}\right] \begin{array}{c} H_1 \\ H_2 \\ H_3 \\ \vdots \\ H_n \\ \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\beta + E_{-\beta}) \\ \frac{1}{\sqrt{2}}(E_\beta - E_{-\beta}) \\ \vdots \end{array}\right.\right.\right.$	(not rendered)	
gilmore-lie-groups_ E06686 (13.4) [conf 0.014]	220	■ 0.014	$\begin{aligned} & \left\{ \left[\frac{h}{p}, \frac{p}{p} \right] \right\} = \left[\frac{h}{p}, \frac{p}{p} \right] = \epsilon \left[\frac{h}{p}, \frac{p}{p} \right] \lim_{\epsilon \rightarrow 0} \epsilon p \rightarrow p' \\ & \left[\frac{h}{p}, \frac{p}{p} \right] = \left[\frac{h}{p}, \frac{p}{p} \right] = \epsilon \left[\frac{h}{p}, \frac{p}{p} \right] \lim_{\epsilon \rightarrow 0} \epsilon^2 (h + p) \rightarrow 0 \end{aligned}$	$\begin{aligned} [h', h'] &= [h, h] \subseteq \begin{aligned} & \left[\frac{h}{p}, \frac{p}{p} \right] = \left[\frac{h}{p}, \frac{p}{p} \right] = \epsilon \left[\frac{h}{p}, \frac{p}{p} \right] \lim_{\epsilon \rightarrow 0} \epsilon p \rightarrow p' \\ & \left[\frac{h}{p}, \frac{p}{p} \right] = \left[\frac{h}{p}, \frac{p}{p} \right] = \epsilon \left[\frac{h}{p}, \frac{p}{p} \right] \lim_{\epsilon \rightarrow 0} \epsilon^2 (h + p) \rightarrow 0 \end{aligned} \end{aligned}$	$\begin{aligned} [h', h'] &= [h, h] \subseteq \begin{aligned} & \left[\frac{h}{p}, \frac{p}{p} \right] = \left[\frac{h}{p}, \frac{p}{p} \right] = \epsilon \left[\frac{h}{p}, \frac{p}{p} \right] \lim_{\epsilon \rightarrow 0} \epsilon p \rightarrow p' \\ & \left[\frac{h}{p}, \frac{p}{p} \right] = \left[\frac{h}{p}, \frac{p}{p} \right] = \epsilon \left[\frac{h}{p}, \frac{p}{p} \right] \lim_{\epsilon \rightarrow 0} \epsilon^2 (h + p) \rightarrow 0 \end{aligned} \end{aligned}$
gilmore-lie-groups_ E08249 [conf 0.031]	84	■ 0.031	$\begin{aligned} f_0(\theta) &= \cos \theta & \text{or} & & f_0(\theta) &= \cos \theta \\ f_1(\theta) &= \sin(\theta)/\theta & & & \theta f_1(\theta) &= \sin(\theta) \\ f_2(\theta) &= (1 - \cos(\theta))/\theta^2 & & & \theta^2 f_2(\theta) &= 1 - \cos(\theta) \end{aligned}$	$\begin{aligned} f_0(\theta) &= \cos \theta & \text{or} & & f_0(\theta) &= \cos \theta \\ f_1(\theta) &= \sin(\theta)/\theta & & & \theta f_1(\theta) &= \sin(\theta) \\ f_2(\theta) &= (1 - \cos(\theta))/\theta^2 & & & \theta^2 f_2(\theta) &= 1 - \cos(\theta) \end{aligned}$	$\begin{aligned} f_0(\theta) &= \cos \theta & \text{or} & & f_0(\theta) &= \cos \theta \\ f_1(\theta) &= \sin(\theta)/\theta & & & \theta f_1(\theta) &= \sin(\theta) \\ f_2(\theta) &= (1 - \cos(\theta))/\theta^2 & & & \theta^2 f_2(\theta) &= 1 - \cos(\theta) \end{aligned}$
gilmore-lie-groups_ E08368 (7.6) [conf 0.032]	114	■ 0.032	$\operatorname{tr} e^{SXS^{-1}} \rightarrow \left\{ \begin{array}{l} 2 \cosh \theta \geq 2 \\ 2 \cos \theta \geq -2 \end{array} \right. \text{ real eigenvalues}$	$\operatorname{tr} e^{SXS^{-1}} \rightarrow \left\{ \begin{array}{l} 2 \cosh \theta \geq 2 \\ 2 \cos \theta \geq -2 \end{array} \right. \text{ imal eigenvalues}$	$\operatorname{tr} e^{SXS^{-1}} \rightarrow \left\{ \begin{array}{l} 2 \cosh \theta \geq 2 \\ 2 \cos \theta \geq -2 \end{array} \right. \text{ real eigenvalues}$

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gilmore-lie-groups_ E09141 [conf 0.052]	56	■ 0.052	$\begin{array}{l} A^{A^t}=I_3 \text{ \& \texttt{\text{quadratic constraints, } } } A \in O(3) \text{ \& } \\ C=0 \text{ \& \texttt{\text{linear constraints } } } B, D \text{ \& } \\ B, \text{ no constraints } \end{array}$	$\begin{array}{l} AA^t = I_3 \\ C = 0 \\ B, D \text{ arbitrary} \\ B, \text{ no constraints} \end{array} \quad \begin{array}{l} \text{quadratic constraints, } A \in O(3) \\ \text{linear constraints} \end{array}$	$\begin{array}{l} AA^t = I_3 \\ C = 0 \\ B, D \text{ arbitrary} \\ B, \text{ no constraints} \end{array} \quad \begin{array}{l} \text{quadratic constraints, } A \in O(3) \\ \text{linear constraints} \\ \text{no constraints} \end{array}$
gilmore-lie-groups_ E09142 (3.25) [conf 0.071]	56	■ 0.071	$\left[\begin{array}{c} x^{\prime} \\ y^{\prime} \\ z^{\prime} \\ 1 \end{array} \right] = \left[\begin{array}{c c} A & t_1 \\ \hline t_2 & 0 \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \\ 1 \end{array} \right] = \left[A \left[\begin{array}{c} x \\ y \\ z \end{array} \right] + t_1 \right]$	$\left[\begin{array}{c} x' \\ y' \\ z' \\ 1 \end{array} \right] = \left[\begin{array}{c c} A & t_1 \\ \hline t_2 & 0 \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \\ 1 \end{array} \right] = \left[A \left[\begin{array}{c} x \\ y \\ z \end{array} \right] + t_1 \right]$	$\left[\begin{array}{c} x' \\ y' \\ z' \\ 1 \end{array} \right] = \left[\begin{array}{c c} A & t_1 \\ \hline t_2 & 0 \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \\ 1 \end{array} \right] = \left[A \left[\begin{array}{c} x \\ y \\ z \end{array} \right] + t_1 \right]$
gilmore-lie-groups_ E09684 (13.2) [conf 0.084]	220	■ 0.084	$\begin{array}{l} \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \\ \mathfrak{h}, \mathfrak{h} \subseteq \mathfrak{h} \\ \mathfrak{h}, \mathfrak{p} \subseteq \mathfrak{p} \end{array} \quad \begin{array}{l} \text{subalgebra} \\ \text{this is important} \\ \text{this is always true} \end{array}$	$\begin{array}{l} \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \\ \mathfrak{h}, \mathfrak{h} \subseteq \mathfrak{h} \\ \mathfrak{h}, \mathfrak{p} \subseteq \mathfrak{p} \end{array} \quad \begin{array}{l} \text{subalgebra} \\ \text{this is important} \\ \text{this is always true} \end{array}$	$\begin{array}{l} \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \\ \mathfrak{h}, \mathfrak{h} \subseteq \mathfrak{h} \\ \mathfrak{h}, \mathfrak{p} \subseteq \mathfrak{p} \end{array} \quad \begin{array}{l} \text{subalgebra} \\ \text{this is important} \\ \text{this is always true} \end{array}$
gilmore-lie-groups_ E09923 (15.57)	286	■ 0.123	$\left. C_k k^0\rangle \right \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \rangle = k\rangle \left \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(C_k)$	$C_k k^0\rangle \left \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle = k\rangle \left \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(C_k)$	$C_k k^0\rangle \left \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle = k\rangle \left \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(C_k)$
gilmore-lie-groups_ E09591 (11.12)	191	■ 0.125	$\begin{array}{l} \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \\ \mathfrak{h}, \mathfrak{h} \subseteq \mathfrak{h} \\ \mathfrak{h}, \mathfrak{p} \subseteq \mathfrak{p} \end{array} \quad \begin{array}{l} \text{subalgebra} \\ \text{this is important} \\ \text{this is always true} \end{array}$	$\begin{array}{l} \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \\ \mathfrak{h}, \mathfrak{h} \subseteq \mathfrak{h} \\ \mathfrak{h}, \mathfrak{p} \subseteq \mathfrak{p} \end{array} \quad \begin{array}{l} \text{subalgebra} \\ \text{this is important} \\ \text{this is always true} \end{array}$	$\begin{array}{l} \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \\ \mathfrak{h}, \mathfrak{h} \subseteq \mathfrak{h} \\ \mathfrak{h}, \mathfrak{p} \subseteq \mathfrak{p} \end{array} \quad \begin{array}{l} \text{subalgebra} \\ \text{this is important} \\ \text{this is always true} \end{array}$

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gilmore-lie-groups_ E00586 (11.6)	189	■ 0.128	$\left.\begin{array}{c} h^1 \\ \vdots \\ h^n \\ e^{+\alpha} \\ e^{-\alpha} \\ e^{+\beta} \\ e^{-\beta} \\ \vdots \end{array}\right\} \begin{array}{c} -1 \\ \\ -1 \\ \ddots \\ -1 \\ -1 \\ -1 \\ \ddots \end{array} \begin{array}{c} iH_1 \\ iH_2 \\ iH_3 \\ \vdots \\ iH_n \\ \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha}) \\ i\frac{1}{\sqrt{2}}(E_\beta + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\beta - E_{-\beta}) \end{array}$	$\begin{bmatrix} h^1 & & & & & & \\ & h^2 & & & & & \\ & & h^3 & & & & \\ & & & \ddots & & & \\ & & & & -1 & & \\ & & & & & -1 & \\ & & & & & & -1 \\ & & & & & & & \ddots \end{bmatrix} \begin{array}{c} iH_1 \\ iH_2 \\ iH_3 \\ \vdots \\ iH_n \\ \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha}) \\ i\frac{1}{\sqrt{2}}(E_\beta + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\beta - E_{-\beta}) \end{array}$	$\begin{bmatrix} h^1 & & & & & & \\ & h^2 & & & & & \\ & & h^3 & & & & \\ & & & \ddots & & & \\ & & & & -1 & & \\ & & & & & -1 & \\ & & & & & & -1 \\ & & & & & & & \ddots \end{bmatrix} \begin{array}{c} iH_1 \\ iH_2 \\ iH_3 \\ \vdots \\ iH_n \\ i\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha}) \\ i\frac{1}{\sqrt{2}}(E_\beta + E_{-\beta}) \\ \frac{1}{\sqrt{2}}(E_\beta - E_{-\beta}) \end{array}$
gilmore-lie-groups_ E00378 (7.16)	118	■ 0.146	$\begin{aligned} & \text{\texttt{\texttt{algebra}}} = \text{\texttt{\texttt{noncompact generators}}} + \text{\texttt{\texttt{compact generators}}} \\ & \text{\texttt{\texttt{EXP}}} \downarrow \\ & \text{\texttt{\texttt{EXP}}} \downarrow \\ & \text{\texttt{\texttt{group}}} = \text{\texttt{\texttt{coset representatives}}} \end{aligned}$	$\begin{aligned} & \text{\texttt{\texttt{algebra}}} = \text{\texttt{\texttt{noncompact generators}}} + \text{\texttt{\texttt{compact generators}}} \\ & \text{\texttt{\texttt{EXP}}} \downarrow \\ & \text{\texttt{\texttt{EXP}}} \downarrow \\ & \text{\texttt{\texttt{group}}} = \text{\texttt{\texttt{coset representatives}}} \end{aligned}$	$\begin{aligned} & \text{\texttt{\texttt{algebra}}} = \text{\texttt{\texttt{noncompact generators}}} + \text{\texttt{\texttt{compact generators}}} \\ & \text{\texttt{\texttt{EXP}}} \downarrow \qquad \text{\texttt{\texttt{EXP}}} \downarrow \qquad \downarrow \qquad \downarrow \text{\texttt{\texttt{EXP}}} \\ & \text{\texttt{\texttt{group}}} = \text{\texttt{\texttt{coset representatives}}} \times \text{\texttt{\texttt{compact subgroup}}} \end{aligned}$
gilmore-lie-groups_ E00687 (13.5)	220	■ 0.160	$\begin{aligned} & \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \quad \mathfrak{p} \rightarrow \mathfrak{p}' = \epsilon \mathfrak{p} \\ & [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \\ & [\mathfrak{h}, \mathfrak{p}] \subseteq \mathfrak{p} \\ & [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{h} + \mathfrak{p} \end{aligned}$	$\begin{aligned} & \mathfrak{g}' = \mathfrak{h} + \mathfrak{p}' \\ & [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \\ & [\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}' \\ & [\mathfrak{p}', \mathfrak{p}'] \subseteq \mathfrak{p}' \end{aligned}$	$\begin{aligned} & \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \quad \mathfrak{p} \rightarrow \mathfrak{p}' = \epsilon \mathfrak{p} \quad \mathfrak{g}' = \mathfrak{h} + \mathfrak{p}' \\ & [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \qquad \longrightarrow \qquad [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \\ & [\mathfrak{h}, \mathfrak{p}] \subseteq \mathfrak{p} \qquad \longrightarrow \qquad [\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}' \\ & [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{h} + \mathfrak{p} \qquad \longrightarrow \qquad [\mathfrak{p}', \mathfrak{p}'] = 0 \end{aligned}$

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gilmore-lie-groups_ EQ0478 (8.10)	146	■ 0.182	$\begin{aligned} X &= \sum y_i Y_i + \sum z_j Z_j \\ \mathfrak{def}(X) &\rightarrow \begin{bmatrix} 0 & +(y_3 - z_3) & -(y_2 - z_2) & +(y_1 + z_1) \\ -(y_3 - z_3) & 0 & +(y_1 - z_1) & +(y_2 + z_2) \\ +(y_2 - z_2) & -(y_1 - z_1) & 0 & +(y_3 + z_3) \\ -(y_1 + z_1) & -(y_2 + z_2) & -(y_3 + z_3) & 0 \end{bmatrix} \\ \mathfrak{Reg}(X) &= \left[\begin{array}{ccc ccc} 0 & -y_3 & +y_2 & & & \\ +y_3 & 0 & -y_1 & & & \\ -y_2 & +y_1 & 0 & & & \\ \hline & & & -z_3 & 0 & +z_1 \\ & & & +z_2 & -z_1 & 0 \end{array} \right] \begin{matrix} Y_1 \\ Y_2 \\ Y_3 \\ Z_1 \\ Z_2 \\ Z_3 \end{matrix} \end{aligned}$	$X = \sum y_i Y_i + \sum z_j Z_j$ $\mathfrak{def}(X) \rightarrow \begin{bmatrix} 0 & +(y_3 - z_3) & -(y_2 - z_2) & +(y_1 + z_1) \\ -(y_3 - z_3) & 0 & +(y_1 - z_1) & +(y_2 + z_2) \\ +(y_2 - z_2) & -(y_1 - z_1) & 0 & +(y_3 + z_3) \\ -(y_1 + z_1) & -(y_2 + z_2) & -(y_3 + z_3) & 0 \end{bmatrix}$ $\mathfrak{Reg}(X) = \left[\begin{array}{ccc ccc} 0 & -y_3 & +y_2 & & & \\ +y_3 & 0 & -y_1 & & & \\ -y_2 & +y_1 & 0 & & & \\ \hline & & & -z_3 & 0 & +z_1 \\ & & & +z_2 & -z_1 & 0 \end{array} \right] \begin{matrix} Y_1 \\ Y_2 \\ Y_3 \\ Z_1 \\ Z_2 \\ Z_3 \end{matrix}$	$X = \sum y_i Y_i + \sum z_j Z_j$ $\mathfrak{def}(X) \rightarrow \begin{bmatrix} 0 & +(y_3 - z_3) & -(y_2 - z_2) & +(y_1 + z_1) \\ -(y_3 - z_3) & 0 & +(y_1 - z_1) & +(y_2 + z_2) \\ +(y_2 - z_2) & -(y_1 - z_1) & 0 & +(y_3 + z_3) \\ -(y_1 + z_1) & -(y_2 + z_2) & -(y_3 + z_3) & 0 \end{bmatrix}$ $\mathfrak{Reg}(X) \rightarrow \left[\begin{array}{ccc ccc} 0 & -y_3 & +y_2 & & & \\ +y_3 & 0 & -y_1 & & & \\ -y_2 & +y_1 & 0 & & & \\ \hline & & & 0 & +z_3 & -z_2 \\ & & & -z_3 & 0 & +z_1 \\ & & & +z_2 & -z_1 & 0 \end{array} \right] \begin{matrix} Y_1 \\ Y_2 \\ Y_3 \\ Z_1 \\ Z_2 \\ Z_3 \end{matrix}$
gilmore-lie-groups_ EQ0097	44	■ 0.191	$\left[\begin{array}{c} x' \\ y' \\ z' \end{array} \right] = \left[\begin{array}{c c} M_1 & a \\ b & \\ \hline a & b \\ M_2 & \end{array} \right] \left[\begin{array}{c c} SO(2) & 0 \\ 0 & 0 \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \end{array} \right]$	$\left[\begin{array}{c} x' \\ y' \\ z' \end{array} \right] = \left[\begin{array}{c c} M_1 & a \\ b & \\ \hline a & b \\ M_2 & \end{array} \right] \left[\begin{array}{c c} SO(2) & 0 \\ 0 & 0 \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \end{array} \right]$	$\left[\begin{array}{c} x' \\ y' \\ z' \end{array} \right] = \left[\begin{array}{c c} M_1 & a \\ b & \\ \hline a & b \\ M_2 & \end{array} \right] \left[\begin{array}{c c} SO(2) & 0 \\ 0 & 0 \\ \hline 0 & 0 \\ 1 & 1 \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \end{array} \right]$
gilmore-lie-groups_ EQ0770 (14.12)	240	■ 0.195	$\begin{aligned} &\text{Klein-Gordon equation} \\ E(n, l) &= \text{Schrödinger equation} \\ \sqrt{1 + (\alpha/N')^2} W(n, l) &= -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2} \quad N = n + l + 1 \\ N' &= n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2} \end{aligned}$	Klein-Gordon equation $E(n, l) = \text{Schrödinger equation}$ $\sqrt{1 + (\alpha/N')^2} W(n, l) = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2} \quad N = n + l + 1$ $N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}$	Klein–Gordon equation $E(n, l) = \frac{m c^2}{\sqrt{1 + (\alpha/N')^2}}$ Schrödinger equation $W(n, l) = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2} \quad N = n + l + 1$

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gilmore-lie-groups_ EQ0384 (7.21)	120	■ 0.217	$\begin{aligned} \mathfrak{so}(3) = & \left\{ \begin{bmatrix} 0 & 0 & -a_2 \\ 0 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{bmatrix} \mid a_1, a_2 \in \mathbb{R} \right\} \\ & \text{EXP} \downarrow \quad \downarrow \text{EXP} \quad \downarrow \text{EXP} \\ & \begin{bmatrix} * & * & -x \\ * & * & y \\ x & -y & z \end{bmatrix} \times \begin{bmatrix} \cos a_3 & \sin a_3 & 0 \\ -\sin a_3 & \cos a_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$	$\mathfrak{so}(3) = \left\{ \begin{bmatrix} 0 & 0 & -a_2 \\ 0 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{bmatrix} \mid a_1, a_2 \in \mathbb{R} \right\}$ $\text{EXP} \downarrow \quad \downarrow \text{EXP} \quad \downarrow \text{EXP}$ $\begin{bmatrix} * & * & -x \\ * & * & y \\ x & -y & z \end{bmatrix} \times \begin{bmatrix} \cos a_3 & \sin a_3 & 0 \\ -\sin a_3 & \cos a_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\mathfrak{so}(3) = \begin{bmatrix} 0 & 0 & -a_2 \\ 0 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & a_3 & 0 \\ -a_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\text{EXP} \downarrow \quad \downarrow \text{EXP} \quad \downarrow \text{EXP}$ $\begin{bmatrix} * & * & -x \\ * & * & y \\ x & -y & z \end{bmatrix} \times \begin{bmatrix} \cos a_3 & \sin a_3 & 0 \\ -\sin a_3 & \cos a_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
gilmore-lie-groups_ EQ0624 (1.20)	24	■ 0.222	$\left. \left(r_1 - r_2 \right) \right\} \xrightarrow{(12)} (r_2 - r_1) = -(r_1 - r_2)$	$(r_1 - r_2) \xrightarrow{I} (r_1 - r_2),$	$(r_1 - r_2) \left\{ \begin{array}{l} \xrightarrow{I} (r_1 - r_2) \\ \xrightarrow{(12)} (r_2 - r_1) = -(r_1 - r_2) \end{array} \right.$
gilmore-lie-groups_ EQ0474 (8.6)	145	■ 0.233	$\begin{aligned} & \eta \left(\hat{n} + \frac{1}{2} \right) + Ra^{\dagger 2} + La^2 + ra^{\dagger} + la + \delta I \\ & \mathfrak{d} \mathfrak{e} \mathfrak{f} = \begin{bmatrix} 0 & l & r & -2\delta \\ 0 & \eta & 2R & -r \\ 0 & -2L & -\eta & l \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ & \mathfrak{R} \mathfrak{e} \mathfrak{g} = \left[\begin{array}{ccc ccc} 0 & -2R & 2L & -r & l & 0 \\ 4L & 2\eta & 0 & 2l & 0 & 0 \\ -4R & 0 & -2\eta & 0 & -2r & 0 \\ \hline & & & \eta & 2L & l \\ & & & -2R & -\eta & -r \\ & & & 0 & 0 & 0 \end{array} \right] \begin{array}{l} \hat{n} + \frac{1}{2} \\ a^{\dagger 2} \\ a^2 \\ a^{\dagger} \\ a \\ I \end{array} \end{aligned}$	(not rendered)	$\eta \left(\hat{n} + \frac{1}{2} \right) + Ra^{\dagger 2} + La^2 + ra^{\dagger} + la + \delta I$ $\mathfrak{d} \mathfrak{e} \mathfrak{f} = \begin{bmatrix} 0 & l & r & -2\delta \\ 0 & \eta & 2R & -r \\ 0 & -2L & -\eta & l \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\mathfrak{R} \mathfrak{e} \mathfrak{g} = \left[\begin{array}{ccc ccc} 0 & -2R & 2L & -r & l & 0 \\ 4L & 2\eta & 0 & 2l & 0 & 0 \\ -4R & 0 & -2\eta & 0 & -2r & 0 \\ \hline & & & \eta & 2L & l \\ & & & -2R & -\eta & -r \\ & & & 0 & 0 & 0 \end{array} \right] \begin{array}{l} \hat{n} + \frac{1}{2} \\ a^{\dagger 2} \\ a^2 \\ a^{\dagger} \\ a \\ I \end{array}$

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gilmore-lie-groups_ EQ0897 (15.31)	281	■ 0.247	$\begin{array}{lll} \text{ (i) } & k \cdot k > 0 & k^0 = (0, 0, 1, 0) \\ \text{ (ii) } & k \cdot k = 0 & k \neq 0 \quad k^0 = (0, 0, 1, +i) \quad \text{ (a)} \\ & & k^0 = (0, 0, 1, -i) \quad \text{ (b)} \\ \text{ (iii) } & k \cdot k < 0 & k^0 = (0, 0, 0, +i) \quad \text{ (a)} \\ & & k^0 = (0, 0, 0, -i) \quad \text{ (b)} \\ \text{ (iv) } & k \cdot k = 0 & k = 0 \quad k^0 = (0, 0, 0, 0) \end{array}$	$\begin{array}{ll} \text{ (i) } & k \cdot k > 0 \quad k^0 = (0, 0, 1, 0) \\ \text{ (ii) } & k \cdot k = 0 \quad k \neq 0 \\ & k^0 = (0, 0, 1, +i) \\ & k^0 = (0, 0, 1, -i) \\ \text{ (iii) } & k \cdot k < 0 \quad k^0 = (0, 0, 0, +i) \\ & k^0 = (0, 0, 0, -i) \\ \text{ (iv) } & k \cdot k = 0 \quad k = 0 \end{array}$	$\begin{array}{ll} \text{ (i) } & k \cdot k > 0 \quad k^0 = (0, 0, 1, 0) \\ \text{ (ii) } & k \cdot k = 0 \quad k \neq 0 \quad k^0 = (0, 0, 1, +i) \quad \text{ (a)} \\ & k^0 = (0, 0, 1, -i) \quad \text{ (b)} \\ \text{ (iii) } & k \cdot k < 0 \quad k^0 = (0, 0, 0, +i) \quad \text{ (a)} \\ & k^0 = (0, 0, 0, -i) \quad \text{ (b)} \\ \text{ (iv) } & k \cdot k = 0 \quad k = 0 \quad k^0 = (0, 0, 0, 0) \end{array}$
gilmore-lie-groups_ EQ0726 (13.44)	228	■ 0.260	$\begin{aligned} & a^{\dagger} a \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} \left[J_3 + \frac{1}{2c^2} \right] \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left[J + M + \left(\frac{1}{2c^2} - J \right) \right] \left \begin{array}{c} J \\ M = n - J \end{array} \right\rangle \rightarrow (n + 0) \left \begin{array}{c} \infty \\ n \end{array} \right\rangle \\ & a^{\dagger} \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} c J_+ \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left \begin{array}{c} J \\ M + 1 \end{array} \right\rangle \sqrt{(J - M)(J + M + 1)c^2} \rightarrow \sqrt{n + 1} \left \begin{array}{c} \infty \\ n + 1 \end{array} \right\rangle \\ & a \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} c J_- \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left \begin{array}{c} J \\ M - 1 \end{array} \right\rangle \sqrt{(J + M)(J - M + 1)c^2} \rightarrow \sqrt{n} \left \begin{array}{c} \infty \\ n - 1 \end{array} \right\rangle \end{aligned}$	$\begin{aligned} & a^{\dagger} a \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} \left(J_3 + \frac{1}{2c^2} \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left[J + M + \left(\frac{1}{2c^2} - J \right) \right] \left \begin{array}{c} J \\ M = n - J \end{array} \right\rangle \rightarrow (n + 0) \left \begin{array}{c} \infty \\ n \end{array} \right\rangle \\ & a^{\dagger} \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} c J_+ \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left \begin{array}{c} J \\ M + 1 \end{array} \right\rangle \sqrt{(J - M)(J + M + 1)c^2} \rightarrow \sqrt{n + 1} \left \begin{array}{c} \infty \\ n + 1 \end{array} \right\rangle \\ & a \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} c J_- \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left \begin{array}{c} J \\ M - 1 \end{array} \right\rangle \sqrt{(J + M)(J - M + 1)c^2} \rightarrow \sqrt{n} \left \begin{array}{c} \infty \\ n - 1 \end{array} \right\rangle \end{aligned}$	$\begin{aligned} & a^{\dagger} a \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} \left(J_3 + \frac{1}{2c^2} \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left[J + M + \left(\frac{1}{2c^2} - J \right) \right] \left \begin{array}{c} J \\ M = n - J \end{array} \right\rangle \rightarrow (n + 0) \left \begin{array}{c} \infty \\ n \end{array} \right\rangle \\ & a^{\dagger} \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} c J_+ \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left \begin{array}{c} J \\ M + 1 \end{array} \right\rangle \sqrt{(J - M)(J + M + 1)c^2} \rightarrow \sqrt{n + 1} \left \begin{array}{c} \infty \\ n + 1 \end{array} \right\rangle \\ & a \left \begin{array}{c} \infty \\ n \end{array} \right\rangle = \lim_{c \rightarrow 0} c J_- \left \begin{array}{c} J \\ M \end{array} \right\rangle \\ & = \lim_{c \rightarrow 0} \left \begin{array}{c} J \\ M - 1 \end{array} \right\rangle \sqrt{(J + M)(J - M + 1)c^2} \rightarrow \sqrt{n} \left \begin{array}{c} \infty \\ n - 1 \end{array} \right\rangle \end{aligned}$

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gilmore-lie-groups_ EQ0778	243	■ 0.264	$\begin{array}{ll} \langle l \rangle \langle l' \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the coordinate representation: } \hbar \sqrt{l(l+1)} \sin \theta \cos \theta & \\ \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the algebraic representation: } \hbar \sqrt{l(l+1)} \cos \theta & \end{array}$ $\begin{array}{ll} \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the coordinate representation: } \hbar \sqrt{l(l+1)} \sin \theta \cos \theta & \\ \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the algebraic representation: } \hbar \sqrt{l(l+1)} \cos \theta & \end{array}$ $\begin{array}{ll} \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the coordinate representation: } \hbar \sqrt{l(l+1)} \sin \theta \cos \theta & \\ \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the algebraic representation: } \hbar \sqrt{l(l+1)} \cos \theta & \end{array}$		$\langle \theta \phi L_- \theta' \phi' \rangle$ $\langle l' L_- l \rangle$ $\langle \theta \phi L_- \theta' \phi' \rangle$
gilmore-lie-groups_ EQ0712 (13.29)	225	■ 0.265	$\begin{array}{ll} \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the coordinate representation: } \hbar \sqrt{l(l+1)} \sin \theta \cos \theta & \\ \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the algebraic representation: } \hbar \sqrt{l(l+1)} \cos \theta & \end{array}$ $\begin{array}{ll} \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the coordinate representation: } \hbar \sqrt{l(l+1)} \sin \theta \cos \theta & \\ \langle l' L_- l \rangle & \text{Matrix element of the angular momentum shift down operator} \\ \text{in the algebraic representation: } \hbar \sqrt{l(l+1)} \cos \theta & \end{array}$	$\begin{bmatrix} h_+ \\ h_- \\ h_3 \\ h_0 \end{bmatrix} = \begin{bmatrix} c & & & \\ & c & & \\ & & 1 & \\ & & & \frac{1}{2c^2} \end{bmatrix} \begin{bmatrix} J_+ \\ J_- \\ J_3 \\ J_0 \end{bmatrix}$	$\begin{bmatrix} h_+ \\ h_- \\ h_3 \\ h_0 \end{bmatrix} = \begin{bmatrix} c & & & \\ & c & & \\ & & 1 & \\ & & & \frac{1}{2c^2} \end{bmatrix} \begin{bmatrix} J_+ \\ J_- \\ J_3 \\ J_0 \end{bmatrix}$
gilmore-lie-groups_ EQ0753	234	■ 0.270	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p} \quad [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}, \quad [\mathfrak{h}, \mathfrak{p}] = \mathfrak{p}, \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{h}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p} \quad [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}, \quad [\mathfrak{h}, \mathfrak{p}] = \mathfrak{p}, \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{h}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p} \quad [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}, \quad [\mathfrak{h}, \mathfrak{p}] = \mathfrak{p}, \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{h}$
gilmore-lie-groups_ EQ0070 (1.65)	34	■ 0.272	$\begin{array}{ll} v_2^3 + v_3^3 = & 27 J_3' = 27 \times 518400 = 1399680 \\ J_3'^3 = & 27 J_3'^3 = 27 \times (-2736)^3 = 552983334912 \end{array}$	$\begin{array}{ll} v_2^3 + v_3^3 = & 27 J_3' = 27 \times 518400 = 1399680 \\ v_2^3 v_3^3 = & -27 J_2'^3 = -27 \times (-2736)^3 = 552983334912 \end{array}$	$\begin{array}{ll} v_2^3 + v_3^3 = & 27 J_3' = 27 \times 518400 = 1399680 \\ v_2^3 v_3^3 = & -27 J_2'^3 = -27 \times (-2736)^3 = 552983334912 \end{array}$

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gilmore-lie-groups_ EQ0311	100	■ 0.278	$\begin{array}{l} \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ll}\}\{\backslash\mathrm{mathfrak{h}},\backslash\mathrm{mathfrak{h}}\}\backslash\mathrm{subse} \\ \mathrm{teq}\backslash\mathrm{mathfrak{h}}\}\&\backslash\mathrm{mathfrak{h}},\backslash\mathrm{mathfrak{h}})\backslash\mathrm{leq}\,0\backslash \\ \backslash\{\backslash\mathrm{left}[\backslash\mathrm{mathfrak{h}},\backslash\mathrm{mathfrak{p}}^{\mathrm{\prime}}\backslash\mathrm{right}]\backslash \\ \mathrm{subse} \\ \mathrm{teq}\backslash\mathrm{mathfrak{p}}^{\mathrm{\prime}}\backslash\mathrm{right}\}\backslash\backslash\{\backslash\mathrm{left}[\backslash\mathrm{mathfrak{p}}^{\mathrm{\prime}}\backslash \\ \mathrm{prime}\},\backslash\mathrm{mathfrak{p}}^{\mathrm{\prime}}\backslash\mathrm{right}]\backslash\mathrm{subse} \\ \mathrm{teq}\backslash\mathrm{mathfrak{h}}\}\&\backslash\mathrm{left}(\backslash\mathrm{mathfrak{h}},\backslash\mathrm{mathfrak{p}}^{\mathrm{\prime}}\backslash \\ \mathrm{prime}\backslash\mathrm{right})=0\backslash\backslash\&\backslash\mathrm{left}(\backslash\mathrm{mathfrak{p}}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash \\ \mathrm{right})\backslash\mathrm{leq}\,0\backslash\mathrm{end}\{\mathrm{array}\} \end{array}$	$\begin{array}{l} [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \quad (\mathfrak{h}, \mathfrak{h}) \leq 0 \\ [\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}' \\ [\mathfrak{p}', \mathfrak{p}'] \subseteq \mathfrak{h} \quad (\mathfrak{h}, \mathfrak{p}') = 0 \\ \quad \quad \quad (\mathfrak{p}', \mathfrak{p}') \leq 0 \end{array}$	$\begin{array}{l} [\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \quad (\mathfrak{h}, \mathfrak{h}) \leq 0 \\ [\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}' \quad \text{and} \quad (\mathfrak{h}, \mathfrak{p}') = 0 \\ [\mathfrak{p}', \mathfrak{p}'] \subseteq \mathfrak{h} \quad (\mathfrak{p}', \mathfrak{p}') \leq 0 \end{array}$
gilmore-lie-groups_ EQ0789 (14.31)	247	■ 0.300	$\begin{array}{l} \backslash\mathrm{begin}\{\mathrm{aligned}\}\&\mathrm{M}_{\mathrm{z}}^{\mathrm{\prime}}\backslash\mathrm{left} \mathrm{N}_{\mathrm{z}}^{\mathrm{\prime}}\backslash\mathrm{right}\backslash\mathrm{rangle}=\mathrm{D}_1\backslash\mathrm{left} \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{c}\}\mathrm{l}+1\backslash\backslash\mathrm{pm} \\ \mathrm{l}\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}\backslash\mathrm{rangle}\quad\mathrm{D}_1=\mathrm{frac}\{1\}\{\mathrm{N}\} \\ \backslash\mathrm{sqrt}\{\mathrm{frac}\{\mathrm{N}^2-(\mathrm{l}+1)^2\}\{2\mathrm{l}+3\}\}\backslash\backslash\&\mathrm{M}_{\mathrm{z}}^{\mathrm{\prime}}\backslash\mathrm{left} \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{c}\}\mathrm{N}^2\backslash\backslash\mathrm{pm} \\ \mathrm{l}\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}\backslash\mathrm{rangle}=\mathrm{pm}\mathrm{D}_2\backslash\mathrm{left} \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{c}\}\mathrm{l}+1\backslash\backslash \\ \mathrm{pm}(\mathrm{l}+1)\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}\backslash\mathrm{rangle}\quad\mathrm{D}_2=\mathrm{frac}\{1\}\{\mathrm{N}\} \\ \backslash\mathrm{sqrt}\{\mathrm{frac}\{2\mathrm{l}+2\}\{2\mathrm{l}+3\}\}\backslash\mathrm{left}[\mathrm{N}^2-(\mathrm{l}+1)^2\backslash\mathrm{right}]\} \\ \backslash\mathrm{end}\{\mathrm{aligned}\} \end{array}$	$\begin{array}{l} M_z' N_{\pm l}^l\rangle = D_1 \left \begin{array}{c} l+1 \\ \pm l \end{array} \right\rangle \quad D_1 = \frac{1}{N} \sqrt{\frac{N^2 - (l+1)^2}{2l+3}} \\ M_{\pm}' \left \begin{array}{c} N^2 \\ \pm l \end{array} \right\rangle = \pm D_2 \left \begin{array}{c} l+1 \\ \pm(l+1) \end{array} \right\rangle \quad D_2 = \frac{1}{N} \sqrt{\frac{2l+2}{2l+3}} [N^2 - (l+1)^2] \end{array}$	$\begin{array}{l} M_z' \left N \begin{array}{c} l \\ \pm l \end{array} \right\rangle = D_1 \left N \begin{array}{c} l+1 \\ \pm l \end{array} \right\rangle \quad D_1 = \frac{1}{N} \sqrt{\frac{N^2 - (l+1)^2}{2l+3}} \\ M_{\pm}' \left N \begin{array}{c} l \\ \pm l \end{array} \right\rangle = \pm D_2 \left N \begin{array}{c} l+1 \\ \pm(l+1) \end{array} \right\rangle \quad D_2 = \frac{1}{N} \sqrt{\frac{2l+2}{2l+3}} [N^2 - (l+1)^2] \end{array}$
gilmore-lie-groups_ EQ0398 (7.34)	124	■ 0.317	$\begin{array}{l} \backslash\mathrm{begin}\{\mathrm{gathered}\}\backslash\mathrm{operatorname}\{\mathrm{EXP}\}\backslash\mathrm{left}[\\ \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{lll}\}\,0\&\mathrm{l}\&\backslash\mathrm{delta}\backslash\backslash\,0\&\,0\&\mathrm{r}\backslash\backslash\,0\& \\ 0\&\,0\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}]=\mathrm{frac}\{\backslash\mathrm{left}[\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\} \\ 1\&\mathrm{l}\&\backslash\mathrm{delta}+\mathrm{frac}\{1\}\{2\}\mathrm{l}\mathrm{r}\backslash\backslash\,0\&\,1\&\mathrm{r}\backslash\backslash\,0\&\,0 \\ \&\,1\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}]\}\{\backslash \}\backslash\backslash\backslash \backslash\backslash\mathrm{e}^{\mathrm{r}^{\mathrm{\prime}}\backslash\mathrm{prime}} \\ \mathrm{a}^{\mathrm{\dagger}}\}\}\mathrm{e}^{\mathrm{\delta}\backslash\mathrm{prime}}\mathrm{I}\}\mathrm{e}^{\mathrm{l}^{\mathrm{\prime}}\backslash\mathrm{prime}}\mathrm{a}\} \\ \backslash\mathrm{rightarrow}\backslash\mathrm{left}[\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\}\,1\&\mathrm{l}^{\mathrm{\prime}}\backslash\mathrm{prime}\}\& \\ \backslash\mathrm{delta}^{\mathrm{\prime}}\backslash\mathrm{prime}\}\backslash\backslash\,0\&\,1\&\mathrm{r}^{\mathrm{\prime}}\backslash\mathrm{prime}\}\backslash\backslash\,0\&\,0\&\,1 \\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}]=\backslash\mathrm{left}[\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\}\,1\&\mathrm{l}^{\mathrm{\prime}}\backslash\mathrm{prime} \\ \mathrm{prime}\}\&\backslash\mathrm{delta}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{prime}\}+\mathrm{l}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{prime}\}\mathrm{r}^{\mathrm{\prime}} \\ \backslash\mathrm{prime}\backslash\mathrm{prime}\}\backslash\backslash\,0\&\,1\&\mathrm{r}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{prime}\}\backslash\backslash\,0\&\,0\& \\ 1\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}]\backslash\mathrm{leftarrow}\mathrm{e}^{\mathrm{l}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{prime}}\mathrm{a}\} \\ \mathrm{e}^{\mathrm{\delta}\backslash\mathrm{prime}\backslash\mathrm{prime}}\mathrm{I}\}\mathrm{e}^{\mathrm{r}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{prime}}\mathrm{a}^{\mathrm{\dagger}}\} \\ \backslash\mathrm{end}\{\mathrm{gathered}\} \end{array}$	$\begin{array}{l} \mathrm{EXP} \left[\begin{array}{ccc} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{array} \right] = \frac{\left[\begin{array}{ccc} 1 & l & \delta + \frac{1}{2}lr \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right]}{\parallel} \\ \parallel \\ e^{r'a^\dagger} e^{\delta'I} e^{l'a} \rightarrow \left[\begin{array}{ccc} 1 & l' & \delta' \\ 0 & 1 & r' \\ 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{ccc} 1 & l'' & \delta'' + l''r'' \\ 0 & 1 & r'' \\ 0 & 0 & 1 \end{array} \right] \leftarrow e^{l''a} e^{\delta''I} e^{r''a^\dagger} \end{array}$	$\begin{array}{l} \mathrm{EXP} \left[\begin{array}{ccc} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{array} \right] = \left[\begin{array}{ccc} 1 & l & \delta + \frac{1}{2}lr \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right] \\ \parallel \\ e^{r'a^\dagger} e^{\delta'I} e^{l'a} \rightarrow \left[\begin{array}{ccc} 1 & l' & \delta' \\ 0 & 1 & r' \\ 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{ccc} 1 & l'' & \delta'' + l''r'' \\ 0 & 1 & r'' \\ 0 & 0 & 1 \end{array} \right] \leftarrow e^{l''a} e^{\delta''I} e^{r''a^\dagger} \end{array}$
gilmore-lie-groups_ EQ0544	170	■ 0.319	$\frac{\mathrm{dim}(\mathrm{operatorname}\{\mathrm{dim}\}(\mathrm{mathfrak{g}}))}{\mathrm{rank}(\mathrm{mathfrak{g}})^2} = \left\{ \begin{array}{lll} \text{ratio} & n \rightarrow \infty & \text{algebra} \\ 2 - \frac{1}{n} & 2 & A_n \\ 2 + \frac{1}{n} & 2 & B_n, C_n \end{array} \right.$	$\frac{\mathrm{dim}(\mathfrak{g})}{\{\mathrm{rank}(\mathfrak{g})\}^2} = \left\{ \begin{array}{lll} \text{ratio} & n \rightarrow \infty & \text{algebra} \\ 1 + \frac{2}{n} & 1 & A_n \\ 2 - \frac{1}{n} & 2 & D_n \\ 2 + \frac{1}{n} & 2 & B_n, C_n \end{array} \right.$	$\frac{\mathrm{dim}(\mathfrak{g})}{\{\mathrm{rank}(\mathfrak{g})\}^2} = \left\{ \begin{array}{lll} \text{ratio} & n \rightarrow \infty & \text{algebra} \\ 1 + \frac{2}{n} & 1 & A_n \\ 2 - \frac{1}{n} & 2 & D_n \\ 2 + \frac{1}{n} & 2 & B_n, C_n \end{array} \right.$

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gilmore-lie-groups_ EQ8581 (9.10)	156	■ 0.333	$\begin{aligned} \mathfrak{def}(Y) &= \frac{1}{2} \left[\begin{array}{cc} i b_3 & b_1 - i b_2 \\ b_1 + i b_2 & -i b_3 \end{array} \right] \\ \mathfrak{Reg}(y) &= \left[\begin{array}{ccc} 0 & -b_3 & -b_2 \\ b_3 & 0 & b_1 \\ -b_2 & b_1 & 0 \end{array} \right] \end{aligned}$	$\mathfrak{def}(Y) = \frac{1}{2} \begin{bmatrix} i b_3 & b_1 - i b_2 \\ b_1 + i b_2 & -i b_3 \end{bmatrix}$ $\mathfrak{Reg}(y) = \begin{bmatrix} 0 & -b_3 & -b_2 \\ b_3 & 0 & b_1 \\ -b_2 & b_1 & 0 \end{bmatrix}$	$\mathfrak{def}(Y) = \frac{1}{2} \begin{bmatrix} i b_3 & b_1 - i b_2 \\ b_1 + i b_2 & -i b_3 \end{bmatrix}$ $\mathfrak{Reg}(y) = \begin{bmatrix} 0 & -b_3 & -b_2 \\ b_3 & 0 & b_1 \\ -b_2 & b_1 & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ8377 (7.15)	118	■ 0.340	$\begin{aligned} &\left[\begin{array}{cc} a & b \\ b & -a \end{array} \right] = \left[\begin{array}{cc} a & b+c \\ b-c & -a \end{array} \right] - \left[\begin{array}{cc} 0 & c \\ -c & 0 \end{array} \right] \\ &\quad \text{EXP} \downarrow \text{EXP} \downarrow \\ &\left[\begin{array}{cc} z+y & x \\ x & z-y \end{array} \right] = \text{SL}(2; \mathbb{R}) \end{aligned}$	$\left[\begin{array}{cc} a & b \\ b & -a \end{array} \right] = \left[\begin{array}{cc} a & b+c \\ b-c & -a \end{array} \right] - \left[\begin{array}{cc} 0 & c \\ -c & 0 \end{array} \right]$ $\text{EXP} \downarrow \text{EXP} \downarrow$ $\left[\begin{array}{cc} z+y & x \\ x & z-y \end{array} \right] = \text{SL}(2; \mathbb{R})$	$\left[\begin{array}{cc} a & b \\ b & -a \end{array} \right] = \left[\begin{array}{cc} a & b+c \\ b-c & -a \end{array} \right] - \left[\begin{array}{cc} 0 & c \\ -c & 0 \end{array} \right]$ $\text{EXP} \downarrow \quad \text{EXP} \downarrow \quad \downarrow \text{EXP} \downarrow \text{EXP}$ $\left[\begin{array}{cc} z+y & x \\ x & z-y \end{array} \right] = \text{SL}(2; \mathbb{R}) \quad / \quad \text{SO}(2)$
gilmore-lie-groups_ EQ8479 (8.11)	147	■ 0.354	$\mathfrak{def} = \frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 - i a_2 \\ a_1 + i a_2 & -a_3 \end{array} \right] \mathfrak{Reg} = \left[\begin{array}{ccc} 0 & -a_3 & +a_2 \\ +a_3 & 0 & -a_1 \\ -a_2 & +a_1 & 0 \end{array} \right] \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$	$\mathfrak{def} = \frac{i}{2} \begin{bmatrix} a_3 & a_1 - i a_2 \\ a_1 + i a_2 & -a_3 \end{bmatrix} \mathfrak{Reg} = \begin{bmatrix} 0 & -a_3 & +a_2 \\ +a_3 & 0 & -a_1 \\ -a_2 & +a_1 & 0 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$	$\mathfrak{def} = \frac{i}{2} \begin{bmatrix} a_3 & a_1 - i a_2 \\ a_1 + i a_2 & -a_3 \end{bmatrix} \mathfrak{Reg} = \begin{bmatrix} 0 & -a_3 & +a_2 \\ +a_3 & 0 & -a_1 \\ -a_2 & +a_1 & 0 \end{bmatrix} \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix}$
gilmore-lie-groups_ EQ8684 (11.25)	193	■ 0.354	$\begin{aligned} &\mathfrak{sp}(n) = \mathfrak{u}(n) + \mathfrak{[sp}(n) - \mathfrak{u}(n)] \\ &\quad \downarrow \quad \downarrow \quad \downarrow \\ &\mathfrak{usp}(2n) = \underbrace{\mathfrak{ou}(2n)}_{\text{real}} + \underbrace{\mathfrak{[usp}(2n) - \mathfrak{ou}(2n)]}_{\text{imaginary}} \\ &\quad \downarrow \quad \downarrow \quad \downarrow \mathfrak{p} \rightarrow i\mathfrak{p} \\ &\mathfrak{sp}(2n; R) = \underbrace{\mathfrak{ou}(2n)}_{\text{real}} + i \underbrace{\mathfrak{[usp}(2n) - \mathfrak{ou}(2n)]}_{\text{real}} \end{aligned}$	$\mathfrak{sp}(n) = \mathfrak{u}(n) + \underbrace{\mathfrak{[sp}(n) - \mathfrak{u}(n)]}_{\downarrow}$ $\mathfrak{usp}(2n) = \underbrace{\mathfrak{ou}(2n)}_{\text{real}} + \underbrace{\mathfrak{[usp}(2n) - \mathfrak{ou}(2n)]}_{\text{imaginary}}$ $\mathfrak{sp}(2n; R) = \underbrace{\mathfrak{ou}(2n)}_{\text{real}} + i \underbrace{\mathfrak{[usp}(2n) - \mathfrak{ou}(2n)]}_{\text{real}}$ $\downarrow \mathfrak{p} \rightarrow i\mathfrak{p}$	$\mathfrak{sp}(n) = \mathfrak{u}(n) + \mathfrak{[sp}(n) - \mathfrak{u}(n)]$ $\downarrow \quad \downarrow \quad \downarrow$ $\mathfrak{usp}(2n) = \underbrace{\mathfrak{ou}(2n)}_{\text{real}} + \underbrace{\mathfrak{[usp}(2n) - \mathfrak{ou}(2n)]}_{\text{imaginary}}$ $\downarrow \quad \downarrow \quad \downarrow \mathfrak{p} \rightarrow i\mathfrak{p}$ $\mathfrak{sp}(2n; R) = \underbrace{\mathfrak{ou}(2n)}_{\text{real}} + i \underbrace{\mathfrak{[usp}(2n) - \mathfrak{ou}(2n)]}_{\text{real}}$
gilmore-lie-groups_ EQ8683 (11.24)	193	■ 0.357	$\begin{aligned} &\mathfrak{u}(n) \longrightarrow \mathfrak{ou}(2n) \\ &\mathfrak{sp}(n) \longrightarrow \mathfrak{usp}(2n) \end{aligned}$	$\begin{aligned} &\mathfrak{u}(n) \longrightarrow \mathfrak{ou}(2n) \\ &\mathfrak{sp}(n) \longrightarrow \mathfrak{usp}(2n) \end{aligned}$	$\begin{aligned} &\mathfrak{u}(n) \longrightarrow \mathfrak{ou}(2n) \\ &\mathfrak{sp}(n) \longrightarrow \mathfrak{usp}(2n) \end{aligned}$

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gilmore-lie-groups_ E00381 (7.19)	119	■ 0.448	$\begin{array}{c} \frac{1}{2} \left[\begin{array}{cc} 0 & ib_1 + b_2 \\ ib_1 - b_2 & 0 \end{array} \right] + \frac{1}{2} \left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array} \right] \\ \downarrow \text{EXP} \quad \downarrow \text{EXP} \\ \left[\begin{array}{cc} e^{+ib_3/2} & 0 \\ 0 & e^{-ib_3/2} \end{array} \right] \end{array}$	$\frac{i}{2} \left[\begin{array}{cc} 0 & ib_1 + b_2 \\ ib_1 - b_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array} \right]$ $[SU(1, 1)/U(1)] \times \left[\begin{array}{cc} e^{+ib_3/2} & 0 \\ 0 & e^{-ib_3/2} \end{array} \right]$	
gilmore-lie-groups_ E00765 (14.11)	239	■ 0.452	$\alpha = \frac{e^2}{\hbar c} = \frac{1}{137.035999796(70)} = 0.0072973525313(38)$	$\alpha = \frac{e^2}{\hbar c} = \frac{1}{137.035999796(70)} = 0.0072973525313(38)$	$\alpha = \frac{e^2}{\hbar c} = \frac{1}{137.035\,999\,796(70)} = 0.007\,297\,352\,531\,3(3\,8)$
gilmore-lie-groups_ E00590 (11.11)	190	■ 0.454	$\frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right]$	$\frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right]$	$\frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right]$
gilmore-lie-groups_ E00890 (15.23)	279	■ 0.463	$\left\langle \Lambda, 0 \right \left \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle = \left \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$	$\left\langle \Lambda, 0 \right \left \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle = \left \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$	$\left\langle \Lambda, 0 \right \left \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle = \left \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$
gilmore-lie-groups_ E00568 (10.9)	177	■ 0.466	$E_8 \pm \mathbf{e}_i \pm \mathbf{e}_j$ $\frac{1}{2} (\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5 \pm \mathbf{e}_6 \pm \mathbf{e}_7 \pm \mathbf{e}_8)$ $1 \leq i \neq j \leq 8$	$E_8 \pm \mathbf{e}_i \pm \mathbf{e}_j$ $\frac{1}{2} (\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5 \pm \mathbf{e}_6 \pm \mathbf{e}_7 \pm \mathbf{e}_8)$ $1 \leq i \neq j \leq 8$	$E_8 \pm \mathbf{e}_i \pm \mathbf{e}_j$ $\frac{1}{2} (\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5 \pm \mathbf{e}_6 \pm \mathbf{e}_7 \pm \mathbf{e}_8)$ $1 \leq i \neq j \leq 8$

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gilmore-lie-groups_ EQ0916 (15.51)	285	■ 0.497	$\left\langle H_{k^0}, 0 \right \left k^0 \right\rangle \left\langle \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right = \left k^0 \right\rangle \left\langle \begin{array}{cc} j & j' \\ v & v' \end{array} \right D_{vv';\mu\mu'}^{jj'}(H_{k^0})$		$\{H_{k^0}, 0\} k^0\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle = k^0\rangle \left \begin{array}{cc} j & j' \\ v & v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(H_{k^0})$
gilmore-lie-groups_ EQ0884 (15.17)	278	■ 0.500	$\Lambda \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle = \left \begin{array}{cc} j & j' \\ v & v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$		$\Lambda \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle = \left \begin{array}{cc} j & j' \\ v & v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$
gilmore-lie-groups_ EQ0308	99	■ 0.509	$\begin{aligned} & \mathfrak{sl}(n; \mathbb{C}) = \text{antihermitian} + \text{hermitian} \\ & \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \end{aligned}$	$\mathfrak{sl}(n; \mathbb{C}) = \text{antihermitian} + \text{hermitian}$ $\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$	$\mathfrak{sl}(n; \mathbb{C}) = \text{antihermitian} + \text{hermitian}$ $\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$
gilmore-lie-groups_ EQ0605 (11.26)	194	■ 0.511	$\begin{aligned} & v_i^{\prime} G_{ij} v_j = (p_1, q_1, p_2, q_2, \dots, p_n, q_n)' \\ & = \sum_{i=1}^n (p_i' q_i - q_i' p_i) \end{aligned}$	$v_i' G_{ij} v_j = (p_1, q_1, p_2, q_2, \dots, p_n, q_n)'$ $= \sum_{i=1}^n (p_i' q_i - q_i' p_i)$	$v_i' G_{ij} v_j = (p_1, q_1, p_2, q_2, \dots, p_n, q_n)'$ $= \sum_{i=1}^n (p_i' q_i - q_i' p_i) \quad (11.26)$
gilmore-lie-groups_ EQ0829 (14.64)	259	■ 0.513	$\begin{aligned} & [rD, r] = +r \quad [a^\dagger a, a^\dagger] = +a^\dagger \\ & [rD, rD^2] = -rD^2 \quad [a^\dagger a, a^\dagger aa] = -a^\dagger aa \\ & [r, rD^2] = -2rD \quad [a^\dagger, a^\dagger aa] = -2a^\dagger a \end{aligned}$	$\begin{aligned} [rD, r] &= +r \quad [a^\dagger a, a^\dagger] = +a^\dagger \\ [rD, rD^2] &= -rD^2 \quad [a^\dagger a, a^\dagger aa] \\ [r, rD^2] &= -2rD \quad [a^\dagger, a^\dagger aa] \\ &= -2a^\dagger a \end{aligned}$	$\begin{aligned} [rD, r] &= +r \quad [a^\dagger a, a^\dagger] = +a^\dagger \\ [rD, rD^2] &= -rD^2 \quad [a^\dagger a, a^\dagger aa] = -a^\dagger aa \\ [r, rD^2] &= -2rD \quad [a^\dagger, a^\dagger aa] = -2a^\dagger a \end{aligned}$

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gilmore-lie-groups_ E00283 (5.22)	93	0.520	$M \mathbb{G} + M^{\dagger} = 0 \quad \begin{array}{l} \mathbb{R} \\ \mathbb{C} \\ \mathbb{Q} \end{array} \quad \begin{array}{l} \mathfrak{o}(n; G) \\ \mathfrak{u}(n; G) \\ \mathfrak{sp}(n; G) \end{array}$	$MG + GM^{\dagger} = 0$	$MG + GM^{\dagger} = 0 \quad \begin{array}{l} \mathbb{R} \\ \mathbb{C} \\ \mathbb{Q} \end{array} \quad \begin{array}{l} \mathfrak{o}(n; G) \\ \mathfrak{u}(n; G) \\ \mathfrak{sp}(n; G) \end{array}$
gilmore-lie-groups_ E00558 (10.3)	174	0.523	$\sum_{\alpha \neq 0} \alpha \cdot \alpha = \text{rank} = l$	$\sum_{\alpha \neq 0} \alpha \cdot \alpha = \text{rank} = l$	$\sum_{\alpha \neq 0} \alpha \cdot \alpha = \text{rank} = l$
gilmore-lie-groups_ E00672	216	0.523	$z \rightarrow z' = \frac{az+b}{cz+d} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2; \mathbb{R}), \quad ad - bc = 1$	$z \rightarrow z' = \frac{az+b}{cz+d} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2; \mathbb{R}), \quad ad - bc = 1$	$z \rightarrow z' = \frac{az+b}{cz+d} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2; \mathbb{R}), \quad ad - bc = 1$
gilmore-lie-groups_ E00500 (9.9)	155	0.530	$\begin{aligned} & \left(\text{Re } g(X) - \lambda I_3 \right) \cdot (-\lambda) = (-\lambda)^3 + (-\lambda)(+a_1^2 + a_2^2 + a_3^2) = 0 \\ & \left(+a_1^2 + a_2^2 + a_3^2 \right) = 0 \quad \& \quad (-\lambda) \left(\lambda^2 + \phi_2(\mathbf{a}) \right) \\ & \phi_2(\mathbf{a}) = +a_1^2 + a_2^2 + a_3^2 \end{aligned}$	$\begin{aligned} \ \Re g(X) - \lambda I_3\ &= (-\lambda)^3 + (-\lambda)(+a_1^2 + a_2^2 + a_3^2) = 0 \\ &= (-\lambda)(\lambda^2 + \phi_2(\mathbf{a})) \\ \phi_2(\mathbf{a}) &= +a_1^2 + a_2^2 + a_3^2 \end{aligned}$	$\begin{aligned} \ \Re g(X) - \lambda I_3\ &= (-\lambda)^3 + (-\lambda)(+a_1^2 + a_2^2 + a_3^2) = 0 \\ &= (-\lambda)(\lambda^2 + \phi_2(\mathbf{a})) \\ \phi_2(\mathbf{a}) &= +a_1^2 + a_2^2 + a_3^2 \end{aligned}$
gilmore-lie-groups_ E00152	59	0.532	$\begin{array}{l} \text{F(n)=F(n-1)+F(n-2)} \\ \text{F(n)} \end{array}$	$\begin{array}{l} F(n) = F(n-1) + F(n-2) \\ \begin{array}{cccccccccc} n & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \dots \\ F(n) & 0 & 1 & 1 & 2 & 3 & 5 & 8 & 13 & \dots \end{array} \end{array}$	$F(n) = F(n-1) + F(n-2)$ $\begin{array}{cccccccccc} n & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \dots \\ F(n) & 0 & 1 & 1 & 2 & 3 & 5 & 8 & 13 & \dots \end{array}$
gilmore-lie-groups_ E00807 (14.47)	253	0.534	$\begin{array}{l} a_1^{\dagger} a_2 \quad (+1, -1, 0, 0) \quad a_1^{\dagger} b_1^{\dagger} \quad (+1, 0, +1, 0) \quad a_1 b_1 \quad (-1, 0, -1, 0) \\ a_2^{\dagger} a_1 \quad (-1, +1, 0, 0) \quad a_1^{\dagger} b_2^{\dagger} \quad (+1, 0, 0, +1) \quad a_1 b_2 \quad (-1, 0, 0, -1) \\ b_1^{\dagger} b_2 \quad (0, 0 + 1, -1) \quad a_2^{\dagger} b_1^{\dagger} \quad (0, +1, +1, 0) \quad a_2 b_1 \quad (0, -1, -1, 0) \\ b_2^{\dagger} b_1 \quad (0, 0, -1, +1) \quad a_2^{\dagger} b_2^{\dagger} \quad (0, +1, 0, +1) \quad a_2 b_2 \quad (0, -1, 0, -1) \end{array}$	$\begin{array}{l} a_1^{\dagger} a_2 \quad (+1, -1, 0, 0) \quad a_1^{\dagger} b_1^{\dagger} \quad (+1, 0, +1, 0) \quad a_1 b_1 \quad (-1, 0, -1, 0) \\ a_2^{\dagger} a_1 \quad (-1, +1, 0, 0) \quad a_1^{\dagger} b_2^{\dagger} \quad (+1, 0, 0, +1) \quad a_1 b_2 \quad (-1, 0, 0, -1) \\ b_1^{\dagger} b_2 \quad (0, 0 + 1, -1) \quad a_2^{\dagger} b_1^{\dagger} \quad (0, +1, +1, 0) \quad a_2 b_1 \quad (0, -1, -1, 0) \\ b_2^{\dagger} b_1 \quad (0, 0, -1, +1) \quad a_2^{\dagger} b_2^{\dagger} \quad (0, +1, 0, +1) \quad a_2 b_2 \quad (0, -1, 0, -1) \end{array}$	$\begin{array}{l} a_1^{\dagger} a_2 \quad (+1, -1, 0, 0) \quad a_1^{\dagger} b_1^{\dagger} \quad (+1, 0, +1, 0) \quad a_1 b_1 \quad (-1, 0, -1, 0) \\ a_2^{\dagger} a_1 \quad (-1, +1, 0, 0) \quad a_1^{\dagger} b_2^{\dagger} \quad (+1, 0, 0, +1) \quad a_1 b_2 \quad (-1, 0, 0, -1) \\ b_1^{\dagger} b_2 \quad (0, 0 + 1, -1) \quad a_2^{\dagger} b_1^{\dagger} \quad (0, +1, +1, 0) \quad a_2 b_1 \quad (0, -1, -1, 0) \\ b_2^{\dagger} b_1 \quad (0, 0, -1, +1) \quad a_2^{\dagger} b_2^{\dagger} \quad (0, +1, 0, +1) \quad a_2 b_2 \quad (0, -1, 0, -1) \end{array}$
gilmore-lie-groups_ E00658	214	0.536	$ds^2 = \frac{(dx_1 dx_2) \begin{bmatrix} 1 + x_2^2 - x_1 x_2 \\ -x_1 x_2 \\ 1 + x_1^2 \end{bmatrix} (dx_1)}{1 + x_1^2 + x_2^2}$	$ds^2 = \frac{(dx_1 dx_2) \begin{bmatrix} 1 + x_2^2 - x_1 x_2 \\ -x_1 x_2 \\ 1 + x_1^2 \end{bmatrix} (dx_1)}{1 + x_1^2 + x_2^2}$	$ds^2 = \frac{(dx_1 dx_2) \begin{bmatrix} 1 + x_2^2 - x_1 x_2 \\ -x_1 x_2 \\ 1 + x_1^2 \end{bmatrix} (dx_1)}{1 + x_1^2 + x_2^2}$

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gilmore-lie-groups_ E09380 (7.18)	119	■ 0.541	$\begin{array}{c} \begin{array}{ccc} \left[\begin{array}{ccc} 0 & 0 & a_2 \\ 0 & 0 & a_1 \\ a_2 & a_1 & 0 \end{array} \right] \\ \text{EXP} \downarrow \end{array} \\ \downarrow \text{SO}(2,1)/\text{SO}(2)] \times \begin{bmatrix} \cos a_3 & \sin a_3 & 0 \\ -\sin a_3 & \cos a_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$	$\begin{array}{c} \begin{array}{ccc} \left[\begin{array}{ccc} 0 & 0 & a_2 \\ 0 & 0 & a_1 \\ a_2 & a_1 & 0 \end{array} \right] \\ \text{EXP} \downarrow \end{array} \\ \downarrow \text{SO}(2,1)/\text{SO}(2)] \times \begin{bmatrix} \cos a_3 & \sin a_3 & 0 \\ -\sin a_3 & \cos a_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$	$\begin{array}{c} \begin{array}{ccc} \left[\begin{array}{ccc} 0 & 0 & a_2 \\ 0 & 0 & a_1 \\ a_2 & a_1 & 0 \end{array} \right] \\ \text{EXP} \downarrow \end{array} \\ \downarrow \text{SO}(2,1)/\text{SO}(2)] \times \begin{bmatrix} \cos a_3 & \sin a_3 & 0 \\ -\sin a_3 & \cos a_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$
gilmore-lie-groups_ E09172	64	■ 0.546	$\{f, g\} = (Df)^t G(Dg) \quad \text{where} \quad G = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix} \quad \text{and} \quad (Dg) = \begin{bmatrix} \partial g / \partial q \\ \partial g / \partial p \end{bmatrix}$	$\{f, g\} = (Df)^t G(Dg) \quad \text{where} \quad G = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix} \quad \text{and} \quad (Dg) = \begin{bmatrix} \partial g / \partial q \\ \partial g / \partial p \end{bmatrix}$	$\{f, g\} = (Df)^t G(Dg) \quad \text{where} \quad G = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix} \quad \text{and} \quad (Dg) = \begin{bmatrix} \partial g / \partial q \\ \partial g / \partial p \end{bmatrix}$
gilmore-lie-groups_ E09180	45	■ 0.551	$X = H(x, y, z, ct) = \begin{bmatrix} ct + z & x - iy \\ x + iy & ct - z \end{bmatrix} = ctI_2 + \sigma \cdot \mathbf{x}$	$X = H(x, y, z, ct) = \begin{bmatrix} ct + z & x - iy \\ x + iy & ct - z \end{bmatrix} = ctI_2 + \sigma \cdot \mathbf{x}$	$X = H(x, y, z, ct) = \begin{bmatrix} ct + z & x - iy \\ x + iy & ct - z \end{bmatrix} = ctI_2 + \sigma \cdot \mathbf{x}$
gilmore-lie-groups_ E09663	214	■ 0.554	$dV = \sqrt{\ g\ } r^{n-1} \sin^{n-2} \phi_2 \sin^{n-3} \phi_3 \cdots \sin^1 \phi_{n-1} \sin^0 \phi_n$	$dV = \sqrt{\ g\ } r^{n-1} \sin^{n-2} \phi_2 \sin^{n-3} \phi_3 \cdots \sin^1 \phi_{n-1} \sin^0 \phi_n$	$dV = \sqrt{\ g\ } r^{n-1} \sin^{n-2} \phi_2 \sin^{n-3} \phi_3 \cdots \sin^1 \phi_{n-1} \sin^0 \phi_n$
gilmore-lie-groups_ E09227 (4.37)	79	■ 0.562	$(X_i, X_j) = \text{Tr} R(X_i) R(X_j) = \sum_r \sum_s R(X_i)_r^s R(X_j)_s^r = \sum_r \sum_s C_{ir}^s C_{js}^r$	$(X_i, X_j) = \text{Tr} R(X_i) R(X_j) = \sum_r \sum_s R(X_i)_r^s R(X_j)_s^r = \sum_r \sum_s C_{ir}^s C_{js}^r$	$(X_i, X_j) = \text{Tr} R(X_i) R(X_j) = \sum_r \sum_s R(X_i)_r^s R(X_j)_s^r = \sum_r \sum_s C_{ir}^s C_{js}^r$
gilmore-lie-groups_ E09724 (13.41)	228	■ 0.562	$\lim_{\substack{c \rightarrow 0 \\ J \rightarrow \infty}} h_3 \left \begin{array}{c} J \\ M \end{array} \right\rangle = n \left \begin{array}{c} \infty \\ n \end{array} \right\rangle$	$\lim_{\substack{c \rightarrow 0 \\ J \rightarrow \infty}} h_3 \left \begin{array}{c} J \\ M \end{array} \right\rangle = n \left \begin{array}{c} \infty \\ n \end{array} \right\rangle$	$\lim_{\substack{c \rightarrow 0 \\ J \rightarrow \infty}} h_3 \left \begin{array}{c} J \\ M \end{array} \right\rangle = n \left \begin{array}{c} \infty \\ n \end{array} \right\rangle$

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gilmore-lie-groups_ EQ0638 (12.10)	208	■ 0.570	$\begin{array}{cc} S\,0(2,1) / S\,0(2) \ \& \ S\,0(3) / S\,0(2) \\ \left[\begin{array}{cc} W & X \\ +X^t & Y \end{array}\right] & \left[\begin{array}{cc} W & X \\ -X^t & Y \end{array}\right] \\ \& \ \left[\begin{array}{cc} W & X \\ -X^t & Y \end{array}\right] & \left[\begin{array}{cc} W & X \\ -X^t & Y \end{array}\right] \\ W^2 = I_2 + \binom{x}{y} (x & y) & W^2 = I_2 - \binom{x}{y} (x & y) \\ Y^2 = I_1 + (x & y) \binom{x}{y} & Y^2 = I_1 - (x & y) \binom{x}{y} \end{array}$	$\begin{array}{cc} SO(2,1)/SO(2) & SO(3)/SO(2) \\ \left[\begin{array}{cc} W & X \\ +X^t & Y \end{array}\right] & \left[\begin{array}{cc} W & X \\ -X^t & Y \end{array}\right] \\ W^2 = I_2 + \binom{x}{y} (x & y) & W^2 = I_2 - \binom{x}{y} (x & y) \\ Y^2 = I_1 + (x & y) \binom{x}{y} & Y^2 = I_1 - (x & y) \binom{x}{y} \end{array}$	$SO(2,1)/SO(2) \qquad SO(3)/SO(2)$ $\left[\begin{array}{cc} W & X \\ +X^t & Y \end{array}\right] \qquad \left[\begin{array}{cc} W & X \\ -X^t & Y \end{array}\right]$ $W^2 = I_2 + \binom{x}{y} (x \ y) \qquad W^2 = I_2 - \binom{x}{y} (x \ y)$ $Y^2 = I_1 + (x \ y) \binom{x}{y} \qquad Y^2 = I_1 - (x \ y) \binom{x}{y}$
gilmore-lie-groups_ EQ0705	224	■ 0.574	$\begin{gathered} \mathcal{C}^2[S\,0(4,1)] = \mathbf{L} \cdot \mathbf{L} - \mathbf{B} \cdot \mathbf{B} + \mathbf{T} \cdot \mathbf{T} - T_{45}^2 \\ \mathcal{C}^2[ISO(3,1)] = 0 - 0 + \nabla \cdot \nabla - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \end{gathered}$	$\mathcal{C}^2[SO(4,1)] = \mathbf{L} \cdot \mathbf{L} - \mathbf{B} \cdot \mathbf{B} + \mathbf{T} \cdot \mathbf{T} - T_{45}^2$ $\mathcal{C}^2[ISO(3,1)] = 0 - 0 + \nabla \cdot \nabla - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$	$\mathcal{C}^2[SO(4,1)] = \mathbf{L} \cdot \mathbf{L} - \mathbf{B} \cdot \mathbf{B} + \mathbf{T} \cdot \mathbf{T} - T_{45}^2$ $\mathcal{C}^2[ISO(3,1)] = 0 - 0 + \nabla \cdot \nabla - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$
gilmore-lie-groups_ EQ0390 (7.26)	123	■ 0.578	$(x, y) \rightarrow \left[\begin{array}{cc} x & y \\ 0 & 1 \end{array}\right] = \left[\begin{array}{cc} e^w & (e^w - 1)z/w \\ 0 & 1 \end{array}\right] \leftarrow (w, z)$	$(x, y) \rightarrow \left[\begin{array}{cc} x & y \\ 0 & 1 \end{array}\right] = \left[\begin{array}{cc} e^w & (e^w - 1)z/w \\ 0 & 1 \end{array}\right] \leftarrow (w, z)$	$(x, y) \rightarrow \left[\begin{array}{cc} x & y \\ 0 & 1 \end{array}\right] = \left[\begin{array}{cc} e^w & (e^w - 1)z/w \\ 0 & 1 \end{array}\right] \leftarrow (w, z)$
gilmore-lie-groups_ EQ0798 (14.41)	251	■ 0.582	$\left[\begin{array}{c} \lambda \mathbf{u} \\ \lambda \end{array}\right]' = \left[\begin{array}{cc} A & B \\ C & D \end{array}\right] \left[\begin{array}{c} \lambda \mathbf{u} \\ \lambda \end{array}\right]$	$\left[\begin{array}{c} \lambda \mathbf{u} \\ \lambda \end{array}\right]' = \left[\begin{array}{cc} A & B \\ C & D \end{array}\right] \left[\begin{array}{c} \lambda \mathbf{u} \\ \lambda \end{array}\right]$	$\left[\begin{array}{c} \lambda \mathbf{u} \\ \lambda \end{array}\right]' = \left[\begin{array}{cc} A & B \\ C & D \end{array}\right] \left[\begin{array}{c} \lambda \mathbf{u} \\ \lambda \end{array}\right]$
gilmore-lie-groups_ EQ1022	319	■ 0.583	$\begin{array}{l} \text{displacements} \\ \text{rotations} \\ \text{boosts} \\ \text{dilations} \end{array} \quad \begin{array}{l} P_i \partial_x, \partial_y, \partial_t \\ L_z \ x \partial_y - y \partial_x \\ B_i \ x \partial_t + t \partial_x, y \partial_t + t \partial_y \\ D_i \ x \partial_x + y \partial_y + t \partial_t, u \partial_u \end{array}$	$\begin{array}{l} \text{displacements} \\ \text{rotations} \\ \text{boosts} \\ \text{dilations} \end{array} \quad \begin{array}{l} P_i \partial_x, \partial_y, \partial_t \\ L_z \ x \partial_y - y \partial_x \\ B_i \ x \partial_t + t \partial_x, y \partial_t + t \partial_y \\ D_i \ x \partial_x + y \partial_y + t \partial_t, u \partial_u \end{array}$	$\begin{array}{l} \text{displacements} \\ \text{rotations} \\ \text{boosts} \\ \text{dilations} \end{array} \quad \begin{array}{l} P_i \partial_x, \partial_y, \partial_t \\ L_z \ x \partial_y - y \partial_x \\ B_i \ x \partial_t + t \partial_x, y \partial_t + t \partial_y \\ D_i \ x \partial_x + y \partial_y + t \partial_t, u \partial_u \end{array}$

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gilmore-lie-groups_ EQ0091	42	■ 0.584	<pre>\begin{aligned} & g\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right) \\ & \circ g\left(y_{\{1\}}, y_{\{2\}}, y_{\{3\}}\right) \\\left[\begin{array}{cc} x_{\{1\}} & x_{\{2\}} \\ x_{\{3\}} & \frac{1+x_{\{2\}} x_{\{3\}}}{x_{\{1\}}} \end{array}\right] \times \left[\begin{array}{cc} y_{\{1\}} & y_{\{2\}} \\ y_{\{3\}} & \frac{1+y_{\{2\}} y_{\{3\}}}{y_{\{1\}}} \end{array}\right] = \left[\begin{array}{cc} z_{\{1\}} & z_{\{2\}} \\ z_{\{3\}} & \frac{1+z_{\{2\}} z_{\{3\}}}{z_{\{1\}}} \end{array}\right] \end{aligned}</pre>	$g\left(x_{1}, x_{2}, x_{3}\right) \circ g\left(y_{1}, y_{2}, y_{3}\right) \\ \left[\begin{array}{cc} x_{1} & x_{2} \\ x_{3} & \frac{1+x_{2} x_{3}}{x_{1}}\end{array}\right] \times\left[\begin{array}{cc} y_{1} & y_{2} \\ y_{3} & \frac{1+y_{2} y_{3}}{y_{1}}\end{array}\right]=\left[\begin{array}{cc} z_{1} & z_{2} \\ z_{3} & \frac{1+z_{2} z_{3}}{z_{1}}\end{array}\right]$	$g\left(x_{1}, x_{2}, x_{3}\right) \circ g\left(y_{1}, y_{2}, y_{3}\right) = g\left(z_{1}, z_{2}, z_{3}\right) \\ \left[\begin{array}{cc} x_{1} & x_{2} \\ x_{3} & \frac{1+x_{2} x_{3}}{x_{1}}\end{array}\right] \times\left[\begin{array}{cc} y_{1} & y_{2} \\ y_{3} & \frac{1+y_{2} y_{3}}{y_{1}}\end{array}\right]=\left[\begin{array}{cc} z_{1} & z_{2} \\ z_{3} & \frac{1+z_{2} z_{3}}{z_{1}}\end{array}\right]$
gilmore-lie-groups_ EQ0946	294	■ 0.584	<pre>g_{\text {flat }}=\left[\begin{array}{c ccc} c^2 & & & \\ \hline & -1 & & \\ & -1 & & \\ & & -1 & \end{array}\right] \quad g_{\text {grav }}=\left[\begin{array}{c ccc} c^2\left(1+\frac{2 \Phi(x)}{c^2}\right) & & & \\ \hline & -1 & & \\ & & -1 & \\ & & & -1 \end{array}\right]</pre>	$g_{\text {flat }}=\left[\begin{array}{c ccc} c^2 & & & \\ \hline & -1 & & \\ & -1 & & \\ & & -1 & \end{array}\right] \quad g_{\text {grav }}=\left[\begin{array}{c ccc} c^2\left(1+\frac{2 \Phi(x)}{c^2}\right) & & & \\ \hline & -1 & & \\ & & -1 & \\ & & & -1 \end{array}\right]$	$g_{\text {flat }}=\left[\begin{array}{c ccc} c^2 & & & \\ \hline & -1 & & \\ & -1 & & \\ & & -1 & \end{array}\right] \quad g_{\text {grav }}=\left[\begin{array}{c ccc} c^2\left(1+\frac{2 \Phi(x)}{c^2}\right) & & & \\ \hline & -1 & & \\ & & -1 & \\ & & & -1 \end{array}\right]$
gilmore-lie-groups_ EQ0639 (12.11)	209	■ 0.594	<pre>\begin{gathered} S 0(2,1) / S 0(2) \quad S 0(3) / S 0(2) \\\text { g }^{*, *}=W^{-2}=\left[I_2+\left(\begin{array}{cc} x & y \\ y & x \end{array}\right)\right]^{-1} \\\text { g }^{*, *}=W^{-2}=\left[I_2-\left(\begin{array}{cc} x & y \\ y & x \end{array}\right)\right]^{-1} \\\text { g }^{*, *}=W^{+2}=\left[\begin{array}{cc} 1+x^2 & +x y \\ +y x & 1+y^2 \end{array}\right] \\\text { g }^{*, *}=W^{+2}=\left[\begin{array}{cc} 1-x^2 & -x y \\ -y x & 1-y^2 \end{array}\right] \end{gathered}</pre>	$SO(2,1) / SO(2) \quad SO(3) / SO(2) \\ g^{*,*}=W^{-2}=\left[I_2+\left(\begin{array}{cc} x & y \\ y & x \end{array}\right)\right]^{-1} \quad g^{*,*}=W^{-2}=\left[I_2-\left(\begin{array}{cc} x & y \\ y & x \end{array}\right)\right]^{-1} \\ g^{*,*}=W^{+2}=\left[\begin{array}{cc} 1+x^2 & +x y \\ +y x & 1+y^2 \end{array}\right] \quad g^{*,*}=W^{+2}=\left[\begin{array}{cc} 1-x^2 & -x y \\ -y x & 1-y^2 \end{array}\right]$	$SO(2,1) / SO(2) \quad SO(3) / SO(2) \\ g^{*,*}=W^{-2}=\left[I_2+\left(\begin{array}{cc} x & y \\ y & x \end{array}\right)\right]^{-1} \quad g^{*,*}=W^{-2}=\left[I_2-\left(\begin{array}{cc} x & y \\ y & x \end{array}\right)\right]^{-1} \\ g^{*,*}=W^{+2}=\left[\begin{array}{cc} 1+x^2 & +x y \\ +y x & 1+y^2 \end{array}\right] \quad g^{*,*}=W^{+2}=\left[\begin{array}{cc} 1-x^2 & -x y \\ -y x & 1-y^2 \end{array}\right]$

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gilmore-lie-groups_ E00019 (1.15)	22	■ 0.596	$\begin{aligned} & \left(z-z_1\right)\left(z-z_2\right) \cdots\left(z-z_n\right)=z^n-I_1 z^{n-1}+I_2 z^{n-2}-\cdots+(-)^n I_n=0 \\ & \left(z-z_1\right) \cdots\left(z-z_n\right)=z^n-I_1 z^{n-1}+I_2 z^{n-2}-\cdots+(-)^n I_n=0 \\ & I_1=\sum_{i=1}^n z_i=z_1+z_2+\cdots+z_n \\ & I_2=\sum_{i<j}^n z_i z_j=z_1 z_2+z_1 z_3+\cdots+z_1 z_n+z_2 z_3+\cdots+z_{n-1} z_n \\ & \vdots \\ & I_n=\sum_{i<j<\cdots<k}^n z_i z_j \cdots z_k=z_1 z_2 \cdots z_n \end{aligned}$	$\begin{aligned} & (z-z_1)(z-z_2)\cdots(z-z_n)=z^n-I_1z^{n-1}+I_2z^{n-2}-\cdots+(-)^nI_n=0 \\ & I_1=\sum_{i=1}^nz_i=z_1+z_2+\cdots+z_n \\ & I_2=\sum_{i<j}^nz_iz_j=z_1z_2+z_1z_3+\cdots+z_1z_n+z_2z_3+\cdots+z_{n-1}z_n \\ & \vdots \\ & I_n=\sum_{i<j<\cdots<k}^nz_iz_j\cdots z_k=z_1z_2\cdots z_n \end{aligned}$	$\begin{aligned} & (z-z_1)(z-z_2)\cdots(z-z_n)=z^n-I_1z^{n-1}+I_2z^{n-2}-\cdots+(-)^nI_n=0 \\ & I_1=\sum_{i=1}^nz_i=z_1+z_2+\cdots+z_n \\ & I_2=\sum_{i<j}^nz_iz_j=z_1z_2+z_1z_3+\cdots+z_1z_n+z_2z_3+\cdots+z_{n-1}z_n \\ & \vdots \\ & I_n=\sum_{i<j<\cdots<k}^nz_iz_j\cdots z_k=z_1z_2\cdots z_n \end{aligned} \quad (1)$
gilmore-lie-groups_ E00567	177	■ 0.603	$\begin{aligned} & E_7 \pm \mathbf{e}_i \pm \mathbf{e}_j \\ & \frac{1}{2} \underbrace{(\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5 \pm \mathbf{e}_6)}_{\text{even number of + signs}} \pm \frac{\sqrt{2}}{4} \mathbf{e}_7 \quad 1 \leq i \neq j \leq 6 \end{aligned}$	$E_7 \pm \mathbf{e}_i \pm \mathbf{e}_j$ $\frac{1}{2} \underbrace{(\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5 \pm \mathbf{e}_6)}_{\text{even number of + signs}} \pm \frac{\sqrt{2}}{4} \mathbf{e}_7 \quad 1 \leq i \neq j \leq 6$	$E_7 \pm \mathbf{e}_i \pm \mathbf{e}_j$ $\frac{1}{2} \underbrace{(\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5 \pm \mathbf{e}_6)}_{\text{even number of + signs}} \pm \frac{\sqrt{2}}{4} \mathbf{e}_7 \quad 1 \leq i \neq j \leq 6$
gilmore-lie-groups_ E00310	99	■ 0.610	$\begin{aligned} & \left[\frac{h}{p}, \frac{h}{p}\right] \subseteq \frac{h}{p} \quad \left(\frac{h}{p}, \frac{h}{p}\right) \leq 0 \\ & \left[\frac{h}{p}, \frac{p}{p}\right] \subseteq \frac{p}{p} \quad \text{and} \quad \left(\frac{h}{p}, \frac{p}{p}\right) = 0 \\ & \left(\frac{h}{p}, \frac{p}{p}\right) \geq 0 \end{aligned}$	$\begin{aligned} & [h, h] \subseteq h \quad (h, h) \leq 0 \\ & [h, p] \subseteq p \quad \text{and} \quad (h, p) = 0 \\ & [p, p] \subseteq h \quad (p, p) \geq 0 \end{aligned}$	$\begin{aligned} & [h, h] \subseteq h \quad (h, h) \leq 0 \\ & [h, p] \subseteq p \quad \text{and} \quad (h, p) = 0 \\ & [p, p] \subseteq h \quad (p, p) \geq 0 \end{aligned}$
gilmore-lie-groups_ E00961 (16.4)	300	■ 0.612	$\begin{aligned} & \text{surface equation} \\ & \text{constraint equation} \quad F(x, y, p) = 0 \\ & p = dy/dx \text{ when } F(x, y, p) = 0 \end{aligned}$	$\begin{aligned} & \text{surface equation} \quad F(x, y, p) = 0 \\ & \text{constraint equation} \quad p = dy/dx \text{ when } F(x, y, p) = 0 \end{aligned}$	$\begin{aligned} & \text{surface equation} \quad F(x, y, p) = 0 \\ & \text{constraint equation} \quad p = dy/dx \text{ when } F(x, y, p) = 0 \end{aligned}$
gilmore-lie-groups_ E00635 (12.7)	208	■ 0.613	$\begin{aligned} & P^{\prime}=\left[\begin{array}{cc} W & X \\ X^{\dagger} & Y \end{array}\right] \\ & W^2=I_n+X X^{\dagger} \\ & Y^2=1+X^{\dagger} X \end{aligned}$	$\begin{aligned} & P^{\prime}=\left[\begin{array}{cc} W & X \\ X^{\dagger} & Y \end{array}\right] \\ & W^2=I_n+X X^{\dagger} \\ & Y^2=1+X^{\dagger} X \end{aligned}$	$\begin{aligned} & P^{\prime}=\left[\begin{array}{cc} W & X \\ X^{\dagger} & Y \end{array}\right] \\ & W^2=I_n+X X^{\dagger} \\ & Y^2=1+X^{\dagger} X \end{aligned}$

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gilmore-lie-groups_ E00600 (11.21)	192	0.619	$\begin{aligned} \mathfrak{g} &= \mathfrak{h} + \mathfrak{p} \longrightarrow \mathfrak{g}' \\ \mathfrak{su}(n) &= \mathfrak{so}(n) + [\mathfrak{su}(n) - \mathfrak{so}(n)] \longrightarrow \mathfrak{sl}(n; \mathbb{R}) \\ \mathfrak{sp}(n) &= \mathfrak{u}(n) + [\mathfrak{sp}(n) - \mathfrak{u}(n)] \longrightarrow \mathfrak{sp}(2n; \mathbb{R}) \end{aligned}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p} \longrightarrow \mathfrak{g}'$ $\mathfrak{su}(n) = \mathfrak{so}(n) + [\mathfrak{su}(n) - \mathfrak{so}(n)] \longrightarrow \mathfrak{sl}(n; \mathbb{R})$ $\mathfrak{sp}(n) = \mathfrak{u}(n) + [\mathfrak{sp}(n) - \mathfrak{u}(n)] \longrightarrow \mathfrak{sp}(2n; \mathbb{R})$	$\begin{aligned} \mathfrak{g} &= \mathfrak{h} + \mathfrak{p} \longrightarrow \mathfrak{g}' \\ \mathfrak{su}(n) &= \mathfrak{so}(n) + [\mathfrak{su}(n) - \mathfrak{so}(n)] \longrightarrow \mathfrak{sl}(n; \mathbb{R}) \\ \mathfrak{sp}(n) &= \mathfrak{u}(n) + [\mathfrak{sp}(n) - \mathfrak{u}(n)] \longrightarrow \mathfrak{sp}(2n; \mathbb{R}) \end{aligned}$
gilmore-lie-groups_ E00411 (7.47)	126	0.625	$\begin{aligned} &\operatorname{EXP}\left(i\theta_{+}^{\prime}\right) \quad \operatorname{EXP}\left(i\theta_{z}^{\prime}\right) \quad \operatorname{EXP}\left(i\theta_{-}^{\prime}\right) \\ &\downarrow \downarrow \\ &\left[\begin{array}{cc} 1 & i\theta_{+}^{\prime} \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} e^{i\theta_{z}^{\prime} / 2} & 0 \\ 0 & e^{-i\theta_{z}^{\prime} / 2} \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ i\theta_{-}^{\prime} & 1 \end{array}\right] \\ &= \left[\begin{array}{cc} e^{i\theta_{z}^{\prime} / 2}-\theta_{+}^{\prime} \theta_{-}^{\prime} e^{-i\theta_{z}^{\prime} / 2} & i\theta_{+}^{\prime} e^{-i\theta_{z}^{\prime} / 2} \\ i\theta_{-}^{\prime} e^{-i\theta_{z}^{\prime} / 2} & e^{-i\theta_{z}^{\prime} / 2} \end{array}\right] \end{aligned}$	$\begin{aligned} &\operatorname{EXP}\left(i\theta_{+}^{\prime} J_{+}\right) \quad \operatorname{EXP}\left(i\theta_{z}^{\prime} J_{z}\right) \quad \operatorname{EXP}\left(i\theta_{-}^{\prime} J_{-}\right) \\ &\downarrow \downarrow \\ &\left[\begin{array}{cc} 1 & i\theta_{+}^{\prime} \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} e^{i\theta_{z}^{\prime} / 2} & 0 \\ 0 & e^{-i\theta_{z}^{\prime} / 2} \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ i\theta_{-}^{\prime} & 1 \end{array}\right] \\ &= \left[\begin{array}{cc} e^{i\theta_{z}^{\prime} / 2}-\theta_{+}^{\prime} \theta_{-}^{\prime} e^{-i\theta_{z}^{\prime} / 2} & i\theta_{+}^{\prime} e^{-i\theta_{z}^{\prime} / 2} \\ i\theta_{-}^{\prime} e^{-i\theta_{z}^{\prime} / 2} & e^{-i\theta_{z}^{\prime} / 2} \end{array}\right] \end{aligned}$	$\begin{aligned} &\operatorname{EXP}\left(i\theta_{+}^{\prime} J_{+}\right) \quad \operatorname{EXP}\left(i\theta_{z}^{\prime} J_{z}\right) \quad \operatorname{EXP}\left(i\theta_{-}^{\prime} J_{-}\right) \\ &\downarrow \downarrow \\ &\left[\begin{array}{cc} 1 & i\theta_{+}^{\prime} \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} e^{i\theta_{z}^{\prime} / 2} & 0 \\ 0 & e^{-i\theta_{z}^{\prime} / 2} \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ i\theta_{-}^{\prime} & 1 \end{array}\right] \\ &= \left[\begin{array}{cc} e^{i\theta_{z}^{\prime} / 2}-\theta_{+}^{\prime} \theta_{-}^{\prime} e^{-i\theta_{z}^{\prime} / 2} & i\theta_{+}^{\prime} e^{-i\theta_{z}^{\prime} / 2} \\ i\theta_{-}^{\prime} e^{-i\theta_{z}^{\prime} / 2} & e^{-i\theta_{z}^{\prime} / 2} \end{array}\right] \end{aligned}$
gilmore-lie-groups_ E00343 (6.28)	106	0.637	$\begin{aligned} \mathcal{X}_a &= -x^{\prime} \partial / \partial x^{\prime} - y^{\prime} \partial / \partial y^{\prime} \\ \mathcal{X}_b &= -\partial / \partial y^{\prime} \end{aligned}$	$\begin{aligned} \mathcal{X}_a &= -x^{\prime} \partial / \partial x^{\prime} - y^{\prime} \partial / \partial y^{\prime} \\ \mathcal{X}_b &= -\partial / \partial y^{\prime} \end{aligned}$	$\begin{aligned} \mathcal{X}_a &= -x^{\prime} \partial / \partial x^{\prime} - y^{\prime} \partial / \partial y^{\prime} \\ \mathcal{X}_b &= -\partial / \partial y^{\prime} \end{aligned}$
gilmore-lie-groups_ E00690 (13.8)	221	0.639	$\begin{aligned} &\left\{ \left[L_3, P_1 \right] \right\} = -P_2 \\ &\left\{ \left[L_3, P_2 \right] \right\} = +P_1 \\ &\left\{ \left[P_1, P_2 \right] \right\} = 0 \end{aligned}$	$\begin{aligned} [L_3, P_1] &= -P_2 \\ [L_3, P_2] &= +P_1 \\ [P_1, P_2] &= 0 \end{aligned}$	$\begin{aligned} [L_3, P_1] &= -P_2 \\ [L_3, P_2] &= +P_1 \\ [P_1, P_2] &= 0 \end{aligned}$
gilmore-lie-groups_ E00736	231	0.640	$\begin{aligned} &\left\langle \begin{array}{c} c \\ m^{\prime} \end{array} \left a J_{\pm} \right \begin{array}{c} c \\ m \end{array} \right\rangle \xrightarrow{\lim} \left(\begin{array}{c} p \\ m^{\prime} \end{array} \left P_{\pm} \right \begin{array}{c} p \\ m \end{array} \right) \\ &a \sqrt{(l \mp m)(l \pm m + 1)} \delta_{m^{\prime}, m \pm 1} \xrightarrow{\lim} p \delta_{m^{\prime}, m \pm 1} \end{aligned}$	$\left\langle \begin{array}{c} c \\ m^{\prime} \end{array} \left a J_{\pm} \right \begin{array}{c} c \\ m \end{array} \right\rangle \xrightarrow{\lim} \left(\begin{array}{c} p \\ m^{\prime} \end{array} \left P_{\pm} \right \begin{array}{c} p \\ m \end{array} \right)$ $a \sqrt{(l \mp m)(l \pm m + 1)} \delta_{m^{\prime}, m \pm 1} \xrightarrow{\lim} p \delta_{m^{\prime}, m \pm 1}$	$\left\langle \begin{array}{c} l \\ m^{\prime} \end{array} \left a J_{\pm} \right \begin{array}{c} l \\ m \end{array} \right\rangle \xrightarrow{\lim} \left\langle \begin{array}{c} p \\ m^{\prime} \end{array} \left P_{\pm} \right \begin{array}{c} p \\ m \end{array} \right\rangle$ $a \sqrt{(l \mp m)(l \pm m + 1)} \delta_{m^{\prime}, m \pm 1} \xrightarrow{\lim} p \delta_{m^{\prime}, m \pm 1}$

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gilmore-lie-groups_ E00873 (15.6)	276	■ 0.640	$\{\Lambda, a\} = \left[\begin{array}{ccc c} & & & x \\ & \Lambda & & y \\ & & & z \\ \hline & & ct & \\ 0 & 0 & 0 & 0 \end{array} \right] 1$	$\{\Lambda, a\} = \left[\begin{array}{ccc c} & & & x \\ & \Lambda & & y \\ & & & z \\ \hline & & ct & \\ 0 & 0 & 0 & 0 \end{array} \right] 1$	$\{\Lambda, a\} = \left[\begin{array}{ccc c} & & & x \\ & \Lambda & & y \\ & & & z \\ \hline & & ct & \\ 0 & 0 & 0 & 0 \end{array} \right] 1$
gilmore-lie-groups_ E00766	240	■ 0.655	$\begin{aligned} & r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0 \\ & r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0 \end{aligned}$	$\begin{aligned} & r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0 \\ & r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0 \end{aligned}$	$\begin{aligned} & r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0 \\ & r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0 \end{aligned}$
gilmore-lie-groups_ E00919	286	■ 0.656	$\begin{aligned} & \begin{aligned} & \left \psi \right\rangle = \sum_{k, \xi} \left k; \xi \right\rangle \left\langle k; \xi \right \psi \rangle \\ & \left \psi \right\rangle = \sum_{k, \mu \mu'} \left k \right\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle \left\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right \psi \rangle \end{aligned} \end{aligned}$	$\begin{aligned} & \left \psi \right\rangle = \sum_{k, \xi} \left k; \xi \right\rangle \left\langle k; \xi \right \psi \rangle \\ & \left \psi \right\rangle = \sum_{k, \mu \mu'} \left k \right\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle \left\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right \psi \rangle \end{aligned}$	$\begin{aligned} & \left \psi \right\rangle = \sum_{k, \xi} \left k; \xi \right\rangle \left\langle k; \xi \right \psi \rangle \\ & \left \psi \right\rangle = \sum_{k, \mu \mu'} \left k \right\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle \left\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right \psi \rangle \end{aligned}$
gilmore-lie-groups_ E00375 (7.13)	117	■ 0.666	$\begin{aligned} & \begin{aligned} & \left[\begin{array}{cc} a & c \\ -c & -a \end{array} \right] \text{EXP} \downarrow \left[\begin{array}{cc} z+y & x \\ -x & z-y \end{array} \right] \times \left[\begin{array}{cc} \cosh b & \sinh b \\ \sinh b & \cosh b \end{array} \right] \\ & + \left[\begin{array}{cc} 0 & b \\ b & 0 \end{array} \right] \downarrow \text{EXP} \end{aligned} \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cc} a & c \\ -c & -a \end{array} \right] \text{EXP} \downarrow \left[\begin{array}{cc} z+y & x \\ -x & z-y \end{array} \right] \times \left[\begin{array}{cc} \cosh b & \sinh b \\ \sinh b & \cosh b \end{array} \right] \\ & + \left[\begin{array}{cc} 0 & b \\ b & 0 \end{array} \right] \downarrow \text{EXP} \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cc} a & c \\ -c & -a \end{array} \right] \text{EXP} \downarrow \left[\begin{array}{cc} z+y & x \\ -x & z-y \end{array} \right] \times \left[\begin{array}{cc} \cosh b & \sinh b \\ \sinh b & \cosh b \end{array} \right] \\ & + \left[\begin{array}{cc} 0 & b \\ b & 0 \end{array} \right] \downarrow \text{EXP} \end{aligned}$
gilmore-lie-groups_ E00149 (3.32)	58	■ 0.672	$\operatorname{Sp}(n) \xrightarrow{\mathbb{Q} \rightarrow 2 \times 2} \operatorname{CU} \operatorname{Sp}(2n) \subset \operatorname{SU}(2n)$	$\operatorname{Sp}(n) \xrightarrow{\mathbb{Q} \rightarrow 2 \times 2} \operatorname{CU} \operatorname{Sp}(2n) \subset \operatorname{SU}(2n)$	$\operatorname{Sp}(n) \xrightarrow{\mathbb{Q} \rightarrow 2 \times 2} \operatorname{CU} \operatorname{Sp}(2n) \subset \operatorname{SU}(2n)$
gilmore-lie-groups_ E01034	321	■ 0.672	$\begin{aligned} & \int \frac{\dim \lambda}{\operatorname{Vol}(G)} \Gamma_{\mu' \nu'}^{\lambda' *} (g) \Gamma_{\mu \nu}^\lambda (g) d\rho(g) = \delta^{\lambda' \lambda} \delta_{\mu' \mu} \delta_{\nu' \nu} \\ & \sum_{\lambda} \sum_{\mu} \sum_{\nu} \frac{\dim \lambda}{\operatorname{Vol}(G)} \Gamma_{\mu \nu}^{\lambda *} (g') \Gamma_{\mu \nu}^\lambda (g) = \delta(g', g) \end{aligned}$	$\begin{aligned} & \int \frac{\dim \lambda}{\operatorname{Vol}(G)} \Gamma_{\mu' \nu'}^{\lambda' *} (g) \Gamma_{\mu \nu}^\lambda (g) d\rho(g) = \delta^{\lambda' \lambda} \delta_{\mu' \mu} \delta_{\nu' \nu} \\ & \sum_{\lambda} \sum_{\mu} \sum_{\nu} \frac{\dim \lambda}{\operatorname{Vol}(G)} \Gamma_{\mu \nu}^{\lambda *} (g') \Gamma_{\mu \nu}^\lambda (g) = \delta(g', g) \end{aligned}$	$\begin{aligned} & \int \frac{\dim \lambda}{\operatorname{Vol}(G)} \Gamma_{\mu' \nu'}^{\lambda' *} (g) \Gamma_{\mu \nu}^\lambda (g) d\rho(g) = \delta^{\lambda' \lambda} \delta_{\mu' \mu} \delta_{\nu' \nu} \\ & \sum_{\lambda} \sum_{\mu} \sum_{\nu} \frac{\dim \lambda}{\operatorname{Vol}(G)} \Gamma_{\mu \nu}^{\lambda *} (g') \Gamma_{\mu \nu}^\lambda (g) = \delta(g', g) \end{aligned}$

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gilmore-lie-groups_ E08817 (14.53)	256	■ 0.675	$\begin{array}{l} \text{\texttt{\begin{array}{lrl} M_{\{1\}}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \text{\texttt{ \& }} M_{\{2\}}=\left[\begin{array}{cccc} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{array}\right] \text{\texttt{ \& }} M_{\{1\}}^t=+M_{\{1\}} \text{\texttt{ \& }} G_{\{1\}}^t=-G_{\{1\}} \text{\texttt{ \& }} M_{\{2\}}^t=-M_{\{2\}} \text{\texttt{ \& }} G_{\{2\}}^t=+G_{\{2\}} \end{array}}} \\ \end{array}$	$M_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad M_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} \quad M_1^t = +M_1 \quad G_1^t = -G_1 \quad M_2^t = -M_2 \quad G_2^t = +G_2$	$M_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad M_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$ $M_1^t = +M_1 \quad G_1^t = -G_1 \quad M_2^t = -M_2 \quad G_2^t = +G_2$
gilmore-lie-groups_ E08335 (6.20)	105	■ 0.676	$\text{\texttt{\begin{array}{c} x^{\prime} \\ 1 \end{array}}} \text{\texttt{\begin{array}{c} 1+d a \text{\texttt{ \& }} d b \\ 0 \text{\texttt{ \& }} 1 \end{array}}} \text{\texttt{\begin{array}{c} x \\ 1 \end{array}}}$	$\begin{bmatrix} x' \\ 1 \end{bmatrix} = \begin{bmatrix} 1+da & db \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}$	$\begin{bmatrix} x' \\ 1 \end{bmatrix} = \begin{bmatrix} 1+da & db \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}$
gilmore-lie-groups_ E09985 (15.39)	282	■ 0.677	$\begin{array}{l} \text{\texttt{\begin{aligned} \{ \left[Y_{\{3\}}, Y_{\{1\}} \right] \} \text{\texttt{ \& }} = -Y_{\{2\}} \\ \{ \left[Y_{\{3\}}, Y_{\{2\}} \right] \} \text{\texttt{ \& }} = +Y_{\{1\}} \\ \{ \left[Y_{\{1\}}, Y_{\{2\}} \right] \} \text{\texttt{ \& }} = 0 \end{aligned}}} \\ \text{\texttt{\operatorname{ISO}(2)}} \end{array}$	$\begin{array}{l} [Y_3, Y_1] = -Y_2 \\ [Y_3, Y_2] = +Y_1 \quad \text{ISO}(2) \\ [Y_1, Y_2] = 0 \end{array}$	$\begin{array}{l} [Y_3, Y_1] = -Y_2 \\ [Y_3, Y_2] = +Y_1 \quad \text{ISO}(2) \\ [Y_1, Y_2] = 0 \end{array}$
gilmore-lie-groups_ E08459 (7.85)	138	■ 0.683	$\text{\texttt{\begin{array}{c} j \\ \theta \end{array}}} \text{\texttt{\begin{array}{c} j \\ -j \end{array}}} \text{\texttt{\begin{array}{c} j \\ -j \end{array}}}$	$\left \begin{array}{c} j \\ \theta \end{array} \right\rangle = SU(2) \left \begin{array}{c} j \\ -j \end{array} \right\rangle$	$\left \begin{array}{c} j \\ \theta \end{array} \right\rangle = SU(2) \left \begin{array}{c} j \\ -j \end{array} \right\rangle$
gilmore-lie-groups_ E08212 (4.22)	76	■ 0.685	$C_{rs}^{\prime t} = (A^{-1})_r^i (A^{-1})_s^j C_{ij}^k A_k^t$	$C_{rs}^{\prime t} = (A^{-1})_r^i (A^{-1})_s^j C_{ij}^k A_k^t$	$C_{rs}^{\prime t} = (A^{-1})_r^i (A^{-1})_s^j C_{ij}^k A_k^t$
gilmore-lie-groups_ E08626 (1.22)	25	■ 0.689	$\text{\texttt{\begin{array}{l} \text{\texttt{\begin{array}{l} 1 & 1 \\ 1 & -1 \end{array}}} \text{\texttt{\begin{array}{l} r_1 \\ r_2 \end{array}}} \text{\texttt{\begin{array}{l} I_1 \\ \pm [I_1^2 - 4I_2]^{1/2} \end{array}}} \end{array}}}$	$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} I_1 \\ \pm [I_1^2 - 4I_2]^{1/2} \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} I_1 \\ \pm [I_1^2 - 4I_2]^{1/2} \end{bmatrix}$
gilmore-lie-groups_ E01837	321	■ 0.697	$\begin{array}{l} \text{\texttt{\begin{aligned} g\rangle\langle g \text{\texttt{ \& }} = \int g\rangle d\rho(g) \langle g \text{\texttt{ \& }} = I \text{\texttt{ \& }} \text{\texttt{\text{in group space}}} \\ \text{\texttt{\begin{array}{c} \lambda \\ \mu\nu \end{array}}} \text{\texttt{\begin{array}{c} \lambda \\ \mu\nu \end{array}}} = \sum_{\lambda} \sum_{\mu} \sum_{\nu} \text{\texttt{\begin{array}{c} \lambda \\ \mu\nu \end{array}}} \text{\texttt{\begin{array}{c} \lambda \\ \mu\nu \end{array}}} = I \text{\texttt{\text{in representation space}}} \end{aligned}}} \\ \end{array}$	$\begin{array}{l} g\rangle\langle g = \int g\rangle d\rho(g) \langle g = I \quad \text{in group space} \\ \left \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \left\langle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right = \sum_{\lambda} \sum_{\mu} \sum_{\nu} \left \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \left\langle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right = I \quad \text{in representation space} \end{array}$	$ g\rangle\langle g = \int g\rangle d\rho(g) \langle g = I \quad \text{in group space}$ $\left \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \left\langle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right = \sum_{\lambda} \sum_{\mu} \sum_{\nu} \left \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \left\langle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right = I \quad \text{in representation space}$
gilmore-lie-groups_ E08272 (5.11)	91	■ 0.699	$\text{\texttt{\begin{array}{l} \text{\texttt{\begin{array}{l} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{array}}} \text{\texttt{\begin{array}{l} X_l \\ X_r \\ X_{\delta} \end{array}}} \end{array}}}$	$\begin{bmatrix} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix} = lX_l + rX_r + \delta X_{\delta}$	$\begin{bmatrix} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix} = lX_l + rX_r + \delta X_{\delta}$

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gilmore-lie-groups_ E09165	63	■ 0.704	$SO(3) \rightarrow \left[\begin{array}{c c} Z_2 & \begin{matrix} x \\ y \end{matrix} \\ \hline -x & -y \end{array} \middle \begin{matrix} Z_1 \\ \hline 1 \end{matrix} \right] \times \left[\begin{array}{cc c} \pm\sqrt{1-z^2} & z & 0 \\ -z & \pm\sqrt{1-z^2} & 0 \\ 0 & 0 & 1 \end{array} \right]$	$SO(3) \rightarrow \left[\begin{array}{c c} Z_2 & \begin{matrix} x \\ y \end{matrix} \\ \hline -x & -y \end{array} \middle \begin{matrix} Z_1 \\ \hline 1 \end{matrix} \right] \times \left[\begin{array}{cc c} \pm\sqrt{1-z^2} & z & 0 \\ -z & \pm\sqrt{1-z^2} & 0 \\ 0 & 0 & 1 \end{array} \right]$	$SO(3) \rightarrow \left[\begin{array}{c c} Z_2 & \begin{matrix} x \\ y \end{matrix} \\ \hline -x & -y \end{array} \middle \begin{matrix} Z_1 \\ \hline 1 \end{matrix} \right] \times \left[\begin{array}{cc c} \pm\sqrt{1-z^2} & z & 0 \\ -z & \pm\sqrt{1-z^2} & 0 \\ 0 & 0 & 1 \end{array} \right]$
gilmore-lie-groups_ E08889	253	■ 0.712	$\begin{array}{l} \begin{array}{c} a_1^{\dagger} a_2 \quad a_2^{\dagger} a_1 \quad \frac{1}{2}(a_1^{\dagger} a_1 - a_2^{\dagger} a_2) \quad \mathfrak{su}(2) \\ b_1^{\dagger} b_2 \quad b_2^{\dagger} b_1 \quad \frac{1}{2}(b_1^{\dagger} b_1 - b_2^{\dagger} b_2) \quad \mathfrak{su}(2) \\ a_i^{\dagger} b_j^{\dagger} \quad a_i b_j \quad \frac{1}{2}(a_i^{\dagger} a_i + b_j^{\dagger} b_j + 1) \quad \mathfrak{su}(1, 1) \end{array} \\ \end{array}$	$\begin{array}{l} a_1^{\dagger} a_2 \quad a_2^{\dagger} a_1 \quad \frac{1}{2}(a_1^{\dagger} a_1 - a_2^{\dagger} a_2) \quad \mathfrak{su}(2) \\ b_1^{\dagger} b_2 \quad b_2^{\dagger} b_1 \quad \frac{1}{2}(b_1^{\dagger} b_1 - b_2^{\dagger} b_2) \quad \mathfrak{su}(2) \\ a_i^{\dagger} b_j^{\dagger} \quad a_i b_j \quad \frac{1}{2}(a_i^{\dagger} a_i + b_j^{\dagger} b_j + 1) \quad \mathfrak{su}(1, 1) \end{array}$	$\begin{array}{l} a_1^{\dagger} a_2 \quad a_2^{\dagger} a_1 \quad \frac{1}{2}(a_1^{\dagger} a_1 - a_2^{\dagger} a_2) \quad \mathfrak{su}(2) \\ b_1^{\dagger} b_2 \quad b_2^{\dagger} b_1 \quad \frac{1}{2}(b_1^{\dagger} b_1 - b_2^{\dagger} b_2) \quad \mathfrak{su}(2) \\ a_i^{\dagger} b_j^{\dagger} \quad a_i b_j \quad \frac{1}{2}(a_i^{\dagger} a_i + b_j^{\dagger} b_j + 1) \quad \mathfrak{su}(1, 1) \end{array}$
gilmore-lie-groups_ E09902 (15.36)	282	■ 0.712	$\begin{array}{l} \left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & ib_1 \\ -\theta_3 & 0 & +\theta_1 & ib_2 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array} \right] \begin{bmatrix} 0 \\ 0 \\ 1 \\ i \end{bmatrix} = \begin{bmatrix} -\theta_2 - b_1 \\ +\theta_1 - b_2 \\ -b_3 \\ -ib_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ \end{array}$	$\begin{bmatrix} 0 & +\theta_3 & -\theta_2 & ib_1 \\ -\theta_3 & 0 & +\theta_1 & ib_2 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ i \end{bmatrix} = \begin{bmatrix} -\theta_2 - b_1 \\ +\theta_1 - b_2 \\ -b_3 \\ -ib_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 & +\theta_3 & -\theta_2 & ib_1 \\ -\theta_3 & 0 & +\theta_1 & ib_2 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ i \end{bmatrix} = \begin{bmatrix} -\theta_2 - b_1 \\ +\theta_1 - b_2 \\ -b_3 \\ -ib_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$
gilmore-lie-groups_ E09470 (8.1)	143	■ 0.714	$\left[Z, X_i \right] = R(Z)_i^j X_j$	$[Z, X_i] = R(Z)_i^j X_j$	$[Z, X_i] = R(Z)_i^j X_j$
gilmore-lie-groups_ E08822 (14.58)	257	■ 0.715	$\begin{array}{l} \Gamma_i = r \mathbf{p}_i \\ \Gamma_4 = \frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r) \\ \Gamma_5 = \frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} + r) \end{array}$	$\begin{array}{l} \Gamma_i = r \mathbf{p}_i \\ \Gamma_4 = \frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r) \\ \Gamma_5 = \frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} + r) \end{array}$	$\begin{array}{l} \Gamma_i = r \mathbf{p}_i \\ \Gamma_4 = \frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r) \\ \Gamma_5 = \frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} + r) \end{array}$
gilmore-lie-groups_ E09578	185	■ 0.715	$L_{ij} = a_i^{\dagger} a_j - a_j^{\dagger} a_i = -L_{ji} \quad Q_{ij} = a_i^{\dagger} a_j + a_j^{\dagger} a_i = +Q_{ji}$	$L_{ij} = a_i^{\dagger} a_j - a_j^{\dagger} a_i = -L_{ji} \quad Q_{ij} = a_i^{\dagger} a_j + a_j^{\dagger} a_i = +Q_{ji}$	$\begin{array}{l} \mathbf{L} \quad \mathbf{Q} \\ L_{ij} = a_i^{\dagger} a_j - a_j^{\dagger} a_i = -L_{ji} \quad Q_{ij} = a_i^{\dagger} a_j + a_j^{\dagger} a_i = +Q_{ji} \end{array}$
gilmore-lie-groups_ E08564	176	■ 0.717	$G_2 + \mathbf{e}_i - \mathbf{e}_j \pm [(\mathbf{e}_i + \mathbf{e}_j) - 2\mathbf{e}_k] \quad 1 \leq i \neq j \neq k \leq 3$	$G_2 + \mathbf{e}_i - \mathbf{e}_j \pm [(\mathbf{e}_i + \mathbf{e}_j) - 2\mathbf{e}_k] \quad 1 \leq i \neq j \neq k \leq 3$	$G_2 + \mathbf{e}_i - \mathbf{e}_j \pm [(\mathbf{e}_i + \mathbf{e}_j) - 2\mathbf{e}_k] \quad 1 \leq i \neq j \neq k \leq 3$
gilmore-lie-groups_ E09699	197	■ 0.718	$w = \alpha + \beta \quad x = \gamma + \delta \quad \text{inner product} \rightarrow 8(\beta^2 + \gamma^2 - \delta^2)$	$w = \alpha + \beta \quad x = \gamma + \delta \quad \text{inner product} \rightarrow 8(\beta^2 + \gamma^2 - \delta^2)$	$w = \alpha + \beta \quad x = \gamma + \delta \quad \text{inner product} \rightarrow 8(\beta^2 + \gamma^2 - \delta^2)$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ E00792 (14.34)	248	■ 0.724	$\begin{gathered} (x, y, z) \in R^3 \rightarrow (w, x, y, z) \in R^4 \\ (p_x, p_y, p_z) \in R^3 \rightarrow (p_w, p_x, p_y, p_z) \in R^4 \end{gathered}$	$(x, y, z) \in R^3 \rightarrow (w, x, y, z) \in R^4$ $(p_x, p_y, p_z) \in R^3 \rightarrow (p_w, p_x, p_y, p_z) \in R^4$	$(x, y, z) \in R^3 \rightarrow (w, x, y, z) \in R^4$ $(p_x, p_y, p_z) \in R^3 \rightarrow (p_w, p_x, p_y, p_z) \in R^4$
gilmore-lie-groups_ E00042 (1.39)	28	■ 0.725	$\begin{aligned} & \quad y^3 - 0y^2 + I_2' y - I_3' = 0 \\ & y - I_3'^{\prime} = 0 \quad \& \quad I_1'^{\prime} = s_1' + s_2' + s_3' = 0 \\ & I_2' = s_1' s_2' + s_1' s_3' + s_2' s_3' = I_2 - (1/3) I_1'^2 \\ & I_3' = s_1' s_2' s_3' = I_3 - (1/3) I_2 I_1 + (2/27) I_1'^3 \end{aligned}$	$y^3 - 0y^2 + I_2' y - I_3' = 0$ $I_1' = s_1' + s_2' + s_3' = 0$ $I_2' = s_1' s_2' + s_1' s_3' + s_2' s_3' = I_2 - (1/3) I_1'^2$ $I_3' = s_1' s_2' s_3' = I_3 - (1/3) I_2 I_1 + (2/27) I_1'^3$	$y^3 - 0y^2 + I_2' y - I_3' = 0$ $I_1' = s_1' + s_2' + s_3' = 0$ $I_2' = s_1' s_2' + s_1' s_3' + s_2' s_3' = I_2 - (1/3) I_1'^2$ $I_3' = s_1' s_2' s_3' = I_3 - (1/3) I_2 I_1 + (2/27) I_1'^3$
gilmore-lie-groups_ E00114 (3.4)	49	■ 0.726	$\begin{array}{l} c_1 = a_1 + i b_1 \in \mathbb{C} \\ c_2 = a_2 + i b_2 \in \mathbb{C} \\ i^2 = -1, \quad j^2 = -1, \quad ij + ji = 0 \\ (c_1, c_2) = c_1 + j c_2 \\ (c_1, c_2) \longrightarrow \begin{bmatrix} c_1 & c_2 \\ -c_2^* & c_1^* \end{bmatrix} \end{array}$	$c_1 = a_1 + i b_1 \in \mathbb{C}$ $c_2 = a_2 + i b_2 \in \mathbb{C}$ $i^2 = -1, \quad j^2 = -1, \quad ij + ji = 0$ $(c_1, c_2) = c_1 + j c_2$ $(c_1, c_2) \longrightarrow \begin{bmatrix} c_1 & c_2 \\ -c_2^* & c_1^* \end{bmatrix}$	$c_1 = a_1 + i b_1 \in \mathbb{C}$ $q = (c_1, c_2) = c_1 + j c_2 \quad c_2 = a_2 + i b_2 \in \mathbb{C}$ $i^2 = -1, \quad j^2 = -1, \quad ij + ji = 0$ $(c_1, c_2) \longrightarrow \begin{bmatrix} c_1 & c_2 \\ -c_2^* & c_1^* \end{bmatrix}$
gilmore-lie-groups_ E00616	198	■ 0.728	$\begin{aligned} & \mathfrak{su}(2) \quad J_3, J_{\pm} \quad [J_3, J_{\pm}] = \pm J_{\pm} \quad [J_+, J_-] = +2J_3 \\ & \mathfrak{su}(1, 1) \quad K_3, K_{\pm} \quad [K_3, K_{\pm}] = \pm K_{\pm} \quad [K_+, K_-] = -2K_3 \end{aligned}$	$\mathfrak{su}(2) \quad J_3, J_{\pm} \quad [J_3, J_{\pm}] = \pm J_{\pm} \quad [J_+, J_-] = +2J_3$ $\mathfrak{su}(1, 1) \quad K_3, K_{\pm} \quad [K_3, K_{\pm}] = \pm K_{\pm} \quad [K_+, K_-] = -2K_3$	$\mathfrak{su}(2) \quad J_3, J_{\pm} \quad [J_3, J_{\pm}] = \pm J_{\pm} \quad [J_+, J_-] = +2J_3$ $\mathfrak{su}(1, 1) \quad K_3, K_{\pm} \quad [K_3, K_{\pm}] = \pm K_{\pm} \quad [K_+, K_-] = -2K_3$
gilmore-lie-groups_ E00612	197	■ 0.729	$\begin{aligned} & \operatorname{Spin}(3) = S U(2) \quad \& \quad \operatorname{Spin}(4) = S U(2) \otimes S U(2) \\ & \operatorname{Spin}(5) = U \operatorname{Sp}(4) \\ & \operatorname{Spin}(6) = S U(4) \end{aligned}$	$\operatorname{Spin}(3) = S U(2)$ $\operatorname{Spin}(4) = S U(2) \otimes S U(2)$ $\operatorname{Spin}(5) = U \operatorname{Sp}(4)$ $\operatorname{Spin}(6) = S U(4)$	$\operatorname{Spin}(3) = S U(2)$ $\operatorname{Spin}(4) = S U(2) \otimes S U(2)$ $\operatorname{Spin}(5) = U \operatorname{Sp}(4)$ $\operatorname{Spin}(6) = S U(4)$

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gilmore-lie-groups_ EQ0788 (14.30)	247	■ 0.732	$\begin{aligned} M_{+}^{\prime} Y_m^l = & \langle \theta \phi \left(\begin{array}{cc} j/2 & j/2 \\ m_a + 1 & m_b \end{array} \right) \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \\ & - \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b + 1 \end{array} \right\rangle \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \right) \end{aligned}$	$M_{+}^{\prime} Y_m^l = \langle \theta \phi \left(\begin{array}{cc} j/2 & j/2 \\ m_a + 1 & m_b \end{array} \right) \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \\ - \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b + 1 \end{array} \right\rangle \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \right)$	$M_{+}^{\prime} Y_m^l = \langle \theta \phi \left(\begin{array}{cc} j/2 & j/2 \\ m_a + 1 & m_b \end{array} \right) \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \\ - \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b + 1 \end{array} \right\rangle \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \right)$
gilmore-lie-groups_ EQ0938	290	■ 0.733	$\begin{aligned} \mathbf{B} + i\mathbf{E} & \rightarrow \mathbf{B}' + i\mathbf{E}' = e^{i\phi}(\mathbf{B} + i\mathbf{E}) \\ \mathbf{J}_m + i\mathbf{J}_e & \rightarrow \mathbf{J}'_m + i\mathbf{J}'_e = e^{i\phi}(\mathbf{J}_m + i\mathbf{J}_e) \end{aligned}$	$\mathbf{B} + i\mathbf{E} \rightarrow \mathbf{B}' + i\mathbf{E}' = e^{i\phi}(\mathbf{B} + i\mathbf{E})$ $\mathbf{J}_m + i\mathbf{J}_e \rightarrow \mathbf{J}'_m + i\mathbf{J}'_e = e^{i\phi}(\mathbf{J}_m + i\mathbf{J}_e)$	$\mathbf{B} + i\mathbf{E} \rightarrow \mathbf{B}' + i\mathbf{E}' = e^{i\phi}(\mathbf{B} + i\mathbf{E})$ $\mathbf{J}_m + i\mathbf{J}_e \rightarrow \mathbf{J}'_m + i\mathbf{J}'_e = e^{i\phi}(\mathbf{J}_m + i\mathbf{J}_e)$
gilmore-lie-groups_ EQ0226 (4.36)	79	■ 0.737	$\begin{aligned} \left(X_i, X_j \right) &= \operatorname{tr} R \left(X_i \right) \left(X_j \right) = \sum_r \sum_s R \left(X_i \right)_r^{s*} R \left(X_j \right)_r^s = \sum_r \sum_s C_{ir}^{s*} C_{jr}^s \\ \left(X_i, X_j \right) &= \operatorname{tr} R \left(X_i \right) \left(X_j \right) = \sum_r \sum_s R \left(X_i \right)_r^{s*} R \left(X_j \right)_r^s = \sum_r \sum_s C_{ir}^{s*} C_{jr}^s \end{aligned}$	$(X_i, X_j) = \operatorname{tr} R(X_i)^\dagger R(X_j) = \sum_r \sum_s R(X_i)_r^{s*} R(X_j)_r^s = \sum_r \sum_s C_{ir}^{s*} C_{jr}^s$	$(X_i, X_j) = \operatorname{tr} R(X_i)^\dagger R(X_j) = \sum_r \sum_s R(X_i)_r^{s*} R(X_j)_r^s = \sum_r \sum_s C_{ir}^{s*} C_{jr}^s$
gilmore-lie-groups_ EQ0723 (13.40)	227	■ 0.737	$\lim_{c \rightarrow 0} \left(-J + \frac{1}{2c^2} \right) = 0 \text{ implies } 2Jc^2 = 1$	$\lim_{c \rightarrow 0} \left(-J + \frac{1}{2c^2} \right) = 0 \text{ implies } 2Jc^2 = 1$	$\lim_{c \rightarrow 0} \left(-J + \frac{1}{2c^2} \right) = 0 \text{ implies } 2Jc^2 = 1$
gilmore-lie-groups_ EQ0683 (13.1)	219	■ 0.743	$C_{rs}^t(0) = \lim_{\epsilon \rightarrow 0} C_{rs}^t(\epsilon)$	$C_{rs}^t(0) = \lim_{\epsilon \rightarrow 0} C_{rs}^t(\epsilon)$	$C_{rs}^{\prime t}(0) = \lim_{\epsilon \rightarrow 0} C_{rs}^{\prime t}(\epsilon)$
gilmore-lie-groups_ EQ0685 (13.3)	220	■ 0.743	$\begin{aligned} \left(\begin{array}{l} \mathbf{h}' \\ \mathbf{p}' \end{array} \right) &= \left[\begin{array}{cc} I_{\dim(\mathbf{h})} & 0 \\ 0 & \epsilon I_{\dim(\mathbf{p})} \end{array} \right] \left[\begin{array}{l} \mathbf{h} \\ \mathbf{p} \end{array} \right] \end{aligned}$	$\left[\begin{array}{l} \mathbf{h}' \\ \mathbf{p}' \end{array} \right] = \left[\begin{array}{cc} I_{\dim(\mathbf{h})} & 0 \\ 0 & \epsilon I_{\dim(\mathbf{p})} \end{array} \right] \left[\begin{array}{l} \mathbf{h} \\ \mathbf{p} \end{array} \right]$	$\left[\begin{array}{l} \mathbf{h}' \\ \mathbf{p}' \end{array} \right] = \left[\begin{array}{cc} I_{\dim(\mathbf{h})} & 0 \\ 0 & \epsilon I_{\dim(\mathbf{p})} \end{array} \right] \left[\begin{array}{l} \mathbf{h} \\ \mathbf{p} \end{array} \right]$
gilmore-lie-groups_ EQ0184	67	■ 0.749	$S_{\{\mathrm{Tot}\}} = \left(\begin{array}{cc} 0 & s_{34} \\ s_{21} & 0 \end{array} \right) + \frac{1}{1 - s_{12}s_{43}} \left[\begin{array}{cc} s_{33}s_{22} & s_{33}s_{112}s_{44} \\ s_{22}s_{43}s_{11} & s_{22}s_{44} \end{array} \right]$	$S_{\mathrm{Tot}} = \left[\begin{array}{cc} 0 & s_{34} \\ s_{21} & 0 \end{array} \right] + \frac{1}{1 - s_{12}s_{43}} \left[\begin{array}{cc} s_{33}s_{22} & s_{33}s_{112}s_{44} \\ s_{22}s_{43}s_{11} & s_{22}s_{44} \end{array} \right]$	$S_{\mathrm{Tot}} = \left[\begin{array}{cc} 0 & s_{34} \\ s_{21} & 0 \end{array} \right] + \frac{1}{1 - s_{12}s_{43}} \left[\begin{array}{cc} s_{33}s_{22} & s_{33}s_{112}s_{44} \\ s_{22}s_{43}s_{11} & s_{22}s_{44} \end{array} \right]$
gilmore-lie-groups_ EQ0618	199	■ 0.749	$\begin{aligned} \left(n_1 + 1, n_2 - 1 \right) & \left J_+ \right n_1, n_2 \rangle = \sqrt{(n_1 + 1)n_2} \\ \left(n_1, n_2 \right) & \left J_- \right n_1 + 1, n_2 - 1 \rangle^* = \sqrt{(n_1 + 1)n_2}^* \end{aligned}$	$\begin{aligned} \langle n_1 + 1, n_2 - 1 J_+ n_1, n_2 \rangle &= \sqrt{(n_1 + 1)n_2} \\ \langle n_1, n_2 J_- n_1 + 1, n_2 - 1 \rangle^* &= \sqrt{(n_1 + 1)n_2}^* \end{aligned}$	$\begin{aligned} \langle n_1 + 1, n_2 - 1 J_+ n_1, n_2 \rangle &= \sqrt{(n_1 + 1)n_2} \\ \langle n_1, n_2 J_- n_1 + 1, n_2 - 1 \rangle^* &= \sqrt{(n_1 + 1)n_2}^* \end{aligned}$

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gilmore-lie-groups_ E08942 (15.72)	291	0.751	$\begin{array}{ll} \operatorname{Re} & +\frac{2}{c} \frac{\partial \mathbf{G}_{\mathbf{e}}}{\partial t}-\nabla \times \mathbf{G}_{\mathbf{m}}=0 \\ \operatorname{Im} & -\frac{2}{c} \frac{\partial \mathbf{G}_{\mathbf{m}}}{\partial t}-\nabla \times \mathbf{G}_{\mathbf{e}}=0 \end{array}$	$\begin{array}{ll} \operatorname{Re} & +\frac{2}{c} \frac{\partial \mathbf{G}_{\mathbf{e}}}{\partial t}-\nabla \times \mathbf{G}_{\mathbf{m}}=0 \\ \operatorname{Im} & -\frac{2}{c} \frac{\partial \mathbf{G}_{\mathbf{m}}}{\partial t}-\nabla \times \mathbf{G}_{\mathbf{e}}=0 \end{array}$	$\begin{array}{ll} \operatorname{Re} & +\frac{2}{c} \frac{\partial \mathbf{G}_{\mathbf{e}}}{\partial t}-\nabla \times \mathbf{G}_{\mathbf{m}}=0 \\ \operatorname{Im} & -\frac{2}{c} \frac{\partial \mathbf{G}_{\mathbf{m}}}{\partial t}-\nabla \times \mathbf{G}_{\mathbf{e}}=0 \end{array}$
gilmore-lie-groups_ E08741	232	0.752	$\begin{array}{l} \lim \epsilon \rightarrow \infty \quad \epsilon J_{\pm} \rightarrow h_{\pm} \quad 2 j \epsilon^2 \rightarrow 1 \\ j \rightarrow +\infty, m \rightarrow -\infty \quad J_3 + \frac{1}{2 \epsilon^2} \rightarrow h_3 \quad \left \begin{array}{c} j \\ m \end{array} \right \rightarrow \\ j+m=n(\text { finite }) \quad \theta \rightarrow \frac{\pi}{2}-\sqrt{2} \epsilon x \end{array}$	$\begin{array}{l} \lim \epsilon \rightarrow \infty \quad \epsilon J_{\pm} \rightarrow h_{\pm} \quad 2 j \epsilon^2 \rightarrow 1 \\ j \rightarrow +\infty, m \rightarrow -\infty \quad J_3 + \frac{1}{2 \epsilon^2} \rightarrow h_3 \quad \left \begin{array}{c} j \\ m \end{array} \right\rangle \rightarrow \left \begin{array}{c} \infty \\ n \end{array} \right\rangle \\ j+m=n(\text { finite }) \quad \theta \rightarrow \frac{\pi}{2}-\sqrt{2} \epsilon x \end{array}$	$\begin{array}{l} \lim \epsilon \rightarrow \infty \quad \epsilon J_{\pm} \rightarrow h_{\pm} \quad 2 j \epsilon^2 \rightarrow 1 \\ j \rightarrow +\infty, m \rightarrow -\infty \quad J_3 + \frac{1}{2 \epsilon^2} \rightarrow h_3 \quad \left \begin{array}{c} j \\ m \end{array} \right\rangle \rightarrow \left \begin{array}{c} \infty \\ n \end{array} \right\rangle \\ j+m=n(\text { finite }) \quad \theta \rightarrow \frac{\pi}{2}-\sqrt{2} \epsilon x \end{array}$
gilmore-lie-groups_ E08884	252	0.753	$\left[\begin{array}{cccc c} 0 & L_3 & -L_2 & M_1 & B_1 \\ -L_3 & 0 & L_1 & M_2 & B_2 \\ L_2 & -L_1 & 0 & M_3 & B_3 \\ -M_1 & -M_2 & -M_3 & 0 & B_4 \\ \hline B_1 & B_2 & B_3 & B_4 & 0 \end{array} \right] \begin{array}{l} + \\ + \\ + \\ + \\ - \end{array}$	$\left[\begin{array}{cccc c} 0 & L_3 & -L_2 & M_1 & B_1 \\ -L_3 & 0 & L_1 & M_2 & B_2 \\ L_2 & -L_1 & 0 & M_3 & B_3 \\ -M_1 & -M_2 & -M_3 & 0 & B_4 \\ \hline B_1 & B_2 & B_3 & B_4 & 0 \end{array} \right] \begin{array}{l} + \\ + \\ + \\ + \\ - \end{array}$	$\left[\begin{array}{cccc c} 0 & L_3 & -L_2 & M_1 & B_1 \\ -L_3 & 0 & L_1 & M_2 & B_2 \\ L_2 & -L_1 & 0 & M_3 & B_3 \\ -M_1 & -M_2 & -M_3 & 0 & B_4 \\ \hline B_1 & B_2 & B_3 & B_4 & 0 \end{array} \right] \begin{array}{l} + \\ + \\ + \\ + \\ - \end{array}$
gilmore-lie-groups_ E08873 (1.68)	34	0.760	$\begin{array}{l} x_1^{\prime}-x_2^{\prime}=x_1-x_2 \\ =\pm 2 i \times 145152 \sqrt{3}=\pm 2 \sqrt{-K_2}=\pm 2 i \sqrt{K_2} \\ =\pm 290304 \sqrt{3} \end{array}$	$\begin{array}{l} x_1^{\prime}-x_2^{\prime}=x_1-x_2 \\ =\pm 2 i \times 145152 \sqrt{3}=\pm 2 \sqrt{-K_2}=\pm 2 i \sqrt{K_2} \\ =\pm 290304 \sqrt{3} \end{array}$	$\begin{array}{l} x_1^{\prime}-x_2^{\prime}=x_1-x_2 \\ =\pm 2 i \times 145152 \sqrt{3}=\pm 2 \sqrt{-K_2}=\pm 2 i \sqrt{K_2} \\ =\pm 290304 \sqrt{3} \end{array}$
gilmore-lie-groups_ E08735	231	0.761	$\begin{array}{l} \lim a \downarrow 0 \quad a J_{\pm} \rightarrow P_{\pm} \quad a^2 l(l+1) \rightarrow p^2 \text { finite } \\ l \uparrow \infty \quad J_3 \rightarrow P_3 \quad \left \begin{array}{c} l \\ m \end{array} \right \rightarrow \left \begin{array}{c} p \\ m \end{array} \right \\ (p / a) \beta=l \beta=x \text { finite } \end{array}$	$\begin{array}{l} \lim a \downarrow 0 \quad a J_{\pm} \rightarrow P_{\pm} \quad a^2 l(l+1) \rightarrow p^2 \text { finite } \\ l \uparrow \infty \quad J_3 \rightarrow P_3 \quad \left \begin{array}{c} l \\ m \end{array} \right \rightarrow \left \begin{array}{c} p \\ m \end{array} \right \\ (p / a) \beta=l \beta=x \text { finite } \end{array}$	$\begin{array}{l} \lim a \downarrow 0 \quad a J_{\pm} \rightarrow P_{\pm} \quad a^2 l(l+1) \rightarrow p^2 \text { finite } \\ l \uparrow \infty \quad J_3 \rightarrow P_3 \quad \left \begin{array}{c} l \\ m \end{array} \right \rightarrow \left \begin{array}{c} p \\ m \end{array} \right \\ (p / a) \beta=l \beta=x \text { finite } \end{array}$
gilmore-lie-groups_ E08397 (7.33)	124	0.762	$\begin{array}{l} e^{r a^{\dagger}+l a+\delta I} \\ \parallel \\ e^{r^{\prime} a^{\dagger}} e^{\delta^{\prime} I} e^{l^{\prime} a}= \\ e^{l^{\prime \prime} a} e^{\delta^{\prime \prime} I} e^{r^{\prime \prime} a^{\dagger}} \end{array}$	$\begin{array}{l} e^{r a^{\dagger}+l a+\delta I} \\ \parallel \\ e^{r^{\prime} a^{\dagger}} e^{\delta^{\prime} I} e^{l^{\prime} a}= \\ e^{l^{\prime \prime} a} e^{\delta^{\prime \prime} I} e^{r^{\prime \prime} a^{\dagger}} \end{array}$	$\begin{array}{l} e^{r a^{\dagger}+l a+\delta I} \\ \parallel \\ e^{r^{\prime} a^{\dagger}} e^{\delta^{\prime} I} e^{l^{\prime} a}= \\ e^{l^{\prime \prime} a} e^{\delta^{\prime \prime} I} e^{r^{\prime \prime} a^{\dagger}} \end{array}$

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gilmore-lie-groups_ E08280 (4.13)	74	0.766	$\begin{aligned} & \operatorname{EXP}(\epsilon X) \operatorname{EXP}(\delta Y) \operatorname{EXP}(-\epsilon X) \operatorname{EXP}(-\delta Y) \\ &= I + \epsilon \delta (XY - YX) = I + \epsilon \delta [X, Y] \end{aligned}$	$\operatorname{EXP}(\epsilon X) \operatorname{EXP}(\delta Y) \operatorname{EXP}(-\epsilon X) \operatorname{EXP}(-\delta Y) = I + \epsilon \delta (XY - YX) = I + \epsilon \delta [X, Y]$	$\begin{aligned} \operatorname{EXP}(\epsilon X) \operatorname{EXP}(\delta Y) \operatorname{EXP}(-\epsilon X) \operatorname{EXP}(-\delta Y) \\ = I + \epsilon \delta (XY - YX) = I + \epsilon \delta [X, Y] \end{aligned}$
gilmore-lie-groups_ E08499 (9.8)	155	0.768	$\begin{aligned} & \operatorname{def}(X) = \frac{1}{2} \begin{bmatrix} ia_3 & i(a_1 - ia_2) \\ i(a_1 + ia_2) & -ia_3 \end{bmatrix} \\ & \operatorname{Reg}(X) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \end{aligned}$	$\begin{aligned} \operatorname{def}(X) &= \frac{1}{2} \begin{bmatrix} ia_3 & i(a_1 - ia_2) \\ i(a_1 + ia_2) & -ia_3 \end{bmatrix} \\ \operatorname{Reg}(X) &= \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \end{aligned}$	$\begin{aligned} \operatorname{def}(X) &= \frac{1}{2} \begin{bmatrix} ia_3 & i(a_1 - ia_2) \\ i(a_1 + ia_2) & -ia_3 \end{bmatrix} \\ \operatorname{Reg}(X) &= \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \end{aligned}$
gilmore-lie-groups_ E0887 (15.20)	279	0.769	$ k\rangle \times \begin{bmatrix} j & j' \\ \mu & \mu' \end{bmatrix}$	$ k\rangle \times \begin{bmatrix} j & j' \\ \mu & \mu' \end{bmatrix}$	$ k\rangle \times \begin{bmatrix} j & j' \\ \mu & \mu' \end{bmatrix}$
gilmore-lie-groups_ E0848 (1.37)	27	0.772	$\begin{aligned} & J_1 = \sum_{i+2j+3k=3} A_{ijk} I_1^i I_2^j I_3^k \\ & J_2 = \sum_{i+2j+3k=6} B_{ijk} I_1^i I_2^j I_3^k \end{aligned}$	$\begin{aligned} J_1 &= \sum_{i+2j+3k=3} A_{ijk} I_1^i I_2^j I_3^k \\ J_2 &= \sum_{i+2j+3k=6} B_{ijk} I_1^i I_2^j I_3^k \end{aligned}$	$\begin{aligned} J_1 &= \sum_{i+2j+3k=3} A_{ijk} I_1^i I_2^j I_3^k \\ J_2 &= \sum_{i+2j+3k=6} B_{ijk} I_1^i I_2^j I_3^k \end{aligned}$
gilmore-lie-groups_ E0841 (6.26)	106	0.772	$\begin{aligned} & \begin{bmatrix} x' & y' \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+da & db \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix} \end{aligned}$	$\begin{bmatrix} x' & y' \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+da & db \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} x' & y' \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+da & db \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}$
gilmore-lie-groups_ E0928 (15.54)	286	0.772	$\begin{aligned} & \left\{ J_3^{(j)} k_3^0 - j k_4^0 I_{2j+1} \right\} \left\langle k^0; \begin{bmatrix} j & 0 \\ m & 0 \end{bmatrix} \middle \psi \right\rangle = 0 \end{aligned}$	$\left\{ J_3^{(j)} k_3^0 - j k_4^0 I_{2j+1} \right\} \left\langle k^0; \begin{bmatrix} j & 0 \\ m & 0 \end{bmatrix} \middle \psi \right\rangle = 0$	$\left\{ J_3^{(j)} k_3^0 - j k_4^0 I_{2j+1} \right\} \left\langle k^0; \begin{bmatrix} j & 0 \\ m & 0 \end{bmatrix} \middle \psi \right\rangle = 0$
gilmore-lie-groups_ E0846 (3.29)	57	0.774	$\begin{aligned} & SO(n) = O(n) \cap SL(n; \mathbb{R}) \\ & SO(p, q) = O(p, q) \cap SL(p+q; \mathbb{R}) \\ & SU(n) = U(n) \cap SL(n; \mathbb{C}) \\ & SU(p, q) = U(p, q) \cap SL(p+q; \mathbb{C}) \end{aligned}$	$\begin{aligned} SO(n) &= O(n) \cap SL(n; \mathbb{R}) \\ SO(p, q) &= O(p, q) \cap SL(p+q; \mathbb{R}) \\ SU(n) &= U(n) \cap SL(n; \mathbb{C}) \\ SU(p, q) &= U(p, q) \cap SL(p+q; \mathbb{C}) \end{aligned}$	$\begin{aligned} SO(n) &= O(n) \cap SL(n; \mathbb{R}) \\ SO(p, q) &= O(p, q) \cap SL(p+q; \mathbb{R}) \\ SU(n) &= U(n) \cap SL(n; \mathbb{C}) \\ SU(p, q) &= U(p, q) \cap SL(p+q; \mathbb{C}) \end{aligned}$

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gilmore-lie-groups_ EQ0725 (13.42)	228	0.776	$\begin{aligned} M=-J+n\rangle &= \frac{(J_+)^n}{[(2J)!n!/(2J-n)!]^{1/2}} \begin{vmatrix} J \\ -J \end{vmatrix} \\ &= \lim_{J \rightarrow \infty} \frac{(cJ_+)^n}{[(2Jc^2)^n n!]^{1/2}} \begin{vmatrix} \infty \\ 0 \end{vmatrix} \\ &= \frac{(a^\dagger)^n}{\sqrt{n!}} \begin{vmatrix} \infty \\ 0 \end{vmatrix} \end{aligned}$	$ M=-J+n\rangle = \frac{(J_+)^n}{[(2J)!n!/(2J-n)!]^{1/2}} \begin{vmatrix} J \\ -J \end{vmatrix}$ $\begin{vmatrix} \infty \\ n \end{vmatrix} = \lim_{J \rightarrow \infty} \frac{(cJ_+)^n}{[(2Jc^2)^n n!]^{1/2}} \begin{vmatrix} \infty \\ 0 \end{vmatrix} = \frac{(a^\dagger)^n}{\sqrt{n!}} \begin{vmatrix} \infty \\ 0 \end{vmatrix}$	$\begin{vmatrix} J \\ M=-J+n \end{vmatrix} = \frac{(J_+)^n}{[(2J)!n!/(2J-n)!]^{1/2}} \begin{vmatrix} J \\ -J \end{vmatrix}$ $\begin{vmatrix} \infty \\ n \end{vmatrix} = \lim_{J \rightarrow \infty} \frac{(cJ_+)^n}{[(2Jc^2)^n n!]^{1/2}} \begin{vmatrix} \infty \\ 0 \end{vmatrix} = \frac{(a^\dagger)^n}{\sqrt{n!}} \begin{vmatrix} \infty \\ 0 \end{vmatrix}$
gilmore-lie-groups_ EQ0118 (3.6)	50	0.791	$\begin{array}{l} m_{i\alpha}=0 \quad p+1 \leq i \leq p+q \\ \quad \quad \quad 1 \leq \alpha \leq p \end{array}$	$m_{i\alpha} = 0 \quad \begin{array}{l} p+1 \leq i \leq p+q \\ 1 \leq \alpha \leq p \end{array}$	$m_{i\alpha} = 0 \quad \begin{array}{l} p+1 \leq i \leq p+q \\ 1 \leq \alpha \leq p \end{array}$
gilmore-lie-groups_ EQ0565	176	0.791	$\begin{aligned} F_4 &\pm \mathbf{e}_i \pm \mathbf{e}_j \\ &\pm 2\mathbf{e}_i \\ &\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \\ 1 \leq i \neq j \leq 4 \end{aligned}$	$F_4 \pm \mathbf{e}_i \pm \mathbf{e}_j \pm 2\mathbf{e}_i \pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \quad 1 \leq i \neq j \leq 4$	$F_4 \pm \mathbf{e}_i \pm \mathbf{e}_j \pm 2\mathbf{e}_i \pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \quad 1 \leq i \neq j \leq 4$
gilmore-lie-groups_ EQ0050 (1.46)	29	0.796	$\begin{aligned} &\left(z^{\{\prime\}}-t_1\right)\left(z^{\{\prime\}}-t_2\right)\left(z^{\{\prime\}}-t_3\right)\left(z^{\{\prime\}}-t_4\right)=z^{\prime 4}-I_1^{\prime} z^{\prime 3}+I_2^{\prime} z^{\prime 2}-I_3^{\prime} z^{\prime}+I_4^{\prime}=0 \\ &I_1^{\prime}=0 \\ &I_2^{\prime}=I_2-\frac{3}{8} I_1^2 \\ &I_3^{\prime}=I_3-\frac{1}{2} I_2 I_1+\frac{1}{8} I_3^3 \\ &I_4^{\prime}=I_4-\frac{1}{4} I_3 I_1+\frac{1}{16} I_2 I_1^2-\frac{3}{4^4} I_1^4 \end{aligned}$	$\begin{aligned} (z'-t_1)(z'-t_2)(z'-t_3)(z'-t_4) &= z'^4 - I_1' z'^3 + I_2' z'^2 - I_3' z' + I_4' = 0 \\ I_1' &= 0 \\ I_2' &= I_2 - \frac{3}{8} I_1^2 \\ I_3' &= I_3 - \frac{1}{2} I_2 I_1 + \frac{1}{8} I_3^3 \\ I_4' &= I_4 - \frac{1}{4} I_3 I_1 + \frac{1}{16} I_2 I_1^2 - \frac{3}{4^4} I_1^4 \end{aligned}$	$\begin{aligned} (z'-t_1)(z'-t_2)(z'-t_3)(z'-t_4) &= z'^4 - I_1' z'^3 + I_2' z'^2 - I_3' z' + I_4' = 0 \\ I_1' &= 0 \\ I_2' &= I_2 - \frac{3}{8} I_1^2 \\ I_3' &= I_3 - \frac{1}{2} I_2 I_1 + \frac{1}{8} I_3^3 \\ I_4' &= I_4 - \frac{1}{4} I_3 I_1 + \frac{1}{16} I_2 I_1^2 - \frac{3}{4^4} I_1^4 \end{aligned}$
gilmore-lie-groups_ EQ0131 (3.17)	54	0.798	$S U(1 ; \mathbb{Q}) \sim S^3 \subset R^4$	$SU(1;\mathbb{Q}) \sim S^3 \subset R^4$	$SU(1;\mathbb{Q}) \sim S^3 \subset R^4$
gilmore-lie-groups_ EQ0695 (13.13)	222	0.798	$\begin{aligned} L_i &= \epsilon_{ijk} x_j \partial_k \\ V_i &= x_i \partial_4 - x_4 \partial_i \quad 1 \leq i \neq j \neq k \leq 3 \end{aligned}$	$L_i = \epsilon_{ijk} x_j \partial_k$ $V_i = x_i \partial_4 - x_4 \partial_i \quad 1 \leq i \neq j \neq k \leq 3$	$L_i = \epsilon_{ijk} x_j \partial_k$ $V_i = x_i \partial_4 - x_4 \partial_i \quad 1 \leq i \neq j \neq k \leq 3$
gilmore-lie-groups_ EQ0363 (7.1)	113	0.799	$\text{Lie groups} \stackrel{\text{ln}}{\rightleftharpoons} \text{Lie algebras}$	$\text{Lie groups} \stackrel{\text{ln}}{\rightleftharpoons} \text{Lie algebras}$	$\text{Lie groups} \stackrel{\text{ln}}{\rightleftharpoons} \text{Lie algebras}$

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gilmore-lie-groups_ E0655 (12.24)	212	0.802	$\begin{aligned} \mathfrak{g}^{\prime} &= \left(\mathfrak{g}_{+,+} + i \mathfrak{g}_{+,-} \right) + \left(i \mathfrak{g}_{-,+} + \mathfrak{g}_{-,-} \right) \\ &= \mathfrak{h}^{\prime} + \mathfrak{p}^{\prime} \end{aligned}$	$\mathfrak{g}^{\prime} = (\mathfrak{g}_{+,+} + i\mathfrak{g}_{+,-}) + (i\mathfrak{g}_{-,+} + \mathfrak{g}_{-,-})$ $= \mathfrak{h}^{\prime} + \mathfrak{p}^{\prime}$	$\mathfrak{g}^{\prime} = (\mathfrak{g}_{+,+} + i\mathfrak{g}_{+,-}) + (i\mathfrak{g}_{-,+} + \mathfrak{g}_{-,-})$ $= \mathfrak{h}^{\prime} + \mathfrak{p}^{\prime}$
gilmore-lie-groups_ E0300	98	0.803	$\begin{aligned} Z &= \left[\begin{array}{c c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c c} A & 0 \\ \hline 0 & D \end{array} \right] + \left[\begin{array}{c c} 0 & B \\ \hline C & 0 \end{array} \right] \\ &= \mathfrak{h} + \mathfrak{p} \end{aligned}$	$Z = \left[\begin{array}{c c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c c} A & 0 \\ \hline 0 & D \end{array} \right] + \left[\begin{array}{c c} 0 & B \\ \hline C & 0 \end{array} \right]$ $= \mathfrak{h} + \mathfrak{p}$	$Z = \left[\begin{array}{c c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c c} A & 0 \\ \hline 0 & D \end{array} \right] + \left[\begin{array}{c c} 0 & B \\ \hline C & 0 \end{array} \right]$ $\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$
gilmore-lie-groups_ E0299	97	0.803	$R(A) = \left[\begin{array}{cc c} 0 & 0 & -b \\ 0 & 0 & +b \\ \hline 0 & 0 & a-c \end{array} \right] \begin{array}{l} X_a \\ X_c \\ X_b \end{array}$	$R(A) = \left[\begin{array}{cc c} 0 & 0 & -b \\ 0 & 0 & +b \\ \hline 0 & 0 & a-c \end{array} \right] \begin{array}{l} X_a \\ X_c \\ X_b \end{array}$	$R(A) = \left[\begin{array}{cc c} 0 & 0 & -b \\ 0 & 0 & +b \\ \hline 0 & 0 & a-c \end{array} \right] \begin{array}{l} X_a \\ X_c \\ X_b \end{array}$
gilmore-lie-groups_ E0346	107	0.803	$\begin{aligned} &\begin{array}{cccc} a_1^{\dagger} a_1 + a_2^{\dagger} a_2 & a_1^{\dagger} a_1 - a_2^{\dagger} a_2 & a_1^{\dagger} a_2 & a_2^{\dagger} a_1 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] & \left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right] & \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] & \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right] \end{array} \\ &\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \quad \left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right] \quad \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] \quad \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right] \end{aligned}$	$\begin{aligned} &\begin{array}{cccc} a_1^{\dagger} a_1 + a_2^{\dagger} a_2 & a_1^{\dagger} a_1 - a_2^{\dagger} a_2 & a_1^{\dagger} a_2 & a_2^{\dagger} a_1 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] & \left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right] & \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] & \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right] \end{array} \\ &\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \quad \left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right] \quad \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] \quad \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right] \end{aligned}$	$\begin{aligned} &\begin{array}{cccc} a_1^{\dagger} a_1 + a_2^{\dagger} a_2 & a_1^{\dagger} a_1 - a_2^{\dagger} a_2 & a_1^{\dagger} a_2 & a_2^{\dagger} a_1 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] & \left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right] & \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] & \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right] \end{array} \\ &\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ &\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \quad \left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right] \quad \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] \quad \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right] \end{aligned}$
gilmore-lie-groups_ E0502 (9.11)	156	0.812	$\begin{aligned} &\left[\begin{array}{c} \operatorname{Reg}(Y) - \lambda I_3 \\ \left(-b_1^2 - b_2^2 + b_3^2 \right) \\ \left(-b_1^2 - b_2^2 + b_3^2 \right) \end{array} \right] = 0 \\ &\left(-b_1^2 - b_2^2 + b_3^2 \right) = 0 \\ &\left(-b_1^2 - b_2^2 + b_3^2 \right) = 0 \end{aligned}$	$\begin{aligned} \ \operatorname{Reg}(Y) - \lambda I_3\ &= (-\lambda)^3 + (-\lambda)(-b_1^2 - b_2^2 + b_3^2) = 0 \\ &= (-\lambda)(\lambda^2 + \phi_2(\mathbf{b})) \\ \phi_2(\mathbf{b}) &= -b_1^2 - b_2^2 + b_3^2 \end{aligned}$	$\begin{aligned} \ \operatorname{Reg}(Y) - \lambda I_3\ &= (-\lambda)^3 + (-\lambda)(-b_1^2 - b_2^2 + b_3^2) = 0 \\ &= (-\lambda)(\lambda^2 + \phi_2(\mathbf{b})) \\ \phi_2(\mathbf{b}) &= -b_1^2 - b_2^2 + b_3^2 \end{aligned}$
gilmore-lie-groups_ E0266 (5.4)	89	0.815	$\begin{aligned} &m_{ij} = 0 \quad p+1 \leq i \leq p+q \\ &\quad \quad \quad 1 \leq j \leq p+q \end{aligned}$	$m_{ij} = 0 \quad p+1 \leq i \leq p+q$ $\quad \quad \quad 1 \leq j \leq p+q$	$m_{ij} = 0 \quad p+1 \leq i \leq p+q$ $\quad \quad \quad 1 \leq j \leq p+q$
gilmore-lie-groups_ E0388 (7.24)	123	0.817	$(x, y) \rightarrow \left[\begin{array}{cc} x & y \\ 0 & 1 \end{array} \right] \quad x > 0$	$(x, y) \rightarrow \left[\begin{array}{cc} x & y \\ 0 & 1 \end{array} \right] \quad x > 0$	$(x, y) \rightarrow \left[\begin{array}{cc} x & y \\ 0 & 1 \end{array} \right] \quad x > 0$
gilmore-lie-groups_ E0581 (11.1)	186	0.817	$\sum_{i=1}^{\operatorname{rank}} h^i H_i + \sum_{\alpha \neq 0} e^{\alpha} E_{\alpha}$	$\sum_{i=1}^{\operatorname{rank}} h^i H_i + \sum_{\alpha \neq 0} e^{\alpha} E_{\alpha}$	$\sum_{i=1}^{\operatorname{rank}} h^i H_i + \sum_{\alpha \neq 0} e^{\alpha} E_{\alpha}$

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gilmore-lie-groups_ E00022 (1.18)	24	■ 0.819	$\begin{array}{l rrc} & I & (12) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & u_1 = r_1 + r_2 \\ \Gamma^2 & 1 & -1 & u_2 = r_1 - r_2 \end{array}$	$\begin{array}{l rrc} & I & (12) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & u_1 = r_1 + r_2 \\ \Gamma^2 & 1 & -1 & u_2 = r_1 - r_2 \end{array}$	$\begin{array}{l rrc} & I & (12) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & u_1 = r_1 + r_2 \\ \Gamma^2 & 1 & -1 & u_2 = r_1 - r_2 \end{array}$
gilmore-lie-groups_ E00208 (4.18)	75	■ 0.820	$\begin{aligned} & \left\{ \left[X_a, X_b \right] \right\} = 2 X_b \\ & \left\{ \left[X_a, X_c \right] \right\} = -2 X_c \\ & \left\{ \left[X_b, X_c \right] \right\} = X_a \end{aligned}$	$\begin{aligned} [X_a, X_b] &= 2X_b \\ [X_a, X_c] &= -2X_c \\ [X_b, X_c] &= X_a \end{aligned}$	$\begin{aligned} [X_a, X_b] &= 2X_b \\ [X_a, X_c] &= -2X_c \\ [X_b, X_c] &= X_a \end{aligned}$
gilmore-lie-groups_ E00334 (6.19)	105	■ 0.820	$\begin{gathered} f(p) \\\downarrow \\ f_S[x(p)] \stackrel{\perp}{=} f_{S'}[x'(p)] \end{gathered}$	$\begin{gathered} f(p) \\\downarrow \\ f_S[x(p)] \stackrel{\perp}{=} f_{S'}[x'(p)] \end{gathered}$	$\begin{gathered} f(p) \\\downarrow \quad \downarrow \\ f_S[x(p)] = f_{S'}[x'(p)] \end{gathered}$
gilmore-lie-groups_ E00660	214	■ 0.824	$\begin{aligned} & ds^2 = \frac{(dr \, d\theta)}{1+r^2} \begin{bmatrix} \frac{1}{1+r^2} & 0 \\ 0 & r^2 \end{bmatrix} \begin{pmatrix} dr \\ d\theta \end{pmatrix} \\ & d\mu = \frac{r \, dr \, d\theta}{\sqrt{1+r^2}} \end{aligned}$	$\begin{aligned} ds^2 &= \frac{(dr d\theta) \begin{bmatrix} \frac{1}{1+r^2} & 0 \\ 0 & r^2 \end{bmatrix} \begin{pmatrix} dr \\ d\theta \end{pmatrix}}{1+r^2} \\ d\mu &= \frac{r dr d\theta}{\sqrt{1+r^2}} \end{aligned}$	$\begin{aligned} ds^2 &= \frac{(dr \, d\theta) \begin{bmatrix} \frac{1}{1+r^2} & 0 \\ 0 & r^2 \end{bmatrix} \begin{pmatrix} dr \\ d\theta \end{pmatrix}}{1+r^2} \\ d\mu &= \frac{r \, dr \, d\theta}{\sqrt{1+r^2}} \end{aligned}$
gilmore-lie-groups_ E00257 (4.61a)	86	■ 0.825	$\left[\begin{array}{ll} \alpha_R + i \alpha_I & \beta_R + i \beta_I \\ \beta_R - i \beta_I & \alpha_R - i \alpha_I \end{array} \right]$	$\left[\begin{array}{ll} \alpha_R + i \alpha_I & \beta_R + i \beta_I \\ \beta_R - i \beta_I & \alpha_R - i \alpha_I \end{array} \right]$	$\left[\begin{array}{ll} \alpha_R + i \alpha_I & \beta_R + i \beta_I \\ \beta_R - i \beta_I & \alpha_R - i \alpha_I \end{array} \right]$
gilmore-lie-groups_ E00891 (15.24)	279	■ 0.826	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right $	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right $	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right $

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gilmore-lie-groups_ EQ0930 (15.65)	288	■ 0.826	$\begin{aligned} & \left[\frac{\partial}{\partial t} \frac{\partial}{\partial x} + \frac{\partial}{\partial y} \frac{\partial}{\partial x} + \frac{\partial}{\partial z} \frac{\partial}{\partial x} \right] \begin{bmatrix} B_1 + iE_1 \\ B_2 + iE_2 \\ B_3 + iE_3 \end{bmatrix} = 0 \\ & -\frac{i}{c} \frac{\partial}{\partial t} (B + iE)_1 + \partial_3 (B + iE)_2 - \partial_2 (B + iE)_3 = 0 \\ & -\partial_3 (B + iE)_1 - \frac{i}{c} \frac{\partial}{\partial t} (B + iE)_2 + \partial_1 (B + iE)_3 = 0 \\ & +\partial_2 (B + iE)_1 - \partial_1 (B + iE)_2 - \frac{i}{c} \frac{\partial}{\partial t} (B + iE)_3 = 0 \end{aligned}$	$\begin{bmatrix} -\frac{i}{c} \frac{\partial}{\partial t} & +\partial_3 & -\partial_2 \\ -\partial_3 & -\frac{i}{c} \frac{\partial}{\partial t} & +\partial_1 \\ +\partial_2 & -\partial_1 & -\frac{i}{c} \frac{\partial}{\partial t} \end{bmatrix} \begin{bmatrix} B_1 + iE_1 \\ B_2 + iE_2 \\ B_3 + iE_3 \end{bmatrix} = 0$ $-\frac{i}{c} \frac{\partial}{\partial t} (B + iE)_1 + \partial_3 (B + iE)_2 - \partial_2 (B + iE)_3 = 0$ $-\partial_3 (B + iE)_1 - \frac{i}{c} \frac{\partial}{\partial t} (B + iE)_2 + \partial_1 (B + iE)_3 = 0$ $+\partial_2 (B + iE)_1 - \partial_1 (B + iE)_2 - \frac{i}{c} \frac{\partial}{\partial t} (B + iE)_3 = 0$	$\begin{bmatrix} -\frac{i}{c} \frac{\partial}{\partial t} & +\partial_3 & -\partial_2 \\ -\partial_3 & -\frac{i}{c} \frac{\partial}{\partial t} & +\partial_1 \\ +\partial_2 & -\partial_1 & -\frac{i}{c} \frac{\partial}{\partial t} \end{bmatrix} \begin{bmatrix} B_1 + iE_1 \\ B_2 + iE_2 \\ B_3 + iE_3 \end{bmatrix} = 0$ $-\frac{i}{c} \frac{\partial}{\partial t} (B + iE)_1 + \partial_3 (B + iE)_2 - \partial_2 (B + iE)_3 = 0$ $-\partial_3 (B + iE)_1 - \frac{i}{c} \frac{\partial}{\partial t} (B + iE)_2 + \partial_1 (B + iE)_3 = 0$ $+\partial_2 (B + iE)_1 - \partial_1 (B + iE)_2 - \frac{i}{c} \frac{\partial}{\partial t} (B + iE)_3 = 0$
gilmore-lie-groups_ EQ0993 (16.36)	309	■ 0.831	$\operatorname{EXP} \left(\lambda \left(x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} - 2p \frac{\partial}{\partial p} \right) \right) \begin{bmatrix} x \\ y \\ p \end{bmatrix} = \begin{bmatrix} e^{\lambda x} x \\ e^{-\lambda y} y \\ e^{-2\lambda p} p \end{bmatrix}$	$\operatorname{EXP} \left(\lambda \left\{ x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} - 2p \frac{\partial}{\partial p} \right\} \right) \begin{bmatrix} x \\ y \\ p \end{bmatrix} = \begin{bmatrix} e^{\lambda x} x \\ e^{-\lambda y} y \\ e^{-2\lambda p} p \end{bmatrix}$	$\operatorname{EXP} \left(\lambda \left\{ x \frac{\partial}{\partial x} - y \frac{\partial}{\partial y} - 2p \frac{\partial}{\partial p} \right\} \right) \begin{bmatrix} x \\ y \\ p \end{bmatrix} = \begin{bmatrix} e^{\lambda x} x \\ e^{-\lambda y} y \\ e^{-2\lambda p} p \end{bmatrix}$
gilmore-lie-groups_ EQ0576 (10.18)	182	■ 0.837	$\begin{aligned} & p \text{ arbitrary, } q=1, B_l, C_l \quad l = p+1 \\ & p=2, q=2, F_4 \end{aligned}$	$p \text{ arbitrary, } q=1, B_l, C_l \quad l = p+1$ $p=2, q=2, F_4$	$p \text{ arbitrary, } q=1, B_l, C_l \quad l = p+1$ $p=2, q=2, F_4$
gilmore-lie-groups_ EQ0277 (5.16)	92	■ 0.839	$\begin{aligned} & \left[\begin{array}{cc} A^t & C^t \\ B^t & D \end{array} \right] \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] + \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] \left[\begin{array}{cc} A & B \\ C & D \end{array} \right] = 0 \\ & \left[\begin{array}{cc} A^t & C^t \\ B^t & D \end{array} \right] \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] + \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] \left[\begin{array}{cc} A & B \\ C & D \end{array} \right] = 0 \end{aligned}$	$\left[\begin{array}{cc} A^t & C^t \\ B^t & D \end{array} \right] \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] + \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] \left[\begin{array}{cc} A & B \\ C & D \end{array} \right] = 0$	$\left[\begin{array}{cc} A^t & C^t \\ B^t & D \end{array} \right] \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] + \left[\begin{array}{cc} I_2 & 0 \\ 0 & -1 \end{array} \right] \left[\begin{array}{cc} A & B \\ C & D \end{array} \right] = 0$
gilmore-lie-groups_ EQ0629 (12.2)	205	■ 0.847	$\begin{aligned} & G/H = \operatorname{EXP}(\mathfrak{p}) \quad ds^2 = g_{\mu, \nu} dx^\mu dx^\nu < 0 \\ & G'/H = \operatorname{EXP}(i\mathfrak{p}) \quad ds^2 = g_{\mu, \nu} dx^\mu dx^\nu > 0 \end{aligned}$	$G/H = \operatorname{EXP}(\mathfrak{p}) \quad ds^2 = g_{\mu, \nu} dx^\mu dx^\nu < 0$ $G'/H = \operatorname{EXP}(i\mathfrak{p}) \quad ds^2 = g_{\mu, \nu} dx^\mu dx^\nu > 0$	$G/H = \operatorname{EXP}(\mathfrak{p}) \quad ds^2 = g_{\mu, \nu} dx^\mu dx^\nu < 0$ $G'/H = \operatorname{EXP}(i\mathfrak{p}) \quad ds^2 = g_{\mu, \nu} dx^\mu dx^\nu > 0$
gilmore-lie-groups_ EQ0217 (4.27)	76	■ 0.848	$Z \stackrel{\text{regular}}{\longrightarrow} R(Z)$	$Z \stackrel{\text{regular}}{\longrightarrow} R(Z)$	$Z \stackrel{\text{regular}}{\longrightarrow} R(Z)$
gilmore-lie-groups_ EQ0543	169	■ 0.848	$\mathcal{C}^2(\mathbf{M}) = (\mathbf{M} + \mathbf{R}) \cdot (\mathbf{M} + \mathbf{R}) - (\mathbf{R}) \cdot (\mathbf{R}) = \mathbf{M} \cdot \mathbf{M} + \mathbf{M} \cdot 2\mathbf{R}$	$\mathcal{C}^2(\mathbf{M}) = (\mathbf{M} + \mathbf{R}) \cdot (\mathbf{M} + \mathbf{R}) - (\mathbf{R}) \cdot (\mathbf{R}) = \mathbf{M} \cdot \mathbf{M} + \mathbf{M} \cdot 2\mathbf{R}$	$\mathcal{C}^2(\mathbf{M}) = (\mathbf{M} + \mathbf{R}) \cdot (\mathbf{M} + \mathbf{R}) - (\mathbf{R}) \cdot (\mathbf{R}) = \mathbf{M} \cdot \mathbf{M} + \mathbf{M} \cdot 2\mathbf{R}$

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gilmore-lie-groups_ EQ0645 (12.14)	210	■ 0.861	$\begin{aligned} & \Gamma_{\mu\nu}^{\sigma} \rightarrow \frac{1}{2} \left(\frac{\partial g_{\mu\sigma}}{\partial x^{\nu}} + \frac{\partial g_{\nu\sigma}}{\partial x^{\mu}} - \frac{\partial g_{\mu\nu}}{\partial x^{\sigma}} \right) \\ & = \frac{1}{2} \left\{ \begin{aligned} & \delta_{\nu\mu} x^{\sigma} + \delta_{\mu\nu} x^{\sigma} - \delta_{\sigma\mu} x^{\nu} \\ & \delta_{\nu\sigma} x^{\mu} + \delta_{\mu\sigma} x^{\nu} - \delta_{\sigma\nu} x^{\mu} \end{aligned} \right\} \\ & \rightarrow \delta_{\mu\nu} x^{\sigma} \quad (\rightarrow 0 \text{ at origin}) \end{aligned}$	$\Gamma_{\mu\nu}^{\sigma} \rightarrow \frac{1}{2} \left(\frac{\partial g_{\mu\sigma}}{\partial x^{\nu}} + \frac{\partial g_{\nu\sigma}}{\partial x^{\mu}} - \frac{\partial g_{\mu\nu}}{\partial x^{\sigma}} \right)$ $= \frac{1}{2} \left\{ \begin{aligned} & \delta_{\nu\mu} x^{\sigma} + \delta_{\mu\nu} x^{\sigma} - \delta_{\sigma\mu} x^{\nu} \\ & \delta_{\nu\sigma} x^{\mu} + \delta_{\mu\sigma} x^{\nu} - \delta_{\sigma\nu} x^{\mu} \end{aligned} \right\}$ $\rightarrow \delta_{\mu\nu} x^{\sigma} \quad (\rightarrow 0 \text{ at origin})$	$\Gamma_{\mu\nu}^{\sigma} \rightarrow \frac{1}{2} \left(\frac{\partial g_{\mu\sigma}}{\partial x^{\nu}} + \frac{\partial g_{\nu\sigma}}{\partial x^{\mu}} - \frac{\partial g_{\mu\nu}}{\partial x^{\sigma}} \right)$ $= \frac{1}{2} \left\{ \begin{aligned} & \delta_{\nu\mu} x^{\sigma} + \delta_{\mu\nu} x^{\sigma} - \delta_{\sigma\mu} x^{\nu} \\ & \delta_{\nu\sigma} x^{\mu} + \delta_{\mu\sigma} x^{\nu} - \delta_{\sigma\nu} x^{\mu} \end{aligned} \right\}$ $\rightarrow \delta_{\mu\nu} x^{\sigma} \quad (\rightarrow 0 \text{ at origin})$
gilmore-lie-groups_ EQ0125	53	■ 0.867	$\begin{array}{ll} A_{i,j} = \delta_{i,j} & 1 \leq i, j \leq p \\ A_{\alpha,\beta} = \delta_{\alpha,\beta} & p+1 \leq \alpha, \beta \leq p+q \\ A_{\alpha,j} = 0 \\ A_{i,\beta} = \text{arbitrary} \end{array}$	$\begin{array}{ll} A_{i,j} = \delta_{i,j} & 1 \leq i, j \leq p \\ A_{\alpha,\beta} = \delta_{\alpha,\beta} & p+1 \leq \alpha, \beta \leq p+q \\ A_{\alpha,j} = 0 \\ A_{i,\beta} = \text{arbitrary} \end{array}$	$\begin{array}{ll} A_{i,j} = \delta_{i,j} & 1 \leq i, j \leq p \\ A_{\alpha,\beta} = \delta_{\alpha,\beta} & p+1 \leq \alpha, \beta \leq p+q \\ A_{\alpha,j} = 0 \\ A_{i,\beta} = \text{arbitrary} \end{array}$
gilmore-lie-groups_ EQ0290 (5.29)	94	■ 0.869	$\mathfrak{su}(2) = \mathfrak{u}(2) \cap \mathfrak{sl}(2; \mathbb{C}) = \frac{i}{2} \left[\begin{array}{cc} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{array} \right]$	$\mathfrak{su}(2) = \mathfrak{u}(2) \cap \mathfrak{sl}(2; \mathbb{C}) = \frac{i}{2} \left[\begin{array}{cc} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{array} \right]$	$\mathfrak{su}(2) = \mathfrak{u}(2) \cap \mathfrak{sl}(2; \mathbb{C}) = \frac{i}{2} \left[\begin{array}{cc} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{array} \right]$
gilmore-lie-groups_ EQ0197 (4.10)	73	■ 0.872	$\text{Lie group} \stackrel{\ln}{\rightleftharpoons} \text{Lie algebra}$	$\text{Lie group} \stackrel{\ln}{\rightleftharpoons} \text{Lie algebra}$	$\text{Lie group} \stackrel{\ln}{\rightleftharpoons} \text{Lie algebra}$
gilmore-lie-groups_ EQ0790 (14.32)	247	■ 0.874	$\begin{array}{l} \mathbf{L} = \mathbf{r} \times \mathbf{p} \quad \mathbf{L} \cdot \mathbf{L} = L^2 \\ \mathbf{M} = \frac{\mathbf{p} \times \mathbf{L}}{m} - K \frac{\mathbf{r}}{r} \quad \mathbf{M} \cdot \mathbf{M} = M^2 = \frac{2E}{m} L^2 + K^2 \\ \mathbf{W} = \frac{\mathbf{p}}{m} L^2 - K \frac{\mathbf{L} \times \mathbf{r}}{r} \quad \mathbf{W} \cdot \mathbf{W} = L^2 M^2 \end{array}$	$\begin{array}{l} \mathbf{L} = \mathbf{r} \times \mathbf{p} \quad \mathbf{L} \cdot \mathbf{L} = L^2 \\ \mathbf{M} = \frac{\mathbf{p} \times \mathbf{L}}{m} - K \frac{\mathbf{r}}{r} \quad \mathbf{M} \cdot \mathbf{M} = M^2 = \frac{2E}{m} L^2 + K^2 \\ \mathbf{W} = \frac{\mathbf{p}}{m} L^2 - K \frac{\mathbf{L} \times \mathbf{r}}{r} \quad \mathbf{W} \cdot \mathbf{W} = L^2 M^2 \end{array}$	$\begin{array}{l} \mathbf{L} = \mathbf{r} \times \mathbf{p} \quad \mathbf{L} \cdot \mathbf{L} = L^2 \\ \mathbf{M} = \frac{\mathbf{p} \times \mathbf{L}}{m} - K \frac{\mathbf{r}}{r} \quad \mathbf{M} \cdot \mathbf{M} = M^2 = \frac{2E}{m} L^2 + K^2 \\ \mathbf{W} = \frac{\mathbf{p}}{m} L^2 - K \frac{\mathbf{L} \times \mathbf{r}}{r} \quad \mathbf{W} \cdot \mathbf{W} = L^2 M^2 \end{array}$
gilmore-lie-groups_ EQ0120 (3.8)	51	■ 0.874	$m_{ij} - \delta_{ij} = 0 \quad p+1 \leq i \leq p+q \quad p+1 \leq j \leq p+q$	$m_{ij} - \delta_{ij} = 0 \quad p+1 \leq i \leq p+q \quad p+1 \leq j \leq p+q$	$m_{ij} - \delta_{ij} = 0 \quad p+1 \leq i \leq p+q \quad p+1 \leq j \leq p+q$
gilmore-lie-groups_ EQ0237 (4.47)	81	■ 0.874	$g(x) = M^{-1}(x)^t g(\text{Id}) M^{-1}(x)$	$g(x) = M^{-1}(x)^t g(\text{Id}) M^{-1}(x)$	$g(x) = M^{-1}(x)^t g(\text{Id}) M^{-1}(x)$
gilmore-lie-groups_ EQ0389 (7.25)	123	■ 0.874	$(w, z) = \text{EXP} \left[\begin{array}{cc} w & z \\ 0 & 1 \end{array} \right] = \left[\begin{array}{cc} e^w & (e^w - 1)z/w \\ 0 & 1 \end{array} \right]$	$(w, z) = \text{EXP} \left[\begin{array}{cc} w & z \\ 0 & 1 \end{array} \right] = \left[\begin{array}{cc} e^w & (e^w - 1)z/w \\ 0 & 1 \end{array} \right]$	$(w, z) = \text{EXP} \left[\begin{array}{cc} w & z \\ 0 & 1 \end{array} \right] = \left[\begin{array}{cc} e^w & (e^w - 1)z/w \\ 0 & 1 \end{array} \right]$

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gilmore-lie-groups_ EQ0973 (16.16)	303	■ 0.874	$\begin{array}{r} \left(\xi(x,y)\frac{\partial}{\partial x}+\eta(x,y)\frac{\partial}{\partial y}\right)R(x,y)=0 \\ \left(\xi(x,y)\frac{\partial}{\partial x}+\eta(x,y)\frac{\partial}{\partial y}+\zeta(x,y)\frac{\partial}{\partial p}\right)T(x,y,p)=0 \end{array}$	$\begin{array}{r} \left(\xi(x,y)\frac{\partial}{\partial x}+\eta(x,y)\frac{\partial}{\partial y}\right)R(x,y)=0 \\ \left(\xi(x,y)\frac{\partial}{\partial x}+\eta(x,y)\frac{\partial}{\partial y}+\zeta(x,y)\frac{\partial}{\partial p}\right)T(x,y,p)=0 \end{array}$	$\begin{array}{r} \left(\xi(x,y)\frac{\partial}{\partial x}+\eta(x,y)\frac{\partial}{\partial y}\right)R(x,y)=0 \\ \left(\xi(x,y)\frac{\partial}{\partial x}+\eta(x,y)\frac{\partial}{\partial y}+\zeta(x,y)\frac{\partial}{\partial p}\right)T(x,y,p)=0 \end{array}$
gilmore-lie-groups_ EQ0471 (8.2)	143	■ 0.876	$S(Z)_r^s=\left(A^{-1}\right)_r^iR(Z)_i^jA_j^s$	$S(Z)_r^s=\left(A^{-1}\right)_r^iR(Z)_i^jA_j^s$	$S(Z)_r^s=\left(A^{-1}\right)_r^iR(Z)_i^jA_j^s$
gilmore-lie-groups_ EQ0869 (15.2)	275	■ 0.878	$\left[\begin{array}{c} x \\ y \\ z \\ ict \end{array}\right]=\left[\begin{array}{c} \Lambda \end{array}\right]\left[\begin{array}{c} x \\ y \\ z \\ ict \end{array}\right]'$	$\left[\begin{array}{c} x \\ y \\ z \\ ict \end{array}\right]=\left[\begin{array}{c} \Lambda \end{array}\right]\left[\begin{array}{c} x \\ y \\ z \\ ict \end{array}\right]'$	$\left[\begin{array}{c} x \\ y \\ z \\ ict \end{array}\right]=\left[\begin{array}{c} \Lambda \end{array}\right]\left[\begin{array}{c} x \\ y \\ z \\ ict \end{array}\right]'$
gilmore-lie-groups_ EQ0385 (7.22)	121	■ 0.887	$\begin{gathered} \mathfrak{su}(2)=\frac{i}{2}\left[\begin{array}{cc} 0 & b_1-i b_2 \\ b_1+i b_2 & 0 \end{array}\right]+\frac{i}{2}\left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array}\right] \\ \downarrow \text{EXP} \\ \left[\begin{array}{cc} z' & i\left(x'-i y'\right) \\ i\left(x'+i y'\right) & z' \end{array}\right] \times\left[\begin{array}{cc} e^{i b_3 / 2} & 0 \\ 0 & e^{-i b_3 / 2} \end{array}\right] \end{gathered}$	$\begin{array}{r} \mathfrak{su}(2)=\frac{i}{2}\left[\begin{array}{cc} 0 & b_1-i b_2 \\ b_1+i b_2 & 0 \end{array}\right]+\frac{i}{2}\left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array}\right] \\ \text{EXP} \downarrow \text{EXP} \\ \downarrow \text{EXP} \\ \left[\begin{array}{cc} z' & i\left(x'-i y'\right) \\ i\left(x'+i y'\right) & z' \end{array}\right] \times\left[\begin{array}{cc} e^{i b_3 / 2} & 0 \\ 0 & e^{-i b_3 / 2} \end{array}\right] \end{array}$	$\begin{array}{r} \mathfrak{su}(2)=\frac{i}{2}\left[\begin{array}{cc} 0 & b_1-i b_2 \\ b_1+i b_2 & 0 \end{array}\right]+\frac{i}{2}\left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array}\right] \\ \text{EXP} \downarrow \quad \quad \downarrow \text{EXP} \quad \quad \downarrow \text{EXP} \\ \left[\begin{array}{cc} z' & i\left(x'-i y'\right) \\ i\left(x'+i y'\right) & z' \end{array}\right] \times\left[\begin{array}{cc} e^{i b_3 / 2} & 0 \\ 0 & e^{-i b_3 / 2} \end{array}\right] \end{array}$
gilmore-lie-groups_ EQ0185	68	■ 0.888	$\left[\begin{array}{c} o_i \\ o_e \end{array}\right]=\left[\begin{array}{cc} A & B \\ C & D \end{array}\right]\left[\begin{array}{c} i_i \\ i_e \end{array}\right] \quad\left[i_i\right]=\Gamma\left[o_i\right]$	$\left[\begin{array}{c} o_i \\ o_e \end{array}\right]=\left[\begin{array}{cc} A & B \\ C & D \end{array}\right]\left[\begin{array}{c} i_i \\ i_e \end{array}\right] \quad\left[i_i\right]=\Gamma\left[o_i\right]$	$\left[\begin{array}{c} o_i \\ o_e \end{array}\right]=\left[\begin{array}{cc} A & B \\ C & D \end{array}\right]\left[\begin{array}{c} i_i \\ i_e \end{array}\right] \quad\left[i_i\right]=\Gamma\left[o_i\right]$
gilmore-lie-groups_ EQ0527 (9.36)	163	■ 0.890	$\sum_{\alpha \neq 0} \alpha_i \alpha_j=\delta_{i j} \quad \text { or } \quad \sum_{\alpha \neq 0} \alpha \cdot \alpha=\text { rank }=l$	$\sum_{\alpha \neq 0} \alpha_i \alpha_j=\delta_{i j} \quad \text { or } \quad \sum_{\alpha \neq 0} \alpha \cdot \alpha=\text { rank }=l$	$\sum_{\alpha \neq 0} \alpha_i \alpha_j=\delta_{i j} \quad \text { or } \quad \sum_{\alpha \neq 0} \alpha \cdot \alpha=\text { rank }=l$
gilmore-lie-groups_ EQ0953	295	■ 0.890	$\begin{aligned} x_1 &=r \cos \theta_2 \\ x_2 &=r \sin \theta_2 \cos \theta_3 \\ &\vdots \\ x_{n-1} &=r \sin \theta_2 \sin \theta_3 \cdots \sin \theta_{n-1} \cos \theta_n \\ x_n &=r \sin \theta_2 \sin \theta_3 \cdots \sin \theta_{n-1} \sin \theta_n \end{aligned}$	$\begin{array}{r} x_1=r \cos \theta_2 \\ x_2=r \sin \theta_2 \cos \theta_3 \\ \vdots \\ x_{n-1}=r \sin \theta_2 \sin \theta_3 \cdots \sin \theta_{n-1} \cos \theta_n \\ x_n=r \sin \theta_2 \sin \theta_3 \cdots \sin \theta_{n-1} \sin \theta_n \end{array}$	$\begin{array}{r} x_1=r \cos \theta_2 \\ x_2=r \sin \theta_2 \cos \theta_3 \\ \vdots \\ x_{n-1}=r \sin \theta_2 \sin \theta_3 \cdots \sin \theta_{n-1} \cos \theta_n \\ x_n=r \sin \theta_2 \sin \theta_3 \cdots \sin \theta_{n-1} \sin \theta_n \end{array}$

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gilmore-lie-groups_ EQ0731 (13.50)	229	0.890	$\begin{aligned} & \lim_{c \rightarrow 0} P_m^l(u=x/\sqrt{l}) \\ &= \lim_{c \rightarrow 0} (-)^n \sqrt{\frac{(2l)!l^{1/2}}{2^{(2l)}l!!}} \sqrt{\frac{1}{2^n n! (2lc^2)^n}} \\ &\quad \times [1-2c^2x^2]^{(-1/2c^2)/2} \frac{d^n}{dx^n} [1-2c^2x^2]^{1/2c^2} \end{aligned}$	$\lim_{c \rightarrow 0} l^{-1/4} P_m^l(u=x/\sqrt{l})$ $= \lim_{c \rightarrow 0} (-)^n \sqrt{\frac{(2l)!l^{1/2}}{2^{(2l)}l!!}} \sqrt{\frac{1}{2^n n! (2lc^2)^n}} \times [1-2c^2x^2]^{(-1/2c^2)/2} \frac{d^n}{dx^n} [1-2c^2x^2]^{1/2c^2}$	$\lim_{c \rightarrow 0} l^{-1/4} P_m^l(u=x/\sqrt{l})$ $= \lim_{c \rightarrow 0} (-)^n \sqrt{\frac{(2l)!l^{1/2}}{2^{(2l)}l!!}} \sqrt{\frac{1}{2^n n! (2lc^2)^n}} \times [1-2c^2x^2]^{(-1/2c^2)/2} \frac{d^n}{dx^n} [1-2c^2x^2]^{1/2c^2}$
gilmore-lie-groups_ EQ0399 (7.35)	124	0.891	$e^{\{r \ a^{\dagger}\}} e^{+\frac{1}{2} \text{lr} I} e^{la} = e^{ra^{\dagger}+la} = e^{la} e^{-\frac{1}{2} \text{lr} I} e^{ra^{\dagger}}$	$e^{ra^{\dagger}} e^{+\frac{1}{2} \text{lr} I} e^{la} = e^{ra^{\dagger}+la} = e^{la} e^{-\frac{1}{2} \text{lr} I} e^{ra^{\dagger}}$	$e^{ra^{\dagger}} e^{+\frac{1}{2} \text{lr} I} e^{la} = e^{ra^{\dagger}+la} = e^{la} e^{-\frac{1}{2} \text{lr} I} e^{ra^{\dagger}}$
gilmore-lie-groups_ EQ0664	214	0.894	$\Delta = \frac{1}{\sqrt{\ g\ }} \partial_{\mu} \sqrt{\ g\ } g^{\mu\nu} \partial_{\nu} \quad \text{where} \quad \partial_{\nu} = \partial/\partial\phi_{\mu}$	$\Delta = \frac{1}{\sqrt{\ g\ }} \partial_{\mu} \sqrt{\ g\ } g^{\mu\nu} \partial_{\nu} \quad \text{where} \quad \partial_{\nu} = \partial/\partial\phi_{\mu}$	$\Delta = \frac{1}{\sqrt{\ g\ }} \partial_{\mu} \sqrt{\ g\ } g^{\mu\nu} \partial_{\nu} \quad \text{where} \quad \partial_{\nu} = \partial/\partial\phi_{\mu}$
gilmore-lie-groups_ EQ0417 (7.53)	127	0.896	$\left\langle \begin{array}{c} j \\ j \end{array} \left J_+^r J_-^s \right \begin{array}{c} j \\ j \end{array} \right\rangle = \frac{(2j)!r!}{(2j-r)!} \delta_{rs}$	$\left\langle \begin{array}{c} j \\ j \end{array} \left J_+^r J_-^s \right \begin{array}{c} j \\ j \end{array} \right\rangle = \frac{(2j)!r!}{(2j-r)!} \delta_{rs}$	$\left\langle \begin{array}{c} j \\ j \end{array} \left J_+^r J_-^s \right \begin{array}{c} j \\ j \end{array} \right\rangle = \frac{(2j)!r!}{(2j-r)!} \delta_{rs}$
gilmore-lie-groups_ EQ0265 (5.3)	89	0.897	$m_{i\alpha} = 0 \quad \begin{array}{c} p+1 \leq i \leq p+q \\ 1 \leq \alpha \leq p \end{array}$	$m_{i\alpha} = 0 \quad \begin{array}{c} p+1 \leq i \leq p+q \\ 1 \leq \alpha \leq p \end{array}$	$m_{i\alpha} = 0 \quad \begin{array}{c} p+1 \leq i \leq p+q \\ 1 \leq \alpha \leq p \end{array}$
gilmore-lie-groups_ EQ0424 (7.60)	130	0.900	$\begin{aligned} & \frac{d}{dt} \left[\begin{array}{ccc} 1 & r^* \frac{(e^{in}-1)}{(in)} & r^* r \left(\frac{(e^{in}-1-in)}{(in)^2} \right) + id \\ 0 & e^{in} & r \frac{(e^{in}-1)}{(in)} \\ 0 & 0 & 1 \end{array} \right] \\ &= -\frac{i}{\hbar} \left[\begin{array}{ccc} 0 & \alpha^*(t) & \delta(t) \\ 0 & \omega(t) & \alpha(t) \\ 0 & 0 & 0 \end{array} \right] \left[\begin{array}{ccc} 1 & r^* \frac{(e^{in}-1)}{(in)} & r^* r \left(\frac{(e^{in}-1-in)}{(in)^2} \right) + id \\ 0 & e^{in} & r \frac{(e^{in}-1)}{(in)} \\ 0 & 0 & 1 \end{array} \right] \end{aligned}$	$\frac{d}{dt} \left[\begin{array}{ccc} 1 & r^* \frac{(e^{in}-1)}{(in)} & r^* r \left(\frac{(e^{in}-1-in)}{(in)^2} \right) + id \\ 0 & e^{in} & r \frac{(e^{in}-1)}{(in)} \\ 0 & 0 & 1 \end{array} \right]$ $= -\frac{i}{\hbar} \left[\begin{array}{ccc} 0 & \alpha^*(t) & \delta(t) \\ 0 & \omega(t) & \alpha(t) \\ 0 & 0 & 0 \end{array} \right] \left[\begin{array}{ccc} 1 & r^* \frac{(e^{in}-1)}{(in)} & r^* r \left(\frac{(e^{in}-1-in)}{(in)^2} \right) + id \\ 0 & e^{in} & r \frac{(e^{in}-1)}{(in)} \\ 0 & 0 & 1 \end{array} \right]$	$\frac{d}{dt} \left[\begin{array}{ccc} 1 & r^* \frac{(e^{in}-1)}{(in)} & r^* r \left(\frac{(e^{in}-1-in)}{(in)^2} \right) + id \\ 0 & e^{in} & r \frac{(e^{in}-1)}{(in)} \\ 0 & 0 & 1 \end{array} \right]$ $= -\frac{i}{\hbar} \left[\begin{array}{ccc} 0 & \alpha^*(t) & \delta(t) \\ 0 & \omega(t) & \alpha(t) \\ 0 & 0 & 0 \end{array} \right] \left[\begin{array}{ccc} 1 & r^* \frac{(e^{in}-1)}{(in)} & r^* r \left(\frac{(e^{in}-1-in)}{(in)^2} \right) + id \\ 0 & e^{in} & r \frac{(e^{in}-1)}{(in)} \\ 0 & 0 & 1 \end{array} \right]$
gilmore-lie-groups_ EQ0391 (7.27)	123	0.900	$\begin{aligned} & x = e^w \\ & y = (e^w - 1)z/w = z \left(1 + \frac{w}{2!} + \frac{w^2}{3!} + \dots \right) \end{aligned}$	$x = e^w$ $y = (e^w - 1)z/w = z \left(1 + \frac{w}{2!} + \frac{w^2}{3!} + \dots \right)$	$x = e^w$ $y = (e^w - 1)z/w = z \left(1 + \frac{w}{2!} + \frac{w^2}{3!} + \dots \right)$

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gilmore-lie-groups_ EQ0436	132	■ 0.905	$\left[\begin{array}{lll} 1 & 0 & B \\ 0 & A & 0 \\ 0 & 0 & 1 \end{array}\right] \rightarrow e^{Aa^\dagger a + BI}$	$\begin{bmatrix} 1 & 0 & B \\ 0 & A & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow e^{Aa^\dagger a + BI}$	$\begin{bmatrix} 1 & 0 & B \\ 0 & A & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow e^{Aa^\dagger a + BI}$
gilmore-lie-groups_ EQ0924 (15.58)	287	■ 0.911	$\begin{aligned} & M^{jj'}(k^0) \left\langle k^0; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \middle \psi \right\rangle = 0 \\ & C_k M^{jj'}(k^0) C_k^{-1} \left\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \middle \psi \right\rangle = 0 \end{aligned}$	$M^{jj'}(k^0) \left\langle k^0; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \middle \psi \right\rangle = 0$ $C_k M^{jj'}(k^0) C_k^{-1} \left\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \middle \psi \right\rangle = 0$	$M^{jj'}(k^0) \left\langle k^0; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \middle \psi \right\rangle = 0$ $C_k M^{jj'}(k^0) C_k^{-1} \left\langle k; \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \middle \psi \right\rangle = 0$
gilmore-lie-groups_ EQ0181	66	■ 0.911	$\left[\begin{array}{l} i_1 \\ o_2 \end{array}\right] = T_1 \left[\begin{array}{l} o_1 \\ i_2 \end{array}\right] \quad \left[\begin{array}{l} i_3 \\ o_4 \end{array}\right] = T_2 \left[\begin{array}{l} o_3 \\ i_4 \end{array}\right]$	$\begin{bmatrix} i_1 \\ o_2 \end{bmatrix} = T_1 \begin{bmatrix} o_1 \\ i_2 \end{bmatrix} \quad \begin{bmatrix} i_3 \\ o_4 \end{bmatrix} = T_2 \begin{bmatrix} o_3 \\ i_4 \end{bmatrix}$	$\begin{bmatrix} i_1 \\ o_2 \end{bmatrix} = T_1 \begin{bmatrix} o_1 \\ i_2 \end{bmatrix} \quad \begin{bmatrix} i_3 \\ o_4 \end{bmatrix} = T_2 \begin{bmatrix} o_3 \\ i_4 \end{bmatrix}$
gilmore-lie-groups_ EQ0279 (5.18)	93	■ 0.919	$\left[\begin{array}{ccc} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ v_1 & v_2 & 0 \end{array}\right]$	$\begin{bmatrix} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ v_1 & v_2 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ v_1 & v_2 & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0803	252	■ 0.923	$\begin{aligned} & \mathbf{B} = \mathbf{M}' u_4 + \mathbf{L} \times \mathbf{u} - \frac{3}{2} i \mathbf{u} = \frac{i}{2} [\mathbf{u}, \mathbf{L}^2 + \mathbf{M}'^2] \\ & B_4 = \mathbf{M}' \cdot \mathbf{u} + \frac{3}{2} i \mathbf{u} = \frac{i}{2} [u_4, \mathbf{L}^2 + \mathbf{M}'^2] \end{aligned}$	$\mathbf{B} = \mathbf{M}' u_4 + \mathbf{L} \times \mathbf{u} - \frac{3}{2} i \mathbf{u} = \frac{i}{2} [\mathbf{u}, \mathbf{L}^2 + \mathbf{M}'^2]$ $B_4 = \mathbf{M}' \cdot \mathbf{u} + \frac{3}{2} i \mathbf{u} = \frac{i}{2} [u_4, \mathbf{L}^2 + \mathbf{M}'^2]$	$\mathbf{B} = \mathbf{M}' u_4 + \mathbf{L} \times \mathbf{u} - \frac{3}{2} i \mathbf{u} = \frac{i}{2} [\mathbf{u}, \mathbf{L}^2 + \mathbf{M}'^2]$ $B_4 = \mathbf{M}' \cdot \mathbf{u} + \frac{3}{2} i \mathbf{u} = \frac{i}{2} [u_4, \mathbf{L}^2 + \mathbf{M}'^2]$
gilmore-lie-groups_ EQ0962 (16.5)	300	■ 0.930	$\left[\begin{array}{c} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial p} \end{array}\right] [p - g(x)] = \begin{bmatrix} * \\ 0 \\ * \end{bmatrix} \quad \frac{\partial}{\partial y} \begin{bmatrix} x \\ y \\ p \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial p} \end{bmatrix} [p - g(x)] = \begin{bmatrix} * \\ 0 \\ * \end{bmatrix} \quad \frac{\partial}{\partial y} \begin{bmatrix} x \\ y \\ p \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial p} \end{bmatrix} [p - g(x)] = \begin{bmatrix} * \\ 0 \\ * \end{bmatrix} \quad \frac{\partial}{\partial y} \begin{bmatrix} x \\ y \\ p \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

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gilmore-lie-groups_ EQ0006 (1.3)	18	■0.944	$\begin{array}{r} I \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \\ \quad (12) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array} \right] \end{array} \quad \begin{array}{r} (123) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right] \\ \quad (23) \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right] \end{array} \quad \begin{array}{r} (321) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \\ \quad (13) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right] \end{array}$	$I \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (123) \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (321) \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ $(12) \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (23) \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (13) \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	
gilmore-lie-groups_ EQ0416 (7.52)	127	■0.946	$\left. \begin{array}{c} j \\ j \end{array} \right e^{\beta' J_-} e^{n' J_z} e^{\alpha' J_+} \left. \begin{array}{c} j \\ j \end{array} \right\rangle = e^{jn'} = (1 + \alpha \beta)^{2j}$	$\left\langle \begin{array}{c} j \\ j \end{array} \left e^{\beta' J_-} e^{n' J_z} e^{\alpha' J_+} \right \begin{array}{c} j \\ j \end{array} \right\rangle = e^{jn'} = (1 + \alpha \beta)^{2j}$	
gilmore-lie-groups_ EQ0490 (8.24)	150	■0.946	$\mathfrak{g} = \underbrace{\left(\hat{n} + \frac{1}{2}, a^{\dagger 2}, a^2 \right)}_{\mathfrak{su}(1,1)} + \underbrace{(a^{\dagger}, a, I)}_{\text{nilpotent}}$	$\mathfrak{g} = \underbrace{\left(\hat{n} + \frac{1}{2}, a^{\dagger 2}, a^2 \right)}_{\mathfrak{su}(1,1)} + \underbrace{(a^{\dagger}, a, I)}_{\text{nilpotent invariant subalgebra}}$	
gilmore-lie-groups_ EQ0628 (12.1)	204	■0.947	$\begin{aligned} \mathfrak{g} &= \mathfrak{h} + \mathfrak{p} \quad (\mathfrak{p}, \mathfrak{p}) < 0 \\ \mathfrak{g}' &= \mathfrak{h} + i\mathfrak{p} \quad (i\mathfrak{p}, i\mathfrak{p}) > 0 \end{aligned}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p} \quad (\mathfrak{p}, \mathfrak{p}) < 0$ $\mathfrak{g}' = \mathfrak{h} + i\mathfrak{p} \quad (i\mathfrak{p}, i\mathfrak{p}) > 0$	
gilmore-lie-groups_ EQ0225 (4.35)	78	■0.948	$(A, A) = \sum_i \sum_j \left A_i^j \right ^2 \geq 0 \quad = 0 \Rightarrow A = 0$	$(A, A) = \sum_i \sum_j \left A_i^j \right ^2 \geq 0 \quad = 0 \Rightarrow A = 0$	
gilmore-lie-groups_ EQ0903 (15.37)	282	■0.950	$\begin{aligned} b_3 &= 0 \\ b_2 &= +\theta_1 \\ b_1 &= -\theta_2 \end{aligned}$	$b_3 = 0$ $b_2 = +\theta_1$ $b_1 = -\theta_2$	
gilmore-lie-groups_ EQ0016 (1.12)	21	■0.951	$\begin{array}{l} g_i \cdot g_j = g_{ij} \\ \downarrow \quad \downarrow \quad \downarrow \\ f(g_i) \times f(g_j) = f(g_i \cdot g_j) \end{array}$	$\begin{array}{ccc} g_i & \cdot & g_j \\ \downarrow & & \downarrow \\ f(g_i) & \times & f(g_j) \end{array} = \begin{array}{ccc} g_i \cdot g_j & & \\ & \downarrow & \\ & f(g_i \cdot g_j) & \end{array}$	
gilmore-lie-groups_ EQ0167	64	■0.951	$\begin{aligned} UT(p, q) &= UT(q, p) & SO(p, q) &= SO(q, p) \\ A(p, q) &= A(q, p) & U(p, q) &= U(q, p) \\ Sp(p, q) &= Sp(q, p) \end{aligned}$	$UT(p, q) = UT(q, p) \quad SO(p, q) = SO(q, p)$ $A(p, q) = A(q, p) \quad U(p, q) = U(q, p)$ $Sp(p, q) = Sp(q, p)$	

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gilmore-lie-groups_ E00211 (4.21)	76	■ 0.951	$X_{\{i\}}=A_{\{i\}}^{\{r\}} Y_{\{r\}}$	$X_i = A_i^r Y_r$	$X_i = A_i^r Y_r$
gilmore-lie-groups_ E00214 (4.24)	76	■ 0.952	$\left[Z, X_{\{i\}} \right] = R(Z)_{\{i\}}^{\{j\}} X_{\{j\}}$	$[Z, X_i] = R(Z)_i^j X_j$	$[Z, X_i] = R(Z)_i^j X_j$
gilmore-lie-groups_ E00433 (7.69)	132	■ 0.952	$e^{\{-\beta M(H)\}} = \mathrm{EXP} \{-\beta \left[\begin{array}{ccc} 0 & \alpha^* & \delta \\ 0 & \hbar\omega & \alpha \\ 0 & 0 & 0 \end{array} \right] \\ \left[\begin{array}{ccc} 0 & \alpha^* & \delta \\ 0 & \hbar\omega & \alpha \\ 0 & 0 & 0 \end{array} \right] = \left[\begin{array}{ccc} 1 & \alpha^* \frac{e^{-\beta\hbar\omega}-1}{\hbar\omega} & \frac{e^{-\beta\hbar\omega}-1+\beta\hbar\omega}{(\hbar\omega)^2} \alpha^* \alpha - \beta\delta \\ 0 & e^{-\beta\hbar\omega} & \alpha \frac{e^{-\beta\hbar\omega}-1}{\hbar\omega} \\ 0 & 0 & 1 \end{array} \right]$		$e^{-\beta M(H)} = \mathrm{EXP} - \beta \left[\begin{array}{ccc} 0 & \alpha^* & \delta \\ 0 & \hbar\omega & \alpha \\ 0 & 0 & 0 \end{array} \right] = \left[\begin{array}{ccc} 1 & \alpha^* \frac{e^{-\beta\hbar\omega}-1}{\hbar\omega} & \frac{e^{-\beta\hbar\omega}-1+\beta\hbar\omega}{(\hbar\omega)^2} \alpha^* \alpha - \beta\delta \\ 0 & e^{-\beta\hbar\omega} & \alpha \frac{e^{-\beta\hbar\omega}-1}{\hbar\omega} \\ 0 & 0 & 1 \end{array} \right]$
gilmore-lie-groups_ E00354	110	■ 0.953	$\left[\begin{array}{c} a \\ a^\dagger \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array} \right] \left[\begin{array}{c} \partial \\ x \end{array} \right]$	$\left[\begin{array}{c} a \\ a^\dagger \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array} \right] \left[\begin{array}{c} \partial \\ x \end{array} \right]$	$\left[\begin{array}{c} a \\ a^\dagger \end{array} \right] = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array} \right] \left[\begin{array}{c} \partial \\ x \end{array} \right]$
gilmore-lie-groups_ E00414 (7.50)	127	■ 0.954	$\left\langle \begin{array}{c} j \\ j \end{array} \left\ J_+^k J_-^k \right\ \begin{array}{c} j \\ j \end{array} \right\rangle$	$\left\langle \begin{array}{c} j \\ j \end{array} \left\ J_+^k J_-^k \right\ \begin{array}{c} j \\ j \end{array} \right\rangle$	$\left\langle \begin{array}{c} j \\ j \end{array} \left\ J_+^k J_-^k \right\ \begin{array}{c} j \\ j \end{array} \right\rangle$
gilmore-lie-groups_ E00918 (15.53)	285	■ 0.954	$\begin{aligned} & D^{\{0\}}(H_{k^0}) = \mathrm{EXP} \left(\theta_3 J_3^{(j')} + \theta_1 \left(J_1^{(j')} - i J_2^{(j')} \right) + \theta_2 \left(J_2^{(j')} + i J_1^{(j')} \right) \right) \\ & = \mathrm{EXP} \left(\theta_3 J_3^{(j')} + (\theta_1 + i\theta_2) \left(J_1^{(j')} - i J_2^{(j')} \right) \right) \\ & = \left[\begin{array}{cccc} e^{ij'\theta_3} & & & \\ * & e^{i(j'-1)\theta_3} & & \\ * & * & \ddots & \\ * & * & * & \ddots \\ * & * & * & * & e^{-ij'\theta_3} \end{array} \right] \end{aligned}$	$D^{0j'}(H_{k^0}) = \mathrm{EXP} \left(\theta_3 J_3^{(j')} + \theta_1 (J_1^{(j')} - i J_2^{(j')}) + \theta_2 (J_2^{(j')} + i J_1^{(j')}) \right)$ $= \mathrm{EXP} \left(\theta_3 J_3^{(j')} + (\theta_1 + i\theta_2) (J_1^{(j')} - i J_2^{(j')}) \right)$ $= \left[\begin{array}{cccc} e^{ij'\theta_3} & & & \\ * & e^{i(j'-1)\theta_3} & & \\ * & * & \ddots & \\ * & * & * & \ddots \\ * & * & * & * & e^{-ij'\theta_3} \end{array} \right]$	$D^{0j'}(H_{k^0}) = \mathrm{EXP} \left(\theta_3 J_3^{(j')} + \theta_1 (J_1^{(j')} - i J_2^{(j')}) + \theta_2 (J_2^{(j')} + i J_1^{(j')}) \right)$ $= \mathrm{EXP} \left(\theta_3 J_3^{(j')} + (\theta_1 + i\theta_2) (J_1^{(j')} - i J_2^{(j')}) \right)$ $= \left[\begin{array}{cccc} e^{ij'\theta_3} & & & \\ * & e^{i(j'-1)\theta_3} & & \\ * & * & \ddots & \\ * & * & * & \ddots \\ * & * & * & * & e^{-ij'\theta_3} \end{array} \right]$
gilmore-lie-groups_ E00215 (4.25)	76	■ 0.955	$\left[Z, Y_{\{r\}} \right] = S(Z)_{\{r\}}^{\{s\}} Y_{\{s\}}$	$[Z, Y_r] = S(Z)_r^s Y_s$	$[Z, Y_r] = S(Z)_r^s Y_s$

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gilmore-lie-groups_ E08145 (3.28)	57	■ 0.955	$\operatorname{det} M=+1 \mathbb{F}=\begin{cases} \mathbb{R} & SL(n,\mathbb{R}) \\ \mathbb{C} & SL(n,\mathbb{C}) \\ \mathbb{Q} & SL(n,\mathbb{Q}) \end{cases}$		
gilmore-lie-groups_ E08138 (3.22)	55	■ 0.957	$\begin{array}{ll} \mathbb{R} & O(n;G) \\ \mathbb{C} & U(n;G) \\ \mathbb{Q} & Sp(n;G) \end{array}$		
gilmore-lie-groups_ E06692 (13.10)	221	■ 0.957	$\begin{aligned} & \frac{1}{R} L_1 = \frac{1}{R} (x^2 \partial_3 - x^3 \partial_2) = \frac{1}{R} (y \partial / \partial z - R \partial / \partial y) \rightarrow -\partial / \partial y = -\partial_2 = -P_2 \\ & \frac{1}{R} L_2 = \frac{1}{R} (x^3 \partial_1 - x^1 \partial_3) = \frac{1}{R} (R \partial / \partial x - x \partial / \partial z) \rightarrow +\partial / \partial x = +\partial_1 = +P_1 \end{aligned}$		
gilmore-lie-groups_ E06631 (12.4)	206	■ 0.957	$\begin{aligned} & \operatorname{SU}(p,q) / \operatorname{S}[U(p) \times U(q)], \operatorname{Sp}(p,q) / \operatorname{Sp}(p) \times \operatorname{Sp}(q) \text{ are} \\ & \left\ \begin{bmatrix} 0 & B \\ B^\dagger & 0 \end{bmatrix} - \lambda I_{p+q} \right\ = \sum_{j=0}^{n=p+q} (-\lambda)^{n-j} \phi_j(B, B^\dagger) \end{aligned}$		
gilmore-lie-groups_ E0531	163	■ 0.958	$\beta - m\alpha \quad \beta - (m-1)\alpha \quad \cdots \quad \beta \quad \beta + \alpha \quad \cdots \quad \beta + n\alpha$		
gilmore-lie-groups_ E08010	19	■ 0.959	$\begin{aligned} & S_2(12) = \{I, (12)\} \\ & S_2(23) = \{I, (23)\} \\ & S_2(13) = \{I, (13)\} \end{aligned}$		
gilmore-lie-groups_ E08140 (3.24)	55	■ 0.960	$\begin{aligned} & \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} A^t & C^t \\ B^t & D \end{bmatrix} = \begin{bmatrix} AA^t & AC^t \\ CA^t & CC^t \end{bmatrix} = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \\ & \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} A^t & C^t \\ B^t & D \end{bmatrix} = \begin{bmatrix} AA^t & AC^t \\ CA^t & CC^t \end{bmatrix} = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$		
gilmore-lie-groups_ E08163	63	■ 0.960	$SO(2) \longrightarrow \begin{bmatrix} \pm\sqrt{1-x^2} & x \\ -x & \pm\sqrt{1-x^2} \end{bmatrix}$		

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gilmore-lie-groups_ EQ0133 (3.19)	54	■ 0.961	$\begin{aligned} & M^t I_{3,1} M = I_{3,1} \quad \& \left[\begin{array}{cc} A^t & C^t \\ B^t & D^t \end{array} \right] \left[\begin{array}{cc} I_3 & 0 \\ 0 & -1 \end{array} \right] \left[\begin{array}{cc} A & B \\ C & D \end{array} \right] \\ & = \left[\begin{array}{cc} A^t A - C^t C & A^t B - C^t D \\ B^t A - D^t C & B^t B - D^t D \end{array} \right] = \left[\begin{array}{cc} I_3 & 0 \\ 0 & -1 \end{array} \right] \end{aligned}$	$M^t I_{3,1} M = I_{3,1}$ $\begin{bmatrix} A^t & C^t \\ B^t & D^t \end{bmatrix} \begin{bmatrix} I_3 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A^t A - C^t C & A^t B - C^t D \\ B^t A - D^t C & B^t B - D^t D \end{bmatrix} = \begin{bmatrix} I_3 & 0 \\ 0 & -1 \end{bmatrix}$	$M^t I_{3,1} M = I_{3,1}$ $\begin{bmatrix} A^t & C^t \\ B^t & D^t \end{bmatrix} \begin{bmatrix} I_3 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A^t A - C^t C & A^t B - C^t D \\ B^t A - D^t C & B^t B - D^t D \end{bmatrix} = \begin{bmatrix} I_3 & 0 \\ 0 & -1 \end{bmatrix}$
gilmore-lie-groups_ EQ0122 (3.10)	52	■ 0.961	$\left[\begin{array}{c cc c} 1 & * & * & * \\ \hline 0 & SU(1,1) & * & \\ 0 & & * & \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$	$\left[\begin{array}{c cc c} 1 & * & * & * \\ \hline 0 & SU(1,1) & * & \\ 0 & & * & \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$	$\left[\begin{array}{c cc c} 1 & * & * & * \\ \hline 0 & SU(1,1) & * & \\ 0 & & * & \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$
gilmore-lie-groups_ EQ0706 (13.23)	224	■ 0.964	$\begin{aligned} & \mathcal{C}^4[SO(5)] = W^\alpha W_\alpha \\ & W^\alpha = \epsilon^{\alpha\beta\gamma\mu\nu} X_{\beta\gamma} X_{\mu\nu} \end{aligned}$	$\mathcal{C}^4[SO(5)] = W^\alpha W_\alpha$ $W^\alpha = \epsilon^{\alpha\beta\gamma\mu\nu} X_{\beta\gamma} X_{\mu\nu}$	$\mathcal{C}^4[SO(5)] = W^\alpha W_\alpha$ $W^\alpha = \epsilon^{\alpha\beta\gamma\mu\nu} X_{\beta\gamma} X_{\mu\nu}$

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gilmore-lie-groups_ E00944	292	■ 0.965	$\begin{aligned} & \left\{ \begin{array}{l} 0 & \partial_y & -\partial_x & 2\partial_z & 0 \\ -\partial_y & 0 & \partial_z & -\partial_x & -\sqrt{3}\partial_x \\ \partial_x & -\partial_z & 0 & -\partial_y & \sqrt{3}\partial_y \\ -2\partial_z & \partial_x & \partial_y & 0 & 0 \\ 0 & \sqrt{3}\partial_x & -\sqrt{3}\partial_y & 0 & 0 \end{array} \right\} \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix} + \frac{2}{c} \frac{\partial}{\partial t} \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \end{pmatrix} = 0 \\ & \left\{ \begin{array}{l} 0 & \partial_y & -\partial_x & 2\partial_z & 0 \\ -\partial_y & 0 & \partial_z & -\partial_x & -\sqrt{3}\partial_x \\ \partial_x & -\partial_z & 0 & -\partial_y & \sqrt{3}\partial_y \\ -2\partial_z & \partial_x & \partial_y & 0 & 0 \\ 0 & \sqrt{3}\partial_x & -\sqrt{3}\partial_y & 0 & 0 \end{array} \right\} \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \end{pmatrix} - \frac{2}{c} \frac{\partial}{\partial t} \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix} = 0 \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{l} 0 & \partial_y & -\partial_x & 2\partial_z & 0 \\ -\partial_y & 0 & \partial_z & -\partial_x & -\sqrt{3}\partial_x \\ \partial_x & -\partial_z & 0 & -\partial_y & \sqrt{3}\partial_y \\ -2\partial_z & \partial_x & \partial_y & 0 & 0 \\ 0 & \sqrt{3}\partial_x & -\sqrt{3}\partial_y & 0 & 0 \end{array} \right\} \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix} + \frac{2}{c} \frac{\partial}{\partial t} \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \end{pmatrix} = 0 \\ & \left\{ \begin{array}{l} 0 & \partial_y & -\partial_x & 2\partial_z & 0 \\ -\partial_y & 0 & \partial_z & -\partial_x & -\sqrt{3}\partial_x \\ \partial_x & -\partial_z & 0 & -\partial_y & \sqrt{3}\partial_y \\ -2\partial_z & \partial_x & \partial_y & 0 & 0 \\ 0 & \sqrt{3}\partial_x & -\sqrt{3}\partial_y & 0 & 0 \end{array} \right\} \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \end{pmatrix} - \frac{2}{c} \frac{\partial}{\partial t} \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix} = 0 \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ccccc} 0 & \partial_y & -\partial_x & 2\partial_z & 0 \\ -\partial_y & 0 & \partial_z & -\partial_x & -\sqrt{3}\partial_x \\ \partial_x & -\partial_z & 0 & -\partial_y & \sqrt{3}\partial_y \\ -2\partial_z & \partial_x & \partial_y & 0 & 0 \\ 0 & \sqrt{3}\partial_x & -\sqrt{3}\partial_y & 0 & 0 \end{array} \right] \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix} + \frac{2}{c} \frac{\partial}{\partial t} \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \end{pmatrix} = 0 \\ & \left[\begin{array}{ccccc} 0 & \partial_y & -\partial_x & 2\partial_z & 0 \\ -\partial_y & 0 & \partial_z & -\partial_x & -\sqrt{3}\partial_x \\ \partial_x & -\partial_z & 0 & -\partial_y & \sqrt{3}\partial_y \\ -2\partial_z & \partial_x & \partial_y & 0 & 0 \\ 0 & \sqrt{3}\partial_x & -\sqrt{3}\partial_y & 0 & 0 \end{array} \right] \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \end{pmatrix} - \frac{2}{c} \frac{\partial}{\partial t} \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{pmatrix} = 0 \end{aligned}$
gilmore-lie-groups_ E00105	47	■ 0.965	$\begin{aligned} & \left\{ \begin{array}{l} X \\ Y \end{array} \right\} \rightarrow \left\{ \begin{array}{l} X' \\ Y' \end{array} \right\} \\ & = \left(\left[\begin{array}{cc} m_{11} & m_{12} \\ m_{21} & m_{22} \end{array} \right] \left[\begin{array}{l} X \\ Y \end{array} \right] + \left[\begin{array}{l} m_{13} \\ m_{23} \end{array} \right] \right) / \left(\left[\begin{array}{cc} m_{31} & m_{32} \end{array} \right] \left[\begin{array}{l} X \\ Y \end{array} \right] + m_{33} \right) \end{aligned}$	$\begin{aligned} & \left[\begin{array}{l} X \\ Y \end{array} \right] \rightarrow \left[\begin{array}{l} X' \\ Y' \end{array} \right] \\ & = \left(\left[\begin{array}{cc} m_{11} & m_{12} \\ m_{21} & m_{22} \end{array} \right] \left[\begin{array}{l} X \\ Y \end{array} \right] + \left[\begin{array}{l} m_{13} \\ m_{23} \end{array} \right] \right) / \left(\left[\begin{array}{cc} m_{31} & m_{32} \end{array} \right] \left[\begin{array}{l} X \\ Y \end{array} \right] + m_{33} \right) \end{aligned}$	$\begin{aligned} & \left[\begin{array}{l} X \\ Y \end{array} \right] \rightarrow \left[\begin{array}{l} X' \\ Y' \end{array} \right] \\ & = \left(\left[\begin{array}{cc} m_{11} & m_{12} \\ m_{21} & m_{22} \end{array} \right] \left[\begin{array}{l} X \\ Y \end{array} \right] + \left[\begin{array}{l} m_{13} \\ m_{23} \end{array} \right] \right) / \left(\left[\begin{array}{cc} m_{31} & m_{32} \end{array} \right] \left[\begin{array}{l} X \\ Y \end{array} \right] + m_{33} \right) \end{aligned}$
gilmore-lie-groups_ E00682	218	■ 0.965	$\begin{aligned} & \left\{ \begin{array}{l} a & b \\ c & d \end{array} \right\} S^{-1} = \left\{ \begin{array}{l} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{array} \right\} \quad S = \frac{1}{\sqrt{2}} \left\{ \begin{array}{l} 1 & -i \\ -i & 1 \end{array} \right\} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{l} a & b \\ c & d \end{array} \right\} S^{-1} = \left\{ \begin{array}{l} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{array} \right\} \quad S = \frac{1}{\sqrt{2}} \left\{ \begin{array}{l} 1 & -i \\ -i & 1 \end{array} \right\} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{l} a & b \\ c & d \end{array} \right\} S^{-1} = \left\{ \begin{array}{l} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{array} \right\} \quad S = \frac{1}{\sqrt{2}} \left\{ \begin{array}{l} 1 & -i \\ -i & 1 \end{array} \right\} \end{aligned}$
gilmore-lie-groups_ E08392 (7.28)	123	■ 0.966	$\begin{aligned} & w = \ln x \\ & z = y \ln(x)/(x-1) \quad z = 0 \text{ for } x = 1 \end{aligned}$	$\begin{aligned} & w = \ln x \\ & z = y \ln(x)/(x-1) \quad z = 0 \text{ for } x = 1 \end{aligned}$	$\begin{aligned} & w = \ln x \\ & z = y \ln(x)/(x-1) \quad z = 0 \text{ for } x = 1 \end{aligned}$

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gilmore-lie-groups_ E08393 (7.29)	123	■ 0.966	$\left[\begin{array}{lll} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{array}\right]=l X_l+r X_r+\delta X_\delta$	$\begin{bmatrix} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix} = lX_l + rX_r + \delta X_\delta$	$\begin{bmatrix} 0 & l & \delta \\ 0 & 0 & r \\ 0 & 0 & 0 \end{bmatrix} = lX_l + rX_r + \delta X_\delta$
gilmore-lie-groups_ E08657	213	■ 0.966	$\operatorname{EXP}\left[\begin{array}{c cc} 0 & t_1 & t_2 \\ t_1 & 0 & 0 \\ t_2 & 0 & 0 \end{array}\right]=\left[\begin{array}{c cc} x_0 & x_1 & x_2 \\ x_1 & * & * \\ x_2 & * & * \end{array}\right] \quad x_0^2-x_1^2-x_2^2=1$	$\operatorname{EXP}\left[\begin{array}{c cc} 0 & t_1 & t_2 \\ t_1 & 0 & 0 \\ t_2 & 0 & 0 \end{array}\right]=\left[\begin{array}{c cc} x_0 & x_1 & x_2 \\ x_1 & * & * \\ x_2 & * & * \end{array}\right] \quad x_0^2-x_1^2-x_2^2=1$	$\operatorname{EXP}\left[\begin{array}{c cc} 0 & t_1 & t_2 \\ t_1 & 0 & 0 \\ t_2 & 0 & 0 \end{array}\right]=\left[\begin{array}{c cc} x_0 & x_1 & x_2 \\ x_1 & * & * \\ x_2 & * & * \end{array}\right] \quad x_0^2-x_1^2-x_2^2=1$
gilmore-lie-groups_ E08157 (3.39)	61	■ 0.967	$l=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right] \quad \mathcal{I}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right] \quad \mathcal{J}=\left[\begin{array}{ll} 0 & -i \\ i & 0 \end{array}\right] \quad \mathcal{K}=\left[\begin{array}{ll} 1 & 0 \\ 0 & -1 \end{array}\right]$	$l=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right] \quad \mathcal{I}=i\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right] \quad \mathcal{J}=i\left[\begin{array}{ll} 0 & -i \\ i & 0 \end{array}\right] \quad \mathcal{K}=i\left[\begin{array}{ll} 1 & 0 \\ 0 & -1 \end{array}\right]$	$l=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right] \quad \mathcal{I}=i\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right] \quad \mathcal{J}=i\left[\begin{array}{ll} 0 & -i \\ i & 0 \end{array}\right] \quad \mathcal{K}=i\left[\begin{array}{ll} 1 & 0 \\ 0 & -1 \end{array}\right]$
gilmore-lie-groups_ E08216 (4.26)	76	■ 0.969	$S_{\{r\}^{\{S\}}(Z)=\left(A^{-1}\right)_r^i R(Z)_i^j A_j^s$	$S_r^S(Z)=\left(A^{-1}\right)_r^i R(Z)_i^j A_j^s$	$S_r^S(Z)=\left(A^{-1}\right)_r^i R(Z)_i^j A_j^s$
gilmore-lie-groups_ E08468 (7.86)	138	■ 0.970	$\left \begin{array}{c} j \\ \theta \end{array}\right\rangle=e^{i\left(\theta_x J_x+\theta_y J_y\right)}\left \begin{array}{c} j \\ -j \end{array}\right\rangle \quad i\left(\theta_x J_x+\theta_y J_y\right)=\frac{i}{2}\left[\begin{array}{cc} 0 & \theta_x-i \theta_y \\ \theta_x+i \theta_y & 0 \end{array}\right]$	$\left \begin{array}{c} j \\ \theta \end{array}\right\rangle=e^{i\left(\theta_x J_x+\theta_y J_y\right)}\left \begin{array}{c} j \\ -j \end{array}\right\rangle \quad i\left(\theta_x J_x+\theta_y J_y\right)=\frac{i}{2}\left[\begin{array}{cc} 0 & \theta_x-i \theta_y \\ \theta_x+i \theta_y & 0 \end{array}\right]$	$\left \begin{array}{c} j \\ \theta \end{array}\right\rangle=e^{i\left(\theta_x J_x+\theta_y J_y\right)}\left \begin{array}{c} j \\ -j \end{array}\right\rangle \quad i\left(\theta_x J_x+\theta_y J_y\right)=\frac{i}{2}\left[\begin{array}{cc} 0 & \theta_x-i \theta_y \\ \theta_x+i \theta_y & 0 \end{array}\right]$
gilmore-lie-groups_ E08662	214	■ 0.970	$\Gamma_{rs}^{t}=\frac{1}{2}g^{tu}\left(\partial_sg_{ru}+\partial_rg_{su}-\partial_ug_{rs}\right)$	$\Gamma_{rs}^{t}=\frac{1}{2}g^{tu}\left(\partial_sg_{ru}+\partial_rg_{su}-\partial_ug_{rs}\right)$	$\Gamma_{rs}^{t}=\frac{1}{2}g^{tu}\left(\partial_sg_{ru}+\partial_rg_{su}-\partial_ug_{rs}\right)$
gilmore-lie-groups_ E08818 (1.14)	22	■ 0.970	$\begin{array}{c} G/H \\ \{I,(12)\} \\ \{I,(12),(321)\} \end{array} \cdot \begin{array}{c} H \\ \{I,(123),(321)\} \\ \{I,(23)\} \end{array} = \begin{array}{c} H \\ \{I,(123),(321),(12),(13),(23)\} \\ \{I,(23),(12),(123),(321),(13)\} \end{array} =$	$\begin{array}{c} G/H \\ \{I,(12)\} \\ \{I,(12),(321)\} \end{array} \cdot \begin{array}{c} H \\ \{I,(123),(321)\} \\ \{I,(23)\} \end{array} = \begin{array}{c} H \\ \{I,(123),(321),(12),(13),(23)\} \\ \{I,(23),(12),(123),(321),(13)\} \end{array} =$	$\begin{array}{c} G/H \\ \{I,(12)\} \\ \{I,(12),(321)\} \end{array} \cdot \begin{array}{c} H \\ \{I,(123),(321)\} \\ \{I,(23)\} \end{array} = \begin{array}{c} G \\ \{I,(123),(321),(12),(13),(23)\} \\ \{I,(23),(12),(123),(321),(13)\} \end{array}$
gilmore-lie-groups_ E08415 (7.51)	127	■ 0.971	$\left\langle\begin{array}{c} j \\ j \end{array}\left e^{\alpha J_+} e^{\beta J_-}\right \begin{array}{c} j \\ j \end{array}\right\rangle=\sum_{rs} \frac{\alpha^r \beta^s}{r ! s !}\left\langle\begin{array}{c} j \\ j \end{array}\left J_+^r J_-^s\right \begin{array}{c} j \\ j \end{array}\right\rangle$	$\left\langle\begin{array}{c} j \\ j \end{array}\left e^{\alpha J_+} e^{\beta J_-}\right \begin{array}{c} j \\ j \end{array}\right\rangle=\sum_{rs} \frac{\alpha^r \beta^s}{r ! s !}\left\langle\begin{array}{c} j \\ j \end{array}\left J_+^r J_-^s\right \begin{array}{c} j \\ j \end{array}\right\rangle$	$\left\langle\begin{array}{c} j \\ j \end{array}\left e^{\alpha J_+} e^{\beta J_-}\right \begin{array}{c} j \\ j \end{array}\right\rangle=\sum_{rs} \frac{\alpha^r \beta^s}{r ! s !}\left\langle\begin{array}{c} j \\ j \end{array}\left J_+^r J_-^s\right \begin{array}{c} j \\ j \end{array}\right\rangle$
gilmore-lie-groups_ E08121 (3.9)	51	■ 0.972	$\left[\begin{array}{c} x' \\ 1 \end{array}\right]=\left[\begin{array}{cc} a & b \\ 0 & 1 \end{array}\right]\left[\begin{array}{c} x \\ 1 \end{array}\right]=\left[\begin{array}{c} ax+b \\ 1 \end{array}\right]$	$\left[\begin{array}{c} x' \\ 1 \end{array}\right]=\left[\begin{array}{cc} a & b \\ 0 & 1 \end{array}\right]\left[\begin{array}{c} x \\ 1 \end{array}\right]=\left[\begin{array}{c} ax+b \\ 1 \end{array}\right]$	$\left[\begin{array}{c} x' \\ 1 \end{array}\right]=\left[\begin{array}{cc} a & b \\ 0 & 1 \end{array}\right]\left[\begin{array}{c} x \\ 1 \end{array}\right]=\left[\begin{array}{c} ax+b \\ 1 \end{array}\right]$

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gilmore-lie-groups_ EQ0661	214	■0.972	$\Delta=g^{\mathrm{r}\,s}\,\partial_r\partial_s-g^{\mathrm{r}\,s}\Gamma_{rs}^{\,\,t}\partial_t$ $\Gamma_{\mathrm{r}\,s}\{ \,\}^{\mathrm{t}}\,\partial_{\mathrm{t}}$	$\Delta = g^{rs}\partial_r\partial_s - g^{rs}\Gamma_{rs}^{\,\,t}\partial_t$	$\Delta = g^{rs}\partial_r\partial_s - g^{rs}\Gamma_{rs}^{\,\,t}\partial_t$
gilmore-lie-groups_ EQ9987 (16.30)	307	■0.973	$\left(\begin{array}{c} R \\\ S \\ T \end{array}\right)=\left(\begin{array}{c} x\,y \\\ \ln(y) \\\ x^2\,p \end{array}\right)\quad\left(\begin{array}{c} x \\\ y \\\ p \end{array}\right)=\left(\begin{array}{c} Re^{-S} \\\ e^S \\\ Te^{2S}/R^2 \end{array}\right)$ $\left(\begin{array}{c} R \\\ S \\ T \end{array}\right)=\left(\begin{array}{c} x\,y \\\ \ln(y) \\\ x^2\,p \end{array}\right)\quad\left(\begin{array}{c} x \\\ y \\\ p \end{array}\right)=\left(\begin{array}{c} Re^{-S} \\\ e^S \\\ Te^{2S}/R^2 \end{array}\right)$	$\left(\begin{array}{c} R \\ S \\ T \end{array}\right) = \left(\begin{array}{c} xy \\ \ln(y) \\ x^2p \end{array}\right) \quad \left(\begin{array}{c} x \\ y \\ p \end{array}\right) = \left(\begin{array}{c} Re^{-S} \\ e^S \\ Te^{2S}/R^2 \end{array}\right)$	$\left(\begin{array}{c} R \\ S \\ T \end{array}\right) = \left(\begin{array}{c} xy \\ \ln(y) \\ x^2p \end{array}\right) \quad \left(\begin{array}{c} x \\ y \\ p \end{array}\right) = \left(\begin{array}{c} Re^{-S} \\ e^S \\ Te^{2S}/R^2 \end{array}\right)$
gilmore-lie-groups_ EQ0674	217	■0.973	$d\,s^{\mathrm{2}}=\left[\begin{array}{ll} d\,x \,\& d\,y \end{array}\right]\left[\begin{array}{cc} \frac{1}{y} \,\& 0 \\ 0 \,\& \frac{1}{y} \end{array}\right]\left[\begin{array}{c} dx \\ dy \end{array}\right]=\frac{d\bar{z}\,dz}{y^2}$ $d\,s^{\mathrm{2}}=\left[\begin{array}{ll} d\,x \,\& d\,y \end{array}\right]\left[\begin{array}{cc} \frac{1}{y} \,\& 0 \\ 0 \,\& \frac{1}{y} \end{array}\right]\left[\begin{array}{c} dx \\ dy \end{array}\right]=\frac{d\bar{z}\,dz}{y^2}$	$ds^2 = \left[\begin{array}{cc} dx & dy \end{array}\right] \left[\begin{array}{cc} \frac{1}{y} & 0 \\ 0 & \frac{1}{y} \end{array}\right] \left[\begin{array}{c} dx \\ dy \end{array}\right] = \frac{d\bar{z}dz}{y^2}$	$ds^2 = \left[\begin{array}{cc} dx & dy \end{array}\right] \left[\begin{array}{cc} \frac{1}{y} & 0 \\ 0 & \frac{1}{y} \end{array}\right] \left[\begin{array}{c} dx \\ dy \end{array}\right] = \frac{d\bar{z}dz}{y^2}$
gilmore-lie-groups_ EQ0218 (4.28)	77	■0.976	$\left[X_{\mathrm{i}},\,X_{\mathrm{j}}\right]=R\left(X_{\mathrm{i}}\right)_{\mathrm{j}}^{\mathrm{k}}X_{\mathrm{k}}$ $X_{\mathrm{k}}=C_{\mathrm{i}\,\mathrm{j}}^{\mathrm{k}}X_{\mathrm{k}}$	$[X_i,X_j] = R(X_i)_j^k X_k = C_{ij}^k X_k$	$[X_i,X_j] = R(X_i)_j^k X_k = C_{ij}^k X_k$
gilmore-lie-groups_ EQ0469	141	■0.976	$\mathrm{mathcal{H}}_{\mathrm{\mathrm{atoms}}}=\epsilon J_z+\frac{\lambda}{\sqrt{N}}\left(J_+\langle a(t)\rangle+J_-\langle a(t)\rangle^*\right)$ $\mathrm{mathcal{H}}_{\mathrm{\mathrm{atoms}}}=\epsilon J_z+\frac{\lambda}{\sqrt{N}}\left(J_+\langle a(t)\rangle+J_-\langle a(t)\rangle^*\right)$	$\mathcal{H}_{\mathrm{atoms}} = \epsilon J_z + \frac{\lambda}{\sqrt{N}} (J_+ \langle a(t) \rangle + J_- \langle a(t) \rangle^*)$	$\mathcal{H}_{\mathrm{atoms}} = \epsilon J_z + \frac{\lambda}{\sqrt{N}} (J_+ \langle a(t) \rangle + J_- \langle a(t) \rangle^*)$
gilmore-lie-groups_ EQ0509 (9.18)	157	■0.976	$M: \sum a^{\mathrm{i}}_{\mathrm{j}} \,\, \}_{\mathrm{j}} \,\, M_{\mathrm{i}}^{\mathrm{j}}$	$M: \sum a^i_j M_i^j$	$M: \sum a^i_j M_i^j$
gilmore-lie-groups_ EQ9906 (15.40)	282	■0.976	$\left(\begin{array}{cccc} 0 \,\& +\theta_3 \,\& -\theta_2 \,\& i b_1 \\ -\theta_3 \,\& 0 \,\& +\theta_1 \,\& i b_2 \\ +\theta_2 \,\& -\theta_1 \,\& 0 \,\& i b_3 \\ -i b_1 \,\& -i b_2 \,\& -i b_3 \,\& 0 \end{array}\right)\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ i \end{array}\right]=\left[\begin{array}{c} -b_1 \\ -b_2 \\ -b_3 \\ 0 \end{array}\right]=\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$ $\left(\begin{array}{cccc} 0 \,\& +\theta_3 \,\& -\theta_2 \,\& i b_1 \\ -\theta_3 \,\& 0 \,\& +\theta_1 \,\& i b_2 \\ +\theta_2 \,\& -\theta_1 \,\& 0 \,\& i b_3 \\ -i b_1 \,\& -i b_2 \,\& -i b_3 \,\& 0 \end{array}\right)\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ i \end{array}\right]=\left[\begin{array}{c} -b_1 \\ -b_2 \\ -b_3 \\ 0 \end{array}\right]=\left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & i b_1 \\ -\theta_3 & 0 & +\theta_1 & i b_2 \\ +\theta_2 & -\theta_1 & 0 & i b_3 \\ -i b_1 & -i b_2 & -i b_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ i \end{array}\right] = \left[\begin{array}{c} -b_1 \\ -b_2 \\ -b_3 \\ 0 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & i b_1 \\ -\theta_3 & 0 & +\theta_1 & i b_2 \\ +\theta_2 & -\theta_1 & 0 & i b_3 \\ -i b_1 & -i b_2 & -i b_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ i \end{array}\right] = \left[\begin{array}{c} -b_1 \\ -b_2 \\ -b_3 \\ 0 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$
gilmore-lie-groups_ EQ0347	108	■0.978	$\begin{aligned} \,\& J_z=\frac{1}{2}\left(a_1^{\dagger}a_1-a_2^{\dagger}a_2\right) \\ a_1-a_2^{\dagger}a_2^{\dagger}a_2\right) \,\& J_x=\frac{1}{2}\left(J_++J_-\right)=\frac{1}{2}\left(a_1^{\dagger}a_2+a_2^{\dagger}a_1\right) \\ \left(J_++J_-\right)=\frac{1}{2}\left(a_1^{\dagger}a_2+a_2^{\dagger}a_1\right) \,\& J_y=\frac{1}{2i}\left(J_+-J_-\right)=\frac{1}{2i}\left(a_1^{\dagger}a_2-a_2^{\dagger}a_1\right) \end{aligned}$ $\begin{aligned} \,\& J_z=\frac{1}{2}\left(a_1^{\dagger}a_1-a_2^{\dagger}a_2\right) \\ a_1-a_2^{\dagger}a_2^{\dagger}a_2\right) \,\& J_x=\frac{1}{2}\left(J_++J_-\right)=\frac{1}{2}\left(a_1^{\dagger}a_2+a_2^{\dagger}a_1\right) \\ \left(J_++J_-\right)=\frac{1}{2}\left(a_1^{\dagger}a_2+a_2^{\dagger}a_1\right) \,\& J_y=\frac{1}{2i}\left(J_+-J_-\right)=\frac{1}{2i}\left(a_1^{\dagger}a_2-a_2^{\dagger}a_1\right) \end{aligned}$	$J_z = \frac{1}{2} \left(a_1^{\dagger} a_1 - a_2^{\dagger} a_2 \right)$ $J_x = \frac{1}{2} (J_+ + J_-) = \frac{1}{2} \left(a_1^{\dagger} a_2 + a_2^{\dagger} a_1 \right)$ $J_y = \frac{1}{2i} (J_+ - J_-) = \frac{1}{2i} \left(a_1^{\dagger} a_2 - a_2^{\dagger} a_1 \right)$	$J_z = \frac{1}{2} (a_1^{\dagger} a_1 - a_2^{\dagger} a_2)$ $J_x = \frac{1}{2} (J_+ + J_-) = \frac{1}{2} (a_1^{\dagger} a_2 + a_2^{\dagger} a_1)$ $J_y = \frac{1}{2i} (J_+ - J_-) = \frac{1}{2i} (a_1^{\dagger} a_2 - a_2^{\dagger} a_1)$
gilmore-lie-groups_ EQ0270 (5.9)	91	■0.979	$\left(\begin{array}{lll} 0 \,\& l \,\& \delta \\ 0 \,\& \eta \,\& r \\ 0 \,\& 0 \,\& 0 \end{array}\right)=\eta X_{\eta}+l X_l+r X_r+\delta X_{\delta}$ $\left(\begin{array}{lll} 0 \,\& l \,\& \delta \\ 0 \,\& \eta \,\& r \\ 0 \,\& 0 \,\& 0 \end{array}\right)=\eta X_{\eta}+l X_l+r X_r+\delta X_{\delta}$	$\left[\begin{array}{ccc} 0 & l & \delta \\ 0 & \eta & r \\ 0 & 0 & 0 \end{array}\right] = \eta X_{\eta} + l X_l + r X_r + \delta X_{\delta}$	$\left[\begin{array}{ccc} 0 & l & \delta \\ 0 & \eta & r \\ 0 & 0 & 0 \end{array}\right] = \eta X_{\eta} + l X_l + r X_r + \delta X_{\delta}$
gilmore-lie-groups_ EQ0435	132	■0.979	$M(S)\left(\begin{array}{ccc} 1 \,\& Z_l \,\& * \\ 0 \,\& * \,\& Z_r \\ 0 \,\& 0 \,\& 1 \end{array}\right)M\left(S^{-1}\right)=\left(\begin{array}{ccc} 1 \,\& 0 \,\& B \\ 0 \,\& A \,\& 0 \\ 0 \,\& 0 \,\& 1 \end{array}\right)$ $M(S)\left(\begin{array}{ccc} 1 \,\& Z_l \,\& * \\ 0 \,\& * \,\& Z_r \\ 0 \,\& 0 \,\& 1 \end{array}\right)M\left(S^{-1}\right)=\left(\begin{array}{ccc} 1 \,\& 0 \,\& B \\ 0 \,\& A \,\& 0 \\ 0 \,\& 0 \,\& 1 \end{array}\right)$	$M(S) \left[\begin{array}{ccc} 1 & Z_l & * \\ 0 & * & Z_r \\ 0 & 0 & 1 \end{array}\right] M(S^{-1}) = \left[\begin{array}{ccc} 1 & 0 & B \\ 0 & A & 0 \\ 0 & 0 & 1 \end{array}\right]$	$M(S) \left[\begin{array}{ccc} 1 & Z_l & * \\ 0 & * & Z_r \\ 0 & 0 & 1 \end{array}\right] M(S^{-1}) = \left[\begin{array}{ccc} 1 & 0 & B \\ 0 & A & 0 \\ 0 & 0 & 1 \end{array}\right]$

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gilmore-lie-groups_ EQ0976 (16.19)	304	■ 0.979	$\begin{array}{lr} \text{surface equation} & F(R,-,T)=0 \\ \text{constraint equation} & dS/dR=f(R,-,T) \end{array}$	surface equation constraint equation $F(R,-,T)=0$ $dS/dR=f(R,-,T)$	surface equation constraint equation $F(R,-,T)=0$ $dS/dR=f(R,-,T)$
gilmore-lie-groups_ EQ0649 (12.18)	210	■ 0.981	$\begin{array}{cc} S_{0(2+1)} / S_{0(2)} & \& S_{0(2,1)} / S_{0(2)} \\ g_{ij} & \rightarrow \delta_{ij} + x^i x^j \end{array}$	$SO(2+1)/SO(2)$ $SO(2,1)/SO(2)$ $g_{ij} \rightarrow \delta_{ij} + x^i x^j \rightarrow g_{ij} \rightarrow \delta_{ij} - x^i x^j$	$SO(2+1)/SO(2)$ $SO(2,1)/SO(2)$ $g_{ij} \rightarrow \delta_{ij} + x^i x^j \rightarrow g_{ij} \rightarrow \delta_{ij} - x^i x^j$
gilmore-lie-groups_ EQ0462 (7.88)	139	■ 0.981	$\left\langle \begin{array}{c} j \\ \theta'_x \theta'_y \end{array} \middle \begin{array}{c} j \\ \theta_x \theta_y \end{array} \right\rangle = \left[\cos\left(\frac{\theta'}{2}\right) \cos\left(\frac{\theta}{2}\right) + e^{i(\phi' - \phi)} \sin\left(\frac{\theta'}{2}\right) \sin\left(\frac{\theta}{2}\right) \right]^{2j}$	$\left\langle \begin{array}{c} j \\ \theta'_x \theta'_y \end{array} \middle \begin{array}{c} j \\ \theta_x \theta_y \end{array} \right\rangle = \left[\cos\left(\frac{\theta'}{2}\right) \cos\left(\frac{\theta}{2}\right) + e^{i(\phi' - \phi)} \sin\left(\frac{\theta'}{2}\right) \sin\left(\frac{\theta}{2}\right) \right]^{2j}$	$\left\langle \begin{array}{c} j \\ \theta'_x \theta'_y \end{array} \middle \begin{array}{c} j \\ \theta_x \theta_y \end{array} \right\rangle = \left[\cos\left(\frac{\theta'}{2}\right) \cos\left(\frac{\theta}{2}\right) + e^{i(\phi' - \phi)} \sin\left(\frac{\theta'}{2}\right) \sin\left(\frac{\theta}{2}\right) \right]^{2j}$
gilmore-lie-groups_ EQ0183	67	■ 0.981	$\left[\begin{array}{l} o_1 \\ o_2 \\ o_3 \\ o_4 \end{array} \right] \left[\begin{array}{cc cc} s_{11} & s_{12} & 0 & 0 \\ s_{21} & s_{22} & 0 & 0 \\ \hline 0 & 0 & s_{33} & s_{34} \\ 0 & 0 & s_{43} & s_{44} \end{array} \right] \left[\begin{array}{l} i_1 \\ i_2 \\ i_3 \\ i_4 \end{array} \right]$	$\left[\begin{array}{l} o_1 \\ o_2 \\ o_3 \\ o_4 \end{array} \right] \left[\begin{array}{cc cc} s_{11} & s_{12} & 0 & 0 \\ s_{21} & s_{22} & 0 & 0 \\ \hline 0 & 0 & s_{33} & s_{34} \\ 0 & 0 & s_{43} & s_{44} \end{array} \right] \left[\begin{array}{l} i_1 \\ i_2 \\ i_3 \\ i_4 \end{array} \right]$	$\left[\begin{array}{l} o_1 \\ o_2 \\ o_3 \\ o_4 \end{array} \right] \left[\begin{array}{cc cc} s_{11} & s_{12} & 0 & 0 \\ s_{21} & s_{22} & 0 & 0 \\ \hline 0 & 0 & s_{33} & s_{34} \\ 0 & 0 & s_{43} & s_{44} \end{array} \right] \left[\begin{array}{l} i_1 \\ i_2 \\ i_3 \\ i_4 \end{array} \right]$
gilmore-lie-groups_ EQ0498 (9.7)	155	■ 0.981	$\left \Re e \, g \left(r a^{\dagger} + l a + \delta I \right) - \lambda I_3 \right = (-\lambda)^3 = 0$	$\ \Re e \, g \left(r a^{\dagger} + l a + \delta I \right) - \lambda I_3 \ = (-\lambda)^3 = 0$	$\ \Re e \, g \left(r a^{\dagger} + l a + \delta I \right) - \lambda I_3 \ = (-\lambda)^3 = 0$
gilmore-lie-groups_ EQ0626	201	■ 0.982	$\begin{array}{l} m=+ j +1,+ j +2,\ldots \\ m=- j -1,- j -2,\ldots \end{array}$	$m=+ j +1,+ j +2,\ldots$ $m=- j -1,- j -2,\ldots$	$m=+ j +1,+ j +2,\ldots$ $m=- j -1,- j -2,\ldots$
gilmore-lie-groups_ EQ0934 (15.69)	288	■ 0.982	$\left[\begin{array}{ccc} +\frac{i}{c} \frac{\partial}{\partial t} & +\partial_3 & -\partial_2 \\ -\partial_3 & +\frac{i}{c} \frac{\partial}{\partial t} & +\partial_1 \\ +\partial_2 & -\partial_1 & +\frac{i}{c} \frac{\partial}{\partial t} \end{array} \right] \left[\begin{array}{l} B_1 - i E_1 \\ B_2 - i E_2 \\ B_3 - i E_3 \end{array} \right] = 0$	$\left[\begin{array}{ccc} +\frac{i}{c} \frac{\partial}{\partial t} & +\partial_3 & -\partial_2 \\ -\partial_3 & +\frac{i}{c} \frac{\partial}{\partial t} & +\partial_1 \\ +\partial_2 & -\partial_1 & +\frac{i}{c} \frac{\partial}{\partial t} \end{array} \right] \left[\begin{array}{l} B_1 - i E_1 \\ B_2 - i E_2 \\ B_3 - i E_3 \end{array} \right] = 0$	$\left[\begin{array}{ccc} +\frac{i}{c} \frac{\partial}{\partial t} & +\partial_3 & -\partial_2 \\ -\partial_3 & +\frac{i}{c} \frac{\partial}{\partial t} & +\partial_1 \\ +\partial_2 & -\partial_1 & +\frac{i}{c} \frac{\partial}{\partial t} \end{array} \right] \left[\begin{array}{l} B_1 - i E_1 \\ B_2 - i E_2 \\ B_3 - i E_3 \end{array} \right] = 0$
gilmore-lie-groups_ EQ0126 (3.13)	53	■ 0.982	$\left[\begin{array}{ll} 1 & a \\ 0 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & b \\ 0 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & a+b \\ 0 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & b \\ 0 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & a \\ 0 & 1 \end{array} \right]$	$\left[\begin{array}{ll} 1 & a \\ 0 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & b \\ 0 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & a+b \\ 0 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & b \\ 0 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & a \\ 0 & 1 \end{array} \right]$	$\left[\begin{array}{ll} 1 & a \\ 0 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & b \\ 0 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & a+b \\ 0 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & b \\ 0 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & a \\ 0 & 1 \end{array} \right]$

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gilmore-lie-groups_ EQ1036	321	■ 0.982	$\begin{aligned} & \int d\rho(g)\left\langle \begin{array}{c} \lambda' \\ \mu'\nu' \end{array} \middle g \right\rangle \left\langle g \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle = \left\langle \begin{array}{c} \lambda' \\ \mu'\nu' \end{array} \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \\ & \sum_{\lambda} \sum_{\mu} \sum_{\nu} \left\langle g' \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \left\langle \begin{array}{c} \lambda \\ \mu\nu \end{array} \middle g \right\rangle = \langle g' g \rangle \end{aligned}$	$\int d\rho(g)\left\langle \begin{array}{c} \lambda' \\ \mu'\nu' \end{array} \middle g \right\rangle \left\langle g \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle = \left\langle \begin{array}{c} \lambda' \\ \mu'\nu' \end{array} \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle$ $\sum_{\lambda} \sum_{\mu} \sum_{\nu} \left\langle g' \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \left\langle \begin{array}{c} \lambda \\ \mu\nu \end{array} \middle g \right\rangle = \langle g' g \rangle$	$\int d\rho(g)\left\langle \begin{array}{c} \lambda' \\ \mu'\nu' \end{array} \middle g \right\rangle \left\langle g \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle = \left\langle \begin{array}{c} \lambda' \\ \mu'\nu' \end{array} \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle$ $\sum_{\lambda} \sum_{\mu} \sum_{\nu} \left\langle g' \middle \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\rangle \left\langle \begin{array}{c} \lambda \\ \mu\nu \end{array} \middle g \right\rangle = \langle g' g \rangle$
gilmore-lie-groups_ EQ0400 (7.36)	124	■ 0.983	$\left[\begin{array}{lll} 0 & l & \delta \\ 0 & \eta & r \\ 0 & 0 & 0 \end{array}\right]=\eta X_{\eta}+l X_l+r X_r+\delta X_{\delta}$	$\left[\begin{array}{ccc} 0 & l & \delta \\ 0 & \eta & r \\ 0 & 0 & 0 \end{array}\right]=\eta X_{\eta}+l X_l+r X_r+\delta X_{\delta}$	$\left[\begin{array}{ccc} 0 & l & \delta \\ 0 & \eta & r \\ 0 & 0 & 0 \end{array}\right]=\eta X_{\eta}+l X_l+r X_r+\delta X_{\delta}$
gilmore-lie-groups_ EQ0634	207	■ 0.983	$\begin{aligned} & dV=\rho(\mathrm{Id})\,dx^1(\mathrm{Id})\wedge dx^2(\mathrm{Id})\wedge\cdots\wedge dx^n(\mathrm{Id}) \\ & =\rho(p)dx^1(p)\wedge dx^2(p)\wedge\cdots\wedge dx^n(p) \\ & \Rightarrow \rho(p)=\ M(p)\ \rho(\mathrm{Id})\sim\sqrt{\det g(p)} \end{aligned}$	$dV=\rho(\mathrm{Id})dx^1(\mathrm{Id})\wedge dx^2(\mathrm{Id})\wedge\cdots\wedge dx^n(\mathrm{Id})$ $=\rho(p)dx^1(p)\wedge dx^2(p)\wedge\cdots\wedge dx^n(p)$ $\Rightarrow \rho(p)=\ M(p)\ \rho(\mathrm{Id})\sim\sqrt{\det g(p)}$	$dV=\rho(\mathrm{Id})dx^1(\mathrm{Id})\wedge dx^2(\mathrm{Id})\wedge\cdots\wedge dx^n(\mathrm{Id})$ $=\rho(p)dx^1(p)\wedge dx^2(p)\wedge\cdots\wedge dx^n(p)$ $\Rightarrow \rho(p)=\ M(p)\ \rho(\mathrm{Id})\sim\sqrt{\det g(p)}$
gilmore-lie-groups_ EQ0656	213	■ 0.984	$\operatorname{det}\left \operatorname{Re} e\,g\left(\mathfrak{p}\right)-\lambda I_3\right =-\lambda\left(\lambda^2+\left(b_1^2+b_2^2\right)\right)=0$	$\det \Re g(\mathfrak{p})-\lambda I_3 =-\lambda\left[\lambda^2+\left(b_1^2+b_2^2\right)\right]=0$	$\det \Re \mathfrak{g}(\mathfrak{p})-\lambda I_3 =-\lambda\left[\lambda^2+\left(b_1^2+b_2^2\right)\right]=0$
gilmore-lie-groups_ EQ0623	201	■ 0.986	$j+\frac{1}{2}=i\beta\quad\beta\text{ real}\quad m=0,\pm 1,\pm 2,\dots$	$j+\frac{1}{2}=i\beta\quad\beta\text{ real}\quad m=0,\pm 1,\pm 2,\dots$	$j+\frac{1}{2}=i\beta\quad\beta\text{ real}\quad m=0,\pm 1,\pm 2,\dots$
gilmore-lie-groups_ EQ0810 (14.48)	254	■ 0.987	$\left[\begin{array}{l} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{array}\right]=M_{KS}\left[\begin{array}{l} q_1 \\ q_2 \\ q_3 \\ q_4 \end{array}\right]=\left[\begin{array}{cccc} q_1 & -q_2 & -q_3 & q_4 \\ q_2 & q_1 & -q_4 & -q_3 \\ q_3 & q_4 & q_1 & q_2 \\ q_4 & -q_3 & q_2 & -q_1 \end{array}\right]\left[\begin{array}{l} q_1 \\ q_2 \\ q_3 \\ q_4 \end{array}\right]$	$\left[\begin{array}{l} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{array}\right]=M_{KS}\left[\begin{array}{l} q_1 \\ q_2 \\ q_3 \\ q_4 \end{array}\right]=\left[\begin{array}{cccc} q_1 & -q_2 & -q_3 & q_4 \\ q_2 & q_1 & -q_4 & -q_3 \\ q_3 & q_4 & q_1 & q_2 \\ q_4 & -q_3 & q_2 & -q_1 \end{array}\right]\left[\begin{array}{l} q_1 \\ q_2 \\ q_3 \\ q_4 \end{array}\right]$	$\left[\begin{array}{l} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{array}\right]=M_{KS}\left[\begin{array}{l} q_1 \\ q_2 \\ q_3 \\ q_4 \end{array}\right]=\left[\begin{array}{cccc} q_1 & -q_2 & -q_3 & q_4 \\ q_2 & q_1 & -q_4 & -q_3 \\ q_3 & q_4 & q_1 & q_2 \\ q_4 & -q_3 & q_2 & -q_1 \end{array}\right]\left[\begin{array}{l} q_1 \\ q_2 \\ q_3 \\ q_4 \end{array}\right]$

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gilmore-lie-groups_ EQ1005 (16.48)	313	■ 0.987	$\begin{aligned} \eta^{\{0\}}(x, y) &= \eta(x, y) \\ \eta^{\{1\}}(x, y, y^{\{1\}}) &= D^{\{0\}}\eta^{\{0\}} - y^{\{1\}}D^{\{0\}}\xi \\ \eta^{\{2\}}(x, y, y^{\{1\}}, y^{\{2\}}) &= D^{\{1\}}\eta^{\{1\}} - y^{\{2\}}D^{\{0\}}\xi \\ \eta^{\{3\}}(x, y, y^{\{1\}}, y^{\{2\}}, y^{\{3\}}) &= D^{\{2\}}\eta^{\{2\}} - y^{\{3\}}D^{\{0\}}\xi \\ &\vdots \\ &\vdots \end{aligned}$	$\begin{aligned} \eta^{(0)}(x, y) &= \eta(x, y) \\ \eta^{(1)}(x, y, y^{(1)}) &= D^{(0)}\eta^{(0)} - y^{(1)}D^{(0)}\xi \\ \eta^{(2)}(x, y, y^{(1)}, y^{(2)}) &= D^{(1)}\eta^{(1)} - y^{(2)}D^{(0)}\xi \\ \eta^{(3)}(x, y, y^{(1)}, y^{(2)}, y^{(3)}) &= D^{(2)}\eta^{(2)} - y^{(3)}D^{(0)}\xi \\ &\vdots \\ &\vdots \end{aligned}$	$\begin{aligned} \eta^{(0)}(x, y) &= \eta(x, y) \\ \eta^{(1)}(x, y, y^{(1)}) &= D^{(0)}\eta^{(0)} - y^{(1)}D^{(0)}\xi \\ \eta^{(2)}(x, y, y^{(1)}, y^{(2)}) &= D^{(1)}\eta^{(1)} - y^{(2)}D^{(0)}\xi \\ \eta^{(3)}(x, y, y^{(1)}, y^{(2)}, y^{(3)}) &= D^{(2)}\eta^{(2)} - y^{(3)}D^{(0)}\xi \\ &\vdots \\ &\vdots \end{aligned}$
gilmore-lie-groups_ EQ0143 (3.26)	56	■ 0.988	$\text{Poincaré group} \left[\begin{array}{c c} 0(3,1) & t \\ \hline 0 & 1 \end{array} \right]$	$\text{Poincaré group} \left[\begin{array}{c c} O(3,1) & t \\ \hline 0 & 1 \end{array} \right]$	$\text{Poincaré group} \left[\begin{array}{c c} O(3,1) & t \\ \hline 0 & 1 \end{array} \right]$
gilmore-lie-groups_ EQ0239 (4.49)	81	■ 0.989	$d\mu(x) = \rho(x)dV(x) = \rho(x)\ M(x)\ dV(\text{Id}) \Rightarrow \rho(x) = \ M(x)\ ^{-1}$	$d\mu(x) = \rho(x)dV(x) = \rho(x)\ M(x)\ dV(\text{Id}) \Rightarrow \rho(x) = \ M(x)\ ^{-1}$	$d\mu(x) = \rho(x)dV(x) = \rho(x)\ M(x)\ dV(\text{Id}) \Rightarrow \rho(x) = \ M(x)\ ^{-1}$
gilmore-lie-groups_ EQ0046	28	■ 0.990	$\begin{aligned} v_2 &= \left\{ -\frac{27}{2}I_3' \pm \frac{1}{2} \left[(27I_3')^2 + 4 \times 27I_2'^3 \right]^{1/2} \right\}^{1/3} \\ v_3 &= \left\{ -\frac{27}{2}I_3' \pm \frac{1}{2} \left[(27I_3')^2 + 4 \times 27I_2'^3 \right]^{1/2} \right\}^{1/3} \end{aligned}$	$v_2 = \left\{ -\frac{27}{2}I_3' \pm \frac{1}{2} \left[(27I_3')^2 + 4 \times 27I_2'^3 \right]^{1/2} \right\}^{1/3}$	$\frac{v_2}{v_3} = \left\{ -\frac{27}{2}I_3' \pm \frac{1}{2} \left[(27I_3')^2 + 4 \times 27I_2'^3 \right]^{1/2} \right\}^{1/3}$
gilmore-lie-groups_ EQ0071 (1.66)	34	■ 0.990	$x^2 - 1399680x + 552983334912 = 0$	$x^2 - 1399680x + 552983334912 = 0$	$x^2 - 1399680x + 552983334912 = 0$
gilmore-lie-groups_ EQ0236 (4.46)	81	■ 0.991	$g_{rs}(x)dx^r dx^s = g_{ij}(\text{Id})d\alpha^i d\alpha^j$	$g_{rs}(x)dx^r dx^s = g_{ij}(\text{Id})d\alpha^i d\alpha^j$	$g_{rs}(x)dx^r dx^s = g_{ij}(\text{Id})d\alpha^i d\alpha^j$
gilmore-lie-groups_ EQ0608	197	■ 0.991	$\text{tr} \Re g^2(wa_1^\dagger a_1 + xa_1^\dagger a_2 + ya_2^\dagger a_1 + za_2^\dagger a_2) = 2(w-z)^2 + 8xy$	$\text{tr} \Re g^2(wa_1^\dagger a_1 + xa_1^\dagger a_2 + ya_2^\dagger a_1 + za_2^\dagger a_2) = 2(w-z)^2 + 8xy$	$\text{tr} \Re g^2(wa_1^\dagger a_1 + xa_1^\dagger a_2 + ya_2^\dagger a_1 + za_2^\dagger a_2) = 2(w-z)^2 + 8xy$
gilmore-lie-groups_ EQ0113 (3.3)	49	■ 0.991	$c = (a, b) = a + ib \quad a \in \mathbb{R}, b \in \mathbb{R}, i^2 = -1$	$(a, b) \longrightarrow \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \quad i = (0, 1) \longrightarrow \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$c = (a, b) = a + ib \quad a \in \mathbb{R}, b \in \mathbb{R}, i^2 = -1$
gilmore-lie-groups_ EQ0115	49	■ 0.991	$q_0 + q_1 \mathcal{I} + q_2 \mathcal{J} + q_3 \mathcal{K} \rightarrow \begin{bmatrix} q_0 + iq_3 & iq_1 + q_2 \\ iq_1 - q_2 & q_0 - iq_3 \end{bmatrix}$	$q_0 + q_1 \mathcal{I} + q_2 \mathcal{J} + q_3 \mathcal{K} \rightarrow \begin{bmatrix} q_0 + iq_3 & iq_1 + q_2 \\ iq_1 - q_2 & q_0 - iq_3 \end{bmatrix}$	$q_0 + q_1 \mathcal{I} + q_2 \mathcal{J} + q_3 \mathcal{K} \rightarrow \begin{bmatrix} q_0 + iq_3 & iq_1 + q_2 \\ iq_1 - q_2 & q_0 - iq_3 \end{bmatrix}$

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gilmore-lie-groups_ EQ0497 (9.6)	155	■ 0.991	$r\ a^{\dagger}+l\ a+\delta I\ \xrightarrow[\text{representation}]{\text{regular}}\left[\begin{array}{ccc}0&l&0\\0&0&0\\0&-r&0\end{array}\right]\begin{array}{l}a^{\dagger}\\I\\a\end{array}$	$ra^{\dagger}+la+\delta I\ \xrightarrow[\text{representation}]{\text{regular}}\left[\begin{array}{ccc}0&l&0\\0&0&0\\0&-r&0\end{array}\right]\begin{array}{l}a^{\dagger}\\I\\a\end{array}$	$ra^{\dagger}+la+\delta I\ \xrightarrow[\text{representation}]{\text{regular}}\left[\begin{array}{ccc}0&l&0\\0&0&0\\0&-r&0\end{array}\right]\begin{array}{l}a^{\dagger}\\I\\a\end{array}$
gilmore-lie-groups_ EQ0512 (9.21)	159	■ 0.991	$\mathcal{C}_2[\mathfrak{so}(n)]=\sum X_{ij}^2$	$\mathcal{C}_2[\mathfrak{so}(n)]=\sum X_{ij}^2$	$\mathcal{C}_2[\mathfrak{so}(n)]=\sum X_{ij}^2$
gilmore-lie-groups_ EQ0261 (4.62b)	87	■ 0.992	$\begin{array}{l}2\ \alpha_1\ &=+m_{11}+\frac{\kappa_R}{\kappa_L}m_{22}\ &2\ \alpha_2\ &=-m_{12}\kappa_R-\kappa_L^{-1}m_{21}\\2\ \beta_1\ &=+m_{11}-\frac{\kappa_R}{\kappa_L}m_{22}\ &2\ \beta_2\ &=+m_{12}\kappa_R-\kappa_L^{-1}m_{21}\end{array}$	$\begin{array}{l}2\alpha_1\ =+m_{11}+\frac{\kappa_R}{\kappa_L}m_{22}\quad 2\alpha_2\ =-m_{12}\kappa_R-\kappa_L^{-1}m_{21}\\2\beta_1\ =+m_{11}-\frac{\kappa_R}{\kappa_L}m_{22}\quad 2\beta_2\ =+m_{12}\kappa_R-\kappa_L^{-1}m_{21}\end{array}$	$\begin{array}{l}2\alpha_1=+m_{11}+\frac{\kappa_R}{\kappa_L}m_{22}\quad 2\alpha_2=-m_{12}\kappa_R-\kappa_L^{-1}m_{21}\\2\beta_1=+m_{11}-\frac{\kappa_R}{\kappa_L}m_{22}\quad 2\beta_2=+m_{12}\kappa_R-\kappa_L^{-1}m_{21}\end{array}$
gilmore-lie-groups_ EQ0035 (1.32)	26	■ 0.992	$\left[\begin{array}{l}v_1\\v_2\\v_3\end{array}\right]=\left[\begin{array}{ccc}1&1&1\\1&\omega&\omega^2\\1&\omega^2&\omega\end{array}\right]\left[\begin{array}{l}s_1\\s_2\\s_3\end{array}\right]=\left[\begin{array}{l}s_1+s_2+s_3\\s_1+\omega s_2+\omega^2 s_3\\s_1+\omega^2 s_2+\omega s_3\end{array}\right]$	$\left[\begin{array}{l}v_1\\v_2\\v_3\end{array}\right]=\left[\begin{array}{ccc}1&1&1\\1&\omega&\omega^2\\1&\omega^2&\omega\end{array}\right]\left[\begin{array}{l}s_1\\s_2\\s_3\end{array}\right]=\left[\begin{array}{l}s_1+s_2+s_3\\s_1+\omega s_2+\omega^2 s_3\\s_1+\omega^2 s_2+\omega s_3\end{array}\right]$	$\left[\begin{array}{l}v_1\\v_2\\v_3\end{array}\right]=\left[\begin{array}{ccc}1&1&1\\1&\omega&\omega^2\\1&\omega^2&\omega\end{array}\right]\left[\begin{array}{l}s_1\\s_2\\s_3\end{array}\right]=\left[\begin{array}{l}s_1+s_2+s_3\\s_1+\omega s_2+\omega^2 s_3\\s_1+\omega^2 s_2+\omega s_3\end{array}\right]$
gilmore-lie-groups_ EQ0878 (15.11)	277	■ 0.992	$\begin{aligned}&\&\left[\mathbf{J}_{ij}^{(1)}\right],\left[\mathbf{J}_{jk}^{(1)}\right]=-\epsilon_{ijk}\mathbf{J}_k^{(1)}\\&\&\left[\mathbf{J}_{ij}^{(2)}\right],\left[\mathbf{J}_{jk}^{(2)}\right]=-\epsilon_{ijk}\mathbf{J}_k^{(2)}\\&\&\left[\mathbf{J}_i^{(1)},\mathbf{J}_j^{(2)}\right]=0\end{aligned}$	$\begin{aligned}&\left[\mathbf{J}_i^{(1)},\mathbf{J}_j^{(1)}\right]=-\epsilon_{ijk}\mathbf{J}_k^{(1)}\\&\left[\mathbf{J}_i^{(2)},\mathbf{J}_j^{(2)}\right]=-\epsilon_{ijk}\mathbf{J}_k^{(2)}\\&\left[\mathbf{J}_i^{(1)},\mathbf{J}_j^{(2)}\right]=0\end{aligned}$	$\begin{aligned}&\left[\mathbf{J}_i^{(1)},\mathbf{J}_j^{(1)}\right]=-\epsilon_{ijk}\mathbf{J}_k^{(1)}\\&\left[\mathbf{J}_i^{(2)},\mathbf{J}_j^{(2)}\right]=-\epsilon_{ijk}\mathbf{J}_k^{(2)}\\&\left[\mathbf{J}_i^{(1)},\mathbf{J}_j^{(2)}\right]=0\end{aligned}$
gilmore-lie-groups_ EQ0668	215	■ 0.992	$M=\left[\begin{array}{cc cc}m_{ij}&&&\\ \hline &0&-\frac{1}{2}&\\ &-\frac{1}{2}&&0\end{array}\right]$	$M=\left[\begin{array}{cc cc}m_{ij}&&&\\ \hline &0&-\frac{1}{2}&\\ &-\frac{1}{2}&&0\end{array}\right]$	$M=\left[\begin{array}{cc cc}m_{ij}&&&\\ \hline &0&-\frac{1}{2}&\\ &-\frac{1}{2}&&0\end{array}\right]$
gilmore-lie-groups_ EQ0743	232	■ 0.993	$l^{1/4}P_{n-l,0}^l\left(\frac{\pi}{2}-\sqrt{2}\epsilon x\right)\xrightarrow{\lim}\psi_n(x)=N_nH_n(x)e^{-x^2/2}$	$l^{1/4}P_{n-l,0}^l\left(\frac{\pi}{2}-\sqrt{2}\epsilon x\right)\xrightarrow{\lim}\psi_n(x)=N_nH_n(x)e^{-x^2/2}$	$l^{1/4}P_{n-l,0}^l\left(\frac{\pi}{2}-\sqrt{2}\epsilon x\right)\xrightarrow{\lim}\psi_n(x)=N_nH_n(x)e^{-x^2/2}$

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gilmore-lie-groups_ EQ0823	257	■ 0.994	$\left[\begin{array}{cccc cc} 0 & J_3 & -J_2 & M_1 & A_1 & \Gamma_1 \\ -J_3 & 0 & J_1 & M_2 & A_2 & \Gamma_2 \\ J_2 & -J_1 & 0 & M_3 & A_3 & \Gamma_3 \\ -M_1 & -M_2 & -M_3 & 0 & A_4 & \Gamma_4 \\ \hline A_1 & A_2 & A_3 & A_4 & 0 & \Gamma_5 \\ \Gamma_1 & \Gamma_2 & \Gamma_3 & \Gamma_4 & -\Gamma_5 & 0 \end{array}\right] +$	$\left[\begin{array}{cccc cc} 0 & J_3 & -J_2 & M_1 & A_1 & \Gamma_1 \\ -J_3 & 0 & J_1 & M_2 & A_2 & \Gamma_2 \\ J_2 & -J_1 & 0 & M_3 & A_3 & \Gamma_3 \\ -M_1 & -M_2 & -M_3 & 0 & A_4 & \Gamma_4 \\ \hline A_1 & A_2 & A_3 & A_4 & 0 & \Gamma_5 \\ \Gamma_1 & \Gamma_2 & \Gamma_3 & \Gamma_4 & -\Gamma_5 & 0 \end{array}\right] +$	
gilmore-lie-groups_ EQ0888 (15.21)	279	■ 0.994	$\begin{aligned} \{I, a\} k \rangle &= k \rangle e^{i k \cdot a} \\ \{I, a\} \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle &= \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle \end{aligned}$	$\{I, a\} k \rangle = k \rangle e^{i k \cdot a}$ $\{I, a\} \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle = \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$	$\{I, a\} k \rangle = k \rangle e^{i k \cdot a}$ $\{I, a\} \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle = \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$
gilmore-lie-groups_ EQ1010 (16.53)	314	■ 0.995	$\int_V k \nabla^2 x ^{2-n} dV = \int_{S=\partial V} k(2-n) \frac{\hat{\mathbf{n}} \cdot d\mathbf{S}}{ x ^{n-1}} = (2-n)kV(S^n) = 1$	$\int_V k \nabla^2 x ^{2-n} dV = \int_{S=\partial V} k(2-n) \frac{\hat{\mathbf{n}} \cdot d\mathbf{S}}{ x ^{n-1}} = (2-n)kV(S^n) = 1$	$\int_V k \nabla^2 x ^{2-n} dV = \int_{S=\partial V} k(2-n) \frac{\hat{\mathbf{n}} \cdot d\mathbf{S}}{ x ^{n-1}} = (2-n)kV(S^n) = 1$
gilmore-lie-groups_ EQ0194 (4.7)	72	■ 0.995	$\begin{aligned} \operatorname{EXP}(X) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left[\begin{array}{cc} a & b \\ c & -a \end{array} \right]^n = I_2 \cosh \theta + X \frac{\sinh \theta}{\theta} \\ \operatorname{EXP}(aX_a + bX_b + cX_c) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left[\begin{array}{cc} a & b \\ c & -a \end{array} \right]^n = I_2 \cosh \theta + X \frac{\sinh \theta}{\theta} \\ \cosh \theta + a \frac{\sinh \theta}{\theta} &= \cosh \theta + a \frac{\sinh \theta}{\theta} \\ c \frac{\sinh \theta}{\theta} &= c \frac{\sinh \theta}{\theta} \end{aligned}$	$\operatorname{EXP}(X) = \operatorname{EXP}(aX_a + bX_b + cX_c) = \sum_{n=0}^{\infty} \frac{1}{n!} \left[\begin{array}{cc} a & b \\ c & -a \end{array} \right]^n = I_2 \cosh \theta + X \frac{\sinh \theta}{\theta}$ $= \left[\begin{array}{cc} \cosh \theta + a \sinh(\theta)/\theta & b \sinh(\theta)/\theta \\ c \sinh(\theta)/\theta & \cosh \theta - a \sinh(\theta)/\theta \end{array} \right]$ $\theta^2 = a^2 + bc$	$\operatorname{EXP}(X) = \operatorname{EXP}(aX_a + bX_b + cX_c) = \sum_{n=0}^{\infty} \frac{1}{n!} \left[\begin{array}{cc} a & b \\ c & -a \end{array} \right]^n = I_2 \cosh \theta + X \frac{\sinh \theta}{\theta}$ $= \left[\begin{array}{cc} \cosh \theta + a \sinh(\theta)/\theta & b \sinh(\theta)/\theta \\ c \sinh(\theta)/\theta & \cosh \theta - a \sinh(\theta)/\theta \end{array} \right]$ $\theta^2 = a^2 + bc$ (4.7)
gilmore-lie-groups_ EQ0549	171	■ 0.995	$M = \left[\begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array}\right] \rightarrow \mathcal{M} = \left[\begin{array}{ccc} 0 & J_3 & -J_2 \\ -J_3 & 0 & J_1 \\ J_2 & -J_1 & 0 \end{array}\right]$	$M = \left[\begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array}\right] \rightarrow \mathcal{M} = \left[\begin{array}{ccc} 0 & J_3 & -J_2 \\ -J_3 & 0 & J_1 \\ J_2 & -J_1 & 0 \end{array}\right]$	$M = \left[\begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array}\right] \rightarrow \mathcal{M} = \left[\begin{array}{ccc} 0 & J_3 & -J_2 \\ -J_3 & 0 & J_1 \\ J_2 & -J_1 & 0 \end{array}\right]$
gilmore-lie-groups_ EQ0688 (13.6)	220	■ 0.995	$\begin{aligned} [L_1, L_2] &= -L_3 \\ [L_2, L_3] &= -L_1 \\ [L_3, L_1] &= -L_2 \end{aligned}$	$[L_1, L_2] = -L_3$ $[L_2, L_3] = -L_1$ $[L_3, L_1] = -L_2$	$[L_1, L_2] = -L_3$ $[L_2, L_3] = -L_1$ $[L_3, L_1] = -L_2$
gilmore-lie-groups_ EQ0210 (4.20)	76	■ 0.996	$[X_i, X_j] + [X_j, X_i] = 0 \quad C_{ij}^k + C_{ji}^k = 0$	$[X_i, X_j] + [X_j, X_i] = 0 \quad C_{ij}^k + C_{ji}^k = 0$	$[X_i, X_j] + [X_j, X_i] = 0 \quad C_{ij}^k + C_{ji}^k = 0$

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gilmore-lie-groups_ E00087 (2.2)	39	■ 0.997	$\begin{aligned} \sum_k \left(\sum_j A_{ij} B_{jk} \right) C_{kl} &\stackrel{?}{=} \sum_j A_{ij} \left(\sum_k B_{jk} C_{kl} \right) \\ \sum_k \sum_j A_{ij} B_{jk} C_{kl} &\stackrel{\text{ok}}{=} \sum_j \sum_k A_{ij} B_{jk} C_{kl} \end{aligned}$	$\sum_k \left(\sum_j A_{ij} B_{jk} \right) C_{kl} \stackrel{?}{=} \sum_j A_{ij} \left(\sum_k B_{jk} C_{kl} \right)$ $\sum_k \sum_j A_{ij} B_{jk} C_{kl} \stackrel{\text{ok}}{=} \sum_j \sum_k A_{ij} B_{jk} C_{kl}$	$\sum_k \left(\sum_j A_{ij} B_{jk} \right) C_{kl} \stackrel{?}{=} \sum_j A_{ij} \left(\sum_k B_{jk} C_{kl} \right)$ $\sum_k \sum_j A_{ij} B_{jk} C_{kl} \stackrel{\text{ok}}{=} \sum_j \sum_k A_{ij} B_{jk} C_{kl}$
gilmore-lie-groups_ E00170	64	■ 0.997	$\frac{\partial}{\partial x^i} \frac{\partial}{\partial x^i} x^{\{i\}} = \frac{\partial}{\partial x^i} \frac{\partial}{\partial y^j} y^{\{j\}} \frac{\partial}{\partial y^j} M_{\{i\}^{\{j\}}}$	$\frac{\partial}{\partial x^i} = \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j} = M_i^j \frac{\partial}{\partial y^j}$	$\frac{\partial}{\partial x^i} = \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j} = M_i^j \frac{\partial}{\partial y^j}$
gilmore-lie-groups_ E00186	68	■ 0.997	$S_{\text{Network}} = D + C \Gamma (I - A \Gamma)^{-1} B$	$S_{\text{Network}} = D + C \Gamma (I - A \Gamma)^{-1} B$	$S_{\text{Network}} = D + C \Gamma (I - A \Gamma)^{-1} B$
gilmore-lie-groups_ E00595 (11.16)	191	■ 0.997	$\begin{aligned} \mathfrak{g} &= \mathfrak{h} + \mathfrak{p} \\ T(\mathfrak{g}) &= T(\mathfrak{h}) + T(\mathfrak{p}) \\ \mathfrak{g} &= (+1)\mathfrak{h} + (-1)\mathfrak{p} \end{aligned}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$ $T(\mathfrak{g}) = T(\mathfrak{h}) + T(\mathfrak{p})$ $\mathfrak{g} = (+1)\mathfrak{h} + (-1)\mathfrak{p}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$ $T(\mathfrak{g}) = T(\mathfrak{h}) + T(\mathfrak{p})$ $\mathfrak{g} = (+1)\mathfrak{h} + (-1)\mathfrak{p}$
gilmore-lie-groups_ E00720 (13.37)	227	■ 0.997	$h_3 \left \begin{array}{c} J \\ M \end{array} \right\rangle = \left(J_3 + \frac{1}{2c^2} J_0 \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle = \left(M + \frac{1}{2c^2} \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle$	$h_3 \left \begin{array}{c} J \\ M \end{array} \right\rangle = \left(J_3 + \frac{1}{2c^2} J_0 \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle = \left(M + \frac{1}{2c^2} \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle$	$h_3 \left \begin{array}{c} J \\ M \end{array} \right\rangle = \left(J_3 + \frac{1}{2c^2} J_0 \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle = \left(M + \frac{1}{2c^2} \right) \left \begin{array}{c} J \\ M \end{array} \right\rangle$
gilmore-lie-groups_ E00927 (15.61)	287	■ 0.997	$\begin{aligned} R(\mathbf{k}) B_z(k_4) \left\{ J_3^{(j)} - j I_{2j+1} \right\} B_z^{-1}(k_4) R^{-1}(\mathbf{k}) \\ = \{ \mathbf{J} \cdot \mathbf{k} - 1 k_4 I_3 \} \left\langle k; \begin{array}{cc} 1 & 0 \\ \mu & 0 \end{array} \middle \psi \right\rangle = 0 \end{aligned}$	$R(\mathbf{k}) B_z(k_4) \left\{ J_3^{(j)} - j I_{2j+1} \right\} B_z^{-1}(k_4) R^{-1}(\mathbf{k})$ $= \{ \mathbf{J} \cdot \mathbf{k} - 1 k_4 I_3 \} \left\langle k; \begin{array}{cc} 1 & 0 \\ \mu & 0 \end{array} \middle \psi \right\rangle = 0$	$R(\mathbf{k}) B_z(k_4) \left\{ J_3^{(j)} - j I_{2j+1} \right\} B_z^{-1}(k_4) R^{-1}(\mathbf{k})$ $= \{ \mathbf{J} \cdot \mathbf{k} - 1 k_4 I_3 \} \left\langle k; \begin{array}{cc} 1 & 0 \\ \mu & 0 \end{array} \middle \psi \right\rangle = 0$
gilmore-lie-groups_ E00914 (15.49)	284	■ 0.997	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle \quad \begin{array}{l} k = \Lambda k^0 \\ k^0 = (0, 0, 1, \pm i) \end{array}$	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle \quad \begin{array}{l} k = \Lambda k^0 \\ k^0 = (0, 0, 1, \pm i) \end{array}$	$ k\rangle \left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle \quad \begin{array}{l} k = \Lambda k^0 \\ k^0 = (0, 0, 1, \pm i) \end{array}$
gilmore-lie-groups_ E00957	296	■ 0.997	$ \mathbf{E} = \frac{q(d)}{d^{n-1}} \quad q(d) = q \times \begin{cases} \left(\frac{d/R}{\sinh(d/R)} \right)^{n-1} \\ 1 \\ \left(\frac{d/R}{\sin(d/R)} \right)^{n-1} \end{cases}$	$ \mathbf{E} = \frac{q(d)}{d^{n-1}} \quad q(d) = q \times \begin{cases} \left(\frac{d/R}{\sinh(d/R)} \right)^{n-1} \\ 1 \\ \left(\frac{d/R}{\sin(d/R)} \right)^{n-1} \end{cases}$	$ \mathbf{E} = \frac{q(d)}{d^{n-1}} \quad q(d) = q \times \begin{cases} \left(\frac{d/R}{\sinh(d/R)} \right)^{n-1} \\ 1 \\ \left(\frac{d/R}{\sin(d/R)} \right)^{n-1} \end{cases}$

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gilmore-lie-groups_ E00094 (2.6)	43	■ 0.998	$\left[\begin{array}{cc} x_1 & x_2 \\ x_3 & (1+x_2x_3)/x_1 \end{array}\right]^{-1} = \left[\begin{array}{cc} (1+x_2x_3)/x_1 & -x_2 \\ -x_3 & x_1 \end{array}\right]$	$\begin{bmatrix} x_1 & x_2 \\ x_3 & (1+x_2x_3)/x_1 \end{bmatrix}^{-1} = \begin{bmatrix} (1+x_2x_3)/x_1 & -x_2 \\ -x_3 & x_1 \end{bmatrix}$	$\begin{bmatrix} x_1 & x_2 \\ x_3 & (1+x_2x_3)/x_1 \end{bmatrix}^{-1} = \begin{bmatrix} (1+x_2x_3)/x_1 & -x_2 \\ -x_3 & x_1 \end{bmatrix}$
gilmore-lie-groups_ E00301	98	■ 0.998	$\begin{aligned} & \{ \mathfrak{h}, \mathfrak{h} \} \subseteq \mathfrak{h} \\ & \{ \mathfrak{h}, \mathfrak{p} \} \subseteq \mathfrak{p} \\ & \{ \mathfrak{p}, \mathfrak{p} \} \subseteq \mathfrak{h} \end{aligned}$	$\begin{aligned} [\mathfrak{h}, \mathfrak{h}] &\subseteq \mathfrak{h} \\ [\mathfrak{h}, \mathfrak{p}] &\subseteq \mathfrak{p} \\ [\mathfrak{p}, \mathfrak{p}] &\subseteq \mathfrak{h} \end{aligned}$	$\begin{aligned} [\mathfrak{h}, \mathfrak{h}] &\subseteq \mathfrak{h} \\ [\mathfrak{h}, \mathfrak{p}] &\subseteq \mathfrak{p} \\ [\mathfrak{p}, \mathfrak{p}] &\subseteq \mathfrak{h} \end{aligned}$
gilmore-lie-groups_ E00937	290	■ 0.998	$\begin{aligned} & \nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = + \frac{4\pi}{c} \mathbf{J}_e \\ & \nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = - \frac{4\pi}{c} \mathbf{J}_m \end{aligned}$	$\begin{aligned} \nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} &= + \frac{4\pi}{c} \mathbf{J}_e \\ \nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} &= - \frac{4\pi}{c} \mathbf{J}_m \end{aligned}$	$\begin{aligned} \nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} &= + \frac{4\pi}{c} \mathbf{J}_e \\ \nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} &= - \frac{4\pi}{c} \mathbf{J}_m \end{aligned}$
gilmore-lie-groups_ E00676	217	■ 0.998	$w \rightarrow w' = \frac{\alpha w + \beta}{\bar{\beta} w + \bar{\alpha}} \quad \begin{bmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{bmatrix} \in SU(1,1), \quad \bar{\alpha}\alpha - \bar{\beta}\beta = 1$	$w \rightarrow w' = \frac{\alpha w + \beta}{\bar{\beta} w + \bar{\alpha}} \quad \begin{bmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{bmatrix} \in SU(1,1), \quad \bar{\alpha}\alpha - \bar{\beta}\beta = 1$	$w \rightarrow w' = \frac{\alpha w + \beta}{\bar{\beta} w + \bar{\alpha}} \quad \begin{bmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{bmatrix} \in SU(1,1), \quad \bar{\alpha}\alpha - \bar{\beta}\beta = 1$
gilmore-lie-groups_ E00063 (1.58)	33	■ 0.998	$\begin{aligned} & w_2^2 = y_2 \rightarrow (-10)^2 = 100 \\ & w_3^2 = y_3 \rightarrow (-4)^2 = 16 \\ & w_4^2 = y_4 \rightarrow (+2)^2 = 4 \end{aligned}$	$\begin{aligned} w_2^2 &= y_2 \rightarrow (-10)^2 = 100 \\ w_3^2 &= y_3 \rightarrow (-4)^2 = 16 \\ w_4^2 &= y_4 \rightarrow (+2)^2 = 4 \end{aligned}$	$\begin{aligned} w_2^2 &= y_2 \rightarrow (-10)^2 = 100 \\ w_3^2 &= y_3 \rightarrow (-4)^2 = 16 \\ w_4^2 &= y_4 \rightarrow (+2)^2 = 4 \end{aligned}$
gilmore-lie-groups_ E00144 (3.27)	56	■ 0.998	$\text{Galilei group} \left[\begin{array}{c c c} O(3) & \mathbf{v} & \mathbf{t} \\ \hline 0 & 1 & t_4 \\ \hline 0 & 0 & 1 \end{array} \right]$	$\text{Galilei group} \left[\begin{array}{c c c} O(3) & \mathbf{v} & \mathbf{t} \\ \hline 0 & 1 & t_4 \\ \hline 0 & 0 & 1 \end{array} \right]$	$\text{Galilei group} \left[\begin{array}{c c c} O(3) & \mathbf{v} & \mathbf{t} \\ \hline 0 & 1 & t_4 \\ \hline 0 & 0 & 1 \end{array} \right]$
gilmore-lie-groups_ E00445	135	■ 0.998	$\begin{aligned} & \langle n' x^4 n \rangle = \frac{d^4}{d\lambda^4} \sum_{p,q,r} \frac{\lambda^{p+2q+r}}{p!q!r!} 2^{-(p/2+2q+r/2)} \langle n' (a^\dagger)^p a^r n \rangle_{\lambda=0} \\ & \rightarrow \sum_{p+2q+r=4} \frac{4!}{p!q!r!} \frac{\langle n' (a^\dagger)^p (a)^r n \rangle}{2^{(p/2+q+r/2)}} \end{aligned}$	$\begin{aligned} \langle n' x^4 n \rangle &= \frac{d^4}{d\lambda^4} \sum_{p,q,r} \frac{\lambda^{p+2q+r}}{p!q!r!} 2^{-(p/2+2q+r/2)} \langle n' (a^\dagger)^p a^r n \rangle_{\lambda=0} \\ &\rightarrow \sum_{p+2q+r=4} \frac{4!}{p!q!r!} \frac{\langle n' (a^\dagger)^p (a)^r n \rangle}{2^{(p/2+q+r/2)}} \end{aligned}$	$\begin{aligned} \langle n' x^4 n \rangle &= \frac{d^4}{d\lambda^4} \sum_{p,q,r} \frac{\lambda^{p+2q+r}}{p!q!r!} 2^{-(p/2+2q+r/2)} \langle n' (a^\dagger)^p a^r n \rangle_{\lambda=0} \\ &\rightarrow \sum_{p+2q+r=4} \frac{4!}{p!q!r!} \frac{\langle n' (a^\dagger)^p (a)^r n \rangle}{2^{(p/2+q+r/2)}} \end{aligned}$
gilmore-lie-groups_ E00452 (7.78)	137	■ 0.998	$S e^{x^j X_j} e^{-\beta a^i X_i} S^{-1} = e^{-\beta d^i(x,a) H_i}$	$S e^{x^j X_j} e^{-\beta a^i X_i} S^{-1} = e^{-\beta d^i(x,a) H_i}$	$S e^{x^j X_j} e^{-\beta a^i X_i} S^{-1} = e^{-\beta d^i(x,a) H_i}$

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gilmore-lie-groups_ EQ0566	176	■ 0.998	$\begin{aligned} & E_{\{6\}} \pm \mathbf{e}_{\{i\}} \pm \mathbf{e}_{\{j\}} \\ & \mathbf{e}_{\{j\}} \quad \& \quad \frac{1}{2} \underbrace{(\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5)}_{\text{even number of + signs}} \pm \frac{\sqrt{3}}{4} \mathbf{e}_6 \quad 1 \leq i \neq j \leq 5 \end{aligned}$	$E_6 \pm \mathbf{e}_i \pm \mathbf{e}_j$ $\frac{1}{2} \underbrace{(\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5)}_{\text{even number of + signs}} \pm \frac{\sqrt{3}}{4} \mathbf{e}_6 \quad 1 \leq i \neq j \leq 5$	$E_6 \pm \mathbf{e}_i \pm \mathbf{e}_j$ $\frac{1}{2} \underbrace{(\pm \mathbf{e}_1 \pm \mathbf{e}_2 \pm \mathbf{e}_3 \pm \mathbf{e}_4 \pm \mathbf{e}_5)}_{\text{even number of + signs}} \pm \frac{\sqrt{3}}{4} \mathbf{e}_6 \quad 1 \leq i \neq j \leq 5$
gilmore-lie-groups_ EQ0632 (12.5)	207	■ 0.998	$d\,x^{\{i\}}(\mathrm{Id})=M^{\{i\}}_{\{\mu\}}\,d\,x^{\{\mu\}}(p)$	$dx^i(\mathrm{Id})=M^i_{\mu}dx^{\mu}(p)$	$dx^i(\mathrm{Id})=M^i_{\mu}dx^{\mu}(p)$
gilmore-lie-groups_ EQ0926 (15.60)	287	■ 0.998	$R(k)(0,0,k,i\,k)=\left(k_{\{1\}},k_{\{2\}},k_{\{3\}},i\,k_{\{4\}}\right)\quad k_{\{1\}}^2+k_{\{2\}}^2+k_{\{3\}}^2=k_{\{4\}}^2=k^2$	$R(k)(0,0,k,ik)=(k_1,k_2,k_3,ik_4) \quad k_1^2+k_2^2+k_3^2=k_4^2=k^2$	$R(k)(0,0,k,ik)=(k_1,k_2,k_3,ik_4) \quad k_1^2+k_2^2+k_3^2=k_4^2=k^2$
gilmore-lie-groups_ EQ0966 (16.9)	301	■ 0.998	$\begin{aligned} & F(x,y,p)=0 \rightarrow F(\bar{x}(\epsilon),\bar{y}(\epsilon),\bar{p}(\epsilon)) \xrightarrow{\epsilon \text{ small}} F(x+\epsilon \xi,y+\epsilon \eta,p+\epsilon \zeta) \\ & \xrightarrow{\epsilon \text{ small}} F(x,y,p)+\epsilon \left(\xi \frac{\partial}{\partial x}+\eta \frac{\partial}{\partial y}+\zeta \frac{\partial}{\partial p}\right) F(x,y,p)+\text{h.o.t.} \end{aligned}$	$F(x,y,p)=0 \rightarrow F(\bar{x}(\epsilon),\bar{y}(\epsilon),\bar{p}(\epsilon)) \xrightarrow{\epsilon \text{ small}} F(x+\epsilon \xi,y+\epsilon \eta,p+\epsilon \zeta)$ $=F(x,y,p)+\epsilon \left(\xi \frac{\partial}{\partial x}+\eta \frac{\partial}{\partial y}+\zeta \frac{\partial}{\partial p}\right) F(x,y,p)+\text{h.o.t.}$	$F(x,y,p)=0 \rightarrow F(\bar{x}(\epsilon),\bar{y}(\epsilon),\bar{p}(\epsilon)) \xrightarrow{\epsilon \text{ small}} F(x+\epsilon \xi,y+\epsilon \eta,p+\epsilon \zeta)$ $=F(x,y,p)+\epsilon \left(\xi \frac{\partial}{\partial x}+\eta \frac{\partial}{\partial y}+\zeta \frac{\partial}{\partial p}\right) F(x,y,p)+\text{h.o.t.} \quad (1)$
gilmore-lie-groups_ EQ1024	319	■ 0.998	$\begin{aligned} & \xi^1=a_1+a_2x+a_3t+a_4xt \\ & \xi^2=2a_2t+a_4t^2+a_5 \\ & \eta=-\frac{1}{2}a_3xu-a_4\left(\frac{1}{2}t+\frac{1}{4}x^2\right)u+a_6u+h(x,t) \end{aligned}$	$\xi^1=a_1+a_2x+a_3t+a_4xt$ $\xi^2=2a_2t+a_4t^2+a_5$ $\eta=-\frac{1}{2}a_3xu-a_4\left(\frac{1}{2}t+\frac{1}{4}x^2\right)u+a_6u+h(x,t)$	$\xi^1=a_1+a_2x+a_3t+a_4xt$ $\xi^2=2a_2t+a_4t^2+a_5$ $\eta=-\frac{1}{2}a_3xu-a_4(\frac{1}{2}t+\frac{1}{4}x^2)u+a_6u+h(x,t)$
gilmore-lie-groups_ EQ0017 (1.13)	21	■ 0.998	$\begin{array}{l} S_3 \rightarrow S_2 \\ I,(123),(321) \rightarrow I \\ (12),(23),(31) \rightarrow (12) \end{array}$	$\begin{array}{ccc} S_3 & \xrightarrow{f} & S_2 \\ I,(123),(321) & \longrightarrow & I \\ (12),(23),(31) & \longrightarrow & (12) \end{array}$	$\begin{array}{ccc} S_3 & \xrightarrow{f} & S_2 \\ I,(123),(321) & \longrightarrow & I \\ (12),(23),(31) & \longrightarrow & (12) \end{array}$
gilmore-lie-groups_ EQ0182	67	■ 0.998	$\left[\begin{array}{c} i_1 \\ o_2 \end{array}\right]=T_{\mathrm{Tot}}\left[\begin{array}{c} o_3 \\ i_4 \end{array}\right] \quad T_{\mathrm{Tot}}=T_1T_2$	$\left[\begin{array}{c} i_1 \\ o_2 \end{array}\right]=T_{\mathrm{Tot}}\left[\begin{array}{c} o_3 \\ i_4 \end{array}\right] \quad T_{\mathrm{Tot}}=T_1T_2$	$\left[\begin{array}{c} i_1 \\ o_2 \end{array}\right]=T_{\mathrm{Tot}}\left[\begin{array}{c} o_3 \\ i_4 \end{array}\right] \quad T_{\mathrm{Tot}}=T_1T_2$

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gilmore-lie-groups_ E00777 (14.20)	243	0.998	$\begin{gathered} \langle \theta \phi L_- \theta' \phi' \rangle \left\langle \theta' \phi' \left \begin{array}{c} l \\ m \end{array} \right\rangle \right. \\ \downarrow \\ \langle \theta \phi L_- \theta' \phi' \rangle \left\langle \theta' \phi' \left \begin{array}{c} l \\ m \end{array} \right\rangle \right. = \left\langle \theta \phi \left \begin{array}{c} l' \\ m' \end{array} \right\rangle \right. \left\langle \begin{array}{c} l' \\ m' \end{array} \left L_- \right \begin{array}{c} l \\ m \end{array} \right\rangle \end{gathered}$	$\langle \theta \phi L_- \theta' \phi' \rangle \left\langle \theta' \phi' \left \begin{array}{c} l \\ m \end{array} \right\rangle \right. = \left\langle \theta \phi \left \begin{array}{c} l' \\ m' \end{array} \right\rangle \right. \left\langle \begin{array}{c} l' \\ m' \end{array} \left L_- \right \begin{array}{c} l \\ m \end{array} \right\rangle$	$\langle \theta \phi L_- \theta' \phi' \rangle \left\langle \theta' \phi' \left \begin{array}{c} l \\ m \end{array} \right\rangle \right. = \left\langle \theta \phi \left \begin{array}{c} l' \\ m' \end{array} \right\rangle \right. \left\langle \begin{array}{c} l' \\ m' \end{array} \left L_- \right \begin{array}{c} l \\ m \end{array} \right\rangle$
gilmore-lie-groups_ E00047 (1.43)	28	0.999	$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{array} \right] \left[\begin{array}{c} s_1 \\ s_2 \\ s_3 \end{array} \right] = \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{array} \right] \left[\begin{array}{c} s_1 \\ s_2 \\ s_3 \end{array} \right] = \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{array} \right] \left[\begin{array}{c} s_1 \\ s_2 \\ s_3 \end{array} \right] = \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array} \right]$
gilmore-lie-groups_ E00132 (3.18)	54	0.999	$\begin{array}{l} \text{orthogonal group } \mathbb{R} \\ \text{unitary group } \mathbb{C} \\ \text{symplectic group } \mathbb{Q} \end{array}$	$G = \begin{array}{l} \mathbb{R} \\ \mathbb{C} \\ \mathbb{Q} \end{array} \begin{array}{l} O(p, q) \\ U(p, q) \\ Sp(p, q) \end{array} \begin{array}{l} \text{orthogonal group} \\ \text{unitary group} \\ \text{symplectic group} \end{array}$	$G = \begin{array}{l} \mathbb{R} \\ \mathbb{C} \\ \mathbb{Q} \end{array} \begin{array}{l} O(p, q) \\ U(p, q) \\ Sp(p, q) \end{array} \begin{array}{l} \text{orthogonal group} \\ \text{unitary group} \\ \text{symplectic group} \end{array}$
gilmore-lie-groups_ E00177	66	0.999	$\left[\begin{array}{c} A_L \\ B_L \end{array} \right] = \left[\begin{array}{cc} t_{11} & t_{12} \\ t_{21} & t_{22} \end{array} \right] \left[\begin{array}{c} A_R \\ B_R \end{array} \right]$	$\left[\begin{array}{c} A_L \\ B_L \end{array} \right] = \left[\begin{array}{cc} t_{11} & t_{12} \\ t_{21} & t_{22} \end{array} \right] \left[\begin{array}{c} A_R \\ B_R \end{array} \right]$	$\left[\begin{array}{c} A_L \\ B_L \end{array} \right] = \left[\begin{array}{cc} t_{11} & t_{12} \\ t_{21} & t_{22} \end{array} \right] \left[\begin{array}{c} A_R \\ B_R \end{array} \right]$
gilmore-lie-groups_ E00234 (4.44)	81	0.999	$dx^r = M(x)^r_i d\alpha^i$	$dx^r = M(x)^r_i d\alpha^i$	$dx^r = M(x)^r_i d\alpha^i$
gilmore-lie-groups_ E00318 (6.3)	102	0.999	$\begin{aligned} & \left[\begin{array}{c} A \\ B \end{array} \right] = \left[\begin{array}{c} A \\ B \end{array} \right] \\ & = A_{ij} B_{rs} \left[\begin{array}{c} b_i^\dagger b_j, b_r^\dagger b_s \end{array} \right] \\ & = A_{ij} B_{rs} \left(b_i^\dagger \delta_{jr} b_s - b_r^\dagger \delta_{si} b_j \right) \\ & = b_i^\dagger A_{ij} B_{js} b_s - b_r^\dagger B_{rs} A_{sj} b_j \\ & = b_i^\dagger [A, B]_{ij} b_j \\ & = \mathcal{C} \end{aligned}$	$\begin{aligned} [A, B] &= \left[b_i^\dagger A_{ij} b_j, b_r^\dagger B_{rs} b_s \right] \\ &= A_{ij} B_{rs} \left[b_i^\dagger b_j, b_r^\dagger b_s \right] \\ &= A_{ij} B_{rs} \left(b_i^\dagger \delta_{jr} b_s - b_r^\dagger \delta_{si} b_j \right) \\ &= b_i^\dagger A_{ij} B_{js} b_s - b_r^\dagger B_{rs} A_{sj} b_j \\ &= b_i^\dagger [A, B]_{ij} b_j \\ &= \mathcal{C} \end{aligned}$	$\begin{aligned} [A, B] &= \left[b_i^\dagger A_{ij} b_j, b_r^\dagger B_{rs} b_s \right] \\ &= A_{ij} B_{rs} \left[b_i^\dagger b_j, b_r^\dagger b_s \right] \\ &= A_{ij} B_{rs} \left(b_i^\dagger \delta_{jr} b_s - b_r^\dagger \delta_{si} b_j \right) \\ &= b_i^\dagger A_{ij} B_{js} b_s - b_r^\dagger B_{rs} A_{sj} b_j \\ &= b_i^\dagger [A, B]_{ij} b_j \\ &= \mathcal{C} \end{aligned}$

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gilmore-lie-groups_ EQ0353	109	■ 0.999	$\left[\begin{array}{c} d \\ d^\dagger \end{array}\right] = \left[\begin{array}{cc} A & B \\ C & D \end{array}\right] \left[\begin{array}{c} a \\ a^\dagger \end{array}\right]$		$\left[\begin{array}{c} d \\ d^\dagger \end{array}\right] = \left[\begin{array}{cc} A & B \\ C & D \end{array}\right] \left[\begin{array}{c} a \\ a^\dagger \end{array}\right]$
gilmore-lie-groups_ EQ0461 (7.87)	139	■ 0.999	$U(\alpha) 0\rangle = e^{i\alpha_+ J_+} \left \begin{array}{c} j \\ -j \end{array} \right\rangle e^{-ij\alpha_z} = \sum_{m=-j}^{m=+j} \frac{(i\alpha_+ J_+)^{j+m}}{(j+m)!} \left \begin{array}{c} j \\ -j \end{array} \right\rangle e^{-ij\alpha_z}$		$U(\alpha) 0\rangle = e^{i\alpha_+ J_+} \left \begin{array}{c} j \\ -j \end{array} \right\rangle e^{-ij\alpha_z} = \sum_{m=-j}^{m=+j} \frac{(i\alpha_+ J_+)^{j+m}}{(j+m)!} \left \begin{array}{c} j \\ -j \end{array} \right\rangle e^{-ij\alpha_z}$
gilmore-lie-groups_ EQ0752	234	■ 0.999	$\begin{aligned} & \mathcal{C}^2 \rightarrow \mathbf{A} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B} \\ & \mathcal{C}^3 \rightarrow \epsilon^{ijkl} X_{ij} A_k B_l \\ & \mathcal{C}^4 \rightarrow \sum_{ij} (\epsilon^{ijkl} A_k B_l)^2 \end{aligned}$		$\begin{aligned} \mathcal{C}^2 &\rightarrow \mathbf{A} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B} \\ \mathcal{C}^3 &\rightarrow \epsilon^{ijkl} X_{ij} A_k B_l \\ \mathcal{C}^4 &\rightarrow \sum_{ij} (\epsilon^{ijkl} A_k B_l)^2 \end{aligned}$
gilmore-lie-groups_ EQ0929 (15.63)	287	■ 0.999	$\langle k x \rangle \left\{ \mathbf{J} \cdot \frac{1}{i} \nabla + 1 \frac{1}{i} \frac{\partial}{\partial(ict)} I_3 \right\} \langle x k \rangle \left\langle k; \begin{array}{cc} 1 & 0 \\ m & 0 \end{array} \middle \psi \right\rangle = 0$		$\langle k x \rangle \left\{ \mathbf{J} \cdot \frac{1}{i} \nabla + 1 \frac{1}{i} \frac{\partial}{\partial(ict)} I_3 \right\} \langle x k \rangle \left\langle k; \begin{array}{cc} 1 & 0 \\ m & 0 \end{array} \middle \psi \right\rangle = 0$
gilmore-lie-groups_ EQ0403 (7.39)	125	■ 0.999	$\begin{aligned} & \eta' = \eta \quad l' = (e^\eta - 1)l/\eta \\ & \delta' = (e^\eta - 1 - \eta)lr/\eta^2 \quad r' = (e^\eta - 1)r/\eta \end{aligned}$		$\begin{aligned} \eta' &= \eta & l' &= (e^\eta - 1)l/\eta \\ \delta' &= (e^\eta - 1 - \eta)lr/\eta^2 & r' &= (e^\eta - 1)r/\eta \end{aligned}$
gilmore-lie-groups_ EQ0067 (1.62)	33	■ 0.999	$y'^3 - J'_1 y'^2 + J'_2 y' - J'_3 = 0$		$\begin{aligned} y'^3 - J'_1 y'^2 + J'_2 y' - J'_3 &= 0 \\ J'_1 &= 0 \\ J'_2 &= -2736 \\ J'_3 &= 51840 \end{aligned}$
gilmore-lie-groups_ EQ0098	44	■ 0.999	$M_1^2 = I_2 + \begin{bmatrix} a \\ b \end{bmatrix} \begin{bmatrix} a & b \end{bmatrix} = \begin{bmatrix} 1+a^2 & ab \\ ba & 1+b^2 \end{bmatrix} \text{ and } M_2^2 = I_1 + \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1+a^2+b^2 \end{bmatrix}$		$M_1^2 = I_2 + \begin{bmatrix} a \\ b \end{bmatrix} \begin{bmatrix} a & b \end{bmatrix} = \begin{bmatrix} 1+a^2 & ab \\ ba & 1+b^2 \end{bmatrix} \text{ and } M_2^2 = I_1 + \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1+a^2+b^2 \end{bmatrix}$
gilmore-lie-groups_ EQ0161	62	■ 0.999	$SU(2) = \left[\begin{array}{cc} c_1 & c_2 \\ -c_2^* & c_1^* \end{array} \right]$		$SU(2) = \left[\begin{array}{cc} c_1 & c_2 \\ -c_2^* & c_1^* \end{array} \right]$

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gilmore-lie-groups_ E00241 (4.51)	82	■ 0.999	$\left[\begin{array}{l} dx \\ dy \\ dz \end{array}\right] = \left[\begin{array}{ccc} 1+x & 0 & y \\ -y & 1+x & 0 \\ z & 0 & \frac{1+yz}{1+x} \end{array}\right] \left[\begin{array}{l} d\alpha^1 \\ d\alpha^2 \\ d\alpha^3 \end{array}\right]$		
gilmore-lie-groups_ E00253 (4.59)	85	■ 0.999	$X = \sum_{ij} a_{ij} X_{ij} \rightarrow \left\{ \begin{array}{l} \mathfrak{def}(X) = \sum a_{ij} \mathfrak{def}(X_{ij}) \\ \mathfrak{reg}(X) = \sum a_{ij} \mathfrak{reg}(X_{ij}) \end{array} \right.$		
gilmore-lie-groups_ E00304	99	■ 0.999	$\begin{aligned} & (\mathfrak{h}, \mathfrak{h}) \leq 0 \\ & (\mathfrak{h}, \mathfrak{p}) = 0 \\ & (\mathfrak{p}, \mathfrak{p}) \geq 0 \end{aligned}$		
gilmore-lie-groups_ E00302 (7.20)	120	■ 0.999	$\mathfrak{so}(3) = \left[\begin{array}{ccc} 0 & a_3 & -a_2 \\ -a_3 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{array}\right] \quad \mathfrak{su}(2) = \frac{i}{2} \left[\begin{array}{cc} b_3 & b_1 - ib_2 \\ b_1 + ib_2 & -b_3 \end{array}\right]$		
gilmore-lie-groups_ E00473 (8.5)	145	■ 0.999	$\mathfrak{Reg}(\eta \hat{n} + la + ra^\dagger + \delta I) = \left[\begin{array}{cccc} 0 & -r & l & 0 \\ 0 & \eta & 0 & l \\ 0 & 0 & -\eta & -r \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \hat{n} \\ a^\dagger \\ a \\ I \end{array}$		
gilmore-lie-groups_ E00524 (9.33)	162	■ 0.999	$\begin{aligned} & [H_i, E_\alpha] = \alpha_i E_\alpha \\ & [\mathbf{H}, E_\alpha] = \alpha E_\alpha \end{aligned}$		
gilmore-lie-groups_ E00540 (9.47)	166	■ 0.999	$\begin{array}{l} \text{normalization} \\ \text{Jacobi} \\ \text{symmetry} \end{array}$		
gilmore-lie-groups_ E00898 (15.32)	281	■ 0.999	$\Lambda = C_k H_{k^0}$		

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gilmore-lie-groups_ E09933 (15.68)	288	■ 0.999	$\begin{aligned} & \nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = 0 \\ & \nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0 \end{aligned}$	$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = 0$ $\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0$	$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = 0$ $\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} = 0$
gilmore-lie-groups_ E09064 (1.59)	33	■ 0.999	$\begin{gathered} y^3 - J_1 y^2 + J_2 y - J_3 = 0 \\ J_1 = (-4)^1 [2I_2'] = (-4)(-30) = 120 \\ J_2 = (-4)^2 [I_2'^2 - 4I_4'] = 16(225 - 4 \times 24) = 2064 \\ J_3 = (-4)^3 [-I_3'^2] = (-64)(-100) = 6400 \end{gathered}$	$y^3 - J_1 y^2 + J_2 y - J_3 = 0$ $J_1 = (-4)^1 [2I_2'] = (-4)(-30) = 120$ $J_2 = (-4)^2 [I_2'^2 - 4I_4'] = 16(225 - 4 \times 24) = 2064$ $J_3 = (-4)^3 [-I_3'^2] = (-64)(-100) = 6400$	$y^3 - J_1 y^2 + J_2 y - J_3 = 0$ $J_1 = (-4)^1 [2I_2'] = (-4)(-30) = 120$ $J_2 = (-4)^2 [I_2'^2 - 4I_4'] = 16(225 - 4 \times 24) = 2064$ $J_3 = (-4)^3 [-I_3'^2] = (-64)(-100) = 6400$
gilmore-lie-groups_ E09297 (5.36)	96	■ 0.999	$\begin{aligned} & X = \left[\begin{array}{cc} 0 & +1 \\ -1 & 0 \end{array} \right] \quad (X, X) = -2 \quad e^{\theta X} = \left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array} \right] \\ & Y = \left[\begin{array}{cc} 0 & +1 \\ +1 & 0 \end{array} \right] \quad (Y, Y) = +2 \quad e^{\theta Y} = \left[\begin{array}{cc} \cosh \theta & \sinh \theta \\ \sinh \theta & \cosh \theta \end{array} \right] \end{aligned}$	$X = \left[\begin{array}{cc} 0 & +1 \\ -1 & 0 \end{array} \right] \quad (X, X) = -2 \quad e^{\theta X} = \left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array} \right]$ $Y = \left[\begin{array}{cc} 0 & +1 \\ +1 & 0 \end{array} \right] \quad (Y, Y) = +2 \quad e^{\theta Y} = \left[\begin{array}{cc} \cosh \theta & \sinh \theta \\ \sinh \theta & \cosh \theta \end{array} \right]$	$X = \left[\begin{array}{cc} 0 & +1 \\ -1 & 0 \end{array} \right] \quad (X, X) = -2 \quad e^{\theta X} = \left[\begin{array}{cc} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{array} \right]$ $Y = \left[\begin{array}{cc} 0 & +1 \\ +1 & 0 \end{array} \right] \quad (Y, Y) = +2 \quad e^{\theta Y} = \left[\begin{array}{cc} \cosh \theta & \sinh \theta \\ \sinh \theta & \cosh \theta \end{array} \right]$
gilmore-lie-groups_ E09670	216	■ 0.999	$\begin{aligned} & L_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu \\ & P_\mu = \partial_\mu \\ & K_\mu = 2x_\mu (x^\nu \partial_\nu) - (x^\nu x_\nu) \partial_\mu = 2x_\mu (x, \partial) - (x, x) \partial_\mu \\ & S = x^\nu \partial_\nu \end{aligned}$	$L_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu$ $P_\mu = \partial_\mu$ $K_\mu = 2x_\mu (x^\nu \partial_\nu) - (x^\nu x_\nu) \partial_\mu = 2x_\mu (x, \partial) - (x, x) \partial_\mu$ $S = x^\nu \partial_\nu$	$L_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu$ $P_\mu = \partial_\mu$ $K_\mu = 2x_\mu (x^\nu \partial_\nu) - (x^\nu x_\nu) \partial_\mu = 2x_\mu (x, \partial) - (x, x) \partial_\mu$ $S = x^\nu \partial_\nu$
gilmore-lie-groups_ E09045 (1.42)	28	■ 1.000	$v_2^3, v_3^3 = -\frac{27}{2} I_3' \pm \frac{1}{2} \left[(27I_3')^2 + 4 \times 27I_2'^3 \right]^{1/2}$	$v_2^3, v_3^3 = -\frac{27}{2} I_3' \pm \frac{1}{2} \left[(27I_3')^2 + 4 \times 27I_2'^3 \right]^{1/2}$	$v_2^3, v_3^3 = -\frac{27}{2} I_3' \pm \frac{1}{2} \left[(27I_3')^2 + 4 \times 27I_2'^3 \right]^{1/2}$
gilmore-lie-groups_ E09048 (1.44)	28	■ 1.000	$\begin{aligned} & \left[\begin{array}{l} s_1 \\ s_2 \\ s_3 \end{array} \right] = \frac{1}{3} \left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{array} \right] \left[\begin{array}{l} I_1 \\ v_2 \\ v_3 \end{array} \right] = \frac{1}{3} \left[\begin{array}{l} v_1 + v_2 + v_3 \\ v_1 + \omega^2 v_2 + \omega v_3 \\ v_1 + \omega v_2 + \omega^2 v_3 \end{array} \right] \end{aligned}$	$\left[\begin{array}{l} s_1 \\ s_2 \\ s_3 \end{array} \right] = \frac{1}{3} \left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{array} \right] \left[\begin{array}{l} I_1 \\ v_2 \\ v_3 \end{array} \right] = \frac{1}{3} \left[\begin{array}{l} v_1 + v_2 + v_3 \\ v_1 + \omega^2 v_2 + \omega v_3 \\ v_1 + \omega v_2 + \omega^2 v_3 \end{array} \right]$	$\left[\begin{array}{l} s_1 \\ s_2 \\ s_3 \end{array} \right] = \frac{1}{3} \left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{array} \right] \left[\begin{array}{l} I_1 \\ v_2 \\ v_3 \end{array} \right] = \frac{1}{3} \left[\begin{array}{l} v_1 + v_2 + v_3 \\ v_1 + \omega^2 v_2 + \omega v_3 \\ v_1 + \omega v_2 + \omega^2 v_3 \end{array} \right]$

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gilmore-lie-groups_ EQ0057 (1.53)	31	■ 1.000	$\left[\begin{array}{l} t_1 \\ t_2 \\ t_3 \\ t_4 \end{array}\right] = \frac{1}{4} \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{array}\right] \left[\begin{array}{c} I_1 \\ w_2 \\ w_3 \\ w_4 \end{array}\right]$		
gilmore-lie-groups_ EQ0066 (1.61)	33	■ 1.000	$\begin{aligned} & \& y_{1}^{\prime}=y_{1}-40 \rightarrow 4-40=-36 \\ & \& y_{2}^{\prime}=y_{2}-40 \rightarrow 16-40=-24 \\ & \& y_{3}^{\prime}=y_{3}-40 \rightarrow 100-40=60 \end{aligned}$		
gilmore-lie-groups_ EQ0075 (1.70)	34	■ 1.000	$\begin{aligned} & \& v_2^3=699840+i 145152 \sqrt{3} \\ & \& v_3^3=699840-i 145152 \sqrt{3} \end{aligned}$		
gilmore-lie-groups_ EQ0077 (1.73)	35	■ 1.000	$\begin{aligned} & \& y_1 \\ & \& y_2 \\ & \& y_3 \end{aligned} = \frac{1}{3} \left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{array}\right] \left[\begin{array}{c} J_1=120 \\ v_2=-36+i 48 \sqrt{3} \\ v_3=-36-i 48 \sqrt{3} \end{array}\right] = \left[\begin{array}{c} 16 \\ 100 \\ 4 \end{array}\right]$		
gilmore-lie-groups_ EQ0086 (2.1)	39	■ 1.000	$A=\left[\begin{array}{ll} \alpha & \beta \\ \gamma & \delta \end{array}\right] \quad \det(A)=\alpha \delta-\beta \gamma=+1$		
gilmore-lie-groups_ EQ0108	47	■ 1.000	$\begin{aligned} & \& -d s^2=d x^2+d y^2-\left(d \sqrt{1+x^2+y^2}\right)^2 \\ & \& =\frac{1}{1+x^2+y^2}\left(\begin{array}{cc} d x & d y \end{array}\right)\left[\begin{array}{cc} 1+y^2 & -x y \\ -y x & 1+x^2 \end{array}\right]\left(\begin{array}{c} d x \\ d y \end{array}\right) \end{aligned}$		
gilmore-lie-groups_ EQ0111 (3.1)	48	■ 1.000	$\left[\begin{array}{cc} a & b \\ 0 & \frac{1}{a} \end{array}\right]$		

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gilmore-lie-groups_ EQ0175	65	■ 1.000	$\psi_L(x) = A_L e^{+ikx} + B_L e^{-ikx} = \begin{bmatrix} e^{+ikx} & e^{-ikx} \end{bmatrix} \begin{bmatrix} A_L \\ B_L \end{bmatrix}$		
gilmore-lie-groups_ EQ0187	68	■ 1.000	$S_{\text{Network}} = D + C\Gamma B + C\Gamma A\Gamma B + C\Gamma A\Gamma A\Gamma B + C\Gamma A\Gamma A\Gamma A\Gamma B + \dots$		
gilmore-lie-groups_ EQ0191 (4.4)	71	■ 1.000	$\begin{gathered} (\delta a, \delta b, \delta c) \rightarrow I_2 + \delta a X_a + \delta b X_b + \delta c X_c = \begin{bmatrix} 1 + \delta a & \delta b \\ \delta c & 1 - \delta a \end{bmatrix} \\ X_a = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \frac{\partial M(a, b, c)}{\partial a} \Big _{(a, b, c) = (0, 0, 0)} \\ X_b = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \frac{\partial M(a, b, c)}{\partial b} \Big _{(a, b, c) = (0, 0, 0)} \\ X_c = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \frac{\partial M(a, b, c)}{\partial c} \Big _{(a, b, c) = (0, 0, 0)} \end{gathered}$		
gilmore-lie-groups_ EQ0213 (4.23)	76	■ 1.000	$C_{ab}^b = -C_{ba}^b = +2 \quad C_{ac}^c = -C_{ca}^c = -2 \quad C_{bc}^a = -C_{cb}^a = +1$		
gilmore-lie-groups_ EQ0222 (4.32)	78	■ 1.000	$L_z = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad P_x = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad P_y = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$		
gilmore-lie-groups_ EQ0235 (4.45)	81	■ 1.000	$ds^2 = g_{ij}(\text{Id}) d\alpha^i d\alpha^j$		
gilmore-lie-groups_ EQ0243 (4.53)	82	■ 1.000	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \xrightarrow[\text{by } (x, y, z)]{\text{left translation}} \begin{bmatrix} \frac{2(1+yz)}{(1+x)^2} & -\frac{z}{(1+x)} & -\frac{y}{(1+x)} \\ -\frac{z}{(1+x)} & 0 & 1 \\ -\frac{y}{(1+x)} & 1 & 0 \end{bmatrix}$		

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gilmore-lie-groups_ E00254 (4.60)	85	■ 1.000	$\operatorname{tr} \frac{\mathrm{d} e f}{\mathrm{d} e f}(X) \frac{\mathrm{d} e f}{\mathrm{d} e f}(X) \leftarrow (X, X) \rightarrow \operatorname{tr} \operatorname{reg}(X) \operatorname{reg}(X)$	$\operatorname{tr} \operatorname{def}(X) \operatorname{def}(X) \leftarrow (X, X) \rightarrow \operatorname{tr} \operatorname{reg}(X) \operatorname{reg}(X)$	$\operatorname{tr} \operatorname{def}(X) \operatorname{def}(X) \leftarrow (X, X) \rightarrow \operatorname{tr} \operatorname{reg}(X) \operatorname{reg}(X)$
gilmore-lie-groups_ E00281 (5.20)	93	■ 1.000	$M^{\dagger} G + G M = 0 \quad G^t = -G \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathfrak{sp}(G; \mathbb{R}) \\ \mathbb{C} & \mathfrak{sp}(G; \mathbb{C}) \end{cases}$	$M^{\dagger} G + G M = 0 \quad G^t = -G \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathfrak{sp}(G; \mathbb{R}) \\ \mathbb{C} & \mathfrak{sp}(G; \mathbb{C}) \end{cases}$	$M^{\dagger} G + G M = 0 \quad G^t = -G \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathfrak{sp}(G; \mathbb{R}) \\ \mathbb{C} & \mathfrak{sp}(G; \mathbb{C}) \end{cases}$
gilmore-lie-groups_ E00329 (6.14)	104	■ 1.000	$\left(\begin{array}{lll} x_1 & x_2 & x_3 \end{array} \right) \left(\begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array} \right) \left(\begin{array}{c} \partial_1 \\ \partial_2 \\ \partial_3 \end{array} \right) = \theta \cdot \mathcal{L}$	$\left(\begin{array}{lll} x_1 & x_2 & x_3 \end{array} \right) \left(\begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array} \right) \left(\begin{array}{c} \partial_1 \\ \partial_2 \\ \partial_3 \end{array} \right) = \theta \cdot \mathcal{L}$	$\left(\begin{array}{lll} x_1 & x_2 & x_3 \end{array} \right) \left(\begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array} \right) \left(\begin{array}{c} \partial_1 \\ \partial_2 \\ \partial_3 \end{array} \right) = \theta \cdot \mathcal{L}$
gilmore-lie-groups_ E00332 (6.17)	105	■ 1.000	$\left(\begin{array}{lll} x_1 & x_2 & 1 \end{array} \right) \left(\begin{array}{ccc} 0 & \theta & 0 \\ -\theta & 0 & 0 \\ t_1 & t_2 & 0 \end{array} \right) \left(\begin{array}{c} \partial_1 \\ \partial_2 \\ 1 \end{array} \right) = \theta \mathcal{L}_z + t_i \mathcal{T}_i$	$\left(\begin{array}{lll} x_1 & x_2 & 1 \end{array} \right) \left(\begin{array}{ccc} 0 & \theta & 0 \\ -\theta & 0 & 0 \\ t_1 & t_2 & 0 \end{array} \right) \left(\begin{array}{c} \partial_1 \\ \partial_2 \\ 1 \end{array} \right) = \theta \mathcal{L}_z + t_i \mathcal{T}_i$	$\left(\begin{array}{lll} x_1 & x_2 & 1 \end{array} \right) \left(\begin{array}{ccc} 0 & \theta & 0 \\ -\theta & 0 & 0 \\ t_1 & t_2 & 0 \end{array} \right) \left(\begin{array}{c} \partial_1 \\ \partial_2 \\ 1 \end{array} \right) = \theta \mathcal{L}_z + t_i \mathcal{T}_i$
gilmore-lie-groups_ E00404 (7.40)	125	■ 1.000	$\langle 0 e^{\eta a^{\dagger} a + r a^{\dagger} + l a} 0 \rangle = \langle 0 e^{r' a^{\dagger}} e^{\eta' a^{\dagger} a + \delta' I} e^{l' a} 0 \rangle$	$\langle 0 e^{\eta a^{\dagger} a + r a^{\dagger} + l a} 0 \rangle = \langle 0 e^{r' a^{\dagger}} e^{\eta' a^{\dagger} a + \delta' I} e^{l' a} 0 \rangle$	$\langle 0 e^{\eta a^{\dagger} a + r a^{\dagger} + l a} 0 \rangle = \langle 0 e^{r' a^{\dagger}} e^{\eta' a^{\dagger} a + \delta' I} e^{l' a} 0 \rangle$
gilmore-lie-groups_ E00443	135	■ 1.000	$e^{r a^{\dagger}} e^{\delta I} e^{l a} = \left(\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{lll} 1 & 0 & \delta \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{lll} 1 & l & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) = \left(\begin{array}{lll} 1 & l & \delta \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right)$	$e^{r a^{\dagger}} e^{\delta I} e^{l a} = \left(\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{lll} 1 & 0 & \delta \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{lll} 1 & l & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) = \left(\begin{array}{lll} 1 & l & \delta \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right)$	$e^{r a^{\dagger}} e^{\delta I} e^{l a} = \left(\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{lll} 1 & 0 & \delta \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \left(\begin{array}{lll} 1 & l & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) = \left(\begin{array}{lll} 1 & l & \delta \\ 0 & 1 & r \\ 0 & 0 & 1 \end{array} \right)$
gilmore-lie-groups_ E00476 (8.8)	146	■ 1.000	$\begin{array}{l} Y_i = \frac{1}{2} (X_{i4} + \frac{1}{2} \epsilon_{irs} X_{rs}) \\ Z_i = \frac{1}{2} (X_{i4} - \frac{1}{2} \epsilon_{irs} X_{rs}) \end{array}$	$\begin{array}{l} Y_i = \frac{1}{2} (X_{i4} + \frac{1}{2} \epsilon_{irs} X_{rs}) \\ Z_i = \frac{1}{2} (X_{i4} - \frac{1}{2} \epsilon_{irs} X_{rs}) \end{array}$	$\begin{array}{l} Y_i = \frac{1}{2} (X_{i4} + \frac{1}{2} \epsilon_{irs} X_{rs}) \\ Z_i = \frac{1}{2} (X_{i4} - \frac{1}{2} \epsilon_{irs} X_{rs}) \end{array}$
gilmore-lie-groups_ E00493 (9.2)	154	■ 1.000	$\left[Z, \sum a^i X_i \right] = \lambda \sum a^j X_j$	$\sum \sum a^i (R(Z)_i^j - \lambda \delta_i^j) X_j = 0$	$\sum \sum a^i (R(Z)_i^j - \lambda \delta_i^j) X_j = 0$

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gilmore-lie-groups_ E08518 (9.27)	161	■ 1.000	$\begin{aligned} & \phi_2(z_1, z_2) = -5z_1^2 \\ & z_1^2 \quad \& \phi_4(z_1, z_2) = 4z_1^4 = 4(-\phi_2(z_1, z_2)/5)^2 \\ & z_1^4 = 4\left(-\phi_2(z_1, z_2)/5\right)^2 \end{aligned}$	$\begin{aligned} \phi_2(z_1, z_2) &= -5z_1^2 \\ \phi_4(z_1, z_2) &= 4z_1^4 = 4(-\phi_2(z_1, z_2)/5)^2 \end{aligned}$	$\begin{aligned} \phi_2(z_1, z_2) &= -5z_1^2 \\ \phi_4(z_1, z_2) &= 4z_1^4 = 4(-\phi_2(z_1, z_2)/5)^2 \end{aligned}$
gilmore-lie-groups_ E08534 (9.41)	164	■ 1.000	$\begin{aligned} & N_{-\alpha, \beta - m\alpha} = 0 \\ & N_{+\alpha, \beta + n\alpha} = 0 \end{aligned}$	$\begin{aligned} N_{-\alpha, \beta - m\alpha} &= 0 \\ N_{+\alpha, \beta + n\alpha} &= 0 \end{aligned}$	$\begin{aligned} N_{-\alpha, \beta - m\alpha}^2 &= 0 \\ N_{+\alpha, \beta + n\alpha}^2 &= 0 \end{aligned}$
gilmore-lie-groups_ E08546	170	■ 1.000	$\frac{\left(\left(A_1, A_2, \ldots, A_r\right)_{\Re e g}\right)_{\left\{f_r\left(\Re e g\right)\right\}}}{f_r\left(\Re e g\right)}=\left(A_1, A_2, \ldots, A_r\right)=\frac{\left(A_1, A_2, \ldots, A_r\right)_{\Gamma}}{f_r\left(\Gamma\right)}$	$\frac{\left(A_1, A_2, \ldots, A_r\right)_{\Re e g}}{f_r\left(\Re e g\right)}=\left(A_1, A_2, \ldots, A_r\right)=\frac{\left(A_1, A_2, \ldots, A_r\right)_{\Gamma}}{f_r\left(\Gamma\right)}$	$\frac{\left(A_1, A_2, \ldots, A_r\right)_{\Re e g}}{f_r\left(\Re e g\right)}=\left(A_1, A_2, \ldots, A_r\right)=\frac{\left(A_1, A_2, \ldots, A_r\right)_{\Gamma}}{f_r\left(\Gamma\right)}$
gilmore-lie-groups_ E08687	197	■ 1.000	$\Re e g\left(w a_1^{\dagger} a_1+x a_1^{\dagger} a_2+y a_2^{\dagger} a_1+z a_2^{\dagger} a_2\right)=\left[\begin{array}{cccc} 0 & -x & y & 0 \\ -y & w-z & 0 & y \\ x & 0 & -w+z & -x \\ 0 & x & -y & 0 \end{array}\right]$	$\Re e g\left(w a_1^{\dagger} a_1+x a_1^{\dagger} a_2+y a_2^{\dagger} a_1+z a_2^{\dagger} a_2\right)=\left[\begin{array}{cccc} 0 & -x & y & 0 \\ -y & w-z & 0 & y \\ x & 0 & -w+z & -x \\ 0 & x & -y & 0 \end{array}\right]$	$\Re e g\left(w a_1^{\dagger} a_1+x a_1^{\dagger} a_2+y a_2^{\dagger} a_1+z a_2^{\dagger} a_2\right)=\left[\begin{array}{cccc} 0 & -x & y & 0 \\ -y & w-z & 0 & y \\ x & 0 & -w+z & -x \\ 0 & x & -y & 0 \end{array}\right]$
gilmore-lie-groups_ E08667	215	■ 1.000	$\begin{aligned} & y^i = s x^i \quad (1 \leq i \leq n) \\ & y^{n+1} = s \\ & y^{n+2} = s(x, x)_m \end{aligned}$	$\begin{aligned} y^i &= s x^i \quad (1 \leq i \leq n) \\ y^{n+1} &= s \\ y^{n+2} &= s(x, x)_m \end{aligned}$	$\begin{aligned} y^i &= s x^i \quad (1 \leq i \leq n) \\ y^{n+1} &= s \\ y^{n+2} &= s(x, x)_m \end{aligned}$
gilmore-lie-groups_ E08675	217	■ 1.000	$s\left(z_1, z_2\right)=2 \tanh ^{-1} \frac{\left \left z_1-z_2\right \right }{\left \left z_1-\bar{z}_2\right \right }=\log \left\{\frac{\left \left z_1-\bar{z}_2\right \right +\left \left z_1-z_2\right \right }{\left \left z_1-\bar{z}_2\right \right -\left \left z_1-z_2\right \right }\right\}$	$s\left(z_1, z_2\right)=2 \tanh ^{-1} \frac{\left \left z_1-z_2\right \right }{\left \left z_1-\bar{z}_2\right \right }=\log \left\{\frac{\left \left z_1-\bar{z}_2\right \right +\left \left z_1-z_2\right \right }{\left \left z_1-\bar{z}_2\right \right -\left \left z_1-z_2\right \right }\right\}$	$s\left(z_1, z_2\right)=2 \tanh ^{-1} \frac{\left \left z_1-z_2\right \right }{\left \left z_1-\bar{z}_2\right \right }=\log \left\{\frac{\left \left z_1-\bar{z}_2\right \right +\left \left z_1-z_2\right \right }{\left \left z_1-\bar{z}_2\right \right -\left \left z_1-z_2\right \right }\right\}$
gilmore-lie-groups_ E08714 (13.31)	226	■ 1.000	$\left[\begin{array}{c} h_3 \\ h_+ \\ h_- \\ h_0 \end{array}\right] \xrightarrow{c \rightarrow 0} \left[\begin{array}{c} \hat{n}+\frac{1}{2} I=\frac{1}{2}\left\{a, a^{\dagger}\right\} \\ a^{\dagger} \\ a \\ I \end{array}\right]$	$\left[\begin{array}{c} h_3 \\ h_+ \\ h_- \\ h_0 \end{array}\right] \xrightarrow{c \rightarrow 0} \left[\begin{array}{c} \hat{n}+\frac{1}{2} I=\frac{1}{2}\left\{a, a^{\dagger}\right\} \\ a^{\dagger} \\ a \\ I \end{array}\right]$	$\left[\begin{array}{c} h_3 \\ h_+ \\ h_- \\ h_0 \end{array}\right] \xrightarrow{c \rightarrow 0} \left[\begin{array}{c} \hat{n}+\frac{1}{2} I=\frac{1}{2}\left\{a, a^{\dagger}\right\} \\ a^{\dagger} \\ a \\ I \end{array}\right]$
gilmore-lie-groups_ E08746	233	■ 1.000	$e^{\alpha a^{\dagger}-\alpha^* a}=e^{\alpha a^{\dagger}} e^{-\frac{1}{2} \alpha^* \alpha I} e^{-\alpha^* a}$	$e^{\alpha a^{\dagger}-\alpha^* a}=e^{\alpha a^{\dagger}} e^{-\frac{1}{2} \alpha^* \alpha I} e^{-\alpha^* a}$	$e^{\alpha a^{\dagger}-\alpha^* a}=e^{\alpha a^{\dagger}} e^{-\frac{1}{2} \alpha^* \alpha I} e^{-\alpha^* a}$
gilmore-lie-groups_ E08769	240	■ 1.000	$\begin{aligned} & E=\frac{m c^2}{\sqrt{1+\left(\alpha / N^{\prime}\right)^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2} \\ & N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, \quad N=n+l+1 \end{aligned}$	$\begin{aligned} E &= \frac{m c^2}{\sqrt{1+\left(\alpha / N^{\prime}\right)^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2} \\ N^{\prime} &= n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, \quad N=n+l+1 \end{aligned}$	$\begin{aligned} E &= \frac{m c^2}{\sqrt{1+\left(\alpha / N^{\prime}\right)^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2} \\ N^{\prime} &= n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, \quad N=n+l+1 \end{aligned}$

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gilmore-lie-groups_ E08882 (14.45)	251	■ 1.000	$\begin{aligned} & \mathbf{u}=2\mathbf{p}_4\left(\mathbf{p}\cdot\mathbf{p}+\mathbf{p}_4^2\right)^{-1}\mathbf{p} \\ & u_4=\left(\mathbf{p}\cdot\mathbf{p}-p_4^2\right)\left(\mathbf{p}\cdot\mathbf{p}+p_4^2\right)^{-1} \end{aligned}$	$\mathbf{u}=2p_4\left(\mathbf{p}\cdot\mathbf{p}+p_4^2\right)^{-1}\mathbf{p}$ $u_4=\left(\mathbf{p}\cdot\mathbf{p}-p_4^2\right)\left(\mathbf{p}\cdot\mathbf{p}+p_4^2\right)^{-1}$	$\mathbf{u}=2p_4(\mathbf{p}\cdot\mathbf{p}+p_4^2)^{-1}\mathbf{p}$ $u_4=(\mathbf{p}\cdot\mathbf{p}-p_4^2)(\mathbf{p}\cdot\mathbf{p}+p_4^2)^{-1}$
gilmore-lie-groups_ E08818 (14.54)	256	■ 1.000	$S\left(8;\mathbb{R}\right)\cap SO\left(4,4\right)=SU\left(2,2\right)\simeq SO\left(4,2\right)$	$Sp(8;\mathbb{R})\cap SO(4,4)=SU(2,2)\simeq SO(4,2)$	$Sp(8;\mathbb{R})\cap SO(4,4)=SU(2,2)\simeq SO(4,2)$
gilmore-lie-groups_ E08833 (14.67)	260	■ 1.000	$e^{\{\theta\}_M}\binom{\Gamma_5}{\Gamma_4}e^{-\theta M_4}=\left[\begin{array}{cc}\cosh\theta&-\sinh\theta\\-\sinh\theta&\cosh\theta\end{array}\right]\binom{\Gamma_5}{\Gamma_4}$	$e^{\theta M_4}\binom{\Gamma_5}{\Gamma_4}e^{-\theta M_4}=\left[\begin{array}{cc}\cosh\theta&-\sinh\theta\\-\sinh\theta&\cosh\theta\end{array}\right]\binom{\Gamma_5}{\Gamma_4}$	$e^{\theta M_4}\binom{\Gamma_5}{\Gamma_4}e^{-\theta M_4}=\left[\begin{array}{cc}\cosh\theta&-\sinh\theta\\-\sinh\theta&\cosh\theta\end{array}\right]\binom{\Gamma_5}{\Gamma_4}$
gilmore-lie-groups_ E08886 (15.19)	278	■ 1.000	$\Lambda T_{\mu\nu}(x)=T_{\mu^{\prime}\nu^{\prime}}(x\Lambda^{-1})\Lambda_{\mu^{\prime}\mu}\Lambda_{\nu^{\prime}\nu}$	$\Lambda T_{\mu\nu}(x)=T_{\mu^{\prime}\nu^{\prime}}(x\Lambda^{-1})\Lambda_{\mu^{\prime}\mu}\Lambda_{\nu^{\prime}\nu}$	$\Lambda T_{\mu\nu}(x)=T_{\mu^{\prime}\nu^{\prime}}(x\Lambda^{-1})\Lambda_{\mu^{\prime}\mu}\Lambda_{\nu^{\prime}\nu}$
gilmore-lie-groups_ E0921 (15.55)	286	■ 1.000	$\left\{J_3^{(j)}k_3^0+jk_4^0I_{2j+1}\right\}\left\langle k^0;\begin{array}{c}0\\0\end{array}\begin{array}{c}j'\\m'\end{array}\middle \psi\right\rangle=0$	$\left\{J_3^{(j)}k_3^0+jk_4^0I_{2j+1}\right\}\left\langle k^0;\begin{array}{c}0\\0\end{array}\begin{array}{c}j'\\m'\end{array}\middle \psi\right\rangle=0$	$\left\{J_3^{(j)}k_3^0+jk_4^0I_{2j+1}\right\}\left\langle k^0;\begin{array}{c}0\\0\end{array}\begin{array}{c}j'\\m'\end{array}\middle \psi\right\rangle=0$
gilmore-lie-groups_ E09946	291	■ 1.000	$\begin{aligned} & \nabla\cdot\mathbf{B}(\mathbf{x},t)=4\pi\rho_m(\mathbf{x},t)+C_m(\mathbf{x}) \\ & \nabla\cdot\mathbf{E}(\mathbf{x},t)=4\pi\rho_e(\mathbf{x},t)+C_e(\mathbf{x}) \end{aligned}$	$\nabla\cdot\mathbf{B}(\mathbf{x},t)=4\pi\rho_m(\mathbf{x},t)+C_m(\mathbf{x})$ $\nabla\cdot\mathbf{E}(\mathbf{x},t)=4\pi\rho_e(\mathbf{x},t)+C_e(\mathbf{x})$	$\nabla\cdot\mathbf{B}(\mathbf{x},t)=4\pi\rho_m(\mathbf{x},t)+C_m(\mathbf{x})$ $\nabla\cdot\mathbf{E}(\mathbf{x},t)=4\pi\rho_e(\mathbf{x},t)+C_e(\mathbf{x})$
gilmore-lie-groups_ E09976 (16.13)	302	■ 1.000	$\xi(x,y)=\sum_{i,j}\xi_{ij}x^iy^j\quad 0\leq i,j,\quad i+j\leq d_\xi$	$\xi(x,y)=\sum_{i,j}\xi_{ij}x^iy^j\quad 0\leq i,j,\quad i+j\leq d_\xi$	$\xi(x,y)=\sum_{i,j}\xi_{ij}x^iy^j\quad 0\leq i,j,\quad i+j\leq d_\xi$
gilmore-lie-groups_ E01009 (16.52)	314	■ 1.000	$\begin{aligned} & X_{ij}=x^i\partial_j-x^j\partial_i \\ & Z=x^i\partial_i+(2-n)u\frac{\partial}{\partial u} \end{aligned}$	$X_{ij}=x^i\partial_j-x^j\partial_i$ $Z=x^i\partial_i+(2-n)u\frac{\partial}{\partial u}$	$X_{ij}=x^i\partial_j-x^j\partial_i$ $Z=x^i\partial_i+(2-n)u\frac{\partial}{\partial u}$
gilmore-lie-groups_ E00001	18	■ 1.000	$\left(g_i\cdot g_j\right)\cdot g_k=g_i\cdot\left(g_j\cdot g_k\right)$	$(g_i\cdot g_j)\cdot g_k=g_i\cdot (g_j\cdot g_k)$	$(g_i\cdot g_j)\cdot g_k=g_i\cdot (g_j\cdot g_k)$
gilmore-lie-groups_ E00002	18	■ 1.000	$g_i\cdot I=g_i=I\cdot g_i$	$g_i\cdot I=g_i=I\cdot g_i$	$g_i\cdot I=g_i=I\cdot g_i$
gilmore-lie-groups_ E00003	18	■ 1.000	$g_i\cdot g_i^{-1}=I=g_i^{-1}\cdot g_i$	$g_i\cdot g_i^{-1}=I=g_i^{-1}\cdot g_i$	$g_i\cdot g_i^{-1}=I=g_i^{-1}\cdot g_i$
gilmore-lie-groups_ E00005 (1.2)	18	■ 1.000	$\left[\begin{array}{c}z_1\\z_2\\\vdots\\z_n\end{array}\right]\longrightarrow\left[\begin{array}{c}z_{i_1}\\z_{i_2}\\\vdots\\z_{i_n}\end{array}\right]$	$\left[\begin{array}{c}z_1\\z_2\\\vdots\\z_n\end{array}\right]\longrightarrow\left[\begin{array}{c}z_{i_1}\\z_{i_2}\\\vdots\\z_{i_n}\end{array}\right]$	$\left[\begin{array}{c}z_1\\z_2\\\vdots\\z_n\end{array}\right]\longrightarrow\left[\begin{array}{c}z_{i_1}\\z_{i_2}\\\vdots\\z_{i_n}\end{array}\right]$

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gilmore-lie-groups_ EQ0007 (1.4)	19	■ 1.000	$\left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array}\right] \xrightarrow{(123)} \left[\begin{array}{l} z_2 \\ z_3 \\ z_1 \end{array}\right]$	$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \xrightarrow{(123)} \begin{bmatrix} z_2 \\ z_3 \\ z_1 \end{bmatrix}$	$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \xrightarrow{(123)} \begin{bmatrix} z_2 \\ z_3 \\ z_1 \end{bmatrix}$
gilmore-lie-groups_ EQ0008 (1.5)	19	■ 1.000	$\left[\begin{array}{l} z_2 \\ z_3 \\ z_1 \end{array}\right] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$	$\begin{bmatrix} z_2 \\ z_3 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$	$\begin{bmatrix} z_2 \\ z_3 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$
gilmore-lie-groups_ EQ0011 (1.7)	20	■ 1.000	$g_i \cdot g_j = g_j \cdot g_i$	$g_i \cdot g_j = g_j \cdot g_i$	$g_i \cdot g_j = g_j \cdot g_i$
gilmore-lie-groups_ EQ0013 (1.9)	20	■ 1.000	$g H_1 g^{-1} = H_2$	$g H_1 g^{-1} = H_2$	$g H_1 g^{-1} = H_2$
gilmore-lie-groups_ EQ0014 (1.10)	20	■ 1.000	$(23) S_2(12)(23)^{-1} = (23)\{I, (12)\}(23)^{-1} = \{I, (13)\} = S_2(13)$	$(23)S_2(12)(23)^{-1} = (23)\{I, (12)\}(23)^{-1} = \{I, (13)\} = S_2(13)$	$(23)S_2(12)(23)^{-1} = (23)\{I, (12)\}(23)^{-1} = \{I, (13)\} = S_2(13)$
gilmore-lie-groups_ EQ0015 (1.11)	20	■ 1.000	$(23) A_3(23)^{-1} = (23)\{I, (123), (321)\}(23)^{-1} = \{I, (321), (123)\} = A_3$	$(23)A_3(23)^{-1} = (23)\{I, (123), (321)\}(23)^{-1} = \{I, (321), (123)\} = A_3$	$(23)A_3(23)^{-1} = (23)\{I, (123), (321)\}(23)^{-1} = \{I, (321), (123)\} = A_3$
gilmore-lie-groups_ EQ0020 (1.16)	23	■ 1.000	$G = G_0 \supset G_1 \supset \dots \supset G_\omega = I$	$G = G_0 \supset G_1 \supset \dots \supset G_\omega = I$	$G = G_0 \supset G_1 \supset \dots \supset G_\omega = I$
gilmore-lie-groups_ EQ0021 (1.17)	24	■ 1.000	$\begin{aligned} &\left(z - r_1\right)\left(z - r_2\right) = z^2 - I_1 z + I_2 = 0 \\ &I_1 = r_1 + r_2 \\ &I_2 = r_1 r_2 \end{aligned}$	$\begin{aligned} (z - r_1)(z - r_2) &= z^2 - I_1 z + I_2 = 0 \\ I_1 &= r_1 + r_2 \\ I_2 &= r_1 r_2 \end{aligned}$	$\begin{aligned} (z - r_1)(z - r_2) &= z^2 - I_1 z + I_2 = 0 \\ I_1 &= r_1 + r_2 \\ I_2 &= r_1 r_2 \end{aligned}$
gilmore-lie-groups_ EQ0023 (1.19)	24	■ 1.000	$\left[\begin{array}{l} u_1 \\ u_2 \end{array}\right] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} r_1 + r_2 \\ r_1 - r_2 \end{bmatrix}$	$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} r_1 + r_2 \\ r_1 - r_2 \end{bmatrix}$	$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} r_1 + r_2 \\ r_1 - r_2 \end{bmatrix}$
gilmore-lie-groups_ EQ0025 (1.21)	24	■ 1.000	$\begin{aligned} &(r_1 - r_2)^2 = r_1^2 - 2r_1 r_2 + r_2^2 \\ &= r_1^2 + 2r_1 r_2 + r_2^2 - 4r_1 r_2 = I_1^2 - 4I_2 = D \end{aligned}$	$\begin{aligned} (r_1 - r_2)^2 &= r_1^2 - 2r_1 r_2 + r_2^2 \\ &= r_1^2 + 2r_1 r_2 + r_2^2 - 4r_1 r_2 = I_1^2 - 4I_2 = D \end{aligned}$	$\begin{aligned} (r_1 - r_2)^2 &= r_1^2 - 2r_1 r_2 + r_2^2 \\ &= r_1^2 + 2r_1 r_2 + r_2^2 - 4r_1 r_2 = I_1^2 - 4I_2 = D \end{aligned}$
gilmore-lie-groups_ EQ0027 (1.23)	25	■ 1.000	$\left[\begin{array}{l} r_1 \\ r_2 \end{array}\right] = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} I_1 \\ \pm \sqrt{D} \end{bmatrix}$	$\begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} I_1 \\ \pm \sqrt{D} \end{bmatrix}$	$\begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} I_1 \\ \pm \sqrt{D} \end{bmatrix}$
gilmore-lie-groups_ EQ0028 (1.24)	25	■ 1.000	$r_1, r_2 = \frac{1}{2} (I_1 \pm \sqrt{D})$	$r_1, r_2 = \frac{1}{2} (I_1 \pm \sqrt{D})$	$r_1, r_2 = \frac{1}{2} (I_1 \pm \sqrt{D})$
gilmore-lie-groups_ EQ0029 (1.25)	25	■ 1.000	$z = x + \frac{1}{2} I_1$	$z = x + \frac{1}{2} I_1$	$z = x + \frac{1}{2} I_1$

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gilmore-lie-groups_ EQ0030 (1.26)	25	■ 1.000	$\begin{aligned} x^2 - I_1 x + I_2 &= x^2 + I_2' & \& \\ = x^2 + I_2' &= 0 \quad \& \quad I_1 = 0 \\ & & \& \quad I_2' = I_2 - \left(\frac{1}{2}I_1\right)^2 \end{aligned}$	$\begin{aligned} x^2 - I_1' x + I_2' &= x^2 + I_2' = 0 \\ I_1' &= 0 \\ I_2' &= I_2 - \left(\frac{1}{2}I_1\right)^2 \end{aligned}$	$\begin{aligned} x^2 - I_1' x + I_2' &= x^2 + I_2' = 0 \\ I_1' &= 0 \\ I_2' &= I_2 - \left(\frac{1}{2}I_1\right)^2 \end{aligned}$
gilmore-lie-groups_ EQ0031 (1.27)	25	■ 1.000	$x = \pm \sqrt{-I_2'}$	$x = \pm \sqrt{-I_2'}$	$x = \pm \sqrt{-I_2'}$
gilmore-lie-groups_ EQ0032 (1.28)	25	■ 1.000	$r_{1,2} = \frac{1}{2} \left(I_1 \pm \sqrt{I_1^2 - 4I_2} \right)$	$r_{1,2} = \frac{1}{2} \left(I_1 \pm \sqrt{I_1^2 - 4I_2} \right)$	$r_{1,2} = \frac{1}{2} \left(I_1 \pm \sqrt{I_1^2 - 4I_2} \right)$
gilmore-lie-groups_ EQ0033 (1.29)	25	■ 1.000	$\begin{aligned} &\& \left(z - s_1 \right) \left(z - s_2 \right) \left(z - s_3 \right) = z^3 - I_1 z^2 + I_2 z - I_3 = 0 \\ &I_1 = s_1 + s_2 + s_3 \\ &I_2 = s_1 s_2 + s_1 s_3 + s_2 s_3 \\ &I_3 = s_1 s_2 s_3 \end{aligned}$	$\begin{aligned} (z - s_1)(z - s_2)(z - s_3) &= z^3 - I_1 z^2 + I_2 z - I_3 = 0 \\ I_1 &= s_1 + s_2 + s_3 \\ I_2 &= s_1 s_2 + s_1 s_3 + s_2 s_3 \\ I_3 &= s_1 s_2 s_3 \end{aligned}$	$\begin{aligned} (z - s_1)(z - s_2)(z - s_3) &= z^3 - I_1 z^2 + I_2 z - I_3 = 0 \\ I_1 &= s_1 + s_2 + s_3 \\ I_2 &= s_1 s_2 + s_1 s_3 + s_2 s_3 \\ I_3 &= s_1 s_2 s_3 \end{aligned}$
gilmore-lie-groups_ EQ0034 (1.31)	26	■ 1.000	$\omega^3 = +1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$	$\omega^3 = +1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$	$\omega^3 = +1 \quad \omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}$
gilmore-lie-groups_ EQ0036 (1.33)	26	■ 1.000	$\begin{aligned} &\& (123)^{-1} v_2 = (321) v_2 = (321) (s_1 + \omega s_2 + \omega^2 s_3) \\ &= s_3 + \omega s_1 + \omega^2 s_2 = \omega (s_1 + \omega s_2 + \omega^2 s_3) = \omega v_2 \end{aligned}$	$\begin{aligned} (123)^{-1} v_2 &= (321) v_2 = (321) (s_1 + \omega s_2 + \omega^2 s_3) \\ &= s_3 + \omega s_1 + \omega^2 s_2 = \omega (s_1 + \omega s_2 + \omega^2 s_3) = \omega v_2 \end{aligned}$	$\begin{aligned} (123)^{-1} v_2 &= (321) v_2 = (321) (s_1 + \omega s_2 + \omega^2 s_3) \\ &= s_3 + \omega s_1 + \omega^2 s_2 = \omega (s_1 + \omega s_2 + \omega^2 s_3) = \omega v_2 \end{aligned}$
gilmore-lie-groups_ EQ0037 (1.34)	27	■ 1.000	$v_1 = I_1$	$v_1 = I_1$	$v_1 = I_1$
gilmore-lie-groups_ EQ0038 (1.35)	27	■ 1.000	$\begin{aligned} &\& (12) \left(v_2 \right)^3 = (12) (s_1 + \omega s_2 + \omega^2 s_3)^3 = (s_2 + \omega s_1 + \omega^2 s_3)^3 \\ &= \omega^3 (s_1 + \omega^2 s_2 + \omega s_3)^3 = (v_3)^3 \\ &(12) (v_3)^3 = (12) (s_1 + \omega^2 s_2 + \omega s_3)^3 = (s_2 + \omega^2 s_1 + \omega s_3)^3 \\ &= \omega^6 (s_1 + \omega s_2 + \omega^2 s_3)^3 = (v_2)^3 \end{aligned}$	$\begin{aligned} (12) (v_2)^3 &= (12) (s_1 + \omega s_2 + \omega^2 s_3)^3 = (s_2 + \omega s_1 + \omega^2 s_3)^3 \\ &= \omega^3 (s_1 + \omega^2 s_2 + \omega s_3)^3 = (v_3)^3 \\ (12) (v_3)^3 &= (12) (s_1 + \omega^2 s_2 + \omega s_3)^3 = (s_2 + \omega^2 s_1 + \omega s_3)^3 \\ &= \omega^6 (s_1 + \omega s_2 + \omega^2 s_3)^3 = (v_2)^3 \end{aligned}$	$\begin{aligned} (12)(v_2)^3 &= (12)(s_1 + \omega s_2 + \omega^2 s_3)^3 = (s_2 + \omega s_1 + \omega^2 s_3)^3 \\ &= \omega^3 (s_1 + \omega^2 s_2 + \omega s_3)^3 = (v_3)^3 \\ (12)(v_3)^3 &= (12)(s_1 + \omega^2 s_2 + \omega s_3)^3 = (s_2 + \omega^2 s_1 + \omega s_3)^3 \\ &= \omega^6 (s_1 + \omega s_2 + \omega^2 s_3)^3 = (v_2)^3 \end{aligned}$

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gilmore-lie-groups_ EQ0039 (1.36)	27	■ 1.000	\begin{aligned} \left(x-v_2^3\right)\left(x-v_3^3\right) &=x^2-J_1 x+J_2=0 \\ &\& J_1=v_2^3+v_3^3 \\ &=v_2^3+v_3^3 \\ &\& J_2=v_2^3 v_3^3 \end{aligned}	$\begin{aligned} \left(x-v_2^3\right)\left(x-v_3^3\right) &=x^2-J_1 x+J_2=0 \\ J_1 &=v_2^3+v_3^3 \\ J_2 &=v_2^3 v_3^3 \end{aligned}$	$\begin{aligned} \left(x-v_2^3\right)\left(x-v_3^3\right) &=x^2-J_1 x+J_2=0 \\ J_1 &=v_2^3+v_3^3 \\ J_2 &=v_2^3 v_3^3 \end{aligned}$
gilmore-lie-groups_ EQ0041 (1.38)	27	■ 1.000	$z=y+\frac{1}{3} I_1$	$z=y+\frac{1}{3} I_1$	$z=y+\frac{1}{3} I_1$
gilmore-lie-groups_ EQ0043 (1.40)	28	■ 1.000	\begin{aligned} &\& J_1=v_2^3+v_3^3=-27 I_3^{\prime} \\ &\& J_2=v_2^3 v_3^3=-27 I_2^{\prime 3} \end{aligned}	$\begin{aligned} J_1 &=v_2^3+v_3^3=-27 I_3^{\prime} \\ J_2 &=v_2^3 v_3^3=-27 I_2^{\prime 3} \end{aligned}$	$\begin{aligned} J_1 &=v_2^3+v_3^3=-27 I_3^{\prime} \\ J_2 &=v_2^3 v_3^3=-27 I_2^{\prime 3} \end{aligned}$
gilmore-lie-groups_ EQ0044 (1.41)	28	■ 1.000	$x^2-\left(-27 I_3^{\prime}\right) x+\left(-27 I_2^{\prime 3}\right)=0$	$x^2-\left(-27 I_3^{\prime}\right) x+\left(-27 I_2^{\prime 3}\right)=0$	$x^2-\left(-27 I_3^{\prime}\right) x+\left(-27 I_2^{\prime 3}\right)=0$
gilmore-lie-groups_ EQ0049 (1.45)	29	■ 1.000	\begin{aligned} \left(z-t_1\right)\left(z-t_2\right)\left(z-t_3\right)\left(z-t_4\right) &=z^4-I_1 z^3+I_2 z^2-I_3 z+I_4=0 \\ &\& I_1=t_1+t_2+t_3+t_4 \\ &=z^4-I_1 z^3+I_2 z^2-I_3 z+I_4=0 \\ &\& I_2=t_1 t_2+t_1 t_3+t_1 t_4+t_2 t_3+t_2 t_4+t_3 t_4 \\ &\& I_3=t_1 t_2 t_3+t_1 t_2 t_4+t_1 t_3 t_4+t_2 t_3 t_4 \\ &\& I_4=t_1 t_2 t_3 t_4 \end{aligned}	$\begin{aligned} \left(z-t_1\right)\left(z-t_2\right)\left(z-t_3\right)\left(z-t_4\right) &=z^4-I_1 z^3+I_2 z^2-I_3 z+I_4=0 \\ I_1 &=t_1+t_2+t_3+t_4 \\ I_2 &=t_1 t_2+t_1 t_3+t_1 t_4+t_2 t_3+t_2 t_4+t_3 t_4 \\ I_3 &=t_1 t_2 t_3+t_1 t_2 t_4+t_1 t_3 t_4+t_2 t_3 t_4 \\ I_4 &=t_1 t_2 t_3 t_4 \end{aligned}$	$\begin{aligned} \left(z-t_1\right)\left(z-t_2\right)\left(z-t_3\right)\left(z-t_4\right) &=z^4-I_1 z^3+I_2 z^2-I_3 z+I_4=0 \\ I_1 &=t_1+t_2+t_3+t_4 \\ I_2 &=t_1 t_2+t_1 t_3+t_1 t_4+t_2 t_3+t_2 t_4+t_3 t_4 \\ I_3 &=t_1 t_2 t_3+t_1 t_2 t_4+t_1 t_3 t_4+t_2 t_3 t_4 \\ I_4 &=t_1 t_2 t_3 t_4 \end{aligned}$
gilmore-lie-groups_ EQ0051	29	■ 1.000	$S_4 \supset A_4 \supset V_4 \supset I$	$S_4 \supset A_4 \supset V_4 \supset I$	$S_4 \supset A_4 \supset V_4 \supset I$
gilmore-lie-groups_ EQ0052 (1.48)	30	■ 1.000	\left[\begin{array}{l} w_1 \\ w_2 \\ w_3 \\ w_4 \end{array}\right]=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{array}\right]\left[\begin{array}{l} t_1 \\ t_2 \\ t_3 \\ t_4 \end{array}\right]=\left[\begin{array}{l} t_1+t_2+t_3+t_4 \\ t_1+t_2-t_3-t_4 \\ t_1-t_2+t_3-t_4 \\ t_1-t_2-t_3+t_4 \end{array}\right]	$\left[\begin{array}{l} w_1 \\ w_2 \\ w_3 \\ w_4 \end{array}\right]=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{array}\right]\left[\begin{array}{l} t_1 \\ t_2 \\ t_3 \\ t_4 \end{array}\right]=\left[\begin{array}{l} t_1+t_2+t_3+t_4 \\ t_1+t_2-t_3-t_4 \\ t_1-t_2+t_3-t_4 \\ t_1-t_2-t_3+t_4 \end{array}\right]$	$\left[\begin{array}{l} w_1 \\ w_2 \\ w_3 \\ w_4 \end{array}\right]=\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{array}\right]\left[\begin{array}{l} t_1 \\ t_2 \\ t_3 \\ t_4 \end{array}\right]=\left[\begin{array}{l} t_1+t_2+t_3+t_4 \\ t_1+t_2-t_3-t_4 \\ t_1-t_2+t_3-t_4 \\ t_1-t_2-t_3+t_4 \end{array}\right]$

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gilmore-lie-groups_ E00053 (1.49)	30	■ 1.000	$\begin{aligned} & \texttt{\backslash begin{aligned} \& w_{2}^{2}=\left(t_{1}^{\backslash prime}+t_{2}^{\backslash prime}-t_{3}^{\backslash prime}-t_{4}^{\backslash prime}\right)^{2}=2^{2} \backslash left(t_{1}^{\backslash prime}+t_{2}^{\backslash prime}\right)^{2}=-4 \backslash left(t_{1}^{\backslash prime}+t_{2}^{\backslash prime}\right)\backslash left(t_{3}^{\backslash prime}+t_{4}^{\backslash prime}\right)\backslash right) \& w_{3}^{2}= \backslash left(t_{1}^{\backslash prime}-t_{2}^{\backslash prime}+t_{3}^{\backslash prime}-t_{4}^{\backslash prime}\right)^{2}=2^{2} \backslash left(t_{1}^{\backslash prime}+t_{3}^{\backslash prime}\right)^{2}=-4 \backslash left(t_{1}^{\backslash prime}+t_{3}^{\backslash prime}\right)(t_{2}^{\backslash prime}+t_{4}^{\backslash prime}) \backslash right) \& w_{4}^{2}= \backslash left(t_{1}^{\backslash prime}-t_{2}^{\backslash prime}-t_{3}^{\backslash prime}+t_{4}^{\backslash prime}\right)^{2}=2^{2} \backslash left(t_{1}^{\backslash prime}+t_{4}^{\backslash prime}\right)^{2}=-4 \backslash left(t_{1}^{\backslash prime}+t_{4}^{\backslash prime}\right)(t_{2}^{\backslash prime}+t_{3}^{\backslash prime}) \backslash right) \& \end{aligned}}$	$\begin{aligned} w_2^2 &= (t'_1 + t'_2 - t'_3 - t'_4)^2 = 2^2 (t'_1 + t'_2)^2 = -4 (t'_1 + t'_2) (t'_3 + t'_4) \\ w_3^2 &= (t'_1 - t'_2 + t'_3 - t'_4)^2 = 2^2 (t'_1 + t'_3)^2 = -4 (t'_1 + t'_3) (t'_2 + t'_4) \\ w_4^2 &= (t'_1 - t'_2 - t'_3 + t'_4)^2 = 2^2 (t'_1 + t'_4)^2 = -4 (t'_1 + t'_4) (t'_2 + t'_3) \end{aligned}$	$\begin{aligned} w_2^2 &= (t'_1 + t'_2 - t'_3 - t'_4)^2 = 2^2 (t'_1 + t'_2)^2 = -4 (t'_1 + t'_2) (t'_3 + t'_4) \\ w_3^2 &= (t'_1 - t'_2 + t'_3 - t'_4)^2 = 2^2 (t'_1 + t'_3)^2 = -4 (t'_1 + t'_3) (t'_2 + t'_4) \\ w_4^2 &= (t'_1 - t'_2 - t'_3 + t'_4)^2 = 2^2 (t'_1 + t'_4)^2 = -4 (t'_1 + t'_4) (t'_2 + t'_3) \end{aligned}$
gilmore-lie-groups_ E00054 (1.50)	31	■ 1.000	$\begin{aligned} & \texttt{\backslash begin{aligned} \backslash left(y-w_{2}^{2}\backslash right) \backslash left(y-w_{3}^{2}\backslash right)\backslash left(y-w_{4}^{2}\backslash right) \& =y^{\backslash 3}-J_{\backslash 1} y^{\backslash 2}+J_{\backslash 2} y-J_{\backslash 3}=0 \& \& J_{\backslash 1} \& =w_{\backslash 2}^{2}+w_{\backslash 3}^{2}+w_{\backslash 4}^{2} \& \& J_{\backslash 2} \& =w_{\backslash 2}^{2} w_{\backslash 3}^{2}+w_{\backslash 2}^{2} w_{\backslash 4}^{2}+w_{\backslash 3}^{2} w_{\backslash 4}^{2} \& \& J_{\backslash 3} \& =w_{\backslash 2}^{2} w_{\backslash 3}^{2} w_{\backslash 4}^{2} \& \& \end{aligned}}$	$\begin{aligned} (y - w_2^2) (y - w_3^2) (y - w_4^2) &= y^3 - J_1 y^2 + J_2 y - J_3 = 0 \\ J_1 &= w_2^2 + w_3^2 + w_4^2 \\ J_2 &= w_2^2 w_3^2 + w_2^2 w_4^2 + w_3^2 w_4^2 \\ J_3 &= w_2^2 w_3^2 w_4^2 \end{aligned}$	$\begin{aligned} (y - w_2^2)(y - w_3^2)(y - w_4^2) &= y^3 - J_1 y^2 + J_2 y - J_3 = 0 \\ J_1 &= w_2^2 + w_3^2 + w_4^2 \\ J_2 &= w_2^2 w_3^2 + w_2^2 w_4^2 + w_3^2 w_4^2 \\ J_3 &= w_2^2 w_3^2 w_4^2 \end{aligned}$
gilmore-lie-groups_ E00055 (1.51)	31	■ 1.000	$\begin{aligned} & \texttt{\backslash begin{aligned} \& J_{\backslash 1}=(-4)^{\backslash 1} \backslash left(2 I_{\backslash 2}^{\backslash prime} \backslash right) \& \& J_{\backslash 2}=(-4)^{\backslash 2} \backslash left(I_{\backslash 2}^{\backslash prime \& 2}-4 I_{\backslash 4}^{\backslash prime} \backslash right) \& \& J_{\backslash 3}=(-4)^{\backslash 3} \backslash left(-I_{\backslash 3}^{\backslash prime \& 2} \backslash right) \& \end{aligned}}$	$\begin{aligned} J_1 &= (-4)^1 (2I'_2) \\ J_2 &= (-4)^2 (I'^2_2 - 4I'_4) \\ J_3 &= (-4)^3 (-I'^2_3) \end{aligned}$	$\begin{aligned} J_1 &= (-4)^1 (2I'_2) \\ J_2 &= (-4)^2 (I'^2_2 - 4I'_4) \\ J_3 &= (-4)^3 (-I'^2_3) \end{aligned}$
gilmore-lie-groups_ E00058 (1.54)	32	■ 1.000	$\texttt{S_{\backslash 5} \supset A_{\backslash 5} \supset I}$	$S_5 \supset A_5 \supset I$	$S_5 \supset A_5 \supset I$
gilmore-lie-groups_ E00059 (1.55)	32	■ 1.000	$\begin{aligned} & \texttt{\backslash begin{aligned} (z+2)(z+1)(z-2)(z-5) \& =z^{\backslash 4}-4 z^{\backslash 3}-9 z^{\backslash 2}+16 z+20=0 \& \& I_{\backslash 1} \& =4 \& \& I_{\backslash 2} \& =-9 \& \& I_{\backslash 3} \& =-16 \& \& I_{\backslash 4} \& =20 \& \end{aligned}}$	$\begin{aligned} (z + 2)(z + 1)(z - 2)(z - 5) &= z^4 - 4z^3 - 9z^2 + 16z + 20 = 0 \\ I_1 &= 4 \\ I_2 &= -9 \\ I_3 &= -16 \\ I_4 &= 20 \end{aligned}$	$\begin{aligned} (z + 2)(z + 1)(z - 2)(z - 5) &= z^4 - 4z^3 - 9z^2 + 16z + 20 = 0 \\ I_1 &= 4 \\ I_2 &= -9 \\ I_3 &= -16 \\ I_4 &= 20 \end{aligned}$
gilmore-lie-groups_ E00060	32	■ 1.000	$\texttt{z=z^{\backslash prime}+\frac{\backslash 1}{\backslash 4} I_{\backslash 1}=z^{\backslash prime}+1}$	$z = z' + \frac{1}{4} I_1 = z' + 1$	$z = z' + \frac{1}{4} I_1 = z' + 1$

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gilmore-lie-groups_ EQ0062 (1.57)	33	■ 1.000	$\left[\begin{array}{l} w_1 \\ w_2 \\ w_3 \\ w_4 \end{array}\right] = \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{array}\right] \left[\begin{array}{l} t_1' \\ t_2' \\ t_3' \\ t_4' \end{array}\right] \rightarrow \left[\begin{array}{l} 0 \\ -10 \\ -4 \\ 2 \end{array}\right]$		$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} t_1' \\ t_2' \\ t_3' \\ t_4' \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -10 \\ -4 \\ 2 \end{bmatrix}$
gilmore-lie-groups_ EQ0065 (1.60)	33	■ 1.000	$\left(y-2^2\right)\left(y-4^2\right)\left(y-10^2\right)=y^3-120y^2+2064y-6400=0$	$(y-2^2)(y-4^2)(y-10^2)=y^3-120y^2+2064y-6400=0$	$(y-2^2)(y-4^2)(y-10^2)=y^3-120y^2+2064y-6400=0$
gilmore-lie-groups_ EQ0068 (1.63)	33	■ 1.000	$\left(y^{\prime}+36\right)\left(y^{\prime}+24\right)\left(y^{\prime}-60\right)=0$	$(y'+36)(y'+24)(y'-60)=0$	$(y'+36)(y'+24)(y'-60)=0$
gilmore-lie-groups_ EQ0074 (1.69)	34	■ 1.000	$\begin{aligned} & \left[\begin{array}{l} x_1 \\ x_2 \end{array}\right] = \frac{1}{2} \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{l} K_1=1399680 \\ 2\sqrt{-K_2}=i \times 290304\sqrt{3} \end{array}\right] \\ & = 699840 \pm i \times 145152\sqrt{3} \end{aligned}$	$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} K_1 = 1399680 \\ 2\sqrt{-K_2} = i \times 290304\sqrt{3} \end{bmatrix} = 699840 \pm i \times 145152\sqrt{3}$	$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} K_1 = 1399680 \\ 2\sqrt{-K_2} = i \times 290304\sqrt{3} \end{bmatrix} = 699840 \pm i \times 145152\sqrt{3}$
gilmore-lie-groups_ EQ0076 (1.71)	35	■ 1.000	$\begin{aligned} & v_2 = -36 + i48\sqrt{3} \\ & v_3 = -36 - i48\sqrt{3} \end{aligned}$	$v_2 = -36 + i48\sqrt{3}$ $v_3 = -36 - i48\sqrt{3}$	$v_2 = -36 + i48\sqrt{3}$ $v_3 = -36 - i48\sqrt{3}$
gilmore-lie-groups_ EQ0078 (1.74)	35	■ 1.000	$\left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \\ x_4 \end{array}\right] = \frac{1}{4} \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{array}\right] \left[\begin{array}{l} I_1=4 \\ w_2=4 \\ w_3=10 \\ w_4=2 \end{array}\right] = \left[\begin{array}{l} 20/4=+5 \\ -4/4=-1 \\ 8/4=+2 \\ -8/4=-2 \end{array}\right]$	$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} I_1 = 4 \\ w_2 = 4 \\ w_3 = 10 \\ w_4 = 2 \end{bmatrix} = \begin{bmatrix} 20/4 = +5 \\ -4/4 = -1 \\ 8/4 = +2 \\ -8/4 = -2 \end{bmatrix}$	$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} I_1 = 4 \\ w_2 = 4 \\ w_3 = 10 \\ w_4 = 2 \end{bmatrix} = \begin{bmatrix} 20/4 = +5 \\ -4/4 = -1 \\ 8/4 = +2 \\ -8/4 = -2 \end{bmatrix}$
gilmore-lie-groups_ EQ0080	36	■ 1.000	$\begin{aligned} & e^{i3\theta} = (e^{i\theta})^3 \\ & \cos(3\theta) + i\sin(3\theta) = (\cos^3(\theta) - 3\cos(\theta)\sin^2(\theta)) + i(3\cos^2(\theta)\sin(\theta) - \sin^3(\theta)) \end{aligned}$	$e^{i3\theta} = (e^{i\theta})^3$ $\cos(3\theta) + i\sin(3\theta) = (\cos^3(\theta) - 3\cos(\theta)\sin^2(\theta)) + i(3\cos^2(\theta)\sin(\theta) - \sin^3(\theta))$	$e^{i3\theta} = (e^{i\theta})^3$ $\cos(3\theta) + i\sin(3\theta) = (\cos^3(\theta) - 3\cos(\theta)\sin^2(\theta)) + i(3\cos^2(\theta)\sin(\theta) - \sin^3(\theta))$
gilmore-lie-groups_ EQ0081	37	■ 1.000	$\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta)$	$\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta)$	$\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta)$
gilmore-lie-groups_ EQ0082	37	■ 1.000	$4\cos^3(\theta) - 3\cos(\theta) - \cos(3\theta) = 0$	$4\cos^3(\theta) - 3\cos(\theta) - \cos(3\theta) = 0$	$4\cos^3(\theta) - 3\cos(\theta) - \cos(3\theta) = 0$
gilmore-lie-groups_ EQ0083	38	■ 1.000	$(g_i \circ g_j) \circ g_k = g_i \circ (g_j \circ g_k)$	$(g_i \circ g_j) \circ g_k = g_i \circ (g_j \circ g_k)$	$(g_i \circ g_j) \circ g_k = g_i \circ (g_j \circ g_k)$
gilmore-lie-groups_ EQ0084	38	■ 1.000	$g_i \circ e = g_i = e \circ g_i$	$g_i \circ e = g_i = e \circ g_i$	$g_i \circ e = g_i = e \circ g_i$
gilmore-lie-groups_ EQ0085	38	■ 1.000	$g_i \circ g_i^{-1} = e = g_i^{-1} \circ g_i$	$g_i \circ g_i^{-1} = e = g_i^{-1} \circ g_i$	$g_i \circ g_i^{-1} = e = g_i^{-1} \circ g_i$

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gilmore-lie-groups_ EQ0088	39	■ 1.000	$e \longrightarrow I_2 = \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right]$	$e \longrightarrow I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$e \longrightarrow I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
gilmore-lie-groups_ EQ0089	39	■ 1.000	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right] \longrightarrow \left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right]^{-1} = \frac{1}{A_{11}A_{22} - A_{12}A_{21}} \begin{bmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{bmatrix}$	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right] \rightarrow \left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right]^{-1} = \frac{1}{A_{11}A_{22} - A_{12}A_{21}} \begin{bmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{bmatrix}$	$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \rightarrow \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^{-1} = \frac{1}{A_{11}A_{22} - A_{12}A_{21}} \begin{bmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{bmatrix}$
gilmore-lie-groups_ EQ0090 (2.3)	40	■ 1.000	$\left(x_1, x_2, x_3 \right) \longrightarrow \left[\begin{array}{c} x_1 \\ x_3 \end{array} \right] \quad x_1 \neq 0$	$(x_1, x_2, x_3) \longrightarrow \begin{bmatrix} x_1 & x_2 \\ x_3 & \frac{1+x_2x_3}{x_1} \end{bmatrix} \quad x_1 \neq 0$	$(x_1, x_2, x_3) \longrightarrow \begin{bmatrix} x_1 & x_2 \\ x_3 & \frac{1+x_2x_3}{x_1} \end{bmatrix} \quad x_1 \neq 0$
gilmore-lie-groups_ EQ0093 (2.5)	42	■ 1.000	$\begin{aligned} & \begin{aligned} & z_1 = \phi_1(x_1, x_2, x_3; y_1, y_2, y_3) = x_1y_1 + x_2y_3 \\ & z_2 = \phi_2(x_1, x_2, x_3; y_1, y_2, y_3) = x_1y_2 + x_2 \frac{1+y_2y_3}{y_1} \\ & z_3 = \phi_3(x_1, x_2, x_3; y_1, y_2, y_3) = x_3y_1 + \frac{1+x_2x_3}{x_1} y_3 \end{aligned} \end{aligned}$	$\begin{aligned} z_1 &= \phi_1(x_1, x_2, x_3; y_1, y_2, y_3) = x_1y_1 + x_2y_3 \\ z_2 &= \phi_2(x_1, x_2, x_3; y_1, y_2, y_3) = x_1y_2 + x_2 \frac{1+y_2y_3}{y_1} \\ z_3 &= \phi_3(x_1, x_2, x_3; y_1, y_2, y_3) = x_3y_1 + \frac{1+x_2x_3}{x_1} y_3 \end{aligned}$	$\begin{aligned} z_1 &= \phi_1(x_1, x_2, x_3; y_1, y_2, y_3) = x_1y_1 + x_2y_3 \\ z_2 &= \phi_2(x_1, x_2, x_3; y_1, y_2, y_3) = x_1y_2 + x_2 \frac{1+y_2y_3}{y_1} \\ z_3 &= \phi_3(x_1, x_2, x_3; y_1, y_2, y_3) = x_3y_1 + \frac{1+x_2x_3}{x_1} y_3 \end{aligned}$
gilmore-lie-groups_ EQ0095 (2.7)	43	■ 1.000	$\begin{aligned} & \begin{aligned} & \psi_1(x_1, x_2, x_3) = y_1 = (1+x_2x_3)/x_1 \\ & \psi_2(x_1, x_2, x_3) = y_2 = -x_2 \\ & \psi_3(x_1, x_2, x_3) = y_3 = -x_3 \end{aligned} \end{aligned}$	$\begin{aligned} \psi_1(x_1, x_2, x_3) &= y_1 = (1+x_2x_3)/x_1 \\ \psi_2(x_1, x_2, x_3) &= y_2 = -x_2 \\ \psi_3(x_1, x_2, x_3) &= y_3 = -x_3 \end{aligned}$	$\begin{aligned} \psi_1(x_1, x_2, x_3) &= y_1 = (1+x_2x_3)/x_1 \\ \psi_2(x_1, x_2, x_3) &= y_2 = -x_2 \\ \psi_3(x_1, x_2, x_3) &= y_3 = -x_3 \end{aligned}$
gilmore-lie-groups_ EQ0099	45	■ 1.000	$\left[\begin{array}{cccc} \pm 1 & 0 & 0 & 0 \\ 0 & \pm 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$	$\begin{bmatrix} \pm 1 & 0 & 0 & 0 \\ 0 & \pm 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \pm 1 & 0 & 0 & 0 \\ 0 & \pm 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
gilmore-lie-groups_ EQ0101	46	■ 1.000	$x \longrightarrow x' = (a, b, c, d) x = \frac{ax+b}{cx+d}$	$x \rightarrow x' = (a, b, c, d)x = \frac{ax+b}{cx+d}$	$x \rightarrow x' = (a, b, c, d)x = \frac{ax+b}{cx+d}$
gilmore-lie-groups_ EQ0102	46	■ 1.000	$(A, B, C, D) = (a', b', c', d') \circ (a, b, c, d)$	$(A, B, C, D) = (a', b', c', d') \circ (a, b, c, d)$	$(A, B, C, D) = (a', b', c', d') \circ (a, b, c, d)$
gilmore-lie-groups_ EQ0103	46	■ 1.000	$\left[\begin{array}{l} y' \\ z' \end{array} \right] = \left[\begin{array}{ll} a & b \\ c & d \end{array} \right] \left[\begin{array}{l} y \\ z \end{array} \right] \Rightarrow x' = \frac{y'}{z'} = \frac{a(y/z) + b}{c(y/z) + d} = \frac{ax+b}{cx+d}$	$\begin{bmatrix} y' \\ z' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} \Rightarrow x' = \frac{y'}{z'} = \frac{a(y/z) + b}{c(y/z) + d} = \frac{ax+b}{cx+d}$	$\begin{bmatrix} y' \\ z' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} \Rightarrow x' = \frac{y'}{z'} = \frac{a(y/z) + b}{c(y/z) + d} = \frac{ax+b}{cx+d}$

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gilmore-lie-groups_ EQ0130 (3.16)	54	■ 1.000	$a_{1}^{2}+b_{1}^{2}+a_{2}^{2}+b_{2}^{2}=1$	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$
gilmore-lie-groups_ EQ0134 (3.20)	55	■ 1.000	$M^{\mathfrak{t}}GM = G \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathrm{Sp}(N, \mathbb{R}) \\ \mathbb{C} & \mathrm{Sp}(N, \mathbb{C}) \end{cases}$	$M^tGM = G \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathrm{Sp}(N, \mathbb{R}) \\ \mathbb{C} & \mathrm{Sp}(N, \mathbb{C}) \end{cases}$	$M^tGM = G \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathrm{Sp}(N, \mathbb{R}) \\ \mathbb{C} & \mathrm{Sp}(N, \mathbb{C}) \end{cases}$
gilmore-lie-groups_ EQ0135	55	■ 1.000	$G=\left[\begin{array}{cc} 0 & I_n \\ -I_n & 0 \end{array}\right]$	$G = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$	$G = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0136	55	■ 1.000	$G=\sum_{\alpha=1}^n \oplus \left[i \sigma_y\right]_{\alpha}$	$G = \sum_{\alpha=1}^n \oplus [i\sigma_y]_{\alpha}$	$G = \sum_{\alpha=1}^n \oplus [i\sigma_y]_{\alpha}$
gilmore-lie-groups_ EQ0137 (3.21)	55	■ 1.000	$\left[\begin{array}{ll} a & c \\ b & d \end{array}\right] \left[\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right] \left[\begin{array}{ll} a & b \\ c & d \end{array}\right] = \left[\begin{array}{cc} 0 & ad-bc \\ bc-ad & 0 \end{array}\right] = \left[\begin{array}{ll} 0 & 1 \\ -1 & 0 \end{array}\right]$	$\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & ad-bc \\ bc-ad & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$\begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 0 & ad-bc \\ bc-ad & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0139 (3.23)	55	■ 1.000	$G=\left[\begin{array}{cc} I_3 & 0 \\ 0 & 0 \end{array}\right]$	$G = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$	$G = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0147 (3.30)	57	■ 1.000	$SU(1,1)=U(1,1) \cap SL(2; \mathbb{C}) \rightarrow \left[\begin{array}{ll} a & b \\ b^* & a^* \end{array}\right]$	$SU(1,1) = U(1,1) \cap SL(2; \mathbb{C}) \rightarrow \begin{bmatrix} a & b \\ b^* & a^* \end{bmatrix}$	$SU(1,1) = U(1,1) \cap SL(2; \mathbb{C}) \rightarrow \begin{bmatrix} a & b \\ b^* & a^* \end{bmatrix}$
gilmore-lie-groups_ EQ0150 (3.33)	58	■ 1.000	$\left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right] \in GL(n; \mathbb{Z})$	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \in GL(n; \mathbb{Z})$	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \in GL(n; \mathbb{Z})$
gilmore-lie-groups_ EQ0151 (3.34)	59	■ 1.000	$\left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right]^n = \left[\begin{array}{cc} F(n+1) & F(n) \\ F(n) & F(n-1) \end{array}\right]$	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n = \begin{bmatrix} F(n+1) & F(n) \\ F(n) & F(n-1) \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n = \begin{bmatrix} F(n+1) & F(n) \\ F(n) & F(n-1) \end{bmatrix}$
gilmore-lie-groups_ EQ0153 (3.35)	59	■ 1.000	$\begin{aligned} & \left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right]^{n+1} = \left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right] \left[\begin{array}{cc} F(n+1) & F(n) \\ F(n) & F(n-1) \end{array}\right] \\ & = \left[\begin{array}{cc} F(n+1)+F(n) & F(n)+F(n-1) \\ F(n+1) & F(n) \end{array}\right] \\ & = \left[\begin{array}{cc} F(n+2) & F(n+1) \\ F(n+1) & F(n) \end{array}\right] \end{aligned}$	$\begin{aligned} & \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{n+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F(n+1) & F(n) \\ F(n) & F(n-1) \end{bmatrix} \\ & = \begin{bmatrix} F(n+1)+F(n) & F(n)+F(n-1) \\ F(n+1) & F(n) \end{bmatrix} \\ & = \begin{bmatrix} F(n+2) & F(n+1) \\ F(n+1) & F(n) \end{bmatrix} \end{aligned}$	$\begin{aligned} & \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{n+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F(n+1) & F(n) \\ F(n) & F(n-1) \end{bmatrix} \\ & = \begin{bmatrix} F(n+1)+F(n) & F(n)+F(n-1) \\ F(n+1) & F(n) \end{bmatrix} \\ & = \begin{bmatrix} F(n+2) & F(n+1) \\ F(n+1) & F(n) \end{bmatrix} \end{aligned}$
gilmore-lie-groups_ EQ0154 (3.36)	60	■ 1.000	$A_n=P_n \cap SL(n; \mathbb{Z})$	$A_n = P_n \cap SL(n; \mathbb{Z})$	$A_n = P_n \cap SL(n; \mathbb{Z})$

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gilmore-lie-groups_ EQ0155 (3.37)	60	■ 1.000	$\begin{array}{l} \pm\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right] \quad \pm\left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right] \quad \pm\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right] \quad \pm\left[\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right] \\ \end{array}$		
gilmore-lie-groups_ EQ0156 (3.38)	60	■ 1.000	$\begin{array}{l} \begin{array}{l} \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \left[\begin{array}{lll} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array}\right] \left[\begin{array}{lll} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right] \\ \left[\begin{array}{lll} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right] \left[\begin{array}{lll} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right] \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right] \end{array} \\ \end{array}$		
gilmore-lie-groups_ EQ0158	61	■ 1.000	$\begin{array}{l} \begin{array}{l} \text{GL}(n; \mathbb{R}) = n^2 \quad \text{SL}(n; \mathbb{R}) = n^2 - 1 \\ \text{GL}(n; \mathbb{C}) = 2n^2 \quad \text{SL}(n; \mathbb{C}) = 2n^2 - 2 \\ \text{GL}(n; \mathbb{Q}) = 4n^2 \end{array} \end{array}$		
gilmore-lie-groups_ EQ0159	61	■ 1.000	$\begin{array}{l} \begin{array}{l} \text{SL}_1(n; \mathbb{C}) \det(M) = e^{i\phi} \quad 2n^2 - 1 \\ \text{SL}_2(n; \mathbb{C}) \det(M) = e^\lambda \quad 2n^2 - 1 \\ \text{SL}_3(n; \mathbb{C}) \det(M) = r \quad 2n^2 - 1 \\ \text{SL}(n; \mathbb{C}) \det(M) = +1 \quad 2n^2 - 2 \end{array} \end{array}$		
gilmore-lie-groups_ EQ0160	62	■ 1.000	$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_\mu(x, t) = -\frac{4\pi}{c} j_\mu$		
gilmore-lie-groups_ EQ0164	63	■ 1.000	$SO(3) \longrightarrow \begin{bmatrix} m_{11} & x & y \\ m_{21} & m_{22} & z \\ m_{31} & m_{32} & m_{33} \end{bmatrix}$		
gilmore-lie-groups_ EQ0166	64	■ 1.000	$E = -E_0 Z^2 \{1 + \delta * (n - l - 1)\} / n^2$		
gilmore-lie-groups_ EQ0168	64	■ 1.000	$G_1 = \begin{bmatrix} I_n & 0 \\ 0 & I_n \end{bmatrix} \quad G_2 = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$		

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gilmore-lie-groups_ E00169	64	■ 1.000	\begin{array}{ccc} G_{\{1\}} & \& G_{\{2\}} & \& G_{\{1\}} \end{array} \text{ { and } } G_{\{2\}} \\\ 0(2\ n\ ;\ \mathbb{R}) & \& \operatorname{Sp}(2\ n\ ;\ \mathbb{R}) & \& 0\ U(2\ n\ ;\ \mathbb{R}) \end{array}	$\begin{array}{ccc} G_1 & G_2 & G_1 \text{ and } G_2 \\ O(2n;\mathbb{R}) & \operatorname{Sp}(2n;\mathbb{R}) & OU(2n;\mathbb{R}) \end{array}$	$\begin{array}{ccc} G_1 & G_2 & G_1 \text{ and } G_2 \\ O(2n;\mathbb{R}) & \operatorname{Sp}(2n;\mathbb{R}) & OU(2n;\mathbb{R}) \end{array}$
gilmore-lie-groups_ E00171	64	■ 1.000	\{f,\ g\}=\sum_{\{k\}}\frac{\partial f}{\partial q_k}\frac{\partial g}{\partial p_k}-\frac{\partial g}{\partial q_k}\frac{\partial f}{\partial p_k}	$\{f,g\}=\sum_k\frac{\partial f}{\partial q_k}\frac{\partial g}{\partial p_k}-\frac{\partial g}{\partial q_k}\frac{\partial f}{\partial p_k}$	$\{f,g\}=\sum_k\frac{\partial f}{\partial q_k}\frac{\partial g}{\partial p_k}-\frac{\partial g}{\partial q_k}\frac{\partial f}{\partial p_k}$
gilmore-lie-groups_ E00173	65	■ 1.000	\left[\begin{array}{l} \partial g/\partial Q \\ \partial g/\partial P \end{array}\right]=\left[\begin{array}{ll} A & B \\ C & D \end{array}\right]\left[\begin{array}{l} \partial g/\partial q \\ \partial g/\partial p \end{array}\right]	$\left[\begin{array}{l} \partial g/\partial Q \\ \partial g/\partial P \end{array}\right]=\left[\begin{array}{ll} A & B \\ C & D \end{array}\right]\left[\begin{array}{l} \partial g/\partial q \\ \partial g/\partial p \end{array}\right]$	$\left[\begin{array}{l} \partial g/\partial Q \\ \partial g/\partial P \end{array}\right]=\left[\begin{array}{ll} A & B \\ C & D \end{array}\right]\left[\begin{array}{l} \partial g/\partial q \\ \partial g/\partial p \end{array}\right]$
gilmore-lie-groups_ E00174	65	■ 1.000	\{u(q,\ p),\ v(q,\ p)\}\leftrightharpoonup\frac{[u(\hat{q},\ \hat{p}),\ v(\hat{q},\ \hat{p})]}{i\hbar}	$\{u(q,p),v(q,p)\}\leftrightarrow\frac{[u(\hat{q},\hat{p}),v(\hat{q},\hat{p})]}{i\hbar}$	$\{u(q,p),v(q,p)\}\leftrightarrow\frac{[u(\hat{q},\hat{p}),v(\hat{q},\hat{p})]}{i\hbar}$
gilmore-lie-groups_ E00176	65	■ 1.000	\left A_{\{L\}}\right ^2-\left B_{\{L\}}\right ^2=\left A_{\{R\}}\right ^2-\left B_{\{R\}}\right ^2	$ A_L ^2- B_L ^2= A_R ^2- B_R ^2$	$ A_L ^2- B_L ^2= A_R ^2- B_R ^2$
gilmore-lie-groups_ E00179	66	■ 1.000	\left[\begin{array}{ll} s_{\{11\}} & s_{\{12\}} \\ s_{\{21\}} & s_{\{22\}} \end{array}\right]=\left[\begin{array}{cc} \frac{1}{t_{11}} & -\frac{t_{12}}{t_{11}} \\ \frac{t_{21}}{t_{11}} & \frac{t_{11}t_{22}-t_{12}t_{21}}{t_{11}} \end{array}\right]	$\left[\begin{array}{ll} s_{11} & s_{12} \\ s_{21} & s_{22} \end{array}\right]=\left[\begin{array}{ll} \frac{1}{t_{11}} & -\frac{t_{12}}{t_{11}} \\ \frac{t_{21}}{t_{11}} & \frac{t_{11}t_{22}-t_{12}t_{21}}{t_{11}} \end{array}\right]$	$\left[\begin{array}{ll} s_{11} & s_{12} \\ s_{21} & s_{22} \end{array}\right]=\left[\begin{array}{ll} \frac{1}{t_{11}} & -\frac{t_{12}}{t_{11}} \\ \frac{t_{21}}{t_{11}} & \frac{t_{11}t_{22}-t_{12}t_{21}}{t_{11}} \end{array}\right]$
gilmore-lie-groups_ E00180	66	■ 1.000	\left[\begin{array}{l} i_{\{2\}} \\ i_{\{3\}} \end{array}\right]=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{l} o_1 \\ o_4 \end{array}\right]	$\left[\begin{array}{l} i_2 \\ i_3 \end{array}\right]=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{l} o_1 \\ o_4 \end{array}\right]$	$\left[\begin{array}{l} i_2 \\ i_3 \end{array}\right]=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{l} o_1 \\ o_4 \end{array}\right]$
gilmore-lie-groups_ E00189 (4.1)	71	■ 1.000	(a,\ b,\ c)\longrightarrow M(a,\ b,\ c)=\left[\begin{array}{cc} 1+a & b \\ c & (1+bc)/(1+a) \end{array}\right]	$(a,b,c)\longrightarrow M(a,b,c)=\left[\begin{array}{cc} 1+a & b \\ c & (1+bc)/(1+a) \end{array}\right]$	$(a,b,c)\longrightarrow M(a,b,c)=\left[\begin{array}{cc} 1+a & b \\ c & (1+bc)/(1+a) \end{array}\right]$
gilmore-lie-groups_ E00190 (4.2)	71	■ 1.000	\begin{gathered} (a,\ b,\ c)\longrightarrow(\delta a,\ \delta b,\ \delta c)\\ \longrightarrow M(\delta a,\ \delta b,\ \delta c)\\ \\\ =\left[\begin{array}{cc} 1+\delta a & \delta b \\ \delta c & (1+\delta b\delta c)/(1+\delta a) \end{array}\right] \end{gathered}	$(a,b,c)\rightarrow(\delta a,\delta b,\delta c)\rightarrow M(\delta a,\delta b,\delta c)\\ =\left[\begin{array}{cc} 1+\delta a & \delta b \\ \delta c & (1+\delta b\delta c)/(1+\delta a) \end{array}\right]$	$(a,b,c)\rightarrow(\delta a,\delta b,\delta c)\rightarrow M(\delta a,\delta b,\delta c)\\ =\left[\begin{array}{cc} 1+\delta a & \delta b \\ \delta c & (1+\delta b\delta c)/(1+\delta a) \end{array}\right]$
gilmore-lie-groups_ E00192 (4.5)	72	■ 1.000	\lim_{\{k\}}\longrightarrow\infty\left(I+\frac{1}{k}X\right)^k=\sum_{n=0}^{\infty}\frac{X^n}{n!}=\operatorname{EXP}(X)	$\lim_{k\rightarrow\infty}\left(I+\frac{1}{k}X\right)^k=\sum_{n=0}^{\infty}\frac{X^n}{n!}=\operatorname{EXP}(X)$	$\lim_{k\rightarrow\infty}\left(I+\frac{1}{k}X\right)^k=\sum_{n=0}^{\infty}\frac{X^n}{n!}=\operatorname{EXP}(X)$
gilmore-lie-groups_ E00193 (4.6)	72	■ 1.000	X=a\ X_{\{a\}}+b\ X_{\{b\}}+c\ X_{\{c\}}=\left[\begin{array}{cc} a & b \\ c & -a \end{array}\right]	$X=aX_a+bX_b+cX_c=\left[\begin{array}{cc} a & b \\ c & -a \end{array}\right]$	$X=aX_a+bX_b+cX_c=\left[\begin{array}{cc} a & b \\ c & -a \end{array}\right]$
gilmore-lie-groups_ E00195 (4.8)	72	■ 1.000	F(X)=f_{\{0\}}(a,\ b,\ c)\ X^{\{0\}}+f_{\{1\}}(a,\ b,\ c)\ X^{\{1\}}	$F(X)=f_0(a,b,c)X^0+f_1(a,b,c)X^1$	$F(X)=f_0(a,b,c)X^0+f_1(a,b,c)X^1$

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gilmore-lie-groups_ EQ0196 (4.9)	73	■ 1.000	$\operatorname{EXP}(X)=f_{\{0\}}\left(\theta^2\right) I_{\{2\}}+f_{\{1\}}\left(\theta^2\right) X$	$\operatorname{EXP}(X)=f_0\left(\theta^2\right) I_2+f_1\left(\theta^2\right) X$	$\operatorname{EXP}(X)=f_0(\theta^2) I_2+f_1(\theta^2) X$
gilmore-lie-groups_ EQ0198 (4.11)	74	■ 1.000	$(I+\epsilon \alpha X)(I+\epsilon \beta X)=I+\epsilon(\alpha X+\beta Y)+\text { higher order terms }$	$(I+\epsilon \alpha X)(I+\epsilon \beta X)=I+\epsilon(\alpha X+\beta Y)+\text { higher order terms }$	$(I+\epsilon \alpha X)(I+\epsilon \beta X)=I+\epsilon(\alpha X+\beta Y)+\text { higher order terms }$
gilmore-lie-groups_ EQ0199 (4.12)	74	■ 1.000	$\text { commutator of } g_{\{1\}} \text { and } g_{\{2\}} \text { is } g_{\{1\}} g_{\{2\}} g_{\{1\}}^{-1} g_{\{2\}}^{-1}$	commutator of g_1 and g_2 is $g_1 g_2 g_1^{-1} g_2^{-1}$	commutator of g_1 and g_2 is $g_1 g_2 g_1^{-1} g_2^{-1}$
gilmore-lie-groups_ EQ0201 (4.14)	74	■ 1.000	$g_{\{1\}} g_{\{2\}} g_{\{1\}}^{-1} g_{\{2\}}^{-1} \in G \Leftrightarrow[X, Y] \in \mathfrak{g}$	$g_1 g_2 g_1^{-1} g_2^{-1} \in G \Leftrightarrow[X, Y] \in \mathfrak{g}$	$g_1 g_2 g_1^{-1} g_2^{-1} \in G \Leftrightarrow[X, Y] \in \mathfrak{g}$
gilmore-lie-groups_ EQ0202 (4.15)	74	■ 1.000	$g_{\{1\}} g_{\{2\}} g_{\{1\}}^{-1} g_{\{2\}}^{-1}=I \Rightarrow[X, Y]=0$	$g_1 g_2 g_1^{-1} g_2^{-1}=I \Rightarrow[X, Y]=0$	$g_1 g_2 g_1^{-1} g_2^{-1}=I \Rightarrow[X, Y]=0$
gilmore-lie-groups_ EQ0203 (4.16)	74	■ 1.000	$g_{\{1\}} H g_{\{1\}}^{-1} \in H \Rightarrow[X, Y] \in \text { Lie algebra of } H$	$g_1 H g_1^{-1} \in H \Rightarrow[X, Y] \in \text { Lie algebra of } H$	$g_1 H g_1^{-1} \in H \Rightarrow[X, Y] \in \text { Lie algebra of } H$
gilmore-lie-groups_ EQ0204 (4.17)	74	■ 1.000	$[X,[Y, Z]]+[Y,[Z, X]]+[Z,[X, Y]]=0$	$[X,[Y, Z]]+[Y,[Z, X]]+[Z,[X, Y]]=0$	$[X,[Y, Z]]+[Y,[Z, X]]+[Z,[X, Y]]=0$
gilmore-lie-groups_ EQ0205	75	■ 1.000	$X \in \mathfrak{g}, \quad Y \in \mathfrak{g}, \quad \alpha X+\beta Y \in \mathfrak{g}$	$X \in \mathfrak{g}, \quad Y \in \mathfrak{g}, \quad \alpha X+\beta Y \in \mathfrak{g}$	$X \in \mathfrak{g}, \quad Y \in \mathfrak{g}, \quad \alpha X+\beta Y \in \mathfrak{g}$
gilmore-lie-groups_ EQ0206	75	■ 1.000	$X \in \mathfrak{g}, \quad Y \in \mathfrak{g}, \quad[X, Y] \in \mathfrak{g}$	$X \in \mathfrak{g}, \quad Y \in \mathfrak{g}, \quad[X, Y] \in \mathfrak{g}$	$X \in \mathfrak{g}, \quad Y \in \mathfrak{g}, \quad[X, Y] \in \mathfrak{g}$
gilmore-lie-groups_ EQ0207	75	■ 1.000	$[X,[Y, Z]]+[Y,[Z, X]]+[Z,[X, Y]]=0$	$[X,[Y, Z]]+[Y,[Z, X]]+[Z,[X, Y]]=0$	$[X,[Y, Z]]+[Y,[Z, X]]+[Z,[X, Y]]=0$
gilmore-lie-groups_ EQ0209 (4.19)	75	■ 1.000	$\left[X_{\{i\}}, X_{\{j\}}\right]=C_{\{i j\}}^{\{k\}} X_{\{k\}}$	$\left[X_i, X_j\right]=C_{i j}^k X_k$	$\left[X_i, X_j\right]=C_{i j}^k X_k$
gilmore-lie-groups_ EQ0219 (4.29)	77	■ 1.000	$R\left(X_{\{i\}}\right)_{\{j\}}^{\{k\}}=C_{\{i j\}}^{\{k\}}$	$R\left(X_i\right)_j^k=C_{i j}^k$	$R\left(X_i\right)_j^k=C_{i j}^k$
gilmore-lie-groups_ EQ0220 (4.30)	77	■ 1.000	$\begin{aligned} R(X) &=R\left(a X_{\{a\}}+b X_{\{b\}}+c X_{\{c\}}\right. \\ &\left.)=a R\left(X_{\{a\}}\right)+b R\left(X_{\{b\}}\right)+c R\right. \\ &\left.\left(X_{\{c\}}\right) \quad \& \quad=\left[\begin{array}{ccc} 0 & -2 b & 2 c \\ -c & 2 a & 0 \\ b & 0 & -2 a \end{array}\right] \end{aligned}$	$R(X)=R\left(a X_a+b X_b+c X_c\right)=a R\left(X_a\right)+b R\left(X_b\right)+c R\left(X_c\right)$ $=\left[\begin{array}{ccc} 0 & -2 b & 2 c \\ -c & 2 a & 0 \\ b & 0 & -2 a \end{array}\right]$	$R(X)=R\left(a X_a+b X_b+c X_c\right)=a R\left(X_a\right)+b R\left(X_b\right)+c R\left(X_c\right)$ $=\left[\begin{array}{ccc} 0 & -2 b & 2 c \\ -c & 2 a & 0 \\ b & 0 & -2 a \end{array}\right]$
gilmore-lie-groups_ EQ0221 (4.31)	78	■ 1.000	$E(2)=\left[\begin{array}{ccc} \cos \theta & \sin \theta & t_1 \\ -\sin \theta & \cos \theta & t_2 \\ 0 & 0 & 1 \end{array}\right]$	$E(2)=\left[\begin{array}{ccc} \cos \theta & \sin \theta & t_1 \\ -\sin \theta & \cos \theta & t_2 \\ 0 & 0 & 1 \end{array}\right]$	$E(2)=\left[\begin{array}{ccc} \cos \theta & \sin \theta & t_1 \\ -\sin \theta & \cos \theta & t_2 \\ 0 & 0 & 1 \end{array}\right]$
gilmore-lie-groups_ EQ0223 (4.33)	78	■ 1.000	$R\left(\theta L_z+t_1 P_x+t_2 P_y\right)=\left[\begin{array}{ccc} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ -t_2 & t_1 & 0 \end{array}\right]$	$R\left(\theta L_z+t_1 P_x+t_2 P_y\right)=\left[\begin{array}{ccc} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ -t_2 & t_1 & 0 \end{array}\right]$	$R\left(\theta L_z+t_1 P_x+t_2 P_y\right)=\left[\begin{array}{ccc} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ -t_2 & t_1 & 0 \end{array}\right]$
gilmore-lie-groups_ EQ0224 (4.34)	78	■ 1.000	$(A, B)=\operatorname{operatorname{tr}} A^{\dagger} B$	$(A, B)=\operatorname{tr} A^{\dagger} B$	$(A, B)=\operatorname{tr} A^{\dagger} B$
gilmore-lie-groups_ EQ0228 (4.38)	79	■ 1.000	$\mathfrak{g}=V_{\{+\}}+V_{\{-\}}+V_{\{0\}}$	$\mathfrak{g}=V_++V_-+V_0$	$\mathfrak{g}=V_++V_-+V_0$

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gilmore-lie-groups_ E00229 (4.39)	80	■ 1.000	$\frac{\mathfrak{g}}{\underset{\text{inner product}}{\text{Cartan-Killing}}}\begin{cases} V_0 & \text{nilpotent invariant subalgebra} \\ V_- & \text{compact subalgebra} \\ V_+ & \text{noncompact operators} \end{cases}$	$\mathfrak{g} \xrightarrow[\text{inner product}]{\text{Cartan-Killing}} \begin{cases} V_0 & \text{nilpotent invariant subalgebra} \\ V_- & \text{compact subalgebra} \\ V_+ & \text{noncompact operators} \end{cases}$	$\mathfrak{g} \xrightarrow[\text{inner product}]{\text{Cartan-Killing}} \begin{cases} V_0 & \text{nilpotent invariant subalgebra} \\ V_- & \text{compact subalgebra} \\ V_+ & \text{noncompact operators} \end{cases}$
gilmore-lie-groups_ E00230 (4.40)	80	■ 1.000	$(X, X) = \operatorname{tr} R(X) R(X) = \operatorname{tr} \begin{bmatrix} 0 & -2b & 2c \\ -c & 2a & 0 \\ b & 0 & -2a \end{bmatrix} = 8(a^2 + bc)$	$(X, X) = \operatorname{tr} R(X) R(X) = \operatorname{tr} \begin{bmatrix} 0 & -2b & 2c \\ -c & 2a & 0 \\ b & 0 & -2a \end{bmatrix} = 8(a^2 + bc)$	$(X, X) = \operatorname{tr} R(X) R(X) = \operatorname{tr} \begin{bmatrix} 0 & -2b & 2c \\ -c & 2a & 0 \\ b & 0 & -2a \end{bmatrix} = 8(a^2 + bc)$
gilmore-lie-groups_ E00232 (4.41)	80	■ 1.000	$8(a^2 + bc) = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$	$8(a^2 + bc) = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$	$8(a^2 + bc) = \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$
gilmore-lie-groups_ E00232 (4.42)	80	■ 1.000	$\begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} X_a \\ X_+ \\ X_- \end{bmatrix}$	$\begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} X_a \\ X_+ \\ X_- \end{bmatrix}$	$\begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} X_a \\ X_+ \\ X_- \end{bmatrix}$
gilmore-lie-groups_ E00233 (4.43)	80	■ 1.000	$(X, X) = \operatorname{tr} \begin{bmatrix} a & b \\ c & -a \end{bmatrix} = 2(a^2 + bc)$	$(X, X) = \operatorname{tr} \begin{bmatrix} a & b \\ c & -a \end{bmatrix} = 2(a^2 + bc)$	$(X, X) = \operatorname{tr} \begin{bmatrix} a & b \\ c & -a \end{bmatrix} = 2(a^2 + bc)$
gilmore-lie-groups_ E00238 (4.48)	81	■ 1.000	$dV(\operatorname{Id}) = d\alpha^1 \wedge d\alpha^2 \wedge \cdots \wedge d\alpha^n$ $dV(x) = dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n = \ M\ d\alpha^1 \wedge d\alpha^2 \wedge \cdots \wedge d\alpha^n$	$dV(\operatorname{Id}) = d\alpha^1 \wedge d\alpha^2 \wedge \cdots \wedge d\alpha^n$ $dV(x) = dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n = \ M\ d\alpha^1 \wedge d\alpha^2 \wedge \cdots \wedge d\alpha^n$	$dV(\operatorname{Id}) = d\alpha^1 \wedge d\alpha^2 \wedge \cdots \wedge d\alpha^n$ $dV(x) = dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n = \ M\ d\alpha^1 \wedge d\alpha^2 \wedge \cdots \wedge d\alpha^n$

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gilmore-lie-groups_ E00240 (4.50)	82	■ 1.000	$\begin{aligned} & \left[\begin{array}{cc} 1+x & y \\ z & \frac{1+yz}{1+x} \end{array} \right] \times \left[\begin{array}{cc} 1+d\alpha^1 & d\alpha^2 \\ d\alpha^3 & 1-d\alpha^1 \end{array} \right] \\ & = \left[\begin{array}{cc} 1+(x+dx) & y+dy \\ z+dz & \frac{1+(y+dy)(z+dz)}{1+(x+dx)} \end{array} \right] \\ & = \left[\begin{array}{cc} (1+x)(1+d\alpha^1)+yd\alpha^3 & (1+x)d\alpha^2+y(1-d\alpha^1) \\ z(1+d\alpha^1)+\frac{(1+yz)d\alpha^3}{(1+x)} & zd\alpha^2+\frac{(1+yz)(1-d\alpha^1)}{(1+x)} \end{array} \right] \end{aligned}$	$\left[\begin{array}{cc} 1+x & y \\ z & \frac{1+yz}{1+x} \end{array} \right] \times \left[\begin{array}{cc} 1+d\alpha^1 & d\alpha^2 \\ d\alpha^3 & 1-d\alpha^1 \end{array} \right] \\ = \left[\begin{array}{cc} 1+(x+dx) & y+dy \\ z+dz & \frac{1+(y+dy)(z+dz)}{1+(x+dx)} \end{array} \right] \\ = \left[\begin{array}{cc} (1+x)(1+d\alpha^1)+yd\alpha^3 & (1+x)d\alpha^2+y(1-d\alpha^1) \\ z(1+d\alpha^1)+\frac{(1+yz)d\alpha^3}{(1+x)} & zd\alpha^2+\frac{(1+yz)(1-d\alpha^1)}{(1+x)} \end{array} \right]$	$\left[\begin{array}{cc} 1+x & y \\ z & \frac{1+yz}{1+x} \end{array} \right] \times \left[\begin{array}{cc} 1+d\alpha^1 & d\alpha^2 \\ d\alpha^3 & 1-d\alpha^1 \end{array} \right] \\ = \left[\begin{array}{cc} 1+(x+dx) & y+dy \\ z+dz & \frac{1+(y+dy)(z+dz)}{1+(x+dx)} \end{array} \right] \\ = \left[\begin{array}{cc} (1+x)(1+d\alpha^1)+yd\alpha^3 & (1+x)d\alpha^2+y(1-d\alpha^1) \\ z(1+d\alpha^1)+\frac{(1+yz)d\alpha^3}{(1+x)} & zd\alpha^2+\frac{(1+yz)(1-d\alpha^1)}{(1+x)} \end{array} \right]$
gilmore-lie-groups_ E00242 (4.52)	82	■ 1.000	$d\mu(x)=\rho(x,y,z)dx\wedge dy\wedge dz=\frac{dx\wedge dy\wedge dz}{1+x}$	$d\mu(x)=\rho(x,y,z)dx\wedge dy\wedge dz=\frac{dx\wedge dy\wedge dz}{1+x}$	$d\mu(x)=\rho(x,y,z)dx\wedge dy\wedge dz=\frac{dx\wedge dy\wedge dz}{1+x}$
gilmore-lie-groups_ E00244	83	■ 1.000	$\left(X^{\prime},X\right)=\frac{1}{2}\left[\left(X^{\prime}+X,X^{\prime}+X\right)-\left(X^{\prime},X^{\prime}\right)-\left(X,X\right)\right]$	$(X',X)=\frac{1}{2}[(X'+X,X'+X)-(X',X')-(X,X)]$	$(X',X)=\frac{1}{2}[(X'+X,X'+X)-(X',X')-(X,X)]$
gilmore-lie-groups_ E00245	83	■ 1.000	$\left(X^{\prime},X\right)=2a^{\prime}a+b^{\prime}c+c^{\prime}b$	$(X',X)=2a'a+b'c+c'b$	$(X',X)=2a'a+b'c+c'b$
gilmore-lie-groups_ E00246 (4.54)	83	■ 1.000	$X=-\sum_{n=1}^{\infty}\frac{(I-Y)^n}{n}$	$X=-\sum_{n=1}^{\infty}\frac{(I-Y)^n}{n}$	$X=-\sum_{n=1}^{\infty}\frac{(I-Y)^n}{n}$
gilmore-lie-groups_ E00247 (4.55)	84	■ 1.000	$\theta\cdot\mathbf{L}=\left[\begin{array}{ccc}0&\theta_3&-\theta_2\\-\theta_3&0&\theta_1\\\theta_2&-\theta_1&0\end{array}\right]=\left[\begin{array}{ccc}0&\theta_{12}&\theta_{13}\\\theta_{21}&0&\theta_{23}\\\theta_{31}&\theta_{32}&0\end{array}\right]=\mathbf{X}$	$\theta\cdot\mathbf{L}=\left[\begin{array}{ccc}0&\theta_3&-\theta_2\\-\theta_3&0&\theta_1\\\theta_2&-\theta_1&0\end{array}\right]=\left[\begin{array}{ccc}0&\theta_{12}&\theta_{13}\\\theta_{21}&0&\theta_{23}\\\theta_{31}&\theta_{32}&0\end{array}\right]=\mathbf{X}$	$\theta\cdot\mathbf{L}=\left[\begin{array}{ccc}0&\theta_3&-\theta_2\\-\theta_3&0&\theta_1\\\theta_2&-\theta_1&0\end{array}\right]=\left[\begin{array}{ccc}0&\theta_{12}&\theta_{13}\\\theta_{21}&0&\theta_{23}\\\theta_{31}&\theta_{32}&0\end{array}\right]=\mathbf{X}$
gilmore-lie-groups_ E00248 (4.56)	84	■ 1.000	$e^{\theta\cdot\mathbf{L}}=I_3f_0(\theta)+\mathbf{X}f_1(\theta)+\mathbf{X}^2f_2(\theta)$	$e^{\theta\cdot\mathbf{L}}=I_3f_0(\theta)+\mathbf{X}f_1(\theta)+\mathbf{X}^2f_2(\theta)$	$e^{\theta\cdot\mathbf{L}}=I_3f_0(\theta)+\mathbf{X}f_1(\theta)+\mathbf{X}^2f_2(\theta)$
gilmore-lie-groups_ E00250 (4.57)	84	■ 1.000	$[X_{ij},X_{rs}]=X_{is}\delta_{jr}+X_{jr}\delta_{is}-X_{ir}\delta_{js}-X_{js}\delta_{ir}$	$[X_{ij},X_{rs}]=X_{is}\delta_{jr}+X_{jr}\delta_{is}-X_{ir}\delta_{js}-X_{js}\delta_{ir}$	$[X_{ij},X_{rs}]=X_{is}\delta_{jr}+X_{jr}\delta_{is}-X_{ir}\delta_{js}-X_{js}\delta_{ir}$
gilmore-lie-groups_ E00251 (4.58)	84	■ 1.000	$[D,[Y,Z]]=[[D,Y],Z]+[Y,[D,Z]]$	$[D,[Y,Z]]=[[D,Y],Z]+[Y,[D,Z]]$	$[D,[Y,Z]]=[[D,Y],Z]+[Y,[D,Z]]$
gilmore-lie-groups_ E00252	84	■ 1.000	$d(f\wedge g)=(df)\wedge g+f\wedge(dg)$	$d(f\wedge g)=(df)\wedge g+f\wedge(dg)$	$d(f\wedge g)=(df)\wedge g+f\wedge(dg)$
gilmore-lie-groups_ E00255	85	■ 1.000	$\mathfrak{g}\rightarrow V_0+V_-+V_+$	$\mathfrak{g}\rightarrow V_0+V_-+V_+$	$\mathfrak{g}\rightarrow V_0+V_-+V_+$
gilmore-lie-groups_ E00256	86	■ 1.000	$[X_{ij},X_{rs}]=+X_{is}g_{jr}+X_{jr}g_{is}-X_{ir}g_{js}+X_{js}g_{ir}$	$[X_{ij},X_{rs}]=+X_{is}g_{jr}+X_{jr}g_{is}-X_{ir}g_{js}+X_{js}g_{ir}$	$[X_{ij},X_{rs}]=+X_{is}g_{jr}+X_{jr}g_{is}-X_{ir}g_{js}+X_{js}g_{ir}$

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gilmore-lie-groups_ EQ0259 (4.61c)	86	■ 1.000	$\left[\begin{array}{cc} i & 0 \\ 0 & -i \end{array}\right] \quad \left[\begin{array}{cc} 0 & i \\ -i & 0 \end{array}\right] \quad \left[\begin{array}{cc} 0 & -1 \\ -1 & 0 \end{array}\right]$		
gilmore-lie-groups_ EQ0260 (4.62a)	86	■ 1.000	$\left[\begin{array}{ll} \alpha_1 + \alpha_2 & \beta_1 + \beta_2 \\ \beta_1 - \beta_2 & \alpha_1 - \alpha_2 \end{array}\right]$		
gilmore-lie-groups_ EQ0262 (4.62c)	87	■ 1.000	$\left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right] \quad \left[\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right] \quad \left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right]$		
gilmore-lie-groups_ EQ0263 (5.1)	88	■ 1.000	$\begin{aligned} &(I + \epsilon A)^\dagger G(I + \epsilon A) = G \\ &G + \epsilon(A^\dagger G + GA) + \mathcal{O}(\epsilon^2) = G \\ &A^\dagger G + GA = 0 \end{aligned}$		
gilmore-lie-groups_ EQ0264 (5.2)	88	■ 1.000	$\begin{aligned} &\operatorname{det}(I + \epsilon A) = 1 + \epsilon \operatorname{tr}(A) + \mathcal{O}(\epsilon^2) = 1 \\ &\operatorname{tr}(A) = 0 \end{aligned}$		
gilmore-lie-groups_ EQ0267 (5.5)	90	■ 1.000	$M(a, b) = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$		
gilmore-lie-groups_ EQ0268 (5.6)	90	■ 1.000	$\left.\frac{\partial M}{\partial a}\right _{(1,0)} = X_a = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \left.\frac{\partial M}{\partial b}\right _{(1,0)} = X_b = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$		
gilmore-lie-groups_ EQ0269 (5.7)	90	■ 1.000	$\begin{bmatrix} 0 & l & r & -2\delta \\ 0 & \eta & 2R & -r \\ 0 & -2L & -\eta & l \\ 0 & 0 & 0 & 0 \end{bmatrix} = \eta X_\eta + RX_R + LX_L + rX_r + lX_l + \delta X_\delta$		
gilmore-lie-groups_ EQ0271 (5.10)	91	■ 1.000	$\begin{aligned} &\left[\begin{array}{l} X_\eta, X_r \\ X_\eta, X_l \\ X_l, X_r \\ X_\eta, X_\delta \end{array}\right] = \begin{bmatrix} +X_r \\ -X_l \\ X_\delta \\ 0 \end{bmatrix} \\ &\left[\begin{array}{l} a^\dagger a, a^\dagger \\ a^\dagger a, a \\ a, a^\dagger \\ a^\dagger a, I \end{array}\right] = \begin{bmatrix} +a^\dagger \\ -a \\ I \\ 0 \end{bmatrix} \end{aligned}$		

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gilmore-lie-groups_ EQ0273 (5.12)	91	■ 1.000	$\begin{aligned} & \left\{ \left[X_l, X_r \right] \right\} = X_{\delta} \\ & \left[a, a^{\dagger} \right] = I \\ & \left[X_l, X_{\delta} \right] = 0 \quad \left[a, I \right] = 0 \\ & \left[X_r, X_{\delta} \right] = 0 \quad \left[a^{\dagger}, I \right] = 0 \end{aligned}$	$\begin{aligned} & \left[X_l, X_r \right] = X_{\delta} \quad \left[a, a^{\dagger} \right] = I \\ & \left[X_l, X_{\delta} \right] = 0 \quad \left[a, I \right] = 0 \\ & \left[X_r, X_{\delta} \right] = 0 \quad \left[a^{\dagger}, I \right] = 0 \end{aligned}$	$\begin{aligned} & \left[X_l, X_r \right] = X_{\delta} \quad \left[a, a^{\dagger} \right] = I \\ & \left[X_l, X_{\delta} \right] = 0 \quad \left[a, I \right] = 0 \\ & \left[X_r, X_{\delta} \right] = 0 \quad \left[a^{\dagger}, I \right] = 0 \end{aligned}$
gilmore-lie-groups_ EQ0275 (5.14)	92	■ 1.000	$\begin{aligned} & \mathfrak{o}(3) = \left\{ \begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array} \right\} \\ & \mathfrak{u}(2) = \frac{1}{2} \left\{ \begin{array}{cc} ix_0 + ix_3 & ix_1 + x_2 \\ ix_1 - x_2 & ix_0 - ix_3 \end{array} \right\} \end{aligned}$	$\begin{aligned} & \mathfrak{o}(3) = \left\{ \begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array} \right\} = \sum_i \theta_i L_i \\ & \mathfrak{u}(2) = \frac{1}{2} \left\{ \begin{array}{cc} ix_0 + ix_3 & ix_1 + x_2 \\ ix_1 - x_2 & ix_0 - ix_3 \end{array} \right\} = \frac{i}{2} \sum_{\mu} x_{\mu} \sigma_{\mu} \end{aligned}$	$\begin{aligned} & \mathfrak{o}(3) = \left\{ \begin{array}{ccc} 0 & \theta_3 & -\theta_2 \\ -\theta_3 & 0 & \theta_1 \\ \theta_2 & -\theta_1 & 0 \end{array} \right\} = \sum_i \theta_i L_i \\ & \mathfrak{u}(2) = \frac{1}{2} \left\{ \begin{array}{cc} ix_0 + ix_3 & ix_1 + x_2 \\ ix_1 - x_2 & ix_0 - ix_3 \end{array} \right\} = \frac{i}{2} \sum_{\mu} x_{\mu} \sigma_{\mu} \end{aligned}$
gilmore-lie-groups_ EQ0278 (5.17)	92	■ 1.000	$\begin{aligned} & A = -A^t \\ & D = -D = 0 \\ & B^t = C \end{aligned}$	$\begin{aligned} & A = -A^t \\ & D = -D = 0 \\ & B^t = C \end{aligned}$	$\begin{aligned} & A = -A^t \\ & D = -D = 0 \\ & B^t = C \end{aligned}$
gilmore-lie-groups_ EQ0280 (5.19)	93	■ 1.000	$\begin{aligned} & \left\{ \begin{array}{cccc} 0 & \theta_3 & -\theta_2 & v_1 \\ -\theta_3 & 0 & \theta_1 & v_2 \\ \theta_2 & -\theta_1 & 0 & v_3 \\ v_1 & v_2 & v_3 & 0 \end{array} \right\} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{cccc} 0 & \theta_3 & -\theta_2 & v_1 \\ -\theta_3 & 0 & \theta_1 & v_2 \\ \theta_2 & -\theta_1 & 0 & v_3 \\ v_1 & v_2 & v_3 & 0 \end{array} \right\} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{cccc} 0 & \theta_3 & -\theta_2 & v_1 \\ -\theta_3 & 0 & \theta_1 & v_2 \\ \theta_2 & -\theta_1 & 0 & v_3 \\ v_1 & v_2 & v_3 & 0 \end{array} \right\} \end{aligned}$
gilmore-lie-groups_ EQ0282 (5.21)	93	■ 1.000	$\begin{aligned} & \operatorname{tr} M = -\operatorname{tr} G^{-1} M^{\dagger} G = -\operatorname{tr} M^{\dagger} = -\operatorname{tr} M^* = 0 \end{aligned}$	$\operatorname{tr} M = -\operatorname{tr} G^{-1} M^{\dagger} G = -\operatorname{tr} M^{\dagger} = -\operatorname{tr} M^* = 0$	$\operatorname{tr} M = -\operatorname{tr} G^{-1} M^{\dagger} G = -\operatorname{tr} M^{\dagger} = -\operatorname{tr} M^* = 0$
gilmore-lie-groups_ EQ0284 (5.23)	93	■ 1.000	$\begin{aligned} & \left\{ \begin{array}{ccc} A & B \\ C & D \end{array} \right\} \left\{ \begin{array}{cc} g & 0 \\ 0 & 0 \end{array} \right\} + \left\{ \begin{array}{cc} g & 0 \\ 0 & 0 \end{array} \right\} \left\{ \begin{array}{cc} A^{\dagger} & C^{\dagger} \\ B^{\dagger} & D^{\dagger} \end{array} \right\} = \left\{ \begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array} \right\} \\ & \Rightarrow Ag + gA^{\dagger} = 0 \quad C = C^{\dagger} = 0 \quad B, D \text{ arbitrary} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{cc} A & B \\ C & D \end{array} \right\} \left\{ \begin{array}{cc} g & 0 \\ 0 & 0 \end{array} \right\} + \left\{ \begin{array}{cc} g & 0 \\ 0 & 0 \end{array} \right\} \left\{ \begin{array}{cc} A^{\dagger} & C^{\dagger} \\ B^{\dagger} & D^{\dagger} \end{array} \right\} = \left\{ \begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array} \right\} \\ & \Rightarrow Ag + gA^{\dagger} = 0 \quad C = C^{\dagger} = 0 \quad B, D \text{ arbitrary} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{cc} A & B \\ C & D \end{array} \right\} \left\{ \begin{array}{cc} g & 0 \\ 0 & 0 \end{array} \right\} + \left\{ \begin{array}{cc} g & 0 \\ 0 & 0 \end{array} \right\} \left\{ \begin{array}{cc} A^{\dagger} & C^{\dagger} \\ B^{\dagger} & D^{\dagger} \end{array} \right\} = \left\{ \begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array} \right\} \\ & \Rightarrow Ag + gA^{\dagger} = 0 \quad C = C^{\dagger} = 0 \quad B, D \text{ arbitrary} \end{aligned}$
gilmore-lie-groups_ EQ0285 (5.24)	93	■ 1.000	$\begin{aligned} & \left\{ \begin{array}{cccc} 0 & \theta_3 & -\theta_2 & t_1 \\ -\theta_3 & 0 & \theta_1 & t_2 \\ \theta_2 & -\theta_1 & 0 & t_3 \\ 0 & 0 & 0 & s_4 \end{array} \right\} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{cccc} 0 & \theta_3 & -\theta_2 & t_1 \\ -\theta_3 & 0 & \theta_1 & t_2 \\ \theta_2 & -\theta_1 & 0 & t_3 \\ 0 & 0 & 0 & s_4 \end{array} \right\} \end{aligned}$	$\begin{aligned} & \left\{ \begin{array}{cccc} 0 & \theta_3 & -\theta_2 & t_1 \\ -\theta_3 & 0 & \theta_1 & t_2 \\ \theta_2 & -\theta_1 & 0 & t_3 \\ 0 & 0 & 0 & s_4 \end{array} \right\} \end{aligned}$

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gilmore-lie-groups_ EQ0286 (5.25)	94	■ 1.000	$\left[\begin{array}{ccccc} 0 & \theta_3 & -\theta_2 & v_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & v_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & v_3 & t_3 \\ v_1 & v_2 & v_3 & 0 & t_4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	$\begin{bmatrix} 0 & \theta_3 & -\theta_2 & v_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & v_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & v_3 & t_3 \\ v_1 & v_2 & v_3 & 0 & t_4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & \theta_3 & -\theta_2 & v_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & v_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & v_3 & t_3 \\ v_1 & v_2 & v_3 & 0 & t_4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0287 (5.26)	94	■ 1.000	$\left[\begin{array}{ccccc} 0 & \theta_3 & -\theta_2 & v_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & v_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & v_3 & t_3 \\ 0 & 0 & 0 & 0 & t_4 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	$\begin{bmatrix} 0 & \theta_3 & -\theta_2 & v_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & v_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & v_3 & t_3 \\ 0 & 0 & 0 & 0 & t_4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & \theta_3 & -\theta_2 & v_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & v_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & v_3 & t_3 \\ 0 & 0 & 0 & 0 & t_4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0288 (5.27)	94	■ 1.000	$\operatorname{tr} M=0 \quad \mathbb{F}=\begin{cases} \mathbb{R} & \mathfrak{sl}(n, \mathbb{R}) \\ \mathbb{C} & \mathfrak{sl}(n, \mathbb{C}) \\ \mathbb{Q} & \mathfrak{sl}(n, \mathbb{Q}) \end{cases}$	$\operatorname{tr} M = 0 \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathfrak{sl}(n, \mathbb{R}) \\ \mathbb{C} & \mathfrak{sl}(n, \mathbb{C}) \\ \mathbb{Q} & \mathfrak{sl}(n, \mathbb{Q}) \end{cases}$	$\operatorname{tr} M = 0 \quad \mathbb{F} = \begin{cases} \mathbb{R} & \mathfrak{sl}(n, \mathbb{R}) \\ \mathbb{C} & \mathfrak{sl}(n, \mathbb{C}) \\ \mathbb{Q} & \mathfrak{sl}(n, \mathbb{Q}) \end{cases}$
gilmore-lie-groups_ EQ0289 (5.28)	94	■ 1.000	$\operatorname{det}(e^M)=e^{\operatorname{tr} M}$	$\det(e^M) = e^{\operatorname{tr} M}$	$\det(e^M) = e^{\operatorname{tr} M}$
gilmore-lie-groups_ EQ0291 (5.30)	95	■ 1.000	$M \in \mathfrak{u}(n) \Rightarrow (M^\dagger)_{ij} = -M_{ji}^*$	$M \in \mathfrak{u}(n) \Rightarrow (M^\dagger)_{ij} = -M_{ji}^*$	$M \in \mathfrak{u}(n) \Rightarrow (M^\dagger)_{ij} = -M_{ji}^*$
gilmore-lie-groups_ EQ0292 (5.31)	95	■ 1.000	$M \in \mathfrak{sp}(n) \Rightarrow (M^\dagger)_{ij} = -M_{ji}^*$	$M \in \mathfrak{sp}(n) \Rightarrow (M^\dagger)_{ij} = -M_{ji}^*$	$M \in \mathfrak{sp}(n) \Rightarrow (M^\dagger)_{ij} = -M_{ji}^*$
gilmore-lie-groups_ EQ0293 (5.32)	95	■ 1.000	$(X, Y)=\operatorname{tr} X^\dagger Y$	$(X, Y) = \operatorname{tr} X^\dagger Y$	$(X, Y) = \operatorname{tr} X^\dagger Y$
gilmore-lie-groups_ EQ0294 (5.33)	96	■ 1.000	$(X', X)=\operatorname{tr} X^\dagger X=2(+\theta'\theta+v_1'v_1+v_2'v_2)$	$(X', X) = \operatorname{tr} X'^\dagger X = 2(+\theta'\theta + v_1'v_1 + v_2'v_2)$	$(X', X) = \operatorname{tr} X'^\dagger X = 2(+\theta'\theta + v_1'v_1 + v_2'v_2)$
gilmore-lie-groups_ EQ0295 (5.34)	96	■ 1.000	$(X, Y)=\operatorname{tr} X Y$	$(X, Y) = \operatorname{tr} XY$	$(X, Y) = \operatorname{tr} XY$
gilmore-lie-groups_ EQ0296 (5.35)	96	■ 1.000	$(X', X)=\operatorname{tr} X' X=2(-\theta'\theta+v_1'v_1+v_2'v_2)$	$(X', X) = \operatorname{tr} X' X = 2(-\theta'\theta + v_1'v_1 + v_2'v_2)$	$(X', X) = \operatorname{tr} X' X = 2(-\theta'\theta + v_1'v_1 + v_2'v_2)$
gilmore-lie-groups_ EQ0298	97	■ 1.000	$A=\left[\begin{array}{l} a & b \\ 0 & c \end{array}\right]=a X_a+b X_b+c X_c$	$A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} = aX_a + bX_b + cX_c$	$A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} = aX_a + bX_b + cX_c$
gilmore-lie-groups_ EQ0302	98	■ 1.000	$(Z, Z)=\operatorname{tr} Z^2=\operatorname{tr} A^2+\operatorname{tr} B C+\operatorname{tr} C B+\operatorname{tr} D^2$	$(Z, Z) = \operatorname{tr} Z^2 = \operatorname{tr} A^2 + \operatorname{tr} BC + \operatorname{tr} CB + \operatorname{tr} D^2$	$(Z, Z) = \operatorname{tr} Z^2 = \operatorname{tr} A^2 + \operatorname{tr} BC + \operatorname{tr} CB + \operatorname{tr} D^2$
gilmore-lie-groups_ EQ0303	98	■ 1.000	$(X, X)=\operatorname{tr}(A^2+D^2) \quad (Y, Y)=\operatorname{tr}(B C+C B)$	$(X, X) = \operatorname{tr}(A^2 + D^2) \quad (Y, Y) = \operatorname{tr}(BC + CB)$	$(X, X) = \operatorname{tr}(A^2 + D^2) \quad (Y, Y) = \operatorname{tr}(BC + CB)$

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gilmore-lie-groups_ EQ0305	99	■ 1.000	$\begin{aligned} & \left(\frac{\mathfrak{h}}{\mathfrak{h}}, \frac{\mathfrak{h}}{\mathfrak{h}} \right) \leq 0 \quad \backslash \\ & \left(\frac{\mathfrak{h}}{\mathfrak{h}}, \frac{\mathfrak{p}}{\mathfrak{p}} \right) = 0 \quad \backslash \backslash \\ & \left(\frac{\mathfrak{p}}{\mathfrak{p}}, \frac{\mathfrak{p}}{\mathfrak{p}} \right) \geq 0 \quad \backslash \end{aligned}$	$\begin{aligned} (\mathfrak{h}, \mathfrak{h}) &\leq 0 \\ (\mathfrak{h}, \mathfrak{p}) &= 0 \\ (\mathfrak{p}, \mathfrak{p}) &\geq 0 \end{aligned}$	$\begin{aligned} (\mathfrak{h}, \mathfrak{h}) &\leq 0 \\ (\mathfrak{h}, \mathfrak{p}) &= 0 \\ (\mathfrak{p}, \mathfrak{p}) &\geq 0 \end{aligned}$
gilmore-lie-groups_ EQ0306	99	■ 1.000	$\begin{aligned} & \frac{\mathfrak{s} \, \mathfrak{l}}{\mathfrak{g}} (n; \mathbb{R}) = A + B \\ & \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \end{aligned}$	$\begin{aligned} \mathfrak{s} \mathfrak{l}(n; \mathbb{R}) &= A + B \\ \mathfrak{g} &= \mathfrak{h} + \mathfrak{p} \end{aligned}$	$\begin{aligned} \mathfrak{s} \mathfrak{l}(n; \mathbb{R}) &= A + B \\ \mathfrak{g} &= \mathfrak{h} + \mathfrak{p} \end{aligned}$
gilmore-lie-groups_ EQ0312	100	■ 1.000	$\begin{aligned} A = & \left\{ \begin{array}{ccc c c} 0 & \theta_3 & -\theta_2 & b_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & b_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & b_3 & t_3 \\ \hline \mu b_1 & \mu b_2 & \mu b_3 & 0 & t_4 \\ \sigma t_1 & \sigma t_2 & \sigma t_3 & -\mu \sigma t_4 & 0 \end{array} \right\} \\ & \frac{1}{2} (A, A) = -(\theta \cdot \theta) + \mu (\mathbf{b} \cdot \mathbf{b}) + \sigma (\mathbf{t} \cdot \mathbf{t}) - \mu \sigma t_4^2 \end{aligned}$	$\begin{aligned} A = & \left[\begin{array}{ccc c c} 0 & \theta_3 & -\theta_2 & b_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & b_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & b_3 & t_3 \\ \hline \mu b_1 & \mu b_2 & \mu b_3 & 0 & t_4 \\ \sigma t_1 & \sigma t_2 & \sigma t_3 & -\mu \sigma t_4 & 0 \end{array} \right] \\ & \frac{1}{2} (A, A) = -(\theta \cdot \theta) + \mu (\mathbf{b} \cdot \mathbf{b}) + \sigma (\mathbf{t} \cdot \mathbf{t}) - \mu \sigma t_4^2 \end{aligned}$	$\begin{aligned} A = & \left[\begin{array}{ccc c c} 0 & \theta_3 & -\theta_2 & b_1 & t_1 \\ -\theta_3 & 0 & \theta_1 & b_2 & t_2 \\ \theta_2 & -\theta_1 & 0 & b_3 & t_3 \\ \hline \mu b_1 & \mu b_2 & \mu b_3 & 0 & t_4 \\ \sigma t_1 & \sigma t_2 & \sigma t_3 & -\mu \sigma t_4 & 0 \end{array} \right] \\ & \frac{1}{2} (A, A) = -(\theta \cdot \theta) + \mu (\mathbf{b} \cdot \mathbf{b}) + \sigma (\mathbf{t} \cdot \mathbf{t}) - \mu \sigma t_4^2 \end{aligned}$
gilmore-lie-groups_ EQ0313	100	■ 1.000	$\begin{aligned} & \left[\begin{array}{l} d \, x \quad \backslash \quad d \, y \quad \backslash \quad d \, z \end{array} \right] \\ & \left[\begin{array}{ccc} m_{11} & 0 & -y \\ 0 & m_{11} & x \\ 0 & m_{21} & m_{22} \end{array} \right] \left[\begin{array}{l} \delta x \\ \delta y \\ \delta z \end{array} \right] \end{aligned}$	$\begin{bmatrix} dx \\ dy \\ dz \end{bmatrix} = \begin{bmatrix} m_{11} & 0 & -y \\ 0 & m_{11} & x \\ 0 & m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta z \end{bmatrix}$	$\begin{bmatrix} dx \\ dy \\ dz \end{bmatrix} = \begin{bmatrix} m_{11} & 0 & -y \\ 0 & m_{11} & x \\ 0 & m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta z \end{bmatrix}$
gilmore-lie-groups_ EQ0314	101	■ 1.000	$\begin{aligned} & \frac{\mathfrak{h}}{\mathfrak{g} + \mathfrak{g}^*} \quad \frac{\mathfrak{p}}{\mathfrak{g} - \mathfrak{g}^*} \quad \backslash \backslash \\ & \frac{\mathfrak{h}}{\mathfrak{g} - \mathfrak{g}^t} \quad \frac{\mathfrak{p}}{\mathfrak{g} + \mathfrak{g}^t} \quad \backslash \backslash \\ & \frac{\mathfrak{h}}{\mathfrak{g} - \mathfrak{g}^\dagger} \quad \frac{\mathfrak{p}}{\mathfrak{g} + \mathfrak{g}^\dagger} \quad \backslash \backslash \end{aligned}$	$\begin{array}{cc} \mathfrak{h} & \mathfrak{p} \\ \hline \mathfrak{g} + \mathfrak{g}^* & \mathfrak{g} - \mathfrak{g}^* \\ \mathfrak{g} - \mathfrak{g}^t & \mathfrak{g} + \mathfrak{g}^t \\ \mathfrak{g} - \mathfrak{g}^\dagger & \mathfrak{g} + \mathfrak{g}^\dagger \end{array}$	$\begin{array}{cc} \mathfrak{h} & \mathfrak{p} \\ \hline \mathfrak{g} + \mathfrak{g}^* & \mathfrak{g} - \mathfrak{g}^* \\ \mathfrak{g} - \mathfrak{g}^t & \mathfrak{g} + \mathfrak{g}^t \\ \mathfrak{g} - \mathfrak{g}^\dagger & \mathfrak{g} + \mathfrak{g}^\dagger \end{array}$
gilmore-lie-groups_ EQ0315	101	■ 1.000	$\begin{aligned} & \frac{\mathfrak{h}}{\mathfrak{h}} \subseteq \frac{\mathfrak{h}}{\mathfrak{h}} \quad \frac{\mathfrak{h}}{\mathfrak{p}} \subseteq \frac{\mathfrak{h}}{\mathfrak{p}} \quad \frac{\mathfrak{p}}{\mathfrak{p}} \subseteq \frac{\mathfrak{h}}{\mathfrak{h}} \\ & \frac{\mathfrak{h}}{\mathfrak{h}} \subseteq \frac{\mathfrak{h}}{\mathfrak{p}} \quad \frac{\mathfrak{h}}{\mathfrak{p}} \subseteq \frac{\mathfrak{p}}{\mathfrak{p}} \quad \frac{\mathfrak{p}}{\mathfrak{p}} \subseteq \frac{\mathfrak{h}}{\mathfrak{h}} \end{aligned}$	$[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \quad [\mathfrak{h}, \mathfrak{p}] \subseteq \mathfrak{p} \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{h}$	$[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h} \quad [\mathfrak{h}, \mathfrak{p}] \subseteq \mathfrak{p} \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{h}$
gilmore-lie-groups_ EQ0316 (6.1)	102	■ 1.000	$\left[b_i, b_j^\dagger \right] = I \, \delta_{ij}$	$\left[b_i, b_j^\dagger \right] = I \delta_{ij}$	$[b_i, b_j^\dagger] = I \delta_{ij}$
gilmore-lie-groups_ EQ0317 (6.2)	102	■ 1.000	$A \rightarrow \mathcal{A} = b^\dagger A b = \sum_i \sum_j b_i^\dagger A_{ij} b_j$	$A \rightarrow \mathcal{A} = b^\dagger A b = \sum_i \sum_j b_i^\dagger A_{ij} b_j$	$A \rightarrow \mathcal{A} = b^\dagger A b = \sum_i \sum_j b_i^\dagger A_{ij} b_j$
gilmore-lie-groups_ EQ0319 (6.4)	103	■ 1.000	$[A, B] = C \quad \Leftrightarrow \quad [\mathcal{A}, \mathcal{B}] = C \quad \mathcal{A} = \sum_{ij} b_i^\dagger A_{ij} b_j$	$[A, B] = C \Leftrightarrow [\mathcal{A}, \mathcal{B}] = C \quad \mathcal{A} = \sum_{ij} b_i^\dagger A_{ij} b_j$	$[A, B] = C \Leftrightarrow [\mathcal{A}, \mathcal{B}] = C \quad \mathcal{A} = \sum_{ij} b_i^\dagger A_{ij} b_j$

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gilmore-lie-groups_ EQ0338 (6.23)	106	■ 1.000	$\mathcal{X}_a=-x^{\prime}\frac{\partial}{\partial x^{\prime}}\frac{\partial}{\partial x^{\prime}}\mathcal{X}_b=-\frac{\partial}{\partial x^{\prime}}\frac{\partial}{\partial x^{\prime}}\mathcal{X}_a$	$\mathcal{X}_a=-x^{\prime}\frac{\partial}{\partial x^{\prime}}\quad\mathcal{X}_b=-\frac{\partial}{\partial x^{\prime}}$	$\mathcal{X}_a=-x^{\prime}\frac{\partial}{\partial x^{\prime}}\quad\mathcal{X}_b=-\frac{\partial}{\partial x^{\prime}}$
gilmore-lie-groups_ EQ0339 (6.24)	106	■ 1.000	$\left[X_a,X_b\right]=X_b\Leftrightarrow\left[\mathcal{X}_a,\mathcal{X}_b\right]=\mathcal{X}_b$	$[X_a,X_b]=X_b\Leftrightarrow[\mathcal{X}_a,\mathcal{X}_b]=\mathcal{X}_b$	$[X_a,X_b]=X_b\Leftrightarrow[\mathcal{X}_a,\mathcal{X}_b]=\mathcal{X}_b$
gilmore-lie-groups_ EQ0340 (6.25)	106	■ 1.000	$G_{S^{\prime}}(x^{\prime},y^{\prime})=G_S(x,y)$	$G_{S^{\prime}}(x^{\prime},y^{\prime})=G_S(x,y)$	$G_{S^{\prime}}(x^{\prime},y^{\prime})=G_S(x,y)$
gilmore-lie-groups_ EQ0342 (6.27)	106	■ 1.000	$\begin{aligned} G_{S^{\prime}}(x^{\prime},y^{\prime})&=G_S(x=(1-da)x^{\prime},y=(1-d)y^{\prime}-db)\\ &=G_S(x^{\prime},y^{\prime})+\left\{da\left(-x^{\prime}\frac{\partial}{\partial x^{\prime}}-y^{\prime}\frac{\partial}{\partial y^{\prime}}\right)+db\left(-\frac{\partial}{\partial y^{\prime}}\right)\right\}G_S(x^{\prime},y^{\prime}) \end{aligned}$	$G_{S^{\prime}}(x^{\prime},y^{\prime})=G_S[x=(1-da)x^{\prime},y=(1-d)y^{\prime}-db]$ $=G_S(x^{\prime},y^{\prime})+\left\{da\left(-x^{\prime}\frac{\partial}{\partial x^{\prime}}-y^{\prime}\frac{\partial}{\partial y^{\prime}}\right)+db\left(-\frac{\partial}{\partial y^{\prime}}\right)\right\}G_S(x^{\prime},y^{\prime})$	$G_{S^{\prime}}(x^{\prime},y^{\prime})=G_S[x=(1-da)x^{\prime},y=(1-d)y^{\prime}-db]$ $=G_S(x^{\prime},y^{\prime})+\left\{da\left(-x^{\prime}\frac{\partial}{\partial x^{\prime}}-y^{\prime}\frac{\partial}{\partial y^{\prime}}\right)+db\left(-\frac{\partial}{\partial y^{\prime}}\right)\right\}G_S(x^{\prime},y^{\prime})$
gilmore-lie-groups_ EQ0344 (6.29)	106	■ 1.000	$\left[X_a,X_b\right]=X_b\Leftrightarrow\left[\mathcal{X}_x,\mathcal{X}_b\right]=\mathcal{X}_b$	$[X_a,X_b]=X_b\Leftrightarrow[\mathcal{X}_x,\mathcal{X}_b]=\mathcal{X}_b$	$[X_a,X_b]=X_b\Leftrightarrow[\mathcal{X}_x,\mathcal{X}_b]=\mathcal{X}_b$
gilmore-lie-groups_ EQ0345 (6.30)	107	■ 1.000	$f_{S^{\prime}}(x(p))=f_S(x^{\prime}(p))$	$f_{S^{\prime}}(x(p))=f_S(x^{\prime}(p))$	$f_{S^{\prime}}(x(p))=f_S(x^{\prime}(p))$
gilmore-lie-groups_ EQ0348	108	■ 1.000	$J^2=\left(\frac{1}{2}\hat{n}\right)\left(\frac{1}{2}\hat{n}+1\right)$	$J^2=\left(\frac{1}{2}\hat{n}\right)\left(\frac{1}{2}\hat{n}+1\right)$	$J^2=\left(\frac{1}{2}\hat{n}\right)\left(\frac{1}{2}\hat{n}+1\right)$
gilmore-lie-groups_ EQ0349	108	■ 1.000	$a_i^{\dagger} n_i\rangle=\sqrt{n_i+1} n_i+1\rangle\quad a_i n_i\rangle=\sqrt{n_i} n_i-1\rangle$	$a_i^{\dagger} n_i\rangle=\sqrt{n_i+1} n_i+1\rangle\quad a_i n_i\rangle=\sqrt{n_i} n_i-1\rangle$	$a_i^{\dagger} n_i\rangle=\sqrt{n_i+1} n_i+1\rangle\quad a_i n_i\rangle=\sqrt{n_i} n_i-1\rangle$
gilmore-lie-groups_ EQ0351 (6.31)	109	■ 1.000	$\langle n_1^{\prime},n_2^{\prime},n_3^{\prime} \hat{b}_i^{\dagger}\hat{b}_j n_1,n_2,n_3\rangle$	$\langle n_1^{\prime},n_2^{\prime},n_3^{\prime} \hat{b}_i^{\dagger}\hat{b}_j n_1,n_2,n_3\rangle$	$\langle n_1^{\prime},n_2^{\prime},n_3^{\prime} \hat{b}_i^{\dagger}\hat{b}_j n_1,n_2,n_3\rangle$
gilmore-lie-groups_ EQ0352 (6.32)	109	■ 1.000	$\langle n_1^{\prime},n_2^{\prime},n_3^{\prime} \mathcal{O} n_1,n_2,n_3\rangle=I_D\delta_{n_1^{\prime},n_1}\delta_{n_2^{\prime},n_2}\delta_{n_3^{\prime},n_3}$	$\langle n_1^{\prime},n_2^{\prime},n_3^{\prime} \mathcal{O} n_1,n_2,n_3\rangle=I_D\delta_{n_1^{\prime},n_1}\delta_{n_2^{\prime},n_2}\delta_{n_3^{\prime},n_3}$	$\langle n_1^{\prime},n_2^{\prime},n_3^{\prime} \mathcal{O} n_1,n_2,n_3\rangle=I_D\delta_{n_1^{\prime},n_1}\delta_{n_2^{\prime},n_2}\delta_{n_3^{\prime},n_3}$
gilmore-lie-groups_ EQ0355	110	■ 1.000	$A^tD^t-B^tC^t=I_n\quad A^tB^t=B^tA^t\quad C^tD^t=D^tC^t$	$AD^t-BC^t=I_n\quad AB^t=BA^t\quad CD^t=DC^t$	$AD^t-BC^t=I_n\quad AB^t=BA^t\quad CD^t=DC^t$
gilmore-lie-groups_ EQ0356	110	■ 1.000	$\mathcal{H}=\hbar\omega\sum_{i=1}^N\left(a_i^{\dagger}a_i+\frac{1}{2}\right)$	$\mathcal{H}=\hbar\omega\sum_{i=1}^N\left(a_i^{\dagger}a_i+\frac{1}{2}\right)$	$\mathcal{H}=\hbar\omega\sum_{i=1}^N\left(a_i^{\dagger}a_i+\frac{1}{2}\right)$
gilmore-lie-groups_ EQ0357	111	■ 1.000	$H_n(x)=e^{x^2}\left(-\frac{d}{dx}\right)^ne^{-x^2}$	$H_n(x)=e^{x^2}\left(-\frac{d}{dx}\right)^ne^{-x^2}$	$H_n(x)=e^{x^2}\left(-\frac{d}{dx}\right)^ne^{-x^2}$
gilmore-lie-groups_ EQ0359	111	■ 1.000	$H_n(x)e^{-x^2/2}=e^{+x^2/2}\left(-\frac{d}{dx}\right)^ne^{-x^2}=\left(x-\frac{d}{dx}\right)^ne^{-x^2/2}$	$H_n(x)e^{-x^2/2}=e^{+x^2/2}\left(-\frac{d}{dx}\right)^ne^{-x^2}=\left(x-\frac{d}{dx}\right)^ne^{-x^2/2}$	$H_n(x)e^{-x^2/2}=e^{+x^2/2}\left(-\frac{d}{dx}\right)^ne^{-x^2}=\left(x-\frac{d}{dx}\right)^ne^{-x^2/2}$

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gilmore-lie-groups_ E08360 (6.33)	111	■ 1.000	$\langle x \mid n \rangle = \frac{1}{\sqrt{2^n n!}} \langle x \mid 0 \rangle = \frac{H_n(x) e^{-x^2/2}}{\sqrt{2^n n!} \sqrt{\pi}} = \psi_n(x)$	$\langle x \mid n \rangle = \frac{(\sqrt{2} a^\dagger)^n}{\sqrt{2^n n!}} \langle x \mid 0 \rangle = \frac{H_n(x) e^{-x^2/2}}{\sqrt{2^n n!} \sqrt{\pi}} = \psi_n(x)$	$\langle x \mid n \rangle = \frac{(\sqrt{2} a^\dagger)^n}{\sqrt{2^n n!}} \langle x \mid 0 \rangle = \frac{H_n(x) e^{-x^2/2}}{\sqrt{2^n n!} \sqrt{\pi}} = \psi_n(x)$
gilmore-lie-groups_ E08361	111	■ 1.000	$H = \sum_{i=1}^n \hbar \omega_i \left(a_i^\dagger a_i + \frac{1}{2} \right) + i \sum_{i < j} \theta_{ij} \left(a_i^\dagger a_j - a_j^\dagger a_i \right) + \sum_{i \leq j} q_{ij} \left(a_i^\dagger a_j + a_j^\dagger a_i \right)$	$H = \sum_{i=1}^n \hbar \omega_i \left(a_i^\dagger a_i + \frac{1}{2} \right) + i \sum_{i < j} \theta_{ij} \left(a_i^\dagger a_j - a_j^\dagger a_i \right) + \sum_{i \leq j} q_{ij} \left(a_i^\dagger a_j + a_j^\dagger a_i \right)$	$H = \sum_{i=1}^n \hbar \omega_i \left(a_i^\dagger a_i + \frac{1}{2} \right) + i \sum_{i < j} \theta_{ij} \left(a_i^\dagger a_j - a_j^\dagger a_i \right) + \sum_{i \leq j} q_{ij} \left(a_i^\dagger a_j + a_j^\dagger a_i \right)$
gilmore-lie-groups_ E08362	111	■ 1.000	$\Gamma_{ij} = \hbar \omega_i \delta_{ij} + (q + i\theta)_{ij}$	$\Gamma_{ij} = \hbar \omega_i \delta_{ij} + (q + i\theta)_{ij}$	$\Gamma_{ij} = \hbar \omega_i \delta_{ij} + (q + i\theta)_{ij}$
gilmore-lie-groups_ E08364 (7.2)	114	■ 1.000	$X = \begin{bmatrix} a & b+c \\ b-c & -a \end{bmatrix} \in \mathfrak{sl}(2; \mathbb{R})$	$X = \begin{bmatrix} a & b+c \\ b-c & -a \end{bmatrix} \in \mathfrak{sl}(2; \mathbb{R})$	$X = \begin{bmatrix} a & b+c \\ b-c & -a \end{bmatrix} \in \mathfrak{sl}(2; \mathbb{R})$
gilmore-lie-groups_ E08365 (7.3)	114	■ 1.000	$\operatorname{tr} \operatorname{EXP}(X) \geq -2$	$\operatorname{tr} \operatorname{EXP}(X) \geq -2$	$\operatorname{tr} \operatorname{EXP}(X) \geq -2$
gilmore-lie-groups_ E08367 (7.5)	114	■ 1.000	$\operatorname{tr} e^X = \operatorname{tr} S e^X S^{-1} = \operatorname{tr} e^{S X S^{-1}}$	$\operatorname{tr} e^X = \operatorname{tr} S e^X S^{-1} = \operatorname{tr} e^{S X S^{-1}}$	$\operatorname{tr} e^X = \operatorname{tr} S e^X S^{-1} = \operatorname{tr} e^{S X S^{-1}}$
gilmore-lie-groups_ E08370 (7.8)	116	■ 1.000	$(0, 0, c) \longrightarrow \begin{bmatrix} \cos c & \sin c \\ -\sin c & \cos c \end{bmatrix}$	$(0, 0, c) \longrightarrow \begin{bmatrix} \cos c & \sin c \\ -\sin c & \cos c \end{bmatrix}$	$(0, 0, c) \longrightarrow \begin{bmatrix} \cos c & \sin c \\ -\sin c & \cos c \end{bmatrix}$
gilmore-lie-groups_ E08371 (7.9)	116	■ 1.000	$(a, b, 0) \longrightarrow \begin{bmatrix} \cosh r + (a/r) \sinh r & (b/r) \sinh r \\ (b/r) \sinh r & \cosh r - (a/r) \sinh r \end{bmatrix} \quad r^2 = a^2 + b^2$	$(a, b, 0) \longrightarrow \begin{bmatrix} \cosh r + (a/r) \sinh r & (b/r) \sinh r \\ (b/r) \sinh r & \cosh r - (a/r) \sinh r \end{bmatrix} \quad r^2 = a^2 + b^2$	$(a, b, 0) \longrightarrow \begin{bmatrix} \cosh r + (a/r) \sinh r & (b/r) \sinh r \\ (b/r) \sinh r & \cosh r - (a/r) \sinh r \end{bmatrix} \quad r^2 = a^2 + b^2$
gilmore-lie-groups_ E08372 (7.10)	116	■ 1.000	$(X, X) = \operatorname{tr} X^2 = 2(a^2 + b^2 - c^2)$	$(X, X) = \operatorname{tr} X^2 = 2(a^2 + b^2 - c^2)$	$(X, X) = \operatorname{tr} X^2 = 2(a^2 + b^2 - c^2)$
gilmore-lie-groups_ E08373 (7.11)	116	■ 1.000	$\begin{bmatrix} a & b \\ b & -a \end{bmatrix} \xrightarrow{\operatorname{EXP}} \begin{bmatrix} z+y & x \\ x & z-y \end{bmatrix} \times \begin{bmatrix} \cos c & \sin c \\ -\sin c & \cos c \end{bmatrix}$	$\begin{bmatrix} a & b \\ b & -a \end{bmatrix} \xrightarrow{\operatorname{EXP}} \begin{bmatrix} z+y & x \\ x & z-y \end{bmatrix} \times \begin{bmatrix} \cos c & \sin c \\ -\sin c & \cos c \end{bmatrix}$	$\begin{bmatrix} a & b \\ b & -a \end{bmatrix} \xrightarrow{\operatorname{EXP}} \begin{bmatrix} z+y & x \\ x & z-y \end{bmatrix} \times \begin{bmatrix} \cos c & \sin c \\ -\sin c & \cos c \end{bmatrix}$
gilmore-lie-groups_ E08374 (7.12)	116	■ 1.000	$z^2 - x^2 - y^2 = 1$	$z^2 - x^2 - y^2 = 1$	$z^2 - x^2 - y^2 = 1$
gilmore-lie-groups_ E08376 (7.14)	117	■ 1.000	$z^2 + x^2 - y^2 = 1$	$z^2 + x^2 - y^2 = 1$	$z^2 + x^2 - y^2 = 1$

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gilmore-lie-groups_ E09379 (7.17)	119	■ 1.000	$\frac{s}{o}(2,1)=\left[\begin{array}{ccc} 0 & a_3 & a_2 \\ & -a_3 & 0 \\ & a_2 & a_1 \end{array}\right] \quad \frac{s}{u}(1,1)=\frac{i}{2}\left[\begin{array}{cc} b_3 & ib_1+b_2 \\ ib_1-b_2 & -b_3 \end{array}\right]$	$\mathfrak{so}(2,1)=\left[\begin{array}{ccc} 0 & a_3 & a_2 \\ -a_3 & 0 & a_1 \\ a_2 & a_1 & 0 \end{array}\right] \quad \mathfrak{su}(1,1)=\frac{i}{2}\left[\begin{array}{cc} b_3 & ib_1+b_2 \\ ib_1-b_2 & -b_3 \end{array}\right]$	$\mathfrak{so}(2,1)=\left[\begin{array}{ccc} 0 & a_3 & a_2 \\ -a_3 & 0 & a_1 \\ a_2 & a_1 & 0 \end{array}\right] \quad \mathfrak{su}(1,1)=\frac{i}{2}\left[\begin{array}{cc} b_3 & ib_1+b_2 \\ ib_1-b_2 & -b_3 \end{array}\right]$
gilmore-lie-groups_ E09383	120	■ 1.000	$\pi(\sin\theta\cos\phi,\sin\theta\sin\phi,\cos\theta)\sim-\pi(\sin\theta\cos\phi,\sin\theta\sin\phi,\cos\theta)$	$\pi(\sin\theta\cos\phi,\sin\theta\sin\phi,\cos\theta)\sim-\pi(\sin\theta\cos\phi,\sin\theta\sin\phi,\cos\theta)$	$\pi(\sin\theta\cos\phi,\sin\theta\sin\phi,\cos\theta)\sim-\pi(\sin\theta\cos\phi,\sin\theta\sin\phi,\cos\theta)$
gilmore-lie-groups_ E09387	121	■ 1.000	$G_1=\bar{G}/D_1\rightarrow\bar{G}/D_{\mathrm{MAX}}\leftarrow\bar{G}/D_2=G_2$	$G_1=\bar{G}/D_1\rightarrow\bar{G}/D_{\mathrm{MAX}}\leftarrow\bar{G}/D_2=G_2$	$G_1=\bar{G}/D_1\rightarrow\bar{G}/D_{\mathrm{MAX}}\leftarrow\bar{G}/D_2=G_2$
gilmore-lie-groups_ E09394 (7.30)	124	■ 1.000	$\left[X_l,X_r\right]=X_\delta \quad \left[X_l,X_\delta\right]=\left[X_r,X_\delta\right]=0$	$\left[X_l,X_r\right]=X_\delta \quad \left[X_l,X_\delta\right]=\left[X_r,X_\delta\right]=0$	$\left[X_l,X_r\right]=X_\delta \quad \left[X_l,X_\delta\right]=\left[X_r,X_\delta\right]=0$
gilmore-lie-groups_ E09395 (7.31)	124	■ 1.000	$\left[a,a^\dagger\right]=I \quad \left[a,I\right]=\left[a^\dagger,I\right]=0$	$\left[a,a^\dagger\right]=I \quad \left[a,I\right]=\left[a^\dagger,I\right]=0$	$\left[a,a^\dagger\right]=I \quad \left[a,I\right]=\left[a^\dagger,I\right]=0$
gilmore-lie-groups_ E09396 (7.32)	124	■ 1.000	$\begin{aligned} &X_l\rightarrow a \\ &X_r\rightarrow a^\dagger \\ &X_\delta\rightarrow I \end{aligned}$	$\begin{aligned} &X_l\rightarrow a \\ &X_r\rightarrow a^\dagger \\ &X_\delta\rightarrow I \end{aligned}$	$\begin{aligned} &X_l\rightarrow a \\ &X_r\rightarrow a^\dagger \\ &X_\delta\rightarrow I \end{aligned}$
gilmore-lie-groups_ E09401 (7.37)	125	■ 1.000	$\begin{aligned} &X_\eta\rightarrow\hat{n} \\ &X_l\rightarrow a \\ &X_r\rightarrow a^\dagger \\ &X_\delta\rightarrow I \end{aligned}$	$\begin{aligned} &X_\eta\rightarrow\hat{n} \\ &X_l\rightarrow a \\ &X_r\rightarrow a^\dagger \\ &X_\delta\rightarrow I \end{aligned}$	$\begin{aligned} &X_\eta\rightarrow\hat{n} \\ &X_l\rightarrow a \\ &X_r\rightarrow a^\dagger \\ &X_\delta\rightarrow I \end{aligned}$
gilmore-lie-groups_ E09402 (7.38)	125	■ 1.000	$\begin{aligned} &\operatorname{EXP}\left(\eta a^\dagger a+r a^\dagger+l a\right)=\operatorname{EXP}\left(r^\prime a^\dagger\right) \operatorname{EXP}\left(\eta^\prime a^\dagger a+\delta^\prime I\right) \operatorname{EXP}\left(l^\prime a\right) \\ &\parallel \\ &\left[\begin{array}{ccc} 1 & \left(e^\eta-1\right) l / \eta & \left(e^\eta-1-\eta\right) l r / \eta^2 \\ 0 & e^\eta & \left(e^\eta-1\right) r / \eta \\ 0 & 0 & 1 \end{array}\right]=\left[\begin{array}{ccc} 1 & l^\prime & \delta^\prime \\ 0 & e^{\eta^\prime} & r^\prime \\ 0 & 0 & 1 \end{array}\right] \end{aligned}$	$\begin{aligned} &\operatorname{EXP}\left(\eta a^\dagger a+r a^\dagger+l a\right)=\operatorname{EXP}\left(r^\prime a^\dagger\right) \operatorname{EXP}\left(\eta^\prime a^\dagger a+\delta^\prime I\right) \operatorname{EXP}\left(l^\prime a\right) \\ &\parallel \\ &\left[\begin{array}{ccc} 1 & \left(e^\eta-1\right) l / \eta & \left(e^\eta-1-\eta\right) l r / \eta^2 \\ 0 & e^\eta & \left(e^\eta-1\right) r / \eta \\ 0 & 0 & 1 \end{array}\right]=\left[\begin{array}{ccc} 1 & l^\prime & \delta^\prime \\ 0 & e^{\eta^\prime} & r^\prime \\ 0 & 0 & 1 \end{array}\right] \end{aligned}$	$\begin{aligned} &\operatorname{EXP}\left(\eta a^\dagger a+r a^\dagger+l a\right)=\operatorname{EXP}\left(r^\prime a^\dagger\right) \operatorname{EXP}\left(\eta^\prime a^\dagger a+\delta^\prime I\right) \operatorname{EXP}\left(l^\prime a\right) \\ &\parallel \\ &\left[\begin{array}{ccc} 1 & \left(e^\eta-1\right) l / \eta & \left(e^\eta-1-\eta\right) l r / \eta^2 \\ 0 & e^\eta & \left(e^\eta-1\right) r / \eta \\ 0 & 0 & 1 \end{array}\right]=\left[\begin{array}{ccc} 1 & l^\prime & \delta^\prime \\ 0 & e^{\eta^\prime} & r^\prime \\ 0 & 0 & 1 \end{array}\right] \end{aligned}$
gilmore-lie-groups_ E09405 (7.41)	125	■ 1.000	$\langle 0 e^{\eta a^\dagger a+r a^\dagger+l a} 0\rangle=e^{\delta^\prime}=\operatorname{EXP}\left(\frac{\left(e^\eta-1-\eta\right) l r}{\eta^2}\right)$	$\langle 0 e^{\eta a^\dagger a+r a^\dagger+l a} 0\rangle=e^{\delta^\prime}=\operatorname{EXP}\left(\frac{\left(e^\eta-1-\eta\right) l r}{\eta^2}\right)$	$\langle 0 e^{\eta a^\dagger a+r a^\dagger+l a} 0\rangle=e^{\delta^\prime}=\operatorname{EXP}\left(\frac{\left(e^\eta-1-\eta\right) l r}{\eta^2}\right)$
gilmore-lie-groups_ E09406 (7.42)	125	■ 1.000	$\left\langle\begin{array}{c} j \\ -j \end{array}\right e^{i \theta \cdot \mathbf{J}}\left \begin{array}{c} j \\ -j \end{array}\right\rangle$	$\left\langle\begin{array}{c} j \\ -j \end{array}\right e^{i \theta \cdot \mathbf{J}}\left \begin{array}{c} j \\ -j \end{array}\right\rangle$	$\left\langle\begin{array}{c} j \\ -j \end{array}\right e^{i \theta \cdot \mathbf{J}}\left \begin{array}{c} j \\ -j \end{array}\right\rangle$

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gilmore-lie-groups_ EQ0407 (7.43)	126	■ 1.000	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta'_+ J_+} e^{i\theta'_z J_z} e^{i\theta'_- J_-} \left \begin{array}{c} j \\ -j \end{array} \right\rangle$	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta'_+ J_+} e^{i\theta'_z J_z} e^{i\theta'_- J_-} \left \begin{array}{c} j \\ -j \end{array} \right\rangle$	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta'_+ J_+} e^{i\theta'_z J_z} e^{i\theta'_- J_-} \left \begin{array}{c} j \\ -j \end{array} \right\rangle$
gilmore-lie-groups_ EQ0408 (7.44)	126	■ 1.000	$e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = (I + i\theta'_- J_- + \dots) \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left \begin{array}{c} j \\ -j \end{array} \right\rangle$	$e^{i\theta'_- J_-} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = (I + i\theta'_- J_- + \dots) \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left \begin{array}{c} j \\ -j \end{array} \right\rangle$	$e^{i\theta'_- J_-} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = (I + i\theta'_- J_- + \dots) \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left \begin{array}{c} j \\ -j \end{array} \right\rangle$
gilmore-lie-groups_ EQ0409 (7.45)	126	■ 1.000	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta'_z J_z} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-ij\theta'_z}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta'_z J_z} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-ij\theta'_z}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = \left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta'_z J_z} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-ij\theta'_z}$
gilmore-lie-groups_ EQ0410 (7.46)	126	■ 1.000	$\begin{aligned} & e^{i\theta \cdot \mathbf{J}} \rightarrow \text{EXP} \frac{i}{2} \left[\begin{array}{cc} \theta_z & \theta_x - i\theta_y \\ \theta_x + i\theta_y & -\theta_z \end{array} \right] \\ & = \left[\begin{array}{cc} \cos(\theta/2) + i(\theta_z/\theta) \sin(\theta/2) & i[(\theta_x - i\theta_y)/\theta] \sin(\theta/2) \\ i[(\theta_x + i\theta_y)/\theta] \sin(\theta/2) & \cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2) \end{array} \right] \end{aligned}$	$e^{i\theta \cdot \mathbf{J}} \rightarrow \text{EXP} \frac{i}{2} \left[\begin{array}{cc} \theta_z & \theta_x - i\theta_y \\ \theta_x + i\theta_y & -\theta_z \end{array} \right] \\ = \left[\begin{array}{cc} \cos(\theta/2) + i(\theta_z/\theta) \sin(\theta/2) & i[(\theta_x - i\theta_y)/\theta] \sin(\theta/2) \\ i[(\theta_x + i\theta_y)/\theta] \sin(\theta/2) & \cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2) \end{array} \right]$	$e^{i\theta \cdot \mathbf{J}} \rightarrow \text{EXP} \frac{i}{2} \left[\begin{array}{cc} \theta_z & \theta_x - i\theta_y \\ \theta_x + i\theta_y & -\theta_z \end{array} \right] \\ = \left[\begin{array}{cc} \cos(\theta/2) + i(\theta_z/\theta) \sin(\theta/2) & i[(\theta_x - i\theta_y)/\theta] \sin(\theta/2) \\ i[(\theta_x + i\theta_y)/\theta] \sin(\theta/2) & \cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2) \end{array} \right]$
gilmore-lie-groups_ EQ0412 (7.48)	126	■ 1.000	$e^{-i\theta'_z/2} = \cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2)$	$e^{-i\theta'_z/2} = \cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2)$	$e^{-i\theta'_z/2} = \cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2)$
gilmore-lie-groups_ EQ0413 (7.49)	126	■ 1.000	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-ij\theta'_z} = (e^{-i\theta'_z/2})^{2j} = [\cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2)]^{2j}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-ij\theta'_z} = (e^{-i\theta'_z/2})^{2j} = [\cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2)]^{2j}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \right e^{i\theta \cdot \mathbf{J}} \left \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-ij\theta'_z} = (e^{-i\theta'_z/2})^{2j} = [\cos(\theta/2) - i(\theta_z/\theta) \sin(\theta/2)]^{2j}$
gilmore-lie-groups_ EQ0418 (7.54)	128	■ 1.000	$H \psi\rangle = i\hbar \frac{\partial}{\partial t} \psi\rangle$	$H \psi\rangle = i\hbar \frac{\partial}{\partial t} \psi\rangle$	$H \psi\rangle = i\hbar \frac{\partial}{\partial t} \psi\rangle$
gilmore-lie-groups_ EQ0419 (7.55)	128	■ 1.000	$ \psi(t+\delta t)\rangle = \left(I - \frac{i}{\hbar} H \delta t\right) \psi(t)\rangle = e^{-\frac{i}{\hbar} H \delta t} \psi(t)\rangle$	$ \psi(t+\delta t)\rangle = \left(I - \frac{i}{\hbar} H \delta t\right) \psi(t)\rangle = e^{-\frac{i}{\hbar} H \delta t} \psi(t)\rangle$	$ \psi(t+\delta t)\rangle = \left(I - \frac{i}{\hbar} H \delta t\right) \psi(t)\rangle = e^{-\frac{i}{\hbar} H \delta t} \psi(t)\rangle$

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gilmore-lie-groups_ E00420 (7.56)	128	■ 1.000	$\begin{aligned} U(t_f, t_i) &= U(t_f, t_f - \delta t) \cdots U(t_i + 2\delta t, t_i + \delta t) U(t_i + \delta t, t_i) \\ &= \prod U(t_i + (n+1)\delta t, t_i + n\delta t) = \int_{t_i}^{t_f} U(\tau) d\tau \\ &= T \int_{t_i}^{t_f} e^{-\frac{i}{\hbar} H(t)} dt \end{aligned}$	$U(t_f, t_i) = U(t_f, t_f - \delta t) \cdots U(t_i + 2\delta t, t_i + \delta t) U(t_i + \delta t, t_i)$ $= \prod U(t_i + (n+1)\delta t, t_i + n\delta t) = \int_{t_i}^{t_f} U(\tau) d\tau$ $= T \int_{t_i}^{t_f} e^{-\frac{i}{\hbar} H(t)} dt$	$U(t_f, t_i) = U(t_f, t_f - \delta t) \cdots U(t_i + 2\delta t, t_i + \delta t) U(t_i + \delta t, t_i)$ $= \prod U(t_i + (n+1)\delta t, t_i + n\delta t) = \int_{t_i}^{t_f} U(\tau) d\tau$ $= T \int_{t_i}^{t_f} e^{-\frac{i}{\hbar} H(t)} dt$
gilmore-lie-groups_ E00421 (7.57)	129	■ 1.000	$H = \epsilon(t) J_z + \alpha(t) J_+ + \alpha^*(t) J_- \xrightarrow{j \rightarrow \frac{1}{2}} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix}$	$H = \epsilon(t) J_z + \alpha(t) J_+ + \alpha^*(t) J_- \xrightarrow{j \rightarrow \frac{1}{2}} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix}$	$H = \epsilon(t) J_z + \alpha(t) J_+ + \alpha^*(t) J_- \xrightarrow{j \rightarrow \frac{1}{2}} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix}$
gilmore-lie-groups_ E00422 (7.58)	129	■ 1.000	$\frac{d}{dt} \begin{bmatrix} a(t) & b(t) \\ -b^*(t) & a^*(t) \end{bmatrix} = -\frac{i}{\hbar} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix} \begin{bmatrix} a(t) & b(t) \\ -b^*(t) & a^*(t) \end{bmatrix}$	$\frac{d}{dt} \begin{bmatrix} a(t) & b(t) \\ -b^*(t) & a^*(t) \end{bmatrix} = -\frac{i}{\hbar} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix} \begin{bmatrix} a(t) & b(t) \\ -b^*(t) & a^*(t) \end{bmatrix}$	$\frac{d}{dt} \begin{bmatrix} a(t) & b(t) \\ -b^*(t) & a^*(t) \end{bmatrix} = -\frac{i}{\hbar} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix} \begin{bmatrix} a(t) & b(t) \\ -b^*(t) & a^*(t) \end{bmatrix}$
gilmore-lie-groups_ E00423 (7.59)	129	■ 1.000	$H = \omega(t) a^\dagger a + \alpha(t) a^\dagger + \alpha^*(t) a + \delta(t) I \rightarrow \begin{bmatrix} 0 & \alpha^*(t) & \delta(t) \\ 0 & \omega(t) & \alpha(t) \\ 0 & 0 & 0 \end{bmatrix}$	$H = \omega(t) a^\dagger a + \alpha(t) a^\dagger + \alpha^*(t) a + \delta(t) I \rightarrow \begin{bmatrix} 0 & \alpha^*(t) & \delta(t) \\ 0 & \omega(t) & \alpha(t) \\ 0 & 0 & 0 \end{bmatrix}$	$H = \omega(t) a^\dagger a + \alpha(t) a^\dagger + \alpha^*(t) a + \delta(t) I \rightarrow \begin{bmatrix} 0 & \alpha^*(t) & \delta(t) \\ 0 & \omega(t) & \alpha(t) \\ 0 & 0 & 0 \end{bmatrix}$
gilmore-lie-groups_ E00425 (7.61)	130	■ 1.000	$\langle \mathcal{O} \rangle = \text{tr } \rho \mathcal{O}$	$\langle \mathcal{O} \rangle = \text{tr } \rho \mathcal{O}$	$\langle \mathcal{O} \rangle = \text{tr } \rho \mathcal{O}$
gilmore-lie-groups_ E00426 (7.62)	130	■ 1.000	$H = \epsilon J_z + \alpha J_+ + \alpha^* J_- \xrightarrow{j \rightarrow \frac{1}{2}} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix}$	$H = \epsilon J_z + \alpha J_+ + \alpha^* J_- \xrightarrow{j \rightarrow \frac{1}{2}} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix}$	$H = \epsilon J_z + \alpha J_+ + \alpha^* J_- \xrightarrow{j \rightarrow \frac{1}{2}} \begin{bmatrix} \frac{1}{2}\epsilon(t) & \alpha(t) \\ \alpha^*(t) & -\frac{1}{2}\epsilon(t) \end{bmatrix}$
gilmore-lie-groups_ E00427 (7.63)	131	■ 1.000	$e^{-\beta H} e^\Lambda \rightarrow \left(I_2 \cosh(\beta H) - \beta \begin{bmatrix} \epsilon/2 & \alpha \\ \alpha^* & -\epsilon/2 \end{bmatrix} \frac{\sinh(\beta H)}{\beta H } \right) \times \left(I_2 \cosh(\Lambda) + \begin{bmatrix} \lambda_3/2 & \lambda \\ \lambda^* & -\lambda_3/2 \end{bmatrix} \right) \frac{\sinh(\Lambda)}{ \Lambda }$	$e^{-\beta H} e^\Lambda \rightarrow \left(I_2 \cosh(\beta H) - \beta \begin{bmatrix} \epsilon/2 & \alpha \\ \alpha^* & -\epsilon/2 \end{bmatrix} \frac{\sinh(\beta H)}{\beta H } \right) \times \left(I_2 \cosh(\Lambda) + \begin{bmatrix} \lambda_3/2 & \lambda \\ \lambda^* & -\lambda_3/2 \end{bmatrix} \right) \frac{\sinh(\Lambda)}{ \Lambda }$	$e^{-\beta H} e^\Lambda \rightarrow \left(I_2 \cosh(\beta H) - \beta \begin{bmatrix} \epsilon/2 & \alpha \\ \alpha^* & -\epsilon/2 \end{bmatrix} \frac{\sinh(\beta H)}{\beta H } \right) \times \left(I_2 \cosh(\Lambda) + \begin{bmatrix} \lambda_3/2 & \lambda \\ \lambda^* & -\lambda_3/2 \end{bmatrix} \right) \frac{\sinh(\Lambda)}{ \Lambda }$

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gilmore-lie-groups_ EQ0428 (7.64)	131	■ 1.000	$\begin{gathered} \operatorname{tr} e^{-\beta H} e^{\Lambda} \rightarrow \\ \Lambda \rightarrow 2 \cosh(\beta H) \cosh(\Lambda) - 2 \frac{H \cdot \Lambda}{\sqrt{H \cdot H} \sqrt{\Lambda \cdot \Lambda}} \sinh(\beta H) \sinh(\Lambda) \end{gathered}$	$\operatorname{tr} e^{-\beta H} e^{\Lambda} \rightarrow 2 \cosh(\beta H) \cosh(\Lambda) - 2 \frac{H \cdot \Lambda}{\sqrt{H \cdot H} \sqrt{\Lambda \cdot \Lambda}} \sinh(\beta H) \sinh(\Lambda)$	$\operatorname{tr} e^{-\beta H} e^{\Lambda} \rightarrow 2 \cosh(\beta H) \cosh(\Lambda) - 2 \frac{H \cdot \Lambda}{\sqrt{H \cdot H} \sqrt{\Lambda \cdot \Lambda}} \sinh(\beta H) \sinh(\Lambda)$
gilmore-lie-groups_ EQ0429 (7.65)	131	■ 1.000	$\operatorname{tr} e^{-\beta H} e^{\lambda \cdot \mathbf{J}} = \operatorname{tr} \left[\begin{array}{cc} e^{+\mu(H, \Lambda)/2} & 0 \\ 0 & e^{-\mu(H, \Lambda)/2} \end{array} \right] = 2 \cosh(\mu(H, \Lambda)/2)$	$\operatorname{tr} e^{-\beta H} e^{\lambda \cdot \mathbf{J}} = \operatorname{tr} \left[\begin{array}{cc} e^{+\mu(H, \Lambda)/2} & 0 \\ 0 & e^{-\mu(H, \Lambda)/2} \end{array} \right] = 2 \cosh(\mu(H, \Lambda)/2)$	$\operatorname{tr} e^{-\beta H} e^{\lambda \cdot \mathbf{J}} = \operatorname{tr} \left[\begin{array}{cc} e^{+\mu(H, \Lambda)/2} & 0 \\ 0 & e^{-\mu(H, \Lambda)/2} \end{array} \right] = 2 \cosh(\mu(H, \Lambda)/2)$
gilmore-lie-groups_ EQ0430 (7.66)	131	■ 1.000	$\chi(H, \Lambda, J) = \frac{\sinh \left(J + \frac{1}{2} \right) \mu(H, \Lambda)}{\sinh \left(\frac{1}{2} \right) \mu(H, \Lambda)}$	$\chi(H, \Lambda, J) = \frac{\sinh \left(J + \frac{1}{2} \right) \mu(H, \Lambda)}{\sinh \left(\frac{1}{2} \right) \mu(H, \Lambda)}$	$\chi(H, \Lambda, J) = \frac{\sinh \left(J + \frac{1}{2} \right) \mu(H, \Lambda)}{\sinh \left(\frac{1}{2} \right) \mu(H, \Lambda)}$
gilmore-lie-groups_ EQ0431 (7.67)	131	■ 1.000	$\langle e^{\Lambda} \rangle = \frac{\chi(H, \Lambda, J)}{\chi(H, 0, J)}$	$\langle e^{\Lambda} \rangle = \frac{\chi(H, \Lambda, J)}{\chi(H, 0, J)}$	$\langle e^{\Lambda} \rangle = \frac{\chi(H, \Lambda, J)}{\chi(H, 0, J)}$
gilmore-lie-groups_ EQ0432 (7.68)	132	■ 1.000	$H = \hbar \omega a^{\dagger} a + \alpha a^{\dagger} + \alpha^* a + \delta I \rightarrow \begin{bmatrix} 0 & \alpha^* & \delta \\ 0 & \hbar \omega & \alpha \\ 0 & 0 & 0 \end{bmatrix}$	$H = \hbar \omega a^{\dagger} a + \alpha a^{\dagger} + \alpha^* a + \delta I \rightarrow \begin{bmatrix} 0 & \alpha^* & \delta \\ 0 & \hbar \omega & \alpha \\ 0 & 0 & 0 \end{bmatrix}$	$H = \hbar \omega a^{\dagger} a + \alpha a^{\dagger} + \alpha^* a + \delta I \rightarrow \begin{bmatrix} 0 & \alpha^* & \delta \\ 0 & \hbar \omega & \alpha \\ 0 & 0 & 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0434	132	■ 1.000	$e^{-\beta M(H)} e^{M(\Lambda)} = \begin{bmatrix} 1 & Z_l & * \\ 0 & * & Z_r \\ 0 & 0 & 1 \end{bmatrix}$	$e^{-\beta M(H)} e^{M(\Lambda)} = \begin{bmatrix} 1 & Z_l & * \\ 0 & * & Z_r \\ 0 & 0 & 1 \end{bmatrix}$	$e^{-\beta M(H)} e^{M(\Lambda)} = \begin{bmatrix} 1 & Z_l & * \\ 0 & * & Z_r \\ 0 & 0 & 1 \end{bmatrix}$
gilmore-lie-groups_ EQ0437	133	■ 1.000	$\operatorname{tr} e^{A a^{\dagger} a + B I} = \frac{e^B}{1 - e^A}$	$\operatorname{tr} e^{A a^{\dagger} a + B I} = \frac{e^B}{1 - e^A}$	$\operatorname{tr} e^{A a^{\dagger} a + B I} = \frac{e^B}{1 - e^A}$
gilmore-lie-groups_ EQ0438	133	■ 1.000	$\log(\chi(H, \Lambda, T)) = B - A - \log(e^{-A} - 1)$	$\log(\chi(H, \Lambda, T)) = B - A - \log(e^{-A} - 1)$	$\log(\chi(H, \Lambda, T)) = B - A - \log(e^{-A} - 1)$

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gilmore-lie-groups_ E08439 (7.70)	133	■ 1.000	$\begin{aligned} -A &= \beta \hbar \omega - \lambda_n \\ B &= \frac{e^{-\beta \hbar \omega} - 1 + \beta \hbar \omega}{(\hbar \omega)^2} \alpha^* \alpha - \beta \delta + d + \frac{e^{\lambda_n} - 1 - \lambda_n}{\lambda_n^2} \lambda^* \lambda \\ &\quad + \frac{e^{-\beta \hbar \omega} - 1}{\hbar \omega} \frac{e^{\lambda_n} - 1}{\lambda_n} (\alpha^* \lambda + \alpha \lambda^*) / (1 - e^{-(\beta \hbar \omega - \lambda_n)}) \\ &\quad + \left[e^{-\beta \hbar \omega} \left(\frac{e^{\lambda_n} - 1}{\lambda_n} \right)^2 \lambda^* \lambda + e^{\lambda_n} \left(\frac{e^{-\beta \hbar \omega} - 1}{\hbar \omega} \right)^2 \alpha^* \alpha \right] / [1 - e^{-(\beta \hbar \omega - \lambda_n)}] \end{aligned}$	$-A = \beta \hbar \omega - \lambda_n$ $B = \frac{e^{-\beta \hbar \omega} - 1 + \beta \hbar \omega}{(\hbar \omega)^2} \alpha^* \alpha - \beta \delta + d + \frac{e^{\lambda_n} - 1 - \lambda_n}{\lambda_n^2} \lambda^* \lambda$ $+ \frac{e^{-\beta \hbar \omega} - 1}{\hbar \omega} \frac{e^{\lambda_n} - 1}{\lambda_n} (\alpha^* \lambda + \alpha \lambda^*) / (1 - e^{-(\beta \hbar \omega - \lambda_n)})$ $+ \left[e^{-\beta \hbar \omega} \left(\frac{e^{\lambda_n} - 1}{\lambda_n} \right)^2 \lambda^* \lambda + e^{\lambda_n} \left(\frac{e^{-\beta \hbar \omega} - 1}{\hbar \omega} \right)^2 \alpha^* \alpha \right] / [1 - e^{-(\beta \hbar \omega - \lambda_n)}]$	$-A = \beta \hbar \omega - \lambda_n$ $B = \frac{e^{-\beta \hbar \omega} - 1 + \beta \hbar \omega}{(\hbar \omega)^2} \alpha^* \alpha - \beta \delta + d + \frac{e^{\lambda_n} - 1 - \lambda_n}{\lambda_n^2} \lambda^* \lambda$ $+ \frac{e^{-\beta \hbar \omega} - 1}{\hbar \omega} \frac{e^{\lambda_n} - 1}{\lambda_n} (\alpha^* \lambda + \alpha \lambda^*) / (1 - e^{-(\beta \hbar \omega - \lambda_n)})$ $+ \left[e^{-\beta \hbar \omega} \left(\frac{e^{\lambda_n} - 1}{\lambda_n} \right)^2 \lambda^* \lambda + e^{\lambda_n} \left(\frac{e^{-\beta \hbar \omega} - 1}{\hbar \omega} \right)^2 \alpha^* \alpha \right] / [1 - e^{-(\beta \hbar \omega - \lambda_n)}]$
gilmore-lie-groups_ E08440	135	■ 1.000	$a^{\dagger} = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right) \quad a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$	$a^{\dagger} = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right) \quad a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$	$a^{\dagger} = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right) \quad a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$
gilmore-lie-groups_ E08441	135	■ 1.000	$\langle n' x^k n \rangle = \frac{d^k}{d\lambda^k} \langle n' e^{\lambda x} n \rangle_{\lambda=0}$	$\langle n' x^k n \rangle = \frac{d^k}{d\lambda^k} \langle n' e^{\lambda x} n \rangle_{\lambda=0}$	$\langle n' x^k n \rangle = \frac{d^k}{d\lambda^k} \langle n' e^{\lambda x} n \rangle_{\lambda=0}$
gilmore-lie-groups_ E08442	135	■ 1.000	$e^{\lambda x} = e^{\lambda(a^{\dagger} + a)/\sqrt{2}} = \text{EXP} \left(\frac{\lambda}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & \lambda/\sqrt{2} & \lambda^2/4 \\ 0 & 1 & \lambda/\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}$	$e^{\lambda x} = e^{\lambda(a^{\dagger} + a)/\sqrt{2}} = \text{EXP} \left(\frac{\lambda}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & \lambda/\sqrt{2} & \lambda^2/4 \\ 0 & 1 & \lambda/\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}$	$e^{\lambda x} = e^{\lambda(a^{\dagger} + a)/\sqrt{2}} = \text{EXP} \left(\frac{\lambda}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & \lambda/\sqrt{2} & \lambda^2/4 \\ 0 & 1 & \lambda/\sqrt{2} \\ 0 & 0 & 1 \end{bmatrix}$
gilmore-lie-groups_ E08444	135	■ 1.000	$e^{\lambda(a^{\dagger} + a)/\sqrt{2}} = e^{\lambda a^{\dagger}/\sqrt{2}} e^{\lambda^2/4} e^{\lambda a/\sqrt{2}}$	$e^{\lambda(a^{\dagger} + a)/\sqrt{2}} = e^{\lambda a^{\dagger}/\sqrt{2}} e^{\lambda^2/4} e^{\lambda a/\sqrt{2}}$	$e^{\lambda(a^{\dagger} + a)/\sqrt{2}} = e^{\lambda a^{\dagger}/\sqrt{2}} e^{\lambda^2/4} e^{\lambda a/\sqrt{2}}$
gilmore-lie-groups_ E08446 (7.71)	136	■ 1.000	$\langle e^{ikx} \rangle = \frac{\text{tr} e^{ikx} e^{-\beta \mathcal{H}}}{\text{tr} e^{-\beta \mathcal{H}}}$	$\langle e^{ikx} \rangle = \frac{\text{tr} e^{ikx} e^{-\beta \mathcal{H}}}{\text{tr} e^{-\beta \mathcal{H}}}$	$\langle e^{ikx} \rangle = \frac{\text{tr} e^{ikx} e^{-\beta \mathcal{H}}}{\text{tr} e^{-\beta \mathcal{H}}}$
gilmore-lie-groups_ E08447 (7.72)	136	■ 1.000	$\text{tr} e^{ikx} e^{-\beta \mathcal{H}} = \sum_{n=0}^{\infty} \langle n e^{ikx} n \rangle e^{-n\beta \hbar \omega} = \sum_{n=0}^{\infty} \langle n e^{ikx} e^{-n\beta \hbar \omega} n \rangle$	$\text{tr} e^{ikx} e^{-\beta \mathcal{H}} = \sum_{n=0}^{\infty} \langle n e^{ikx} n \rangle e^{-n\beta \hbar \omega} = \sum_{n=0}^{\infty} \langle n e^{ikx} e^{-n\beta \hbar \omega} n \rangle$	$\text{tr} e^{ikx} e^{-\beta \mathcal{H}} = \sum_{n=0}^{\infty} \langle n e^{ikx} n \rangle e^{-n\beta \hbar \omega} = \sum_{n=0}^{\infty} \langle n e^{ikx} e^{-n\beta \hbar \omega} n \rangle$
gilmore-lie-groups_ E08448 (7.73)	136	■ 1.000	$\text{tr} e^{ikx} e^{-\beta \hbar \omega a^{\dagger} a} = \text{tr} e^{-\beta \hbar \omega a^{\dagger} a} e^{\delta} = e^{\delta} \text{tr} e^{-\beta \hbar \omega a^{\dagger} a}$	$\text{tr} e^{ikx} e^{-\beta \hbar \omega a^{\dagger} a} = \text{tr} e^{-\beta \hbar \omega a^{\dagger} a} e^{\delta} = e^{\delta} \text{tr} e^{-\beta \hbar \omega a^{\dagger} a}$	$\text{tr} e^{ikx} e^{-\beta \hbar \omega a^{\dagger} a} = \text{tr} e^{-\beta \hbar \omega a^{\dagger} a} e^{\delta} = e^{\delta} \text{tr} e^{-\beta \hbar \omega a^{\dagger} a}$

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gilmore-lie-groups_ E00449 (7.74)	136	■ 1.000	$M(S) M\left(e^{\mathrm{i} \, k \, x}\right) M\left(e^{-\beta \hbar \omega \, \hbar \omega \, \hat{n}}\right) M\left(S^{-1}\right)=M\left(e^{-\beta \hbar \omega \, \hbar \omega \, \hat{n}}\right) M\left(e^{\delta}\right)$	$M(S)M\left(e^{ikx}\right)M\left(e^{-\beta\hbar\omega\hat{n}}\right)M\left(S^{-1}\right)=M\left(e^{-\beta\hbar\omega'\hat{n}}\right)M\left(e^{\delta}\right)$	$M(S) M(e^{ikx}) M(e^{-\beta\hbar\omega\hat{n}}) M(S^{-1}) = M(e^{-\beta\hbar\omega'\hat{n}}) M(e^{\delta})$
gilmore-lie-groups_ E00450 (7.76)	136	■ 1.000	$\left\langle e^{\mathrm{i} \, k \, x}\right\rangle=e^{\delta} \quad \delta=-\frac{1}{2} k^2 \coth \left(\frac{1}{2} \beta \hbar \omega\right)$	$\left\langle e^{ikx}\right\rangle=e^{\delta} \quad \delta=-\frac{1}{2}k^2\coth\left(\frac{1}{2}\beta\hbar\omega\right)$	$\langle e^{ikx}\rangle=e^{\delta} \quad \delta=-\frac{1}{2}k^2\coth\left(\frac{1}{2}\beta\hbar\omega\right)$
gilmore-lie-groups_ E00451 (7.77)	137	■ 1.000	$\left\langle e^{x^i X_i}\right\rangle=\frac{\mathrm{tr} e^{R\left(x^i X_i\right)} e^{-\beta \mathcal{H}}}{\mathrm{tr} e^{-\beta \mathcal{H}}} \rightarrow \frac{\mathrm{tr} R\left(e^{x^i X_i}\right) R\left(e^{-\beta a^i X_i}\right)}{\mathrm{tr} R\left(e^{-\beta a^i X_i}\right)}$	$\left\langle e^{x^i X_i}\right\rangle=\frac{\mathrm{tr} R\left(e^{x^i X_i}\right) e^{-\beta \mathcal{H}}}{\mathrm{tr} e^{-\beta \mathcal{H}}} \rightarrow \frac{\mathrm{tr} R\left(e^{x^i X_i}\right) R\left(e^{-\beta a^i X_i}\right)}{\mathrm{tr} R\left(e^{-\beta a^i X_i}\right)}$	$\langle e^{x^i X_i}\rangle=\frac{\mathrm{tr} e^{R(x^i X_i)} e^{-\beta \mathcal{H}}}{\mathrm{tr} e^{-\beta \mathcal{H}}} \rightarrow \frac{\mathrm{tr} R(e^{x^i X_i}) R(e^{-\beta a^i X_i})}{\mathrm{tr} R(e^{-\beta a^i X_i})}$
gilmore-lie-groups_ E00453 (7.79)	137	■ 1.000	$\left\langle e^{x^i X_i}\right\rangle=\frac{\mathrm{tr} R\left(e^{-\beta d^i(x, a) H_i}\right)}{\mathrm{tr} R\left(e^{-\beta d^i(0, a) H_i}\right)}$	$\left\langle e^{x^i X_i}\right\rangle=\frac{\mathrm{tr} R\left(e^{-\beta d^i(x, a) H_i}\right)}{\mathrm{tr} R\left(e^{-\beta d^i(0, a) H_i}\right)}$	$\langle e^{x^i X_i}\rangle=\frac{\mathrm{tr} R(e^{-\beta d^i(x, a) H_i})}{\mathrm{tr} R(e^{-\beta d^i(0, a) H_i})}$
gilmore-lie-groups_ E00454 (7.80)	137	■ 1.000	$U(\alpha) 0\rangle=e^{\left(\alpha a^{\dagger}-\alpha^* a\right)} 0\rangle=e^{\beta a^{\dagger}} e^{\delta I} e^{\beta' a} 0\rangle$	$U(\alpha) 0\rangle=e^{\left(\alpha a^{\dagger}-\alpha^* a\right)} 0\rangle=e^{\beta a^{\dagger}} e^{\delta I} e^{\beta' a} 0\rangle$	$U(\alpha) 0\rangle=e^{\left(\alpha a^{\dagger}-\alpha^* a\right)} 0\rangle=e^{\beta a^{\dagger}} e^{\delta I} e^{\beta' a} 0\rangle$
gilmore-lie-groups_ E00455 (7.81)	138	■ 1.000	$\mathrm{EXP}\left[\begin{array}{ccc} 1 & -\alpha^* & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{array}\right]=\left[\begin{array}{ccc} 1 & -\alpha^* & -\alpha^* \alpha / 2 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{array}\right]$	$\mathrm{EXP}\left[\begin{array}{ccc} 1 & -\alpha^* & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{array}\right]=\left[\begin{array}{ccc} 1 & -\alpha^* & -\alpha^* \alpha / 2 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{array}\right]$	$\mathrm{EXP}\left[\begin{array}{ccc} 1 & -\alpha^* & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{array}\right]=\left[\begin{array}{ccc} 1 & -\alpha^* & -\alpha^* \alpha / 2 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{array}\right]$
gilmore-lie-groups_ E00457 (7.83)	138	■ 1.000	$e^{\left(\alpha a^{\dagger}-\alpha^* a\right)} 0\rangle=e^{\alpha a^{\dagger}} e^{-\alpha^* \alpha I / 2} e^{-\alpha^* a} 0\rangle=\sum \frac{\left(\alpha a^{\dagger}\right)^n}{n!} 0\rangle e^{-\alpha^* \alpha / 2}$	$e^{\left(\alpha a^{\dagger}-\alpha^* a\right)} 0\rangle=e^{\alpha a^{\dagger}} e^{-\alpha^* \alpha I / 2} e^{-\alpha^* a} 0\rangle=\sum \frac{\left(\alpha a^{\dagger}\right)^n}{n!} 0\rangle e^{-\alpha^* \alpha / 2}$	$e^{\left(\alpha a^{\dagger}-\alpha^* a\right)} 0\rangle=e^{\alpha a^{\dagger}} e^{-\alpha^* \alpha I / 2} e^{-\alpha^* a} 0\rangle=\sum \frac{\left(\alpha a^{\dagger}\right)^n}{n!} 0\rangle e^{-\alpha^* \alpha / 2}$
gilmore-lie-groups_ E00458 (7.84)	138	■ 1.000	$ \alpha\rangle=U(\alpha) 0\rangle=e^{-\alpha^* \alpha / 2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} n\rangle$	$ \alpha\rangle=U(\alpha) 0\rangle=e^{-\alpha^* \alpha / 2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} n\rangle$	$ \alpha\rangle=U(\alpha) 0\rangle=e^{-\alpha^* \alpha / 2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} n\rangle$
gilmore-lie-groups_ E00463	139	■ 1.000	$\left(E J_3+p J_1-Z\right) u\rangle=0$	$(E J_3+p J_1-Z) u\rangle=0$	$(E J_3+p J_1-Z) u\rangle=0$
gilmore-lie-groups_ E00464	139	■ 1.000	$\begin{aligned} \mathrm{EXP}(i \beta J_y)_{m n}=D_{m n}^j(\beta)=P_{m n}^j(z)=\frac{(-)^{j-m}}{2^j(j-n)!}\left[\frac{(j-n)!(j+m)!}{(j+n)!(j-m)!}\right]^{1 / 2} \\ \times(1+z)^{-(m+n) / 2}(1-z)^{-(m-n) / 2}\left(\frac{d}{d z}\right)^{j-m}\left[(1-z)^{j-n}(1+z)^{j+n}\right] \end{aligned}$	$\mathrm{EXP}(i \beta J_y)_{m n}=D_{m n}^j(\beta)=P_{m n}^j(z)=\frac{(-)^{j-m}}{2^j(j-n)!}\left[\frac{(j-n)!(j+m)!}{(j+n)!(j-m)!}\right]^{1 / 2} \\ \times(1+z)^{-(m+n) / 2}(1-z)^{-(m-n) / 2}\left(\frac{d}{d z}\right)^{j-m}\left[(1-z)^{j-n}(1+z)^{j+n}\right]$	$\mathrm{EXP}(i \beta J_y)_{m n}=D_{m n}^j(\beta)=P_{m n}^j(z)=\frac{(-)^{j-m}}{2^j(j-n)!}\left[\frac{(j-n)!(j+m)!}{(j+n)!(j-m)!}\right]^{1 / 2} \\ \times(1+z)^{-(m+n) / 2}(1-z)^{-(m-n) / 2}\left(\frac{d}{d z}\right)^{j-m}\left[(1-z)^{j-n}(1+z)^{j+n}\right]$

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gilmore-lie-groups_ EQ0465	140	■ 1.000	$\left[\begin{array}{cc} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{array}\right] \xrightarrow{2\rightarrow 1} \left[\begin{array}{cc} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{array}\right]$		$\left[\begin{array}{cc} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{array}\right] \xrightarrow{2\rightarrow 1} \left[\begin{array}{cc} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{array}\right]$
gilmore-lie-groups_ EQ0466	140	■ 1.000	$\mathcal{H} = \sum_{i=1}^N \frac{\epsilon}{2} \sigma_z^{(i)} + \hbar\omega a^\dagger a + \frac{\lambda}{\sqrt{N}} \sum_{i=1}^N \sigma_+^{(i)} a + \sigma_-^{(i)} a^\dagger$		$\mathcal{H} = \sum_{i=1}^N \frac{\epsilon}{2} \sigma_z^{(i)} + \hbar\omega a^\dagger a + \frac{\lambda}{\sqrt{N}} \sum_{i=1}^N \sigma_+^{(i)} a + \sigma_-^{(i)} a^\dagger$
gilmore-lie-groups_ EQ0467	140	■ 1.000	$\left[\sigma_z^{(i)}, \sigma_{\pm}^{(j)}\right] = \pm \sigma_{\pm}^{(i)} \delta_{ij} \quad \left[\sigma_+^{(i)}, \sigma_-^{(j)}\right] = \sigma_z^{(i)} \delta_{ij}$		$\left[\sigma_z^{(i)}, \sigma_{\pm}^{(j)}\right] = \pm \sigma_{\pm}^{(i)} \delta_{ij} \quad \left[\sigma_+^{(i)}, \sigma_-^{(j)}\right] = \sigma_z^{(i)} \delta_{ij}$
gilmore-lie-groups_ EQ0468	141	■ 1.000	$\mathcal{H}_{\text{field}} = \epsilon \langle J_z(t) \rangle + \hbar\omega a^\dagger a + \frac{\lambda}{\sqrt{N}} (\langle J_+(t) \rangle a + \langle J_-(t) \rangle a^\dagger)$		$\mathcal{H}_{\text{field}} = \epsilon \langle J_z(t) \rangle + \hbar\omega a^\dagger a + \frac{\lambda}{\sqrt{N}} (\langle J_+(t) \rangle a + \langle J_-(t) \rangle a^\dagger)$
gilmore-lie-groups_ EQ0472 (8.4)	144	■ 1.000	$\text{Re } e^{g\left(l + r a^\dagger + \delta I\right)} = \left[\begin{array}{ccc} 0 & 0 & l \\ 0 & 0 & -r \\ 0 & 0 & 0 \end{array}\right] \begin{array}{c} a^\dagger \\ a \\ I \end{array}$		$\Re g(la + ra^\dagger + \delta I) = \left[\begin{array}{ccc} 0 & 0 & l \\ 0 & 0 & -r \\ 0 & 0 & 0 \end{array}\right] \begin{array}{c} a^\dagger \\ a \\ I \end{array}$
gilmore-lie-groups_ EQ0475 (8.7)	145	■ 1.000	$\left[X_{ij}, X_{rs}\right] = X_{is}\delta_{jr} + X_{jr}\delta_{is} - X_{ir}\delta_{js} - X_{js}\delta_{ir}$		$[X_{ij}, X_{rs}] = X_{is}\delta_{jr} + X_{jr}\delta_{is} - X_{ir}\delta_{js} - X_{js}\delta_{ir}$
gilmore-lie-groups_ EQ0477 (8.9)	146	■ 1.000	$\begin{aligned} \left\{ \left[Y_i, Y_j\right] \right\} &= -\epsilon_{ijk} Y_k \\ \left\{ \left[Z_i, Z_j\right] \right\} &= +\epsilon_{ijk} Z_k \\ \left\{ \left[Y_i, Z_j\right] \right\} &= 0 \end{aligned}$	$\begin{aligned} [Y_i, Y_j] &= -\epsilon_{ijk} Y_k \\ [Z_i, Z_j] &= +\epsilon_{ijk} Z_k \\ [Y_i, Z_j] &= 0 \end{aligned}$	$\begin{aligned} [Y_i, Y_j] &= -\epsilon_{ijk} Y_k \\ [Z_i, Z_j] &= +\epsilon_{ijk} Z_k \\ [Y_i, Z_j] &= 0 \end{aligned}$
gilmore-lie-groups_ EQ0480 (8.12)	147	■ 1.000	$\left[X_i, X_j\right] = 0$	$[X_i, X_j] = 0$	$[X_i, X_j] = 0$
gilmore-lie-groups_ EQ0481 (8.13)	147	■ 1.000	$\begin{array}{l} \left\{ \left[Z, X_i\right] \right\} = R(Z)_i^j X_j \\ R(Z)_i^j = 0 \text{ unless } j > i \text{ nilpotent} \\ j \geq i \text{ solvable} \end{array}$	$\begin{array}{l} [Z, X_i] = R(Z)_i^j X_j \\ R(Z)_i^j = 0 \text{ unless } j > i \text{ nilpotent} \\ j \geq i \text{ solvable} \end{array}$	$\begin{array}{l} [Z, X_i] = R(Z)_i^j X_j \\ R(Z)_i^j = 0 \text{ unless } j > i \text{ nilpotent} \\ j \geq i \text{ solvable} \end{array}$

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gilmore-lie-groups_ EQ0482 (8.14)	147	■ 1.000	$\begin{array}{c} V_1 \text{ \texttt{\text{spanned by}} } X_n, \\ X_{n-1}, X_{n-2}, \ldots, X_2, X_1 \\ \cup \\ V_2 \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2}, \ldots, X_2 \\ \cup \\ \vdots \quad \quad \quad \vdots \\ \cup \\ V_{n-2} \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2} \\ \cup \\ V_{n-1} \text{ \texttt{\text{spanned by}} } X_n, X_{n-1} \\ \cup \\ V_n \text{ \texttt{\text{spanned by}} } X_n \end{array}$	$\begin{array}{c} V_1 \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2}, \ldots, X_2, X_1 \\ \cup \\ V_2 \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2}, \ldots, X_2 \\ \cup \quad \vdots \\ \vdots \quad \vdots \\ \cup \\ V_{n-2} \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2} \\ \cup \\ V_{n-1} \text{ \texttt{\text{spanned by}} } X_n, X_{n-1} \\ \cup \\ V_n \text{ \texttt{\text{spanned by}} } X_n \end{array}$	$\begin{array}{c} V_1 \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2}, \ldots, X_2, X_1 \\ \cup \\ V_2 \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2}, \ldots, X_2 \\ \cup \\ \vdots \quad \quad \quad \vdots \\ \cup \\ V_{n-2} \text{ \texttt{\text{spanned by}} } X_n, X_{n-1}, X_{n-2} \\ \cup \\ V_{n-1} \text{ \texttt{\text{spanned by}} } X_n, X_{n-1} \\ \cup \\ V_n \text{ \texttt{\text{spanned by}} } X_n \end{array}$
gilmore-lie-groups_ EQ0483 (8.17)	148	■ 1.000	$X=[Y, \ Z]$	$X = [Y, Z]$	$X = [Y, Z]$
gilmore-lie-groups_ EQ0484 (8.18)	149	■ 1.000	$\mathfrak{g}=\left(V_{-}+V_{+}\right)+V_{0}$	$\mathfrak{g} = (V_- + V_+) + V_0$	$\mathfrak{g} = (V_- + V_+) + V_0$
gilmore-lie-groups_ EQ0485 (8.19)	149	■ 1.000	$\mathfrak{g}^{\prime}=\left(V_{-}^{\prime}+V_{+}^{\prime}\right)+V_{0}^{\prime}$	$\mathfrak{g}' = (V_-^{\prime} + V_+^{\prime}) + V_0^{\prime}$	$\mathfrak{g}' = (V_-^{\prime} + V_+^{\prime}) + V_0^{\prime}$
gilmore-lie-groups_ EQ0487 (8.21)	150	■ 1.000	$(X, \ X)=-40 \ R \ L+10 \ \eta^2$	$(X, X) = -40RL + 10\eta^2$	$(X, X) = -40RL + 10\eta^2$
gilmore-lie-groups_ EQ0488 (8.22)	150	■ 1.000	$\eta\left(a^{\dagger}a+\frac{1}{2}\right)+Ra^{\dagger 2}+La^2\longrightarrow\left[\begin{array}{ccc} 0 & -2R & 2L \\ 4L & 2\eta & 0 \\ -4R & 0 & -2\eta \end{array}\right]\begin{array}{c} \hat{n}+\frac{1}{2} \\ a^{\dagger 2} \\ a^2 \end{array}$	$\eta\left(a^{\dagger}a+\frac{1}{2}\right)+Ra^{\dagger 2}+La^2\longrightarrow\left[\begin{array}{ccc} 0 & -2R & 2L \\ 4L & 2\eta & 0 \\ -4R & 0 & -2\eta \end{array}\right]\begin{array}{c} \hat{n}+\frac{1}{2} \\ a^{\dagger 2} \\ a^2 \end{array}$	$\eta(a^{\dagger}a+\frac{1}{2})+Ra^{\dagger 2}+La^2\longrightarrow\left[\begin{array}{ccc} 0 & -2R & 2L \\ 4L & 2\eta & 0 \\ -4R & 0 & -2\eta \end{array}\right]\begin{array}{c} \hat{n}+\frac{1}{2} \\ a^{\dagger 2} \\ a^2 \end{array}$
gilmore-lie-groups_ EQ0489 (8.23)	150	■ 1.000	$(X, \ X)^{\prime}=-32 \ R \ L+8 \ \eta^2$	$(X, X)' = -32RL + 8\eta^2$	$(X, X)' = -32RL + 8\eta^2$
gilmore-lie-groups_ EQ0491 (8.26)	152	■ 1.000	$\begin{array}{l} X_{ij} = x^i \partial_j - x^j \partial_i \\ Y_i = 2t \frac{\partial}{\partial x^i} - x^i u \frac{\partial}{\partial u} \\ Z = 2t \frac{\partial}{\partial t} + x^i \frac{\partial}{\partial x^i} - nu \frac{\partial}{\partial u} \end{array}$	$\begin{array}{l} X_{ij} = x^i \partial_j - x^j \partial_i \\ Y_i = 2t \frac{\partial}{\partial x^i} - x^i u \frac{\partial}{\partial u} \\ Z = 2t \frac{\partial}{\partial t} + x^i \frac{\partial}{\partial x^i} - nu \frac{\partial}{\partial u} \end{array}$	$\begin{array}{l} X_{ij} = x^i \partial_j - x^j \partial_i \\ Y_i = 2t \frac{\partial}{\partial x^i} - x^i u \frac{\partial}{\partial u} \\ Z = 2t \frac{\partial}{\partial t} + x^i \frac{\partial}{\partial x^i} - nu \frac{\partial}{\partial u} \end{array}$
gilmore-lie-groups_ EQ0492 (9.1)	154	■ 1.000	$[Z, \ X]=\lambda X$	$[Z, X] = \lambda X$	$[Z, X] = \lambda X$
gilmore-lie-groups_ EQ0494 (9.3)	154	■ 1.000	$\left\ R(Z)-\lambda I_N\right\ =0$	$\ R(Z) - \lambda I_N\ = 0$	$\ R(Z) - \lambda I_N\ = 0$

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gilmore-lie-groups_ E08495 (9.4)	154	■ 1.000	$\sum_{j=0}^N (-\lambda)^{N-j} \phi_j(Z) = 0$	$\sum_{j=0}^N (-\lambda)^{N-j} \phi_j(Z) = 0$	$\sum_{j=0}^N (-\lambda)^{N-j} \phi_j(Z) = 0$
gilmore-lie-groups_ E08496 (9.5)	154	■ 1.000	$\phi_j(Z) \rightarrow \phi_j(z^i)$	$\phi_j(Z) \rightarrow \phi_j(z^i)$	$\phi_j(Z) \rightarrow \phi_j(z^i)$
gilmore-lie-groups_ E08503 (9.12)	156	■ 1.000	$\left \frac{d}{d\lambda} f(X) - \lambda I_2 \right = \lambda^2 + \left(\frac{1}{2} \right)^2 (a_1^2 + a_2^2 + a_3^2) = 0$	$\ \mathfrak{d} f(X) - \lambda I_2 \ = \lambda^2 + \left(\frac{1}{2} \right)^2 (a_1^2 + a_2^2 + a_3^2) = 0$	$\ \mathfrak{d} f(X) - \lambda I_2 \ = \lambda^2 + \left(\frac{1}{2} \right)^2 (a_1^2 + a_2^2 + a_3^2) = 0$
gilmore-lie-groups_ E08504 (9.13)	156	■ 1.000	$\left \frac{d}{d\lambda} f(Y) - \lambda I_2 \right = \lambda^2 + \left(\frac{1}{2} \right)^2 (-b_1^2 - b_2^2 + b_3^2) = 0$	$\ \mathfrak{d} f(Y) - \lambda I_2 \ = \lambda^2 + \left(\frac{1}{2} \right)^2 (-b_1^2 - b_2^2 + b_3^2) = 0$	$\ \mathfrak{d} f(Y) - \lambda I_2 \ = \lambda^2 + \left(\frac{1}{2} \right)^2 (-b_1^2 - b_2^2 + b_3^2) = 0$
gilmore-lie-groups_ E08505 (9.14)	157	■ 1.000	$z^i \rightarrow X_i \quad \phi_j(z^i) \rightarrow \phi_j(X_i)$	$z^i \rightarrow X_i \quad \phi_j(z^i) \rightarrow \phi_j(X_i)$	$z^i \rightarrow X_i \quad \phi_j(z^i) \rightarrow \phi_j(X_i)$
gilmore-lie-groups_ E08506 (9.15)	157	■ 1.000	$[\phi_j(X_i), X_k] = 0$	$[\phi_j(X_i), X_k] = 0$	$[\phi_j(X_i), X_k] = 0$
gilmore-lie-groups_ E08507 (9.16)	157	■ 1.000	$\phi_2(\mathbf{a}) = +a_1^2 + a_2^2 + a_3^2 \rightarrow \phi_2(X) = +X_1^2 + X_2^2 + X_3^2$	$\phi_2(\mathbf{a}) = +a_1^2 + a_2^2 + a_3^2 \rightarrow \phi_2(X) = +X_1^2 + X_2^2 + X_3^2$	$\phi_2(\mathbf{a}) = +a_1^2 + a_2^2 + a_3^2 \rightarrow \phi_2(X) = +X_1^2 + X_2^2 + X_3^2$
gilmore-lie-groups_ E08508 (9.17)	157	■ 1.000	$\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2 \rightarrow \phi_2(Y) = -Y_1^2 - Y_2^2 + Y_3^2$	$\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2 \rightarrow \phi_2(Y) = -Y_1^2 - Y_2^2 + Y_3^2$	$\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2 \rightarrow \phi_2(Y) = -Y_1^2 - Y_2^2 + Y_3^2$
gilmore-lie-groups_ E08510 (9.19)	159	■ 1.000	$\left \frac{d}{d\lambda} f(X) - \lambda I_n \right = \sum_{j=0}^n (-\lambda)^{n-j} \phi_j(X) = 0$	$\ \mathfrak{d} f(X) - \lambda I_n \ = \sum_{j=0}^n (-\lambda)^{n-j} \phi_j(X) = 0$	$\ \mathfrak{d} f(X) - \lambda I_n \ = \sum_{j=0}^n (-\lambda)^{n-j} \phi_j(X) = 0$
gilmore-lie-groups_ E08511 (9.20)	159	■ 1.000	$\phi_j(X) = \phi_j(-X) = (-)^j \phi_j(X)$	$\phi_j(X) = \phi_j(-X) = (-)^j \phi_j(X)$	$\phi_j(X) = \phi_j(-X) = (-)^j \phi_j(X)$
gilmore-lie-groups_ E08513 (9.22)	160	■ 1.000	$\mathcal{C}_4[\mathfrak{so}(5)] = \sum_{i=1}^5 v_i^2$	$\mathcal{C}_4[\mathfrak{so}(5)] = \sum_{i=1}^5 v_i^2$	$\mathcal{C}_4[\mathfrak{so}(5)] = \sum_{i=1}^5 v_i^2$
gilmore-lie-groups_ E08514 (9.23)	160	■ 1.000	$v^5 = \epsilon^{ijkl5} X_{ij} X_{kl} \sim X_{12} X_{34} - X_{13} X_{24} + X_{14} X_{23}$	$v^5 = \epsilon^{ijkl5} X_{ij} X_{kl} \sim X_{12} X_{34} - X_{13} X_{24} + X_{14} X_{23}$	$v^5 = \epsilon^{ijkl5} X_{ij} X_{kl} \sim X_{12} X_{34} - X_{13} X_{24} + X_{14} X_{23}$
gilmore-lie-groups_ E08515 (9.24)	160	■ 1.000	$\mathcal{C}_4[\mathfrak{so}(4)] = (\epsilon^{ijkl} X_{ij} X_{kl})^2 \sim (X_{12} X_{34} - X_{13} X_{24} + X_{14} X_{23})^2$	$\mathcal{C}_4[\mathfrak{so}(4)] = (\epsilon^{ijkl} X_{ij} X_{kl})^2 \sim (X_{12} X_{34} - X_{13} X_{24} + X_{14} X_{23})^2$	$\mathcal{C}_4[\mathfrak{so}(4)] = (\epsilon^{ijkl} X_{ij} X_{kl})^2 \sim (X_{12} X_{34} - X_{13} X_{24} + X_{14} X_{23})^2$
gilmore-lie-groups_ E08516 (9.25)	160	■ 1.000	$Z = z_1 \left(\hat{n} + \frac{1}{2} \right) + z_2 I$	$Z = z_1 \left(\hat{n} + \frac{1}{2} \right) + z_2 I$	$Z = z_1 \left(\hat{n} + \frac{1}{2} \right) + z_2 I$
gilmore-lie-groups_ E08519 (9.28)	161	■ 1.000	$\phi_2(z_1, z_2) = -z_1^2$	$\phi_2(z_1, z_2) = -z_1^2$	$\phi_2(z_1, z_2) = -z_1^2$
gilmore-lie-groups_ E08520 (9.29)	161	■ 1.000	$\left[\hat{n} + \frac{1}{2}, X_{(k,0)} \right] = k X_{(k,0)}$ $[I, X_{(k,0)}] = 0 X_{(k,0)}$	$\left[\hat{n} + \frac{1}{2}, X_{(k,0)} \right] = k X_{(k,0)}$ $[I, X_{(k,0)}] = 0 X_{(k,0)}$	$\left[\hat{n} + \frac{1}{2}, X_{(k,0)} \right] = k X_{(k,0)}$ $[I, X_{(k,0)}] = 0 X_{(k,0)}$
gilmore-lie-groups_ E08521 (9.30)	161	■ 1.000	$[X_{(k,0)}, X_{(l,0)}] \sim X_{(k+l,0)}$	$[X_{(k,0)}, X_{(l,0)}] \sim X_{(k+l,0)}$	$[X_{(k,0)}, X_{(l,0)}] \sim X_{(k+l,0)}$
gilmore-lie-groups_ E08522 (9.31)	162	■ 1.000	$[V_\alpha, V_\beta] \subset V_{\alpha+\beta}$	$[V_\alpha, V_\beta] \subset V_{\alpha+\beta}$	$[V_\alpha, V_\beta] \subset V_{\alpha+\beta}$
gilmore-lie-groups_ E08523 (9.32)	162	■ 1.000	$[H_i, H_j] = 0 \quad 1 \leq i, j \leq l$	$[H_i, H_j] = 0 \quad 1 \leq i, j \leq l$	$[H_i, H_j] = 0 \quad 1 \leq i, j \leq l$

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gilmore-lie-groups_ E08525 (9.34)	162	■ 1.000	$\left[E_{\{\alpha\}}, E_{\{-\alpha\}}\right]=\alpha^i H_{\{i\}}$	$[E_{\alpha}, E_{-\alpha}] = \alpha^i H_i$	$[E_{\alpha}, E_{-\alpha}] = \alpha^i H_i$
gilmore-lie-groups_ E08526 (9.35)	162	■ 1.000	$(\alpha, \beta)=\alpha_{\{i\}} \beta^{\{i\}}=\alpha^j \beta_j$ $\beta_{\{j\}}=\alpha_{\{i\}} h^{\{i j\}} \beta_j$	$(\alpha, \beta) = \alpha_i \beta^i = \alpha^j \beta_j = \alpha_i h^{ij} \beta_j$	$(\alpha, \beta) = \alpha_i \beta^i = \alpha^j \beta_j = \alpha_i h^{ij} \beta_j$
gilmore-lie-groups_ E08528	163	■ 1.000	$\left[E_{\{\alpha\}}, E_{\{\beta\}}\right] \rightarrow$ $\begin{cases} 0 & \alpha+\beta \text { not a root } \\ N_{\alpha, \beta} E_{\alpha+\beta} & \alpha+\beta \text { a root } \\ \alpha \cdot \mathbf{H} & \alpha+\beta=0 \end{cases}$	$[E_{\alpha}, E_{\beta}] \rightarrow \begin{cases} 0 & \alpha+\beta \text { not a root } \\ N_{\alpha, \beta} E_{\alpha+\beta} & \alpha+\beta \text { a root } \\ \alpha \cdot \mathbf{H} & \alpha+\beta=0 \end{cases}$	$[E_{\alpha}, E_{\beta}] \rightarrow \begin{cases} 0 & \alpha+\beta \text { not a root } \\ N_{\alpha, \beta} E_{\alpha+\beta} & \alpha+\beta \text { a root } \\ \alpha \cdot \mathbf{H} & \alpha+\beta=0 \end{cases}$
gilmore-lie-groups_ E08529 (9.37)	163	■ 1.000	$\left[\left[E_{\{\alpha\}}, E_{\{\beta\}}\right], E_{\{\gamma\}}\right. \\ \left.\left.\right]+\left[\left[E_{\{\beta\}}, E_{\{\gamma\}}\right], E_{\{\alpha\}}\right. \\ \left.\left.\right]+\left[\left[E_{\{\gamma\}}, E_{\{\alpha\}}\right], E_{\{\beta\}}\right]=0$	$[[E_{\alpha}, E_{\beta}], E_{\gamma}] + [[E_{\beta}, E_{\gamma}], E_{\alpha}] + [[E_{\gamma}, E_{\alpha}], E_{\beta}] = 0$	$[[E_{\alpha}, E_{\beta}], E_{\gamma}] + [[E_{\beta}, E_{\gamma}], E_{\alpha}] + [[E_{\gamma}, E_{\alpha}], E_{\beta}] = 0$
gilmore-lie-groups_ E08532 (9.39)	163	■ 1.000	$\begin{aligned} & \beta-(m+1) \alpha \quad \& \quad \beta+(n+1) \alpha \\ & \alpha \end{aligned}$	$\beta-(m+1) \alpha$ $\beta+(n+1) \alpha$	$\beta-(m+1) \alpha$ $\beta+(n+1) \alpha$
gilmore-lie-groups_ E08533 (9.40)	163	■ 1.000	$N_{\{\alpha, \beta+(k-1) \alpha\}}^2=N_{\{\alpha, \beta+k \alpha\}}^2+\alpha \cdot(\beta+k \alpha)$	$N_{\alpha, \beta+(k-1) \alpha}^2=N_{\alpha, \beta+k \alpha}^2+\alpha \cdot(\beta+k \alpha)$	$N_{\alpha, \beta+(k-1) \alpha}^2=N_{\alpha, \beta+k \alpha}^2+\alpha \cdot(\beta+k \alpha)$
gilmore-lie-groups_ E08535 (9.42)	164	■ 1.000	$N_{\{\alpha, \beta+(k-1) \alpha\}}^2=(n-k+1)\left(\alpha \cdot \beta+\frac{1}{2}(n+k) \alpha \cdot \alpha\right)$	$N_{\alpha, \beta+(k-1) \alpha}^2=(n-k+1)\left(\alpha \cdot \beta+\frac{1}{2}(n+k) \alpha \cdot \alpha\right)$	$N_{\alpha, \beta+(k-1) \alpha}^2=(n-k+1)\left(\alpha \cdot \beta+\frac{1}{2}(n+k) \alpha \cdot \alpha\right)$
gilmore-lie-groups_ E08536 (9.43)	164	■ 1.000	$N_{\{\alpha, \beta-(m+1) \alpha\}}^2=(n+m+1)\left(\alpha \cdot \beta+\frac{1}{2}(n-m) \alpha \cdot \alpha\right)=0$	$N_{\alpha, \beta-(m+1) \alpha}^2=(n+m+1)\left(\alpha \cdot \beta+\frac{1}{2}(n-m) \alpha \cdot \alpha\right)=0$	$N_{\alpha, \beta-(m+1) \alpha}^2=(n+m+1)\left(\alpha \cdot \beta+\frac{1}{2}(n-m) \alpha \cdot \alpha\right)=0$
gilmore-lie-groups_ E08537 (9.44)	164	■ 1.000	$-n \leq \frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}=-n+m \leq m$	$-n \leq \frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}=-n+m \leq m$	$-n \leq \frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}=-n+m \leq m$
gilmore-lie-groups_ E08538 (9.45)	164	■ 1.000	$\beta^{\prime}=\beta-2 \frac{\beta \cdot \alpha}{\alpha \cdot \alpha} \alpha$	$\beta' = \beta - 2 \frac{\beta \cdot \alpha}{\alpha \cdot \alpha} \alpha$	$\beta' = \beta - 2 \frac{\beta \cdot \alpha}{\alpha \cdot \alpha} \alpha$
gilmore-lie-groups_ E08539 (9.46)	165	■ 1.000	$\begin{aligned} & \left\{\left[H_{\{i\}}, H_{\{j\}}\right]\right\} \&=0 \& \& \backslash \\ & \backslash\left\{\left[\mathbf{H}, E_{\{\alpha\}}\right]\right\} \&=\alpha E_{\{\alpha\}} \\ & \& \backslash\left\{\left[E_{\{\alpha\}}, E_{\{\beta\}}\right]\right\} \& \\ & =\alpha \cdot \mathbf{H} \& \& \alpha+\beta=0 \backslash \& =N_{\{\alpha, \beta\}} E_{\{\alpha+\beta\}} \& \& \alpha+\beta \neq 0, \text { a root } \\ & \backslash \& =0 \& \& \alpha+\beta \text { not a root } \end{aligned}$	$\begin{aligned} & \left[H_i, H_j\right]=0 \\ & \left[\mathbf{H}, E_{\alpha}\right]=\alpha E_{\alpha} \\ & \left[E_{\alpha}, E_{\beta}\right]=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ & \quad =N_{\alpha, \beta} E_{\alpha+\beta} \quad \alpha+\beta \neq 0, \text { a root } \\ & \quad =0 \quad \alpha+\beta \text { not a root } \end{aligned}$	$\begin{aligned} & \left[H_i, H_j\right]=0 \\ & \left[\mathbf{H}, E_{\alpha}\right]=\alpha E_{\alpha} \\ & \left[E_{\alpha}, E_{\beta}\right]=\alpha \cdot \mathbf{H} \quad \alpha+\beta=0 \\ & \quad =N_{\alpha, \beta} E_{\alpha+\beta} \quad \alpha+\beta \neq 0, \text { a root } \\ & \quad =0 \quad \alpha+\beta \text { not a root } \end{aligned}$
gilmore-lie-groups_ E08541 (9.48)	166	■ 1.000	$\left(\mathbf{e}_{\{i\}}, \mathbf{e}_{\{j\}}\right)=\frac{1}{12} \delta_{i j}$	$(\mathbf{e}_i, \mathbf{e}_j) = \frac{1}{12} \delta_{ij}$	$(\mathbf{e}_i, \mathbf{e}_j) = \frac{1}{12} \delta_{ij}$

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gilmore-lie-groups_ E08542	169	■ 1.000	$\begin{array}{cccc} b_{\{i\}}^{\dagger} b_{\{j\}} & b_{\{i\}}^{\dagger} b_{\{j\}} + \frac{1}{2} \delta_{ij} & b_{\{i\}}^{\dagger} b_{\{j\}}^{\dagger} & b_{\{i\}}^{\dagger} b_{\{j\}} \\ b_{\{i\}}^{\dagger} b_{\{j\}} + \frac{1}{2} \delta_{ij} & b_{\{i\}}^{\dagger} b_{\{j\}}^{\dagger} & b_{\{i\}}^{\dagger} b_{\{j\}} & b_{\{i\}}^{\dagger} b_{\{j\}} \\ b_{\{i\}}^{\dagger} b_{\{j\}}^{\dagger} & b_{\{i\}}^{\dagger} b_{\{j\}} & b_{\{i\}}^{\dagger} b_{\{j\}} & b_{\{i\}}^{\dagger} b_{\{j\}} \\ b_{\{i\}}^{\dagger} b_{\{j\}} & b_{\{i\}}^{\dagger} b_{\{j\}} + \frac{1}{2} \delta_{ij} & b_{\{i\}}^{\dagger} b_{\{j\}}^{\dagger} & b_{\{i\}}^{\dagger} b_{\{j\}} \end{array}$	$\begin{array}{cccc} b_i^\dagger b_j & b_i^\dagger b_j + \frac{1}{2} \delta_{ij} & b_i^\dagger b_j^\dagger & b_i b_j \\ f_i^\dagger f_j & f_i^\dagger f_j + \frac{1}{2} \delta_{ij} & f_i^\dagger f_j^\dagger & f_i f_j \\ x^i \partial_j & x^i \partial_j + \frac{1}{2} \delta_{ij} & x^i x^j & \partial_i \partial_j \end{array}$	$\begin{array}{cccc} b_i^\dagger b_j & b_i^\dagger b_j + \frac{1}{2} \delta_{ij} & b_i^\dagger b_j^\dagger & b_i b_j \\ f_i^\dagger f_j & f_i^\dagger f_j + \frac{1}{2} \delta_{ij} & f_i^\dagger f_j^\dagger & f_i f_j \\ x^i \partial_j & x^i \partial_j + \frac{1}{2} \delta_{ij} & x^i x^j & \partial_i \partial_j \end{array}$
gilmore-lie-groups_ E08547	170	■ 1.000	$\frac{f_r(\Gamma)}{f_r(\text{def})} = \frac{\text{tr}(\Gamma(A))^r}{\text{tr}(\text{def}(A))^r} = \frac{\dim(\Gamma) \mathcal{C}^r(\Gamma)}{\dim(\text{def}) \mathcal{C}^r(\text{def})}$	$\frac{f_r(\Gamma)}{f_r(\text{def})} = \frac{\text{tr}(\Gamma(A))^r}{\text{tr}(\text{def}(A))^r} = \frac{\dim(\Gamma) \mathcal{C}^r(\Gamma)}{\dim(\text{def}) \mathcal{C}^r(\text{def})}$	$\frac{f_r(\Gamma)}{f_r(\text{def})} = \frac{\text{tr}(\Gamma(A))^r}{\text{tr}(\text{def}(A))^r} = \frac{\dim(\Gamma) \mathcal{C}^r(\Gamma)}{\dim(\text{def}) \mathcal{C}^r(\text{def})}$
gilmore-lie-groups_ E08548	170	■ 1.000	$f_{\{2\}}(j) = \frac{1}{6} \{(2j)(2j+1)(2j+2)\} f_{\{2\}} \left(j = \frac{1}{2} \right)$	$f_2(j) = \frac{1}{6} \{(2j)(2j+1)(2j+2)\} f_2 \left(j = \frac{1}{2} \right)$	$f_2(j) = \frac{1}{6} \{(2j)(2j+1)(2j+2)\} f_2(j = \frac{1}{2})$
gilmore-lie-groups_ E08550	171	■ 1.000	$f(X) = f_{\{0\}} I_{\{n\}} + f_{\{1\}} X^{\{1\}} + f_{\{2\}} X^{\{2\}} + \dots + f_{\{n-1\}} X^{\{n-1\}} = \sum_{j=0}^{n-1} f_j X^j$	$f(X) = f_0 I_n + f_1 X^1 + f_2 X^2 + \dots + f_{n-1} X^{n-1} = \sum_{j=0}^{j=n-1} f_j X^j$	$f(X) = f_0 I_n + f_1 X^1 + f_2 X^2 + \dots + f_{n-1} X^{n-1} = \sum_{j=0}^{j=n-1} f_j X^j$
gilmore-lie-groups_ E08551	171	■ 1.000	$\left[\begin{array}{cccccc} 1 & \lambda_1 & \lambda_1^2 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \lambda_2^2 & \dots & \lambda_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \lambda_n^2 & \dots & \lambda_n^{n-1} \end{array} \right] \left[\begin{array}{c} f_0 \\ f_1 \\ \vdots \\ f_{n-1} \end{array} \right] = \left[\begin{array}{c} f(\lambda_1) \\ f(\lambda_2) \\ \vdots \\ f(\lambda_n) \end{array} \right]$	$\left[\begin{array}{ccccc} 1 & \lambda_1 & \lambda_1^2 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \lambda_2^2 & \dots & \lambda_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \lambda_n^2 & \dots & \lambda_n^{n-1} \end{array} \right] \left[\begin{array}{c} f_0 \\ f_1 \\ \vdots \\ f_{n-1} \end{array} \right] = \left[\begin{array}{c} f(\lambda_1) \\ f(\lambda_2) \\ \vdots \\ f(\lambda_n) \end{array} \right]$	$\left[\begin{array}{ccccc} 1 & \lambda_1 & \lambda_1^2 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \lambda_2^2 & \dots & \lambda_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \lambda_n^2 & \dots & \lambda_n^{n-1} \end{array} \right] \left[\begin{array}{c} f_0 \\ f_1 \\ \vdots \\ f_{n-1} \end{array} \right] = \left[\begin{array}{c} f(\lambda_1) \\ f(\lambda_2) \\ \vdots \\ f(\lambda_n) \end{array} \right]$
gilmore-lie-groups_ E08552	172	■ 1.000	$\left[\begin{array}{ccc} 1 & \frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 0 \\ 1 & -1 & 1 \end{array} \right] \left[\begin{array}{ccc} 1 & \frac{3}{2} & \left(\frac{3}{2}\right)^2 \\ 1 & \frac{1}{2} & \left(\frac{1}{2}\right)^2 \\ 1 & -\frac{1}{2} & \left(-\frac{1}{2}\right)^2 \end{array} \right] \left[\begin{array}{ccc} \left(\frac{3}{2}\right)^3 & \left(\frac{3}{2}\right)^3 & \left(\frac{3}{2}\right)^3 \\ \left(\frac{1}{2}\right)^3 & \left(\frac{1}{2}\right)^3 & \left(\frac{1}{2}\right)^3 \\ \left(-\frac{1}{2}\right)^3 & \left(-\frac{1}{2}\right)^3 & \left(-\frac{1}{2}\right)^3 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & \frac{3}{2} & \left(\frac{3}{2}\right)^2 \\ 1 & \frac{1}{2} & \left(\frac{1}{2}\right)^2 \\ 1 & -\frac{1}{2} & \left(-\frac{1}{2}\right)^2 \end{array} \right] \left[\begin{array}{ccc} \left(\frac{3}{2}\right)^3 & \left(\frac{3}{2}\right)^3 & \left(\frac{3}{2}\right)^3 \\ \left(\frac{1}{2}\right)^3 & \left(\frac{1}{2}\right)^3 & \left(\frac{1}{2}\right)^3 \\ \left(-\frac{1}{2}\right)^3 & \left(-\frac{1}{2}\right)^3 & \left(-\frac{1}{2}\right)^3 \end{array} \right]$	$\left[\begin{array}{ccc} 1 & \frac{3}{2} & \left(\frac{3}{2}\right)^2 \\ 1 & \frac{1}{2} & \left(\frac{1}{2}\right)^2 \\ 1 & -\frac{1}{2} & \left(-\frac{1}{2}\right)^2 \end{array} \right] \left[\begin{array}{ccc} \left(\frac{3}{2}\right)^3 & \left(\frac{3}{2}\right)^3 & \left(\frac{3}{2}\right)^3 \\ \left(\frac{1}{2}\right)^3 & \left(\frac{1}{2}\right)^3 & \left(\frac{1}{2}\right)^3 \\ \left(-\frac{1}{2}\right)^3 & \left(-\frac{1}{2}\right)^3 & \left(-\frac{1}{2}\right)^3 \end{array} \right]$

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gilmore-lie-groups_ E08553	172	■ 1.000	$\left[\begin{array}{ccccc} 1 & 2 & 4 & 8 & 16 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & -2 & 4 & -8 & 16 \end{array}\right] \left[\begin{array}{c} f_0 \\ (i\phi)^1 f_1 \\ (i\phi)^2 f_2 \\ (i\phi)^3 f_3 \\ (i\phi)^4 f_4 \end{array}\right] = \left[\begin{array}{c} e^{2i\phi} \\ e^{i\phi} \\ 1 \\ e^{-i\phi} \\ e^{-2i\phi} \end{array}\right]$	$\left[\begin{array}{ccccc} 1 & 2 & 4 & 8 & 16 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & -2 & 4 & -8 & 16 \end{array}\right] \left[\begin{array}{c} f_0 \\ (i\phi)^1 f_1 \\ (i\phi)^2 f_2 \\ (i\phi)^3 f_3 \\ (i\phi)^4 f_4 \end{array}\right] = \left[\begin{array}{c} e^{2i\phi} \\ e^{i\phi} \\ 1 \\ e^{-i\phi} \\ e^{-2i\phi} \end{array}\right]$	$\begin{bmatrix} 1 & 2 & 4 & 8 & 16 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & -2 & 4 & -8 & 16 \end{bmatrix} \begin{bmatrix} f_0 \\ (i\phi)^1 f_1 \\ (i\phi)^2 f_2 \\ (i\phi)^3 f_3 \\ (i\phi)^4 f_4 \end{bmatrix} = \begin{bmatrix} e^{2i\phi} \\ e^{i\phi} \\ 1 \\ e^{-i\phi} \\ e^{-2i\phi} \end{bmatrix}$
gilmore-lie-groups_ E08554	172	■ 1.000	$\begin{array}{l} j=\frac{1}{2} \quad e^X = \cos\left(\frac{\phi}{2}\right) I_2 + \frac{\sin(\phi/2)}{\phi/2} X \\ l=1 \quad e^X = I_3 + \frac{\sin(\phi)}{\phi} X + \frac{1-\cos(\phi)}{\phi^2} X^2 \end{array}$	$j = \frac{1}{2} \quad e^X = \cos\left(\frac{\phi}{2}\right) I_2 + \frac{\sin(\phi/2)}{\phi/2} X$ $l = 1 \quad e^X = I_3 + \frac{\sin(\phi)}{\phi} X + \frac{1-\cos(\phi)}{\phi^2} X^2$	$j = \frac{1}{2} \quad e^X = \cos\left(\frac{\phi}{2}\right) I_2 + \frac{\sin(\phi/2)}{\phi/2} X$ $l = 1 \quad e^X = I_3 + \frac{\sin(\phi)}{\phi} X + \frac{1-\cos(\phi)}{\phi^2} X^2$
gilmore-lie-groups_ E08555	172	■ 1.000	$\operatorname{tr} X^2 = -\frac{j(j+1)(2j+1)}{3} \phi^2$	$\operatorname{tr} X^2 = -\frac{j(j+1)(2j+1)}{3} \phi^2$	$\operatorname{tr} X^2 = -\frac{j(j+1)(2j+1)}{3} \phi^2$
gilmore-lie-groups_ E08557 (10.2)	174	■ 1.000	$N_{\alpha, \beta+k\alpha}^2 = (n-k)(m+k+1)(\alpha \cdot \alpha)/2$	$N_{\alpha, \beta+k\alpha}^2 = (n-k)(m+k+1)(\alpha \cdot \alpha)/2$	$N_{\alpha, \beta+k\alpha}^2 = (n-k)(m+k+1)(\alpha \cdot \alpha)/2$
gilmore-lie-groups_ E08559 (10.4)	174	■ 1.000	$\frac{2(\alpha \cdot \beta)}{\alpha \cdot \alpha} \text{ is an integer}$ $\beta' = \beta - \frac{2(\alpha \cdot \beta)}{\alpha \cdot \alpha} \alpha \text{ is a root}$	$\frac{2(\alpha \cdot \beta)}{\alpha \cdot \alpha} \text{ is an integer}$ $\beta' = \beta - \frac{2(\alpha \cdot \beta)}{\alpha \cdot \alpha} \alpha \text{ is a root}$	$\frac{2(\alpha \cdot \beta)}{\alpha \cdot \alpha} \text{ is an integer}$ $\beta' = \beta - \frac{2(\alpha \cdot \beta)}{\alpha \cdot \alpha} \alpha \text{ is a root}$
gilmore-lie-groups_ E08560 (10.5)	174	■ 1.000	$0 \leq \cos^2(\alpha, \beta) = \left(\frac{\alpha \cdot \beta}{\alpha \cdot \alpha}\right) \left(\frac{\alpha \cdot \beta}{\beta \cdot \beta}\right) = \frac{n}{2} \frac{n'}{2} \leq 1$	$0 \leq \cos^2(\alpha, \beta) = \left(\frac{\alpha \cdot \beta}{\alpha \cdot \alpha}\right) \left(\frac{\alpha \cdot \beta}{\beta \cdot \beta}\right) = \frac{n}{2} \frac{n'}{2} \leq 1$	$0 \leq \cos^2(\alpha, \beta) = \left(\frac{\alpha \cdot \beta}{\alpha \cdot \alpha}\right) \left(\frac{\alpha \cdot \beta}{\beta \cdot \beta}\right) = \frac{n}{2} \frac{n'}{2} \leq 1$
gilmore-lie-groups_ E08561 (10.6)	175	■ 1.000	$\begin{array}{l} A_{l-1} \quad +\mathbf{e}_i - \mathbf{e}_j \quad 1 \leq i \neq j \leq l \quad l-1 \geq 1 \\ D_l \quad \pm \mathbf{e}_i \pm \mathbf{e}_j \quad 1 \leq i \neq j \leq l \quad l > 3 \\ B_l \quad \pm \mathbf{e}_i \pm \mathbf{e}_j, \pm \mathbf{e}_i \quad 1 \leq i \neq j \leq l \quad l > 2 \\ C_l \quad \pm \mathbf{e}_i \pm \mathbf{e}_j, \pm 2\mathbf{e}_i \quad 1 \leq i \neq j \leq l \quad l > 1 \end{array}$	$\begin{array}{l} A_{l-1} \quad +\mathbf{e}_i - \mathbf{e}_j \quad 1 \leq i \neq j \leq l \quad l-1 \geq 1 \\ D_l \quad \pm \mathbf{e}_i \pm \mathbf{e}_j \quad 1 \leq i \neq j \leq l \quad l > 3 \\ B_l \quad \pm \mathbf{e}_i \pm \mathbf{e}_j, \pm \mathbf{e}_i \quad 1 \leq i \neq j \leq l \quad l > 2 \\ C_l \quad \pm \mathbf{e}_i \pm \mathbf{e}_j, \pm 2\mathbf{e}_i \quad 1 \leq i \neq j \leq l \quad l > 1 \end{array}$	$\begin{array}{lll} A_{l-1} & +\mathbf{e}_i - \mathbf{e}_j & 1 \leq i \neq j \leq l \quad l-1 \geq 1 \\ D_l & \pm \mathbf{e}_i \pm \mathbf{e}_j & 1 \leq i \neq j \leq l \quad l > 3 \\ B_l & \pm \mathbf{e}_i \pm \mathbf{e}_j, \pm \mathbf{e}_i & 1 \leq i \neq j \leq l \quad l > 2 \\ C_l & \pm \mathbf{e}_i \pm \mathbf{e}_j, \pm 2\mathbf{e}_i & 1 \leq i \neq j \leq l \quad l > 1 \end{array}$
gilmore-lie-groups_ E08562 (10.7)	175	■ 1.000	$A_1 = B_1 = C_1$ $B_2 = C_2$ $A_3 = D_3$	$A_1 = B_1 = C_1$ $B_2 = C_2$ $A_3 = D_3$	$A_1 = B_1 = C_1$ $B_2 = C_2$ $A_3 = D_3$

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gilmore-lie-groups_ EQ0563 (10.8)	175	■ 1.000	$D_{\{2\}}=A_{\{1\}}+A_{\{1\}}$	$D_2 = A_1 + A_1$	$D_2 = A_1 + A_1$
gilmore-lie-groups_ EQ0569 (10.10)	177	■ 1.000	$\begin{array}{l} A_{l-1} \quad SU(l), SL(l;\mathbb{R}), SU(p,q) \\ D_l \quad SO(2l), SO(p,q) \\ B_l \quad SO(2l+1), SO(p,q) \\ C_l \quad Sp(l), Sp(p,q) \end{array} \quad \begin{array}{l} p+q=l \\ p+q=2l \\ p+q=2l+1 \\ p+q=l \end{array}$	$\begin{array}{l} A_{l-1} \quad SU(l), SL(l;\mathbb{R}), SU(p,q) \\ D_l \quad SO(2l), SO(p,q) \\ B_l \quad SO(2l+1), SO(p,q) \\ C_l \quad Sp(l), Sp(p,q) \end{array} \quad \begin{array}{l} p+q=l \\ p+q=2l \\ p+q=2l+1 \\ p+q=l \end{array}$	$\begin{array}{l} A_{l-1} \quad SU(l), SL(l;\mathbb{R}), SU(p,q) \\ D_l \quad SO(2l), SO(p,q) \\ B_l \quad SO(2l+1), SO(p,q) \\ C_l \quad Sp(l), Sp(p,q) \end{array} \quad \begin{array}{l} p+q=l \\ p+q=2l \\ p+q=2l+1 \\ p+q=l \end{array}$
gilmore-lie-groups_ EQ0570 (10.12)	180	■ 1.000	$\left(\alpha_i, \alpha_j\right)=-\sqrt{n_{ij} / 4}$	$\left(\alpha_i, \alpha_j\right)=-\sqrt{n_{ij} / 4}$	$\left(\alpha_i, \alpha_j\right)=-\sqrt{n_{ij} / 4}$
gilmore-lie-groups_ EQ0571 (10.13)	180	■ 1.000	$\sum\left(\mathbf{u} \cdot \mathbf{v}\right)^2 \leq 1$	$\sum(\mathbf{u} \cdot \mathbf{v}_i)^2 \leq 1$	$\sum(\mathbf{u} \cdot \mathbf{v}_i)^2 \leq 1$
gilmore-lie-groups_ EQ0572 (10.14)	181	■ 1.000	$\left(\sum \mathbf{u}_i, \sum \mathbf{u}_j\right)=n+2 \sum \sum \mathbf{u}_i \cdot \mathbf{u}_j \geq 0$	$\left(\sum \mathbf{u}_i, \sum \mathbf{u}_j\right)=n+2 \sum \sum \mathbf{u}_i \cdot \mathbf{u}_j \geq 0$	$\left(\sum \mathbf{u}_i, \sum \mathbf{u}_j\right)=n+2 \sum \sum \mathbf{u}_i \cdot \mathbf{u}_j \geq 0$
gilmore-lie-groups_ EQ0573 (10.15)	181	■ 1.000	$\sum\left(\mathbf{u} \cdot \mathbf{v}\right)^2=\sum n_{i} / 4<1$	$\sum(\mathbf{u} \cdot \mathbf{v}_i)^2=\sum n_i / 4 < 1$	$\sum(\mathbf{u} \cdot \mathbf{v}_i)^2=\sum n_i / 4 < 1$
gilmore-lie-groups_ EQ0574 (10.16)	181	■ 1.000	$\mathbf{u}=\sum_{i=1}^p i \mathbf{u}_i \quad \mathbf{v}=\sum_{j=1}^q j \mathbf{v}_j$	$\mathbf{u}=\sum_{i=1}^p i \mathbf{u}_i \quad \mathbf{v}=\sum_{j=1}^q j \mathbf{v}_j$	$\mathbf{u}=\sum_{i=1}^p i \mathbf{u}_i \quad \mathbf{v}=\sum_{j=1}^q j \mathbf{v}_j$
gilmore-lie-groups_ EQ0575 (10.17)	182	■ 1.000	$\left(1+\frac{1}{p}\right)\left(1+\frac{1}{q}\right)>2$	$\left(1+\frac{1}{p}\right)\left(1+\frac{1}{q}\right)>2$	$\left(1+\frac{1}{p}\right)\left(1+\frac{1}{q}\right)>2$
gilmore-lie-groups_ EQ0577 (10.19)	182	■ 1.000	$\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$	$\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$	$\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$
gilmore-lie-groups_ EQ0579	185	■ 1.000	$\begin{aligned} & \left\{ \left[L_{ij}, L_{rs} \right] \right\} = \left[L_{ij}, \left[L_{rs}, L_{ij} \right] \right] + \left[L_{rs}, \left[L_{ij}, L_{rs} \right] \right] \\ & \left[L_{ij}, \left[L_{rs}, L_{ij} \right] \right] + \left[L_{rs}, \left[L_{ij}, L_{rs} \right] \right] = \left[L_{ij}, L_{rs} \right] \\ & \left[L_{ij}, L_{rs} \right] = \left[L_{ij}, L_{rs} \right] \end{aligned}$	$\begin{aligned} [\mathbf{L}, \mathbf{L}] &= \mathbf{L} [L_{ij}, L_{rs}] = +\delta_{jr} L_{is} + \delta_{is} L_{jr} - \delta_{ir} L_{js} - \delta_{js} L_{ir} \\ [\mathbf{L}, \mathbf{Q}] &= \mathbf{Q} [L_{ij}, Q_{rs}] = +\delta_{jr} Q_{is} - \delta_{is} Q_{jr} - \delta_{ir} Q_{js} + \delta_{js} Q_{ir} \\ [\mathbf{Q}, \mathbf{Q}] &= \mathbf{L} [Q_{ij}, Q_{rs}] = +\delta_{jr} L_{is} + \delta_{is} L_{jr} + \delta_{ir} L_{js} + \delta_{js} L_{ir} \end{aligned}$	$\begin{aligned} [\mathbf{L}, \mathbf{L}] &= \mathbf{L} \left[L_{ij}, L_{rs} \right] = +\delta_{jr} L_{is} + \delta_{is} L_{jr} - \delta_{ir} L_{js} - \delta_{js} L_{ir} \\ [\mathbf{L}, \mathbf{Q}] &= \mathbf{Q} \left[L_{ij}, Q_{rs} \right] = +\delta_{jr} Q_{is} - \delta_{is} Q_{jr} - \delta_{ir} Q_{js} + \delta_{js} Q_{ir} \\ [\mathbf{Q}, \mathbf{Q}] &= \mathbf{L} \left[Q_{ij}, Q_{rs} \right] = +\delta_{jr} L_{is} + \delta_{is} L_{jr} + \delta_{ir} L_{js} + \delta_{js} L_{ir} \end{aligned}$
gilmore-lie-groups_ EQ0580	185	■ 1.000	$\hat{N}=\sum_{i=1}^3 a_i^{\dagger} a_i \quad \hat{Q}_{ij}=Q_{ij}-\frac{2}{3} \hat{N} \delta_{ij}$	$\hat{N}=\sum_{i=1}^3 a_i^{\dagger} a_i \quad \hat{Q}_{ij}=Q_{ij}-\frac{2}{3} \hat{N} \delta_{ij}$	$\hat{N}=\sum_{i=1}^3 a_i^{\dagger} a_i \quad \hat{Q}_{ij}=Q_{ij}-\frac{2}{3} \hat{N} \delta_{ij}$
gilmore-lie-groups_ EQ0585 (11.5)	189	■ 1.000	$i h^i H_i+i e^{\alpha} \frac{1}{\sqrt{2}}\left(E_{\alpha}+E_{-\alpha}\right)+e^{-\alpha} \frac{1}{\sqrt{2}}\left(E_{\alpha}-E_{-\alpha}\right)$	$i h^i H_i+i e^{\alpha} \frac{1}{\sqrt{2}}\left(E_{\alpha}+E_{-\alpha}\right)+e^{-\alpha} \frac{1}{\sqrt{2}}\left(E_{\alpha}-E_{-\alpha}\right)$	$i h^i H_i+i e^{\alpha} \frac{1}{\sqrt{2}}\left(E_{\alpha}+E_{-\alpha}\right)+e^{-\alpha} \frac{1}{\sqrt{2}}\left(E_{\alpha}-E_{-\alpha}\right)$
gilmore-lie-groups_ EQ0588 (11.9)	190	■ 1.000	$\text { - dimension } \leq \chi=\text { character } \leq+\text { rank }$	$\text { - dimension } \leq \chi=\text { character } \leq+\text { rank }$	$\text { - dimension } \leq \chi=\text { character } \leq+\text { rank }$

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gilmore-lie-groups_ E09592 (11.13)	191	■ 1.000	$\chi\!\left(\!\!\mathfrak{g}^{\!\prime}\!\right)\!=\!\operatorname{dim}\!\left(\!\!\mathfrak{p}^{\!\prime}\!\right)\!-\!\operatorname{dim}\!\left(\!\!\mathfrak{h}\!\right)\!=\!\operatorname{dim}\!\left(\!\!\mathfrak{g}\!\right)\!-\!2\operatorname{dim}\!\left(\!\!\mathfrak{h}\!\right)\!=\!2\operatorname{dim}\!\left(\!\!\mathfrak{p}^{\!\prime}\!\right)\!-\!\operatorname{dim}\!\left(\!\!\mathfrak{g}\!\right)$	$\chi(\mathfrak{g}') = \dim(\mathfrak{p}') - \dim(\mathfrak{h}) = \dim(\mathfrak{g}) - 2 \dim(\mathfrak{h}) = 2 \dim(\mathfrak{p}') - \dim(\mathfrak{g})$	$\chi(\mathfrak{g}') = \dim(\mathfrak{p}') - \dim(\mathfrak{h}) = \dim(\mathfrak{g}) - 2 \dim(\mathfrak{h}) = 2 \dim(\mathfrak{p}') - \dim(\mathfrak{g})$
gilmore-lie-groups_ E09593 (11.14)	191	■ 1.000	$\mathfrak{su}(2) \rightarrow \mathfrak{su}(1,1) : \quad \frac{i}{2} \begin{bmatrix} a_3 & 0 \\ 0 & -a_3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{bmatrix}$	$\mathfrak{su}(2) \rightarrow \mathfrak{su}(1,1) : \quad \frac{i}{2} \begin{bmatrix} a_3 & 0 \\ 0 & -a_3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{bmatrix}$	$\mathfrak{su}(2) \rightarrow \mathfrak{su}(1,1) : \quad \frac{i}{2} \begin{bmatrix} a_3 & 0 \\ 0 & -a_3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{bmatrix}$
gilmore-lie-groups_ E09594 (11.15)	191	■ 1.000	$T^2=I$	$T^2 = I$	$T^2 = I$
gilmore-lie-groups_ E09596 (11.17)	191	■ 1.000	$(\mathfrak{h},\mathfrak{p})=(T^2\mathfrak{h},\mathfrak{p})=(T\mathfrak{h},T\mathfrak{p})=(\mathfrak{h},-\mathfrak{p})=- (\mathfrak{h},\mathfrak{p})=0$	$(\mathfrak{h},\mathfrak{p}) = (T^2\mathfrak{h},\mathfrak{p}) = (T\mathfrak{h},T\mathfrak{p}) = (\mathfrak{h},-\mathfrak{p}) = -(\mathfrak{h},\mathfrak{p}) = 0$	$(\mathfrak{h},\mathfrak{p}) = (T^2\mathfrak{h},\mathfrak{p}) = (T\mathfrak{h},T\mathfrak{p}) = (\mathfrak{h},-\mathfrak{p}) = -(\mathfrak{h},\mathfrak{p}) = 0$
gilmore-lie-groups_ E09597 (11.18)	192	■ 1.000	$\mathfrak{u}(n;\mathbb{F}) \quad \begin{bmatrix} A_p & 0 \\ 0 & A_q \end{bmatrix} + \begin{bmatrix} 0 & B \\ -B^\dagger & 0 \end{bmatrix}$	$\mathfrak{u}(n;\mathbb{F}) \quad \begin{bmatrix} A_p & 0 \\ 0 & A_q \end{bmatrix} + \begin{bmatrix} 0 & B \\ -B^\dagger & 0 \end{bmatrix}$	$\mathfrak{u}(n;\mathbb{F}) \quad \begin{bmatrix} A_p & 0 \\ 0 & A_q \end{bmatrix} + \begin{bmatrix} 0 & B \\ -B^\dagger & 0 \end{bmatrix}$
gilmore-lie-groups_ E09598 (11.19)	192	■ 1.000	$\mathfrak{u}(p,q;\mathbb{F}) \quad \begin{bmatrix} A_p & 0 \\ 0 & A_q \end{bmatrix} + \begin{bmatrix} 0 & B \\ +B^\dagger & 0 \end{bmatrix}$	$\mathfrak{u}(p,q;\mathbb{F}) \quad \begin{bmatrix} A_p & 0 \\ 0 & A_q \end{bmatrix} + \begin{bmatrix} 0 & B \\ +B^\dagger & 0 \end{bmatrix}$	$\mathfrak{u}(p,q;\mathbb{F}) \quad \begin{bmatrix} A_p & 0 \\ 0 & A_q \end{bmatrix} + \begin{bmatrix} 0 & B \\ +B^\dagger & 0 \end{bmatrix}$
gilmore-lie-groups_ E09602 (11.23)	193	■ 1.000	$\begin{aligned} &\alpha+i\beta \longrightarrow \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix} \\ &\alpha+\mathcal{I}\beta+\mathcal{J}\gamma+\mathcal{K}\delta \longrightarrow \begin{bmatrix} \alpha+i\delta & i\beta+\gamma \\ i\beta-\gamma & \alpha-i\delta \end{bmatrix} \end{aligned}$	$\alpha+i\beta \longrightarrow \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}$ $\alpha+\mathcal{I}\beta+\mathcal{J}\gamma+\mathcal{K}\delta \longrightarrow \begin{bmatrix} \alpha+i\delta & i\beta+\gamma \\ i\beta-\gamma & \alpha-i\delta \end{bmatrix}$	$\alpha+i\beta \longrightarrow \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}$ $\alpha+\mathcal{I}\beta+\mathcal{J}\gamma+\mathcal{K}\delta \longrightarrow \begin{bmatrix} \alpha+i\delta & i\beta+\gamma \\ i\beta-\gamma & \alpha-i\delta \end{bmatrix}$
gilmore-lie-groups_ E09610	197	■ 1.000	$w\,a_1^{\dagger}a_1+x\,a_1^{\dagger}a_2+y\,a_2^{\dagger}a_1+z\,a_2^{\dagger}a_2=\alpha\left(a_1^{\dagger}a_1+a_2^{\dagger}a_2\right)$	$wa_1^\dagger a_1 + xa_1^\dagger a_2 + ya_2^\dagger a_1 + za_2^\dagger a_2 = \alpha \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right)$	$wa_1^\dagger a_1 + xa_1^\dagger a_2 + ya_2^\dagger a_1 + za_2^\dagger a_2 = \alpha (a_1^\dagger a_1 + a_2^\dagger a_2)$
gilmore-lie-groups_ E09611	197	■ 1.000	$+\beta\left(a_1^{\dagger}a_1-a_2^{\dagger}a_2\right)+\gamma\left(a_1^{\dagger}a_2+a_2^{\dagger}a_1\right)+\delta\left(a_1^{\dagger}a_2-a_2^{\dagger}a_1\right)$	$+\beta \left(a_1^\dagger a_1 - a_2^\dagger a_2 \right) + \gamma \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right) + \delta \left(a_1^\dagger a_2 - a_2^\dagger a_1 \right)$	$+\beta (a_1^\dagger a_1 - a_2^\dagger a_2) + \gamma (a_1^\dagger a_2 + a_2^\dagger a_1) + \delta (a_1^\dagger a_2 - a_2^\dagger a_1)$
gilmore-lie-groups_ E09613	198	■ 1.000	$\mathcal{C}^2=\sum H_i^2+\sum E_{\alpha}E_{-\alpha}+E_{-\alpha}E_{\alpha}$	$\mathcal{C}^2 = \sum H_i^2 + \sum E_\alpha E_{-\alpha} + E_{-\alpha} E_\alpha$	$\mathcal{C}^2 = \sum H_i^2 + \sum E_\alpha E_{-\alpha} + E_{-\alpha} E_\alpha$
gilmore-lie-groups_ E09614	198	■ 1.000	$H_i\left n_1,n_2,\ldots,n_r\right\rangle=n_i\left n_1,n_2,\ldots,n_r\right\rangle$	$H_i n_1,n_2,\dots,n_r\rangle = n_i n_1,n_2,\dots,n_r\rangle$	$H_i n_1,n_2,\dots,n_r\rangle = n_i n_1,n_2,\dots,n_r\rangle$
gilmore-lie-groups_ E09615	198	■ 1.000	$\mathcal{C}^2\mid\text{state}\rangle=-l(l+n-2)\mid\text{state}\rangle$	$\mathcal{C}^2 \mid \text{state} \rangle = -l(l+n-2) \mid \text{state} \rangle$	$\mathcal{C}^2 \text{state} \rangle = -l(l+n-2) \text{state} \rangle$

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gilmore-lie-groups_ E0640	209	■ 1.000	$\Gamma_{\mu\nu}^{\sigma}=\frac{1}{2}g^{\sigma\alpha}\left(\frac{\partial g_{\mu\alpha}}{\partial x^{\nu}}+\frac{\partial g_{\nu\alpha}}{\partial x^{\mu}}-\frac{\partial g_{\mu\nu}}{\partial x^{\alpha}}\right)$	$\Gamma_{\mu\nu}^{\sigma}=\frac{1}{2}g^{\sigma\alpha}\left(\frac{\partial g_{\mu\alpha}}{\partial x^{\nu}}+\frac{\partial g_{\nu\alpha}}{\partial x^{\mu}}-\frac{\partial g_{\mu\nu}}{\partial x^{\alpha}}\right)$	$\Gamma_{\mu\nu}^{\sigma}=\frac{1}{2}g^{\sigma\alpha}\left(\frac{\partial g_{\mu\alpha}}{\partial x^{\nu}}+\frac{\partial g_{\nu\alpha}}{\partial x^{\mu}}-\frac{\partial g_{\mu\nu}}{\partial x^{\alpha}}\right)$
gilmore-lie-groups_ E0641	209	■ 1.000	$R_{\{\sigma,\alpha\}\{\beta\}}^{\mu}=\frac{\partial\Gamma_{\sigma\beta}^{\mu}}{\partial x^{\alpha}}-\frac{\partial\Gamma_{\sigma\alpha}^{\mu}}{\partial x^{\beta}}+\Gamma_{\rho\alpha}^{\mu}\Gamma_{\sigma\beta}^{\rho}-\Gamma_{\rho\beta}^{\mu}\Gamma_{\sigma\alpha}^{\rho}$	$R_{\sigma,\alpha\beta}^{\mu}=\frac{\partial\Gamma_{\sigma\beta}^{\mu}}{\partial x^{\alpha}}-\frac{\partial\Gamma_{\sigma\alpha}^{\mu}}{\partial x^{\beta}}+\Gamma_{\rho\alpha}^{\mu}\Gamma_{\sigma\beta}^{\rho}-\Gamma_{\rho\beta}^{\mu}\Gamma_{\sigma\alpha}^{\rho}$	$R_{\sigma,\alpha\beta}^{\mu}=\frac{\partial\Gamma_{\sigma\beta}^{\mu}}{\partial x^{\alpha}}-\frac{\partial\Gamma_{\sigma\alpha}^{\mu}}{\partial x^{\beta}}+\Gamma_{\rho\alpha}^{\mu}\Gamma_{\sigma\beta}^{\rho}-\Gamma_{\rho\beta}^{\mu}\Gamma_{\sigma\alpha}^{\rho}$
gilmore-lie-groups_ E0642	209	■ 1.000	$R_{\{\sigma\}\{\beta\}}=R_{\{\sigma,\mu\}\{\beta\}}^{\mu}$	$R_{\sigma\beta}=R_{\sigma,\mu\beta}^{\mu}$	$R_{\sigma\beta}=R_{\sigma,\mu\beta}^{\mu}$
gilmore-lie-groups_ E0643 (12.12)	209	■ 1.000	$R=g^{\sigma\beta}R_{\sigma\beta}$	$R=g^{\sigma\beta}R_{\sigma\beta}$	$R=g^{\sigma\beta}R_{\sigma\beta}$
gilmore-lie-groups_ E0644 (12.13)	210	■ 1.000	$g_{ij}=W^{-2}=\left[I_N-XX^t\right]_{ij}^{-1}\simeq\left[I_N+XX^t\right]_{ij}\rightarrow\delta_{ij}+x^ix^j$	$g_{ij}=W^{-2}=\left[I_N-XX^t\right]_{ij}^{-1}\simeq\left[I_N+XX^t\right]_{ij}\rightarrow\delta_{ij}+x^ix^j$	$g_{ij}=W^{-2}=\left[I_N-XX^t\right]_{ij}^{-1}\simeq\left[I_N+XX^t\right]_{ij}\rightarrow\delta_{ij}+x^ix^j$
gilmore-lie-groups_ E0646 (12.15)	210	■ 1.000	$R_{\{\sigma,\alpha\}\{\beta\}}^{\mu}\rightarrow\frac{\partial}{\partial x^{\alpha}}(\delta_{\sigma\beta}x^{\mu})-\frac{\partial}{\partial x^{\beta}}(\delta_{\sigma\alpha}x^{\mu})=\delta_{\sigma\beta}\delta_{\alpha}^{\mu}-\delta_{\sigma\alpha}\delta_{\beta}^{\mu}$	$R_{\sigma,\alpha\beta}^{\mu}\rightarrow\frac{\partial}{\partial x^{\alpha}}(\delta_{\sigma\beta}x^{\mu})-\frac{\partial}{\partial x^{\beta}}(\delta_{\sigma\alpha}x^{\mu})=\delta_{\sigma\beta}\delta_{\alpha}^{\mu}-\delta_{\sigma\alpha}\delta_{\beta}^{\mu}$	$R_{\sigma,\alpha\beta}^{\mu}\rightarrow\frac{\partial}{\partial x^{\alpha}}(\delta_{\sigma\beta}x^{\mu})-\frac{\partial}{\partial x^{\beta}}(\delta_{\sigma\alpha}x^{\mu})=\delta_{\sigma\beta}\delta_{\alpha}^{\mu}-\delta_{\sigma\alpha}\delta_{\beta}^{\mu}$
gilmore-lie-groups_ E0647 (12.16)	210	■ 1.000	$R_{\{\sigma\}\{\beta\}}=R_{\{\sigma,\mu\}\{\beta\}}^{\mu}=\delta_{\sigma\beta}-\delta_{\sigma\mu}\delta_{\beta\mu}=N\delta_{\sigma\beta}-\delta_{\sigma\beta}$	$R_{\sigma\beta}=R_{\sigma,\mu\beta}^{\mu}=\delta_{\sigma\beta}\delta_{\mu\mu}-\delta_{\sigma\mu}\delta_{\beta\mu}=N\delta_{\sigma\beta}-\delta_{\sigma\beta}$	$R_{\sigma\beta}=R_{\sigma,\mu\beta}^{\mu}=\delta_{\sigma\beta}\delta_{\mu\mu}-\delta_{\sigma\mu}\delta_{\beta\mu}=N\delta_{\sigma\beta}-\delta_{\sigma\beta}$
gilmore-lie-groups_ E0648 (12.17)	210	■ 1.000	$R=g^{\sigma\beta}R_{\sigma\beta}\rightarrow\delta^{\sigma\beta}(N-1)\delta_{\sigma\beta}=N(N-1)$	$R=g^{\sigma\beta}R_{\sigma\beta}\rightarrow\delta^{\sigma\beta}(N-1)\delta_{\sigma\beta}=N(N-1)$	$R=g^{\sigma\beta}R_{\sigma\beta}\rightarrow\delta^{\sigma\beta}(N-1)\delta_{\sigma\beta}=N(N-1)$
gilmore-lie-groups_ E0650 (12.19)	211	■ 1.000	$\mathfrak{g}^{\prime}=\mathfrak{h}^{\prime}+\mathfrak{p}^{\prime}$	$\mathfrak{g}''=\mathfrak{h}''+\mathfrak{p}''$	$\mathfrak{g}''=\mathfrak{h}''+\mathfrak{p}''$
gilmore-lie-groups_ E0651 (12.20)	211	■ 1.000	$\begin{aligned} &\&\left[\mathfrak{h}^{\prime}\right]\subseteq\mathfrak{h}''\\ &\&\left[\mathfrak{h}^{\prime},\mathfrak{p}^{\prime}\right]\subseteq\mathfrak{p}''\\ &\&\left[\mathfrak{p}^{\prime},\mathfrak{p}^{\prime}\right]\subseteq\mathfrak{h}'' \end{aligned}$	$\begin{aligned} &[\mathfrak{h}'',\mathfrak{h}'']\subseteq\mathfrak{h}''\\ &[\mathfrak{h}'',\mathfrak{p}'']\subseteq\mathfrak{p}''\\ &[\mathfrak{p}'',\mathfrak{p}'']\subseteq\mathfrak{h}'' \end{aligned}$	$\begin{aligned} &[\mathfrak{h}'',\mathfrak{h}'']\subseteq\mathfrak{h}''\\ &[\mathfrak{h}'',\mathfrak{p}'']\subseteq\mathfrak{p}''\\ &[\mathfrak{p}'',\mathfrak{p}'']\subseteq\mathfrak{h}'' \end{aligned}$
gilmore-lie-groups_ E0652 (12.21)	211	■ 1.000	$P^{\prime}=\operatorname{EXP}\left(\mathfrak{p}^{\prime}\right)=G^{\prime} / H^{\prime}$	$P''=\operatorname{EXP}\left(\mathfrak{p}''\right)=G'' / H''$	$P''=\operatorname{EXP}\left(\mathfrak{p}''\right)=G'' / H''$
gilmore-lie-groups_ E0654 (12.23)	211	■ 1.000	$\mathfrak{g}^{\prime}=\left(\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-}\right)+i\left(\mathfrak{g}_{-,-}+\mathfrak{g}_{-,+}\right)$	$\mathfrak{g}'=(\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-})+i(\mathfrak{g}_{-,-}+\mathfrak{g}_{-,+})$	$\mathfrak{g}'=(\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-})+i(\mathfrak{g}_{-,-}+\mathfrak{g}_{-,+})$

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gilmore-lie-groups_ EQ0696 (13.14)	223	■ 1.000	$\begin{aligned} \{\left[L_i, L_j\right] \} &= -\epsilon_{ijk} L_k \\ \{\left[L_i, V_j\right] \} &= -\epsilon_{ijk} V_k \\ \{\left[V_i, V_j\right] \} &= -\epsilon_{ijk} L_k \end{aligned}$	$\begin{aligned} [L_i, L_j] &= -\epsilon_{ijk} L_k \\ [L_i, V_j] &= -\epsilon_{ijk} V_k \\ [V_i, V_j] &= -\epsilon_{ijk} L_k \end{aligned}$	$\begin{aligned} [L_i, L_j] &= -\epsilon_{ijk} L_k \\ [L_i, V_j] &= -\epsilon_{ijk} V_k \\ [V_i, V_j] &= -\epsilon_{ijk} L_k \end{aligned}$
gilmore-lie-groups_ EQ0697 (13.15)	223	■ 1.000	$-P_i = \lim_{R \rightarrow \infty} \frac{1}{R} V_i = \lim_{R \rightarrow \infty} \frac{1}{R} (x_i \partial_4 - x_4 \partial_i) = -\partial_i$	$-P_i = \lim_{R \rightarrow \infty} \frac{1}{R} V_i = \lim_{R \rightarrow \infty} \frac{1}{R} (x_i \partial_4 - x_4 \partial_i) = -\partial_i$	$-P_i = \lim_{R \rightarrow \infty} \frac{1}{R} V_i = \lim_{R \rightarrow \infty} \frac{1}{R} (x_i \partial_4 - x_4 \partial_i) = -\partial_i$
gilmore-lie-groups_ EQ0698 (13.16)	223	■ 1.000	$\begin{aligned} \{\left[L_i, L_j\right] \} &= -\epsilon_{ijk} L_k \\ \{\left[L_i, P_j\right] \} &= -\epsilon_{ijk} P_k \\ \{\left[P_i, P_j\right] \} &= 0 \end{aligned}$	$\begin{aligned} [L_i, L_j] &= -\epsilon_{ijk} L_k \\ [L_i, P_j] &= -\epsilon_{ijk} P_k \\ [P_i, P_j] &= 0 \end{aligned}$	$\begin{aligned} [L_i, L_j] &= -\epsilon_{ijk} L_k \\ [L_i, P_j] &= -\epsilon_{ijk} P_k \\ [P_i, P_j] &= 0 \end{aligned}$
gilmore-lie-groups_ EQ0699 (13.17)	223	■ 1.000	$\begin{aligned} \mathcal{C}_1^2[ISO(3)] &= \lim_{R \rightarrow \infty} (1/R^2) (\mathbf{L} \cdot \mathbf{L} + \mathbf{V} \cdot \mathbf{V}) \\ &= \lim_{R \rightarrow \infty} [(\mathbf{L}/R) \cdot (\mathbf{L}/R) + (\mathbf{V}/R) \cdot (\mathbf{V}/R)] \\ &= 0 + \mathbf{P} \cdot \mathbf{P} = \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \end{aligned}$	$\begin{aligned} \mathcal{C}_1^2[ISO(3)] &= \lim_{R \rightarrow \infty} (1/R^2) (\mathbf{L} \cdot \mathbf{L} + \mathbf{V} \cdot \mathbf{V}) \\ &= \lim_{R \rightarrow \infty} [(\mathbf{L}/R) \cdot (\mathbf{L}/R) + (\mathbf{V}/R) \cdot (\mathbf{V}/R)] \\ &= 0 + \mathbf{P} \cdot \mathbf{P} = \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \end{aligned}$	$\begin{aligned} \mathcal{C}_1^2[ISO(3)] &= \lim_{R \rightarrow \infty} (1/R^2) (\mathbf{L} \cdot \mathbf{L} + \mathbf{V} \cdot \mathbf{V}) \\ &= \lim_{R \rightarrow \infty} [(\mathbf{L}/R) \cdot (\mathbf{L}/R) + (\mathbf{V}/R) \cdot (\mathbf{V}/R)] \\ &= 0 + \mathbf{P} \cdot \mathbf{P} = \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \end{aligned}$
gilmore-lie-groups_ EQ0700 (13.18)	223	■ 1.000	$\mathcal{C}_2^2[SO(4)] = \epsilon^{ijkl} X_{ij} X_{kl} \rightarrow 8 \mathbf{L} \cdot \mathbf{V}$	$\mathcal{C}_2^2[SO(4)] = \epsilon^{ijkl} X_{ij} X_{kl} \rightarrow 8 \mathbf{L} \cdot \mathbf{V}$	$\mathcal{C}_2^2[SO(4)] = \epsilon^{ijkl} X_{ij} X_{kl} \rightarrow 8 \mathbf{L} \cdot \mathbf{V}$
gilmore-lie-groups_ EQ0701 (13.19)	223	■ 1.000	$\begin{aligned} \mathcal{C}_2^2[ISO(3)]/8 &= \lim_{R \rightarrow \infty} (1/R) (\mathbf{L} \cdot \mathbf{V}) \\ &= \lim_{R \rightarrow \infty} [\mathbf{L} \cdot (\mathbf{V}/R)] = -\mathbf{L} \cdot \mathbf{P} \end{aligned}$	$\begin{aligned} \mathcal{C}_2^2[ISO(3)]/8 &= \lim_{R \rightarrow \infty} (1/R) (\mathbf{L} \cdot \mathbf{V}) \\ &= \lim_{R \rightarrow \infty} [\mathbf{L} \cdot (\mathbf{V}/R)] = -\mathbf{L} \cdot \mathbf{P} \end{aligned}$	$\begin{aligned} \mathcal{C}_2^2[ISO(3)]/8 &= \lim_{R \rightarrow \infty} (1/R) (\mathbf{L} \cdot \mathbf{V}) \\ &= \lim_{R \rightarrow \infty} [\mathbf{L} \cdot (\mathbf{V}/R)] = -\mathbf{L} \cdot \mathbf{P} \end{aligned}$
gilmore-lie-groups_ EQ0702 (13.20)	224	■ 1.000	$x^2 + y^2 + z^2 - (ct)^2$	$x^2 + y^2 + z^2 - (ct)^2$	$x^2 + y^2 + z^2 - (ct)^2$
gilmore-lie-groups_ EQ0703 (13.21)	224	■ 1.000	$(x - x')^2 + (y - y')^2 + (z - z')^2 - (ct - ct')^2$	$(x - x')^2 + (y - y')^2 + (z - z')^2 - (ct - ct')^2$	$(x - x')^2 + (y - y')^2 + (z - z')^2 - (ct - ct')^2$
gilmore-lie-groups_ EQ0707 (13.24)	224	■ 1.000	$\begin{aligned} \lim_{R \rightarrow \infty} (1/R) W^\alpha &\rightarrow \lim_{R \rightarrow \infty} \epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} [(1/R) X_{\mu 5}] \\ &= \epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial/\partial x^\mu) \end{aligned}$	$\begin{aligned} \lim_{R \rightarrow \infty} (1/R) W^\alpha &\rightarrow \lim_{R \rightarrow \infty} \epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} [(1/R) X_{\mu 5}] \\ &= \epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial/\partial x^\mu) \end{aligned}$	$\begin{aligned} \lim_{R \rightarrow \infty} (1/R) W^\alpha &\rightarrow \lim_{R \rightarrow \infty} \epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} [(1/R) X_{\mu 5}] \\ &= \epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial/\partial x^\mu) \end{aligned}$

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gilmore-lie-groups_ EQ0708 (13.25)	225	■ 1.000	$P_{\mu}=(0,0,0, m\ c)$	$P_{\mu}=(0,0,0,mc)$	$P_{\mu}=(0,0,0,mc)$
gilmore-lie-groups_ EQ0709 (13.26)	225	■ 1.000	$W^{\alpha}=\epsilon^{\alpha\beta\gamma\mu}\ X_{\beta\gamma}\ X_{\mu}\ X_{\alpha}\ m\ c=L_{\alpha}mc$	$W^{\alpha}=\epsilon^{\alpha\beta\gamma\mu}X_{\beta\gamma}mc=L_{\alpha}mc$	$W^{\alpha}=\epsilon^{\alpha\beta\gamma\mu}X_{\beta\gamma}mc=L_{\alpha}mc$
gilmore-lie-groups_ EQ0710 (13.27)	225	■ 1.000	$W^{\alpha}\ W_{\alpha}=(\mathbf{L}\cdot\mathbf{L})(\mathbf{P}\cdot\mathbf{P})$	$W^{\alpha}W_{\alpha}=(\mathbf{L}\cdot\mathbf{L})(\mathbf{P}\cdot\mathbf{P})$	$W^{\alpha}W_{\alpha}=(\mathbf{L}\cdot\mathbf{L})(\mathbf{P}\cdot\mathbf{P})$
gilmore-lie-groups_ EQ0711 (13.28)	225	■ 1.000	$\begin{aligned} \left\{J_3, J_{\pm}\right\} &= \pm J_{\pm} \\ \left\{J_{+}, J_{-}\right\} &= 2J_3 \\ \left\{J_0, \mathbf{J}\right\} &= 0 \end{aligned}$	$\begin{aligned} [J_3, J_{\pm}] &= \pm J_{\pm} \\ [J_{+}, J_{-}] &= 2J_3 \\ [J_0, \mathbf{J}] &= 0 \end{aligned}$	$\begin{aligned} [J_3, J_{\pm}] &= \pm J_{\pm} \\ [J_{+}, J_{-}] &= 2J_3 \\ [J_0, \mathbf{J}] &= 0 \end{aligned}$
gilmore-lie-groups_ EQ0713 (13.30)	226	■ 1.000	$\begin{aligned} \left\{h_3, h_{\pm}\right\} &= \pm h_{\pm} \\ \left\{h_{+}, h_{-}\right\} &= 2c^2h_3 - h_0 \\ \left\{h_0, \mathbf{h}\right\} &= 0 \end{aligned}$	$\begin{aligned} [h_3, h_{\pm}] &= \pm h_{\pm} \\ [h_{+}, h_{-}] &= 2c^2h_3 - h_0 \\ [h_0, \mathbf{h}] &= 0 \end{aligned}$	$\begin{aligned} [h_3, h_{\pm}] &= \pm h_{\pm} \\ [h_{+}, h_{-}] &= 2c^2h_3 - h_0 \\ [h_0, \mathbf{h}] &= 0 \end{aligned}$
gilmore-lie-groups_ EQ0715 (13.32)	226	■ 1.000	$\begin{aligned} \mathcal{C}^1 &= J_0 \\ \mathcal{C}^2 &= J_3^2 + \frac{1}{2}(J_{+}J_{-} + J_{-}J_{+}) \end{aligned}$	$\begin{aligned} \mathcal{C}^1 &= J_0 \\ \mathcal{C}^2 &= J_3^2 + \frac{1}{2}(J_{+}J_{-} + J_{-}J_{+}) \end{aligned}$	$\begin{aligned} \mathcal{C}^1 &= J_0 \\ \mathcal{C}^2 &= J_3^2 + \frac{1}{2}(J_{+}J_{-} + J_{-}J_{+}) \end{aligned}$
gilmore-lie-groups_ EQ0716 (13.33)	226	■ 1.000	$\begin{aligned} \lim_{c\rightarrow 0} \mathcal{C}^2 &= \lim_{c\rightarrow 0} \left(h_3 - \frac{1}{2c^2}h_0 \right)^2 + \frac{1}{2}[(cJ_{+})(cJ_{-}) + (cJ_{-})(cJ_{+})] \\ &= \lim_{c\rightarrow 0} c^2h_3^2 - \frac{1}{2}(h_3h_0 + h_0h_3) + c^2\left(-\frac{1}{2c^2}h_0\right)^2 \\ &\quad + \frac{1}{2}[(h_{+})(h_{-}) + (h_{-})(h_{+})] \end{aligned}$	$\begin{aligned} \lim_{c\rightarrow 0} c^2\mathcal{C}^2 &= \lim_{c\rightarrow 0} c^2\left(h_3 - \frac{1}{2c^2}h_0\right)^2 + \frac{1}{2}[(cJ_{+})(cJ_{-}) + (cJ_{-})(cJ_{+})] \\ &= \lim_{c\rightarrow 0} c^2h_3^2 - \frac{1}{2}(h_3h_0 + h_0h_3) + c^2\left(-\frac{1}{2c^2}h_0\right)^2 \\ &\quad + \frac{1}{2}[(h_{+})(h_{-}) + (h_{-})(h_{+})] \end{aligned}$	$\begin{aligned} \lim_{c\rightarrow 0} c^2\mathcal{C}^2 &= \lim_{c\rightarrow 0} c^2\left(h_3 - \frac{1}{2c^2}h_0\right)^2 + \frac{1}{2}[(cJ_{+})(cJ_{-}) + (cJ_{-})(cJ_{+})] \\ &= \lim_{c\rightarrow 0} c^2h_3^2 - \frac{1}{2}(h_3h_0 + h_0h_3) + c^2\left(-\frac{1}{2c^2}h_0\right)^2 \\ &\quad + \frac{1}{2}[(h_{+})(h_{-}) + (h_{-})(h_{+})] \end{aligned}$
gilmore-lie-groups_ EQ0717 (13.34)	226	■ 1.000	$\begin{aligned} \lim_{c\rightarrow 0} \mathcal{C}^2[U(2)] &= \lim_{c\rightarrow 0} \left(h_0/2c \right)^2 \rightarrow \mathcal{C}^2[H_4] = -\frac{1}{2}\left[\left(\hat{n} + \frac{1}{2}I\right)I + I\left(\hat{n} + \frac{1}{2}I\right)\right] \\ &\quad + \frac{1}{2}(aa^{\dagger} + a^{\dagger}a) \end{aligned}$	$\begin{aligned} \lim_{c\rightarrow 0} c^2\mathcal{C}^2[U(2)] &= \lim_{c\rightarrow 0} (h_0/2c)^2 \rightarrow \mathcal{C}^2[H_4] = -\frac{1}{2}\left[\left(\hat{n} + \frac{1}{2}I\right)I + I\left(\hat{n} + \frac{1}{2}I\right)\right] \\ &\quad + \frac{1}{2}(aa^{\dagger} + a^{\dagger}a) \end{aligned}$	$\begin{aligned} \lim_{c\rightarrow 0} c^2\mathcal{C}^2[U(2)] &= \lim_{c\rightarrow 0} (h_0/2c)^2 \rightarrow \mathcal{C}^2[H_4] = -\frac{1}{2}\left[\left(\hat{n} + \frac{1}{2}I\right)I + I\left(\hat{n} + \frac{1}{2}I\right)\right] \\ &\quad + \frac{1}{2}(aa^{\dagger} + a^{\dagger}a) \end{aligned} \tag{13.3}$

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gilmore-lie-groups_ EQ0718 (13.35)	227	■ 1.000	$\begin{aligned} i \, \theta_{\mu} J_{\mu} &= \frac{1}{2} \theta e^{-i \phi} J_{+} - \frac{1}{2} \theta e^{+i \phi} J_{-} + i \theta_3 J_3 + i \theta_0 J_0 \\ \theta e^{-i \phi} J_{+} &= \frac{\theta}{2c} e^{-i \phi} h_{+} - \frac{\theta}{2c} e^{+i \phi} h_{-} + i \theta_3 h_3 + i \left(\theta_0 - \frac{\theta_3}{2c^2} \right) h_0 \end{aligned}$	$i \theta_{\mu} J_{\mu} = \frac{1}{2} \theta e^{-i \phi} J_{+} - \frac{1}{2} \theta e^{+i \phi} J_{-} + i \theta_3 J_3 + i \theta_0 J_0$ $= \frac{\theta}{2c} e^{-i \phi} h_{+} - \frac{\theta}{2c} e^{+i \phi} h_{-} + i \theta_3 h_3 + i \left(\theta_0 - \frac{\theta_3}{2c^2} \right) h_0$	$i \theta_{\mu} J_{\mu} = \frac{1}{2} \theta e^{-i \phi} J_{+} - \frac{1}{2} \theta e^{+i \phi} J_{-} + i \theta_3 J_3 + i \theta_0 J_0$ $= \frac{\theta}{2c} e^{-i \phi} h_{+} - \frac{\theta}{2c} e^{+i \phi} h_{-} + i \theta_3 h_3 + i \left(\theta_0 - \frac{\theta_3}{2c^2} \right) h_0$
gilmore-lie-groups_ EQ0719 (13.36)	227	■ 1.000	$\begin{aligned} &\lim_{c \rightarrow 0} \left(\theta e^{-i \phi} J_{+} + \frac{\theta}{2c} e^{-i \phi} J_{+} \right) = \alpha \\ &\lim_{c \rightarrow 0} \left(-\theta e^{+i \phi} J_{-} - \frac{\theta}{2c} e^{+i \phi} J_{-} \right) = -\alpha^* \end{aligned}$	$\lim_{c \rightarrow 0} \left(\theta e^{-i \phi} J_{+} + \frac{\theta}{2c} e^{-i \phi} J_{+} \right) = \alpha$ $\lim_{c \rightarrow 0} \left(-\theta e^{+i \phi} J_{-} - \frac{\theta}{2c} e^{+i \phi} J_{-} \right) = -\alpha^*$	$\lim_{c \rightarrow 0} \left(\theta e^{-i \phi} J_{+} + \frac{\theta}{2c} e^{-i \phi} J_{+} \right) = \alpha$ $\lim_{c \rightarrow 0} \left(-\theta e^{+i \phi} J_{-} - \frac{\theta}{2c} e^{+i \phi} J_{-} \right) = -\alpha^*$
gilmore-lie-groups_ EQ0721 (13.38)	227	■ 1.000	$n = J + M$	$n = J + M$	$n = J + M$
gilmore-lie-groups_ EQ0722 (13.39)	227	■ 1.000	$\lim_{c \rightarrow 0} \left(M + \frac{1}{2c^2} \right) = \lim_{c \rightarrow 0} \left(n - J + \frac{1}{2c^2} \right)$	$\lim_{c \rightarrow 0} \left(M + \frac{1}{2c^2} \right) = \lim_{c \rightarrow 0} \left(n - J + \frac{1}{2c^2} \right)$	$\lim_{c \rightarrow 0} \left(M + \frac{1}{2c^2} \right) = \lim_{c \rightarrow 0} \left(n - J + \frac{1}{2c^2} \right)$
gilmore-lie-groups_ EQ0727 (13.46)	229	■ 1.000	$e^{\left(\zeta J_{+} - \zeta^* J_{-} \right)} = e^{\tau J_{+}} e^{\ln(1 + \tau^* \tau) J_3} e^{-\tau^* J_{-}}$ $\zeta \tan \zeta = \tau$	$e^{(\zeta J_{+} - \zeta^* J_{-})} = e^{\tau J_{+}} e^{\ln(1 + \tau^* \tau) J_3} e^{-\tau^* J_{-}}$ $\frac{\zeta}{ \zeta } \tan \zeta = \tau$	$e^{(\zeta J_{+} - \zeta^* J_{-})} = e^{\tau J_{+}} e^{\ln(1 + \tau^* \tau) J_3} e^{-\tau^* J_{-}}$ $\frac{\zeta}{ \zeta } \tan \zeta = \tau$
gilmore-lie-groups_ EQ0728 (13.47)	229	■ 1.000	$e^{\left(\alpha a^{\dagger} - \alpha^* a \right)} = e^{\alpha a^{\dagger}} e^{-\frac{1}{2} \alpha^* \alpha I} e^{-\alpha^* a}$ $\alpha = \lim_{c \rightarrow 0} \zeta / c$	$e^{(\alpha a^{\dagger} - \alpha^* a)} = e^{\alpha a^{\dagger}} e^{-\frac{1}{2} \alpha^* \alpha I} e^{-\alpha^* a}$ $\alpha = \lim_{c \rightarrow 0} \zeta / c$	$e^{(\alpha a^{\dagger} - \alpha^* a)} = e^{\alpha a^{\dagger}} e^{-\frac{1}{2} \alpha^* \alpha I} e^{-\alpha^* a}$ $\alpha = \lim_{c \rightarrow 0} \zeta / c$
gilmore-lie-groups_ EQ0729 (13.48)	229	■ 1.000	$Y_m^l(\theta, \phi) = \frac{e^{im\phi}}{\sqrt{2\pi}} P_m^l(\cos \theta)$ $Y_{-m}^l(\theta, \phi) = (-)^m Y_{+m}^l(\theta, \phi)^*$	$Y_m^l(\theta, \phi) = \frac{e^{im\phi}}{\sqrt{2\pi}} P_m^l(\cos \theta)$ $Y_{-m}^l(\theta, \phi) = (-)^m Y_{+m}^l(\theta, \phi)^*$	$Y_m^l(\theta, \phi) = \frac{e^{im\phi}}{\sqrt{2\pi}} P_m^l(\cos \theta)$ $Y_{-m}^l(\theta, \phi) = (-)^m Y_{+m}^l(\theta, \phi)^*$
gilmore-lie-groups_ EQ0730 (13.49)	229	■ 1.000	$P_m^l(u) = (-)^{l+m} \frac{1}{2^l l!} \sqrt{\frac{2l+1}{2}} \sqrt{\frac{(l-m)!}{(l+m)!}} (1-u^2)^{+m/2} \frac{d^{l+m}}{du^{l+m}} (1-u^2)^l$	$P_m^l(u) = (-)^{l+m} \frac{1}{2^l l!} \sqrt{\frac{2l+1}{2}} \sqrt{\frac{(l-m)!}{(l+m)!}} (1-u^2)^{+m/2} \frac{d^{l+m}}{du^{l+m}} (1-u^2)^l$	$P_m^l(u) = (-)^{l+m} \frac{1}{2^l l!} \sqrt{\frac{2l+1}{2}} \sqrt{\frac{(l-m)!}{(l+m)!}} (1-u^2)^{+m/2} \frac{d^{l+m}}{du^{l+m}} (1-u^2)^l$
gilmore-lie-groups_ EQ0732 (13.51)	230	■ 1.000	$\lim_{c \rightarrow 0} l^{-1/4} P_m^l(u = x/\sqrt{l}) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} e^{x^2/2} \left(-\frac{d}{dx} \right)^n e^{-x^2} = \psi_n(x)$	$\lim_{c \rightarrow 0} l^{-1/4} P_m^l(u = x/\sqrt{l}) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} e^{x^2/2} \left(-\frac{d}{dx} \right)^n e^{-x^2} = \psi_n(x)$	$\lim_{c \rightarrow 0} l^{-1/4} P_m^l(u = x/\sqrt{l}) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} e^{x^2/2} \left(-\frac{d}{dx} \right)^n e^{-x^2} = \psi_n(x)$
gilmore-lie-groups_ EQ0733 (13.52)	230	■ 1.000	$\psi_n(x) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} H_n(x) e^{-x^2/2}$	$\psi_n(x) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} H_n(x) e^{-x^2/2}$	$\psi_n(x) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} H_n(x) e^{-x^2/2}$

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gilmore-lie-groups_ EQ0734 (13.53)	230	■ 1.000	$\begin{aligned} \delta_{mm'} &= \int_{-1}^{+1} P_m^l(u) P_{m'}^l(u) du \\ &\rightarrow \lim_{l \rightarrow \infty} \int_{-\sqrt{l}}^{+\sqrt{l}} \left(\frac{1}{l^{1/4}} P_m^l(x/\sqrt{l}) \right) \left(\frac{1}{l^{1/4}} P_{m'}^l(x/\sqrt{l}) \right) d(u\sqrt{l}) \\ &\rightarrow \int_{-\infty}^{+\infty} \psi_n(x) \psi_{n'}(x) dx = \delta_{nn'} \end{aligned}$	$\delta_{mm'} = \int_{-1}^{+1} P_m^l(u) P_{m'}^l(u) du$ $\rightarrow \lim_{l \rightarrow \infty} \int_{-\sqrt{l}}^{+\sqrt{l}} \left(\frac{1}{l^{1/4}} P_m^l(x/\sqrt{l}) \right) \left(\frac{1}{l^{1/4}} P_{m'}^l(x/\sqrt{l}) \right) d(u\sqrt{l})$ $\rightarrow \int_{-\infty}^{+\infty} \psi_n(x) \psi_{n'}(x) dx = \delta_{nn'}$	$\delta_{mm'} = \int_{-1}^{+1} P_m^l(u) P_{m'}^l(u) du$ $\rightarrow \lim_{l \rightarrow \infty} \int_{-\sqrt{l}}^{+\sqrt{l}} \left(\frac{1}{l^{1/4}} P_m^l(x/\sqrt{l}) \right) \left(\frac{1}{l^{1/4}} P_{m'}^l(x/\sqrt{l}) \right) d(u\sqrt{l})$ $\rightarrow \int_{-\infty}^{+\infty} \psi_n(x) \psi_{n'}(x) dx = \delta_{nn'}$
gilmore-lie-groups_ EQ0737	231	■ 1.000	$\lim_{l \rightarrow \infty} P_{mn}^l(\cos(x/l)) = (-)^{m-n} J_{m-n}(x)$	$\lim_{l \rightarrow \infty} P_{mn}^l(\cos(x/l)) = (-)^{m-n} J_{m-n}(x)$	$\lim_{l \rightarrow \infty} P_{mn}^l(\cos(x/l)) = (-)^{m-n} J_{m-n}(x)$
gilmore-lie-groups_ EQ0738	231	■ 1.000	$\lim_{l \rightarrow \infty} \sqrt{\frac{2\pi}{l}} Y_m^l(\beta = x/l) \rightarrow J_m(x)$	$\lim_{l \rightarrow \infty} \sqrt{\frac{2\pi}{l}} Y_m^l(\beta = x/l) \rightarrow J_m(x)$	$\lim_{l \rightarrow \infty} \sqrt{\frac{2\pi}{l}} Y_m^l(\beta = x/l) \rightarrow J_m(x)$
gilmore-lie-groups_ EQ0739	231	■ 1.000	$\lim_{l \rightarrow \infty} P^l(\cos(\beta = x/l)) \rightarrow J_0(x)$	$\lim_{l \rightarrow \infty} P^l(\cos(\beta = x/l)) \rightarrow J_0(x)$	$\lim_{l \rightarrow \infty} P^l(\cos(\beta = x/l)) \rightarrow J_0(x)$
gilmore-lie-groups_ EQ0740	231	■ 1.000	$e^{\alpha J_+} Y_m^l(\theta, \phi) = \sum_{k \geq 0} A_k^l Y_{m+k}^l(\theta, \phi) = Y_m^l(\theta', \phi')$	$e^{\alpha J_+} Y_m^l(\theta, \phi) = \sum_{k \geq 0} A_k^l Y_{m+k}^l(\theta, \phi) = Y_m^l(\theta', \phi')$	$e^{\alpha J_+} Y_m^l(\theta, \phi) = \sum_{k \geq 0} A_k^l Y_{m+k}^l(\theta, \phi) = Y_m^l(\theta', \phi')$
gilmore-lie-groups_ EQ0744	232	■ 1.000	$\begin{aligned} \lim_{\epsilon \rightarrow 0} \epsilon (2J_3) &\rightarrow P \quad P' = \partial_x \\ \lim_{\epsilon \rightarrow 0} \epsilon (J_+) &\rightarrow T \quad T' = \partial_t \\ \lim_{\epsilon \rightarrow 0} (J_-) &\rightarrow V \quad V' = t \partial_x \end{aligned}$	$\lim_{\epsilon \rightarrow 0} \epsilon (2J_3) \rightarrow P \quad P' = \partial_x$ $\lim_{\epsilon \rightarrow 0} \epsilon (J_+) \rightarrow T \quad T' = \partial_t$ $\lim_{\epsilon \rightarrow 0} (J_-) \rightarrow V \quad V' = t \partial_x$	$\lim_{\epsilon \rightarrow 0} \epsilon (2J_3) \rightarrow P \quad P' = \partial_x$ $\lim_{\epsilon \rightarrow 0} \epsilon (J_+) \rightarrow T \quad T' = \partial_t$ $\lim_{\epsilon \rightarrow 0} (J_-) \rightarrow V \quad V' = t \partial_x$
gilmore-lie-groups_ EQ0745	233	■ 1.000	$e^{\zeta J_+ - \zeta^* J_-} = e^{\tau J_+} e^{\log(1 + \tau^* \tau) J_3} e^{-\tau^* J_-}$	$e^{\zeta J_+ - \zeta^* J_-} = e^{\tau J_+} e^{\log(1 + \tau^* \tau) J_3} e^{-\tau^* J_-}$	$e^{\zeta J_+ - \zeta^* J_-} = e^{\tau J_+} e^{\log(1 + \tau^* \tau) J_3} e^{-\tau^* J_-}$
gilmore-lie-groups_ EQ0747	233	■ 1.000	$\begin{aligned} e^{\theta \cdot J} &\rightarrow \begin{bmatrix} \cosh(\theta/2) + (\theta_z/\theta) \sinh(\theta/2) & (\theta_x - i\theta_y)/\theta \sinh(\theta/2) \\ (\theta_x + i\theta_y)/\theta \sinh(\theta/2) & \cosh(\theta/2) - (\theta_z/\theta) \sinh(\theta/2) \end{bmatrix} \\ e^{-\beta \mathcal{H}} &\rightarrow \begin{bmatrix} e^{-\beta \epsilon/2} & 0 \\ 0 & e^{+\beta \epsilon/2} \end{bmatrix} \end{aligned}$	$e^{\theta \cdot J} \rightarrow \begin{bmatrix} \cosh(\theta/2) + (\theta_z/\theta) \sinh(\theta/2) & (\theta_x - i\theta_y)/\theta \sinh(\theta/2) \\ (\theta_x + i\theta_y)/\theta \sinh(\theta/2) & \cosh(\theta/2) - (\theta_z/\theta) \sinh(\theta/2) \end{bmatrix}$ $e^{-\beta \mathcal{H}} \rightarrow \begin{bmatrix} e^{-\beta \epsilon/2} & 0 \\ 0 & e^{+\beta \epsilon/2} \end{bmatrix}$	$e^{\theta \cdot J} \rightarrow \begin{bmatrix} \cosh(\theta/2) + (\theta_z/\theta) \sinh(\theta/2) & (\theta_x - i\theta_y)/\theta \sinh(\theta/2) \\ (\theta_x + i\theta_y)/\theta \sinh(\theta/2) & \cosh(\theta/2) - (\theta_z/\theta) \sinh(\theta/2) \end{bmatrix}$ $e^{-\beta \mathcal{H}} \rightarrow \begin{bmatrix} e^{-\beta \epsilon/2} & 0 \\ 0 & e^{+\beta \epsilon/2} \end{bmatrix}$
gilmore-lie-groups_ EQ0748	233	■ 1.000	$2 \cosh(\theta/2) \cosh(\beta \epsilon/2) - 2(\theta_z/\theta) \sinh(\theta/2) \sinh(\beta \epsilon/2) \quad (= 2 \cosh(\psi/2))$	$2 \cosh(\theta/2) \cosh(\beta \epsilon/2) - 2(\theta_z/\theta) \sinh(\theta/2) \sinh(\beta \epsilon/2) \quad (= 2 \cosh(\psi/2))$	$2 \cosh(\theta/2) \cosh(\beta \epsilon/2) - 2(\theta_z/\theta) \sinh(\theta/2) \sinh(\beta \epsilon/2) \quad (= 2 \cosh(\psi/2))$

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gilmore-lie-groups_ E0749	233	■ 1.000	$\left\langle e^{\mathrm{i} \theta J} \right\rangle = \frac{\sinh \psi / \sinh (\psi / 2)}{\sinh \beta \epsilon / \sinh (\beta \epsilon / 2)}$	$\langle e^{\theta \cdot J} \rangle = (\sinh \psi / \sinh (\psi / 2)) / (\sinh \beta \epsilon / \sinh (\beta \epsilon / 2))$	$\langle e^{\theta \cdot J} \rangle = (\sinh \psi / \sinh (\psi / 2)) / (\sinh \beta \epsilon / \sinh (\beta \epsilon / 2))$
gilmore-lie-groups_ E0750	233	■ 1.000	$\left\langle e^{\mathrm{i} \theta J} \right\rangle = \frac{\sinh ((2 j+1) \psi / 2) / \sinh (\psi / 2)}{\sinh ((2 j+1) \beta \epsilon / 2) / \sinh (\beta \epsilon / 2)}$	$\langle e^{\theta \cdot J} \rangle = \frac{\sinh ((2 j+1) \psi / 2) / \sinh (\psi / 2)}{\sinh ((2 j+1) \beta \epsilon / 2) / \sinh (\beta \epsilon / 2)}$	$\langle e^{\theta \cdot J} \rangle = \frac{\sinh ((2 j+1) \psi / 2) / \sinh (\psi / 2)}{\sinh ((2 j+1) \beta \epsilon / 2) / \sinh (\beta \epsilon / 2)}$
gilmore-lie-groups_ E0751	233	■ 1.000	$\left\langle e^{\mathrm{i} \theta J} \right\rangle \xrightarrow{j \rightarrow \infty} \sinh (j \psi) / \sinh (j \beta \epsilon)$	$\langle e^{\theta \cdot J} \rangle \xrightarrow{j \rightarrow \infty} \sinh (j \psi) / \sinh (j \beta \epsilon)$	$\langle e^{\theta \cdot J} \rangle \xrightarrow{j \rightarrow \infty} \sinh (j \psi) / \sinh (j \beta \epsilon)$
gilmore-lie-groups_ E0755	234	■ 1.000	$g(\mathfrak{p}')_{\alpha, \beta} = \lim_{\epsilon \rightarrow 0} (Y_{\alpha}, Y_{\beta}) / \epsilon^2 = (X_{\alpha}, X_{\beta})$	$g(\mathfrak{p}')_{\alpha, \beta} = \lim_{\epsilon \rightarrow 0} (Y_{\alpha}, Y_{\beta}) / \epsilon^2 = (X_{\alpha}, X_{\beta})$	$g(\mathfrak{p}')_{\alpha, \beta} = \lim_{\epsilon \rightarrow 0} (Y_{\alpha}, Y_{\beta}) / \epsilon^2 = (X_{\alpha}, X_{\beta})$
gilmore-lie-groups_ E0756 (14.1)	237	■ 1.000	$\langle S \mid \psi \rangle = \langle S \mid S' \rangle \langle S' \mid \psi \rangle$	$\langle S \mid \psi \rangle = \langle S \mid S' \rangle \langle S' \mid \psi \rangle$	$\langle S \mid \psi \rangle = \langle S \mid S' \rangle \langle S' \mid \psi \rangle$
gilmore-lie-groups_ E0757	237	■ 1.000	the rest of the universe looks the same	<i>therestoftheuniverselooksthesame</i>	the rest of the universe looks the same
gilmore-lie-groups_ E0758 (14.2)	237	■ 1.000	$\langle S \mid \psi' \rangle = \langle S' \mid \psi \rangle$	$\langle S \mid \psi' \rangle = \langle S' \mid \psi \rangle$	$\langle S \mid \psi' \rangle = \langle S' \mid \psi \rangle$
gilmore-lie-groups_ E0759 (14.3)	237	■ 1.000	$E^2 - (\mathbf{p} c)^2 = (m c^2)^2$	$E^2 - (\mathbf{p} c)^2 = (m c^2)^2$	$E^2 - (\mathbf{p} c)^2 = (m c^2)^2$
gilmore-lie-groups_ E0760 (14.4)	238	■ 1.000	$\left\{ E^2 - (m c^2)^2 + 2 E \left(\frac{e^2}{r} \right) + \left(\frac{e^2}{r} \right)^2 - (-i \hbar c \nabla)^2 \right\} \psi(x) = 0$	$\left\{ E^2 - (m c^2)^2 + 2 E \left(\frac{e^2}{r} \right) + \left(\frac{e^2}{r} \right)^2 - (-i \hbar c \nabla)^2 \right\} \psi(x) = 0$	$\left\{ E^2 - (m c^2)^2 + 2 E \left(\frac{e^2}{r} \right) + \left(\frac{e^2}{r} \right)^2 - (-i \hbar c \nabla)^2 \right\} \psi(x) = 0$
gilmore-lie-groups_ E0761 (14.5)	238	■ 1.000	$\left\{ \frac{\mathbf{p} \cdot \mathbf{p}}{2 m} - \frac{e^2}{r} - W \right\} \psi(\mathbf{x}) = \left\{ -\frac{\hbar^2}{2 m} \nabla^2 - \frac{e^2}{r} - W \right\} \psi(\mathbf{x}) = 0$	$\left\{ \frac{\mathbf{p} \cdot \mathbf{p}}{2 m} - \frac{e^2}{r} - W \right\} \psi(\mathbf{x}) = \left\{ -\frac{\hbar^2}{2 m} \nabla^2 - \frac{e^2}{r} - W \right\} \psi(\mathbf{x}) = 0$	$\left\{ \frac{\mathbf{p} \cdot \mathbf{p}}{2 m} - \frac{e^2}{r} - W \right\} \psi(\mathbf{x}) = \left\{ -\frac{\hbar^2}{2 m} \nabla^2 - \frac{e^2}{r} - W \right\} \psi(\mathbf{x}) = 0$
gilmore-lie-groups_ E0762 (14.6)	239	■ 1.000	$\nabla^2 = \left(\frac{1}{r} \frac{\partial}{\partial r} r \right)^2 + \frac{\mathcal{L}^2(S^2)}{r^2}$	$\nabla^2 = \left(\frac{1}{r} \frac{\partial}{\partial r} r \right)^2 + \frac{\mathcal{L}^2(S^2)}{r^2}$	$\nabla^2 = \left(\frac{1}{r} \frac{\partial}{\partial r} r \right)^2 + \frac{\mathcal{L}^2(S^2)}{r^2}$
gilmore-lie-groups_ E0763 (14.8)	239	■ 1.000	$\psi(r, \theta, \phi) \rightarrow \frac{1}{r} R(r) Y_m^l(\theta, \phi)$	$\psi(r, \theta, \phi) \rightarrow \frac{1}{r} R(r) Y_m^l(\theta, \phi)$	$\psi(r, \theta, \phi) \rightarrow \frac{1}{r} R(r) Y_m^l(\theta, \phi)$
gilmore-lie-groups_ E0764 (14.9)	239	■ 1.000	$\left(\frac{d^2}{d r^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) = 0$	$\left(\frac{d^2}{d r^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) = 0$	$\left(\frac{d^2}{d r^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) = 0$

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gilmore-lie-groups_ EQ0767	240	■ 1.000	$\begin{aligned} & \& f \simeq e^{-2\lambda r} \text{ if series does not terminate } \\ & \& \simeq e^{+1\lambda r} \text{ if series does terminate } (\lambda < 0) \end{aligned}$	$f \simeq e^{-2\lambda r} \text{ if series does not terminate}$ $\simeq e^{+1\lambda r} \text{ if series does terminate } (\lambda < 0)$	$f \simeq e^{-2\lambda r} \text{ if series does not terminate}$ $\simeq e^{+1\lambda r} \text{ if series does terminate } (\lambda < 0)$
gilmore-lie-groups_ EQ0768	240	■ 1.000	$\begin{aligned} & 2\lambda(n+\gamma)+B=0 \text{ or } \\ & n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2\sqrt{-C}} \end{aligned}$	$2\lambda(n+\gamma)+B=0 \text{ or}$ $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2\sqrt{-C}}$	$2\lambda(n+\gamma)+B=0 \text{ or}$ $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2\sqrt{-C}}$
gilmore-lie-groups_ EQ0771 (14.13)	241	■ 1.000	$g_{\{i\}} H g_{\{i\}}^{-1} = H \quad \text{or} \quad g_{\{i\}} H = H g_{\{i\}}$	$g_i H g_i^{-1} = H \quad \text{or} \quad g_i H = H g_i$	$g_i H g_i^{-1} = H \quad \text{or} \quad g_i H = H g_i$
gilmore-lie-groups_ EQ0772 (14.14)	242	■ 1.000	$H\left(g_{\{i\}} \psi\rangle\right)=\left(H g_{\{i\}}\right) \psi\rangle$ $\psi\rangle=\left(g_{\{i\}} H\right) \psi\rangle=g_{\{i\}}\left(H \psi\rangle\right)=g_{\{i\}}\left(E \psi\rangle\right)=E\left(g_{\{i\}} \psi\rangle\right)$	$H\left(g_i \psi\right)=\left(H g_i\right) \psi\rangle=\left(g_i H\right) \psi\rangle=g_i\left(H \psi\right)=g_i\left(E \psi\right)=E\left(g_i \psi\right)$	$H\left(g_i \psi\right)=\left(H g_i\right) \psi\rangle=\left(g_i H\right) \psi\rangle=g_i\left(H \psi\right)=g_i\left(E \psi\right)=E\left(g_i \psi\right)$
gilmore-lie-groups_ EQ0773 (14.15)	242	■ 1.000	$R(\theta)=e^{\epsilon_{ijk}\theta_ix_j\partial_k}=e^{i\theta\cdot\mathbf{L}/\hbar}$	$R(\theta)=e^{\epsilon_{ijk}\theta_ix_j\partial_k}=e^{i\theta\cdot\mathbf{L}/\hbar}$	$R(\theta)=e^{\epsilon_{ijk}\theta_ix_j\partial_k}=e^{i\theta\cdot\mathbf{L}/\hbar}$
gilmore-lie-groups_ EQ0774 (14.16)	242	■ 1.000	$\left[L_{\{i\}},L_{\{j\}}\right]=i\hbar\epsilon_{ijk}L_{\{k\}}$	$[L_i,L_j]=i\hbar\epsilon_{ijk}L_k$	$[L_i,L_j]=i\hbar\epsilon_{ijk}L_k$
gilmore-lie-groups_ EQ0775 (14.18)	242	■ 1.000	$\begin{aligned} & \left\{\left[L_z,L_{\pm}\right]\right\}=\pm\hbar L_{\pm} \\ & \left\{\left[L_{\pm},L_z\right]\right\}=\mp\hbar L_{\pm} \end{aligned}$	$[L_z,L_{\pm}]=\pm\hbar L_{\pm}$ $[L_{\pm},L_z]=\mp\hbar L_{\pm}$	$[L_z,L_{\pm}]=\pm\hbar L_{\pm}$ $[L_{\pm},L_z]=\mp\hbar L_{\pm}$
gilmore-lie-groups_ EQ0779 (14.21)	244	■ 1.000	$L_{+}Y_m^l(\theta,\phi)=Y_{m+1}^l(\theta,\phi)\sqrt{(l-m)(l+m+1)}$	$L_{+}Y_m^l(\theta,\phi)=Y_{m+1}^l(\theta,\phi)\sqrt{(l-m)(l+m+1)}$	$L_{+}Y_m^l(\theta,\phi)=Y_{m+1}^l(\theta,\phi)\sqrt{(l-m)(l+m+1)}$
gilmore-lie-groups_ EQ0781 (14.23)	245	■ 1.000	$\mathbf{M}=\frac{\mathbf{p}\times\mathbf{L}}{m}-K\frac{\mathbf{r}}{r}$	$\mathbf{M}=\frac{\mathbf{p}\times\mathbf{L}}{m}-K\frac{\mathbf{r}}{r}$	$\mathbf{M}=\frac{\mathbf{p}\times\mathbf{L}}{m}-K\frac{\mathbf{r}}{r}$
gilmore-lie-groups_ EQ0782 (14.24)	245	■ 1.000	$\hat{\mathbf{M}}=\frac{\hat{\mathbf{p}}\times\hat{\mathbf{L}}-\hat{\mathbf{L}}\times\hat{\mathbf{p}}}{2m}-K\frac{\hat{\mathbf{r}}}{r}$	$\hat{\mathbf{M}}=\frac{\hat{\mathbf{p}}\times\hat{\mathbf{L}}-\hat{\mathbf{L}}\times\hat{\mathbf{p}}}{2m}-K\frac{\hat{\mathbf{r}}}{r}$	$\hat{\mathbf{M}}=\frac{\hat{\mathbf{p}}\times\hat{\mathbf{L}}-\hat{\mathbf{L}}\times\hat{\mathbf{p}}}{2m}-K\frac{\hat{\mathbf{r}}}{r}$
gilmore-lie-groups_ EQ0783 (14.25)	245	■ 1.000	$\begin{aligned} & \left\{\left[L_{\{i\}},L_{\{j\}}\right]\right\}=i\hbar\epsilon_{ijk}L_{\{k\}} \\ & \left\{\left[L_{\{i\}},M_{\{j\}}\right]\right\}=i\hbar\epsilon_{ijk}M_{\{k\}} \\ & \left\{\left[M_{\{i\}},M_{\{j\}}\right]\right\}=-\frac{2H}{m}i\hbar\epsilon_{ijk}L_{\{k\}} \end{aligned}$	$[L_i,L_j]=i\hbar\epsilon_{ijk}L_k$ $[L_i,M_j]=i\hbar\epsilon_{ijk}M_k$ $[M_i,M_j]=\left(-\frac{2H}{m}\right)i\hbar\epsilon_{ijk}L_k$	$[L_i,L_j]=i\hbar\epsilon_{ijk}L_k$ $[L_i,M_j]=i\hbar\epsilon_{ijk}M_k$ $[M_i,M_j]=\left(-\frac{2H}{m}\right)i\hbar\epsilon_{ijk}L_k$
gilmore-lie-groups_ EQ0784 (14.26)	246	■ 1.000	$\begin{aligned} & \mathbf{L}\cdot\mathbf{M}=\mathbf{M}\cdot\mathbf{L}=0 \\ & \mathbf{M}\cdot\mathbf{M}=\frac{2H}{m}\left(\mathbf{L}\cdot\mathbf{L}+\hbar^2\right)+K^2 \end{aligned}$	$\mathbf{L}\cdot\mathbf{M}=\mathbf{M}\cdot\mathbf{L}=0$ $\mathbf{M}\cdot\mathbf{M}=\frac{2H}{m}\left(\mathbf{L}\cdot\mathbf{L}+\hbar^2\right)+K^2$	$\mathbf{L}\cdot\mathbf{M}=\mathbf{M}\cdot\mathbf{L}=0$ $\mathbf{M}\cdot\mathbf{M}=\frac{2H}{m}\left(\mathbf{L}\cdot\mathbf{L}+\hbar^2\right)+K^2$

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gilmore-lie-groups_ E00785 (14.27)	246	■ 1.000	$\begin{aligned} \{\left[L_i, L_j\right] \} &= i \hbar \epsilon_{ijk} L_k \\ \{\left[\epsilonpsilon_{ij k} L_k\right] \} &= i \hbar \epsilon_{ijk} M_k^{\prime} \\ \{\left[M_i^{\prime}, M_j^{\prime}\right] \} &= i \hbar \epsilon_{ijk} L_k \end{aligned}$	$\begin{aligned} [L_i, L_j] &= i \hbar \epsilon_{ijk} L_k \\ [L_i, M_j^{\prime}] &= i \hbar \epsilon_{ijk} M_k^{\prime} \\ [M_i^{\prime}, M_j^{\prime}] &= i \hbar \epsilon_{ijk} L_k \end{aligned}$	$\begin{aligned} [L_i, L_j] &= i \hbar \epsilon_{ijk} L_k \\ [L_i, M_j^{\prime}] &= i \hbar \epsilon_{ijk} M_k^{\prime} \\ [M_i^{\prime}, M_j^{\prime}] &= i \hbar \epsilon_{ijk} L_k \end{aligned}$
gilmore-lie-groups_ E00786 (14.28)	246	■ 1.000	$\begin{aligned} &\mathbf{A}=\frac{1}{2}\left(\mathbf{L}+\mathbf{M}^{\prime}\right) \\ &\mathbf{B}=\frac{1}{2}\left(\mathbf{L}-\mathbf{M}^{\prime}\right) \end{aligned}$	$\begin{aligned} \mathbf{A} &= \frac{1}{2}(\mathbf{L} + \mathbf{M}') \\ \mathbf{B} &= \frac{1}{2}(\mathbf{L} - \mathbf{M}') \end{aligned}$	$\begin{aligned} \mathbf{A} &= \frac{1}{2}(\mathbf{L} + \mathbf{M}') \\ \mathbf{B} &= \frac{1}{2}(\mathbf{L} - \mathbf{M}') \end{aligned}$
gilmore-lie-groups_ E00791 (14.33)	248	■ 1.000	$p_x^2+\left(p_y-\frac{m}{L}M\right)^2=\left(\frac{m}{L}K\right)^2$	$p_x^2+\left(p_y-\frac{mM}{L}\right)^2=\left(\frac{mK}{L}\right)^2$	$p_x^2+\left(p_y-\frac{mM}{L}\right)^2=\left(\frac{mK}{L}\right)^2$
gilmore-lie-groups_ E00793 (14.35)	248	■ 1.000	$\hat{\mathbf{u}}\stackrel{T}{=} \frac{\mathbf{p}\cdot\mathbf{p}-p_0^2}{\mathbf{p}\cdot\mathbf{p}+p_0^2}\hat{\mathbf{w}}+\frac{2p_0}{\mathbf{p}\cdot\mathbf{p}+p_0^2}\mathbf{p}$	$\hat{\mathbf{u}}\stackrel{T}{=} \frac{\mathbf{p}\cdot\mathbf{p}-p_0^2}{\mathbf{p}\cdot\mathbf{p}+p_0^2}\hat{\mathbf{w}}+\frac{2p_0}{\mathbf{p}\cdot\mathbf{p}+p_0^2}\mathbf{p}$	$\hat{\mathbf{u}}\stackrel{T}{=} \frac{\mathbf{p}\cdot\mathbf{p}-p_0^2}{\mathbf{p}\cdot\mathbf{p}+p_0^2}\hat{\mathbf{w}}+\frac{2p_0}{\mathbf{p}\cdot\mathbf{p}+p_0^2}\mathbf{p}$
gilmore-lie-groups_ E00794 (14.36)	248	■ 1.000	$\text{circle in } R^3 \xrightarrow{T} \text{circle in } S^3 \xrightarrow{SO(4)} \text{circle in } S^3 \xrightarrow{T^{-1}} \text{circle in } R^3$	$\text{circle in } R^3 \xrightarrow{T} \text{circle in } S^3 \xrightarrow{SO(4)} \text{circle in } S^3 \xrightarrow{T^{-1}} \text{circle in } R^3$	$\text{circle in } R^3 \xrightarrow{T} \text{circle in } S^3 \xrightarrow{SO(4)} \text{circle in } S^3 \xrightarrow{T^{-1}} \text{circle in } R^3$
gilmore-lie-groups_ E00795 (14.37)	249	■ 1.000	$(\mathbf{y},\mathbf{y})_N-z_1z_2=(\lambda\mathbf{x},\lambda\mathbf{x})_N-\lambda\left[\lambda(\mathbf{x},\mathbf{x})_N\right]=0$	$(\mathbf{y},\mathbf{y})_N-z_1z_2=(\lambda\mathbf{x},\lambda\mathbf{x})_N-\lambda\left[\lambda(\mathbf{x},\mathbf{x})_N\right]=0$	$(\mathbf{y},\mathbf{y})_N-z_1z_2=(\lambda\mathbf{x},\lambda\mathbf{x})_N-\lambda\left[\lambda(\mathbf{x},\mathbf{x})_N\right]=0$
gilmore-lie-groups_ E00796 (14.39)	250	■ 1.000	$\begin{aligned} &\hat{\mathbf{u}}=\frac{\frac{1}{2}(p_0^2-\mathbf{p}\cdot\mathbf{p})}{\frac{1}{2}(p_0^2+\mathbf{p}\cdot\mathbf{p})}\mathbf{w}+\frac{p_0\mathbf{p}}{\frac{1}{2}(p_0^2+\mathbf{p}\cdot\mathbf{p})} \quad E<0 \\ &\hat{\mathbf{u}}=\frac{\frac{1}{2}(p_0^2+\mathbf{p}\cdot\mathbf{p})}{\frac{1}{2}(p_0^2-\mathbf{p}\cdot\mathbf{p})}\mathbf{w}+\frac{p_0\mathbf{p}}{\frac{1}{2}(p_0^2-\mathbf{p}\cdot\mathbf{p})} \quad E>0 \end{aligned}$	$\begin{aligned} \hat{\mathbf{u}} &= \frac{\frac{1}{2}(p_0^2 - \mathbf{p} \cdot \mathbf{p})}{\frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p})} \mathbf{w} + \frac{p_0 \mathbf{p}}{\frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p})} E < 0 \\ \hat{\mathbf{u}} &= \frac{\frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p})}{\frac{1}{2}(p_0^2 - \mathbf{p} \cdot \mathbf{p})} \mathbf{w} + \frac{p_0 \mathbf{p}}{\frac{1}{2}(p_0^2 - \mathbf{p} \cdot \mathbf{p})} E > 0 \end{aligned}$	$\begin{aligned} \hat{\mathbf{u}} &= \frac{\frac{1}{2}(p_0^2 - \mathbf{p} \cdot \mathbf{p})}{\frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p})} \mathbf{w} + \frac{p_0 \mathbf{p}}{\frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p})} \quad E < 0 \\ \hat{\mathbf{u}} &= \frac{\frac{1}{2}(p_0^2 + \mathbf{p} \cdot \mathbf{p})}{\frac{1}{2}(p_0^2 - \mathbf{p} \cdot \mathbf{p})} \mathbf{w} + \frac{p_0 \mathbf{p}}{\frac{1}{2}(p_0^2 - \mathbf{p} \cdot \mathbf{p})} \quad E > 0 \end{aligned}$
gilmore-lie-groups_ E00797 (14.40)	250	■ 1.000	$\mathbf{u}^t G \mathbf{u} = u_0^2 \pm \sum_{i=1}^3 u_i^2 \quad \begin{aligned} &+ \text{ for } E < 0 \\ &- \text{ for } E > 0 \end{aligned}$	$\mathbf{u}^t G \mathbf{u} = u_0^2 \pm \sum_{i=1}^3 u_i^2 \quad \begin{aligned} &+ \text{ for } E < 0 \\ &- \text{ for } E > 0 \end{aligned}$	$\mathbf{u}^t G \mathbf{u} = u_0^2 \pm \sum_{i=1}^3 u_i^2 \quad \begin{aligned} &+ \text{ for } E < 0 \\ &- \text{ for } E > 0 \end{aligned}$
gilmore-lie-groups_ E00799 (14.42)	251	■ 1.000	$\mathbf{u}^{\prime}=\frac{A(\lambda \mathbf{u})+B \lambda}{C(\lambda \mathbf{u})+D \lambda}$	$\mathbf{u}^{\prime}=\frac{A(\lambda \mathbf{u})+B \lambda}{C(\lambda \mathbf{u})+D \lambda}$	$\mathbf{u}^{\prime}=\frac{A(\lambda \mathbf{u})+B \lambda}{C(\lambda \mathbf{u})+D \lambda}$

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gilmore-lie-groups_ EQ0800 (14.43)	251	■ 1.000	$\left(\mathbf{u}^{\prime}\right)^t G \mathbf{u}^{\prime}-1=\frac{(A \lambda \mathbf{u}+B \lambda)^t G(A \lambda \mathbf{u}+B \lambda)-(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}{(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}$	$\left(\mathbf{u}^{\prime}\right)^t G \mathbf{u}^{\prime}-1=\frac{(A \lambda \mathbf{u}+B \lambda)^t G(A \lambda \mathbf{u}+B \lambda)-(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}{(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}$	$\left(\mathbf{u}^{\prime}\right)^t G \mathbf{u}^{\prime}-1=\frac{(A \lambda \mathbf{u}+B \lambda)^t G(A \lambda \mathbf{u}+B \lambda)-(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}{(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}$
gilmore-lie-groups_ EQ0801 (14.44)	251	■ 1.000	$\left(\mathbf{u}^{\prime}\right)^t G \mathbf{u}^{\prime}-1=\frac{(A \lambda \mathbf{u}+B \lambda)^t G(A \lambda \mathbf{u}+B \lambda)-(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}{(C \lambda \mathbf{u}+D \lambda)^t(C \lambda \mathbf{u}+D \lambda)}$	$\left(\mathbf{u}^{\prime}, \mathbf{u}^{\prime}\right)_N-1=\frac{(\mathbf{u}, \mathbf{u})_N-1}{(C \mathbf{u}+D)^t(C \mathbf{u}+D)}$	$\left(\mathbf{u}^{\prime}, \mathbf{u}^{\prime}\right)_N-1=\frac{(\mathbf{u}, \mathbf{u})_N-1}{(C \mathbf{u}+D)^t(C \mathbf{u}+D)}$
gilmore-lie-groups_ EQ0805	252	■ 1.000	$\begin{aligned} & a_{1}^{\dagger} a_{2} \left m_{1}, m_{2} ; n_{1}, n_{2} \right\rangle = \left m_{1}+1, m_{2}-1 ; n_{1} n_{2} \right\rangle \sqrt{m_{1}+1} \sqrt{m_{2}} \\ & b_{1}^{\dagger} b_{1} \left m_{1}, m_{2} ; n_{1}, n_{2} \right\rangle = \left m_{1}, m_{2} ; n_{1} n_{2} \right\rangle (\sqrt{n_{1}})^2 \end{aligned}$	$\begin{aligned} & a_{1}^{\dagger} a_{2} \left m_{1}, m_{2} ; n_{1}, n_{2} \right\rangle = \left m_{1}+1, m_{2}-1 ; n_{1} n_{2} \right\rangle \sqrt{m_{1}+1} \sqrt{m_{2}} \\ & b_{1}^{\dagger} b_{1} \left m_{1}, m_{2} ; n_{1}, n_{2} \right\rangle = \left m_{1}, m_{2} ; n_{1} n_{2} \right\rangle (\sqrt{n_{1}})^2 \end{aligned}$	$\begin{aligned} & a_{1}^{\dagger} a_{2} \left m_{1}, m_{2} ; n_{1}, n_{2} \right\rangle = \left m_{1}+1, m_{2}-1 ; n_{1} n_{2} \right\rangle \sqrt{m_{1}+1} \sqrt{m_{2}} \\ & b_{1}^{\dagger} b_{1} \left m_{1}, m_{2} ; n_{1}, n_{2} \right\rangle = \left m_{1}, m_{2} ; n_{1} n_{2} \right\rangle (\sqrt{n_{1}})^2 \end{aligned}$
gilmore-lie-groups_ EQ0806 (14.46)	253	■ 1.000	$\begin{array}{l} \text{operators} \\ \text{subalgebra} \\ \text{number} \end{array} \begin{array}{cccc} a_i^{\dagger} a_j & b_i^{\dagger} b_j & a_i^{\dagger} b_j^{\dagger} & a_i b_j \\ u(2) & u(2) & & \\ 4 & 4 & 4 & 4 \end{array}$	$\begin{array}{l} \text{operators} \\ \text{subalgebra} \\ \text{number} \end{array} \begin{array}{cccc} a_i^{\dagger} a_j & b_i^{\dagger} b_j & a_i^{\dagger} b_j^{\dagger} & a_i b_j \\ u(2) & u(2) & & \\ 4 & 4 & 4 & 4 \end{array}$	$\begin{array}{l} \text{operators} \\ \text{subalgebra} \\ \text{number} \end{array} \begin{array}{cccc} a_i^{\dagger} a_j & b_i^{\dagger} b_j & a_i^{\dagger} b_j^{\dagger} & a_i b_j \\ u(2) & u(2) & & \\ 4 & 4 & 4 & 4 \end{array}$
gilmore-lie-groups_ EQ0808	253	■ 1.000	$\left(a_{1}^{\dagger} a_{1}+a_{2}^{\dagger} a_{2}\right)-\left(b_{1}^{\dagger} b_{1}+b_{2}^{\dagger} b_{2}\right)$	$\left(a_{1}^{\dagger} a_{1}+a_{2}^{\dagger} a_{2}\right)-\left(b_{1}^{\dagger} b_{1}+b_{2}^{\dagger} b_{2}\right)$	$\left(a_{1}^{\dagger} a_{1}+a_{2}^{\dagger} a_{2}\right)-\left(b_{1}^{\dagger} b_{1}+b_{2}^{\dagger} b_{2}\right)$
gilmore-lie-groups_ EQ0811	254	■ 1.000	$\left(P_{1}, P_{2}, P_{3}, P_{4}\right)^t=\frac{1}{2 R} M_{K S}\left(p_{1}, p_{2}, p_{3}, p_{4}\right)^t$	$\left(P_{1}, P_{2}, P_{3}, P_{4}\right)^t=\frac{1}{2 R} M_{K S}\left(p_{1}, p_{2}, p_{3}, p_{4}\right)^t$	$\left(P_{1}, P_{2}, P_{3}, P_{4}\right)^t=\frac{1}{2 R} M_{K S}\left(p_{1}, p_{2}, p_{3}, p_{4}\right)^t$
gilmore-lie-groups_ EQ0812 (14.49)	254	■ 1.000	$\zeta=-2 R P_4=\left(q_1 p_4-q_4 p_1\right)+\left(q_3 p_2-q_2 p_3\right)=0$	$\zeta=-2 R P_4=\left(q_1 p_4-q_4 p_1\right)+\left(q_3 p_2-q_2 p_3\right)=0$	$\zeta=-2 R P_4=\left(q_1 p_4-q_4 p_1\right)+\left(q_3 p_2-q_2 p_3\right)=0$
gilmore-lie-groups_ EQ0813 (14.50)	255	■ 1.000	$\frac{P^2}{2 m}-\frac{e^2}{R}=E \xrightarrow{\times R} \frac{R P^2}{2 m}-e^2=E R \xrightarrow{K S} \frac{p^2}{8 m}-e^2=E q^2$	$\frac{P^2}{2 m}-\frac{e^2}{R}=E \xrightarrow{\times R} \frac{R P^2}{2 m}-e^2=E R \xrightarrow{K S} \frac{p^2}{8 m}-e^2=E q^2$	$\frac{P^2}{2 m}-\frac{e^2}{R}=E \xrightarrow{\times R} \frac{R P^2}{2 m}-e^2=E R \xrightarrow{K S} \frac{p^2}{8 m}-e^2=E q^2$
gilmore-lie-groups_ EQ0814 (14.51)	255	■ 1.000	$\frac{p^2}{2 m}-4 E q^2=4 e^2$	$\frac{p^2}{2 m}-4 E q^2=4 e^2$	$\frac{p^2}{2 m}-4 E q^2=4 e^2$

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gilmore-lie-groups_ E08815 (14.52)	255	■1.000	$\begin{aligned} & \left(Q_1, Q_2, Q_3, Q_4 \right) \left[\begin{array}{cccc} 0 & \theta_3 & -\theta_2 & * \\ -\theta_3 & 0 & \theta_1 & * \\ \theta_2 & -\theta_1 & 0 & * \\ -* & -* & -* & 0 \end{array} \right] \left[\begin{array}{c} P_1 \\ P_2 \\ P_3 \\ P_4 \end{array} \right] \\ & \frac{1}{2} (q_1, q_2, q_3, q_4) \left[\begin{array}{cccc} 0 & \theta_3 & -\theta_2 & \theta_1 \\ -\theta_3 & 0 & \theta_1 & \theta_2 \\ \theta_2 & -\theta_1 & 0 & \theta_3 \\ -\theta_1 & -\theta_2 & -\theta_3 & 0 \end{array} \right] \left[\begin{array}{c} p_1 \\ p_2 \\ p_3 \\ p_4 \end{array} \right] \end{aligned}$	$(Q_1, Q_2, Q_3, Q_4) \left[\begin{array}{cccc} 0 & \theta_3 & -\theta_2 & * \\ -\theta_3 & 0 & \theta_1 & * \\ \theta_2 & -\theta_1 & 0 & * \\ -* & -* & -* & 0 \end{array} \right] \left[\begin{array}{c} P_1 \\ P_2 \\ P_3 \\ P_4 \end{array} \right]$ $\frac{1}{2} (q_1, q_2, q_3, q_4) \left[\begin{array}{cccc} 0 & \theta_3 & -\theta_2 & \theta_1 \\ -\theta_3 & 0 & \theta_1 & \theta_2 \\ \theta_2 & -\theta_1 & 0 & \theta_3 \\ -\theta_1 & -\theta_2 & -\theta_3 & 0 \end{array} \right] \left[\begin{array}{c} p_1 \\ p_2 \\ p_3 \\ p_4 \end{array} \right]$	$(Q_1, Q_2, Q_3, Q_4) \left[\begin{array}{cccc} 0 & \theta_3 & -\theta_2 & * \\ -\theta_3 & 0 & \theta_1 & * \\ \theta_2 & -\theta_1 & 0 & * \\ -* & -* & -* & 0 \end{array} \right] \left[\begin{array}{c} P_1 \\ P_2 \\ P_3 \\ P_4 \end{array} \right]$ $\frac{1}{2} (q_1, q_2, q_3, q_4) \left[\begin{array}{cccc} 0 & \theta_3 & -\theta_2 & \theta_1 \\ -\theta_3 & 0 & \theta_1 & \theta_2 \\ \theta_2 & -\theta_1 & 0 & \theta_3 \\ -\theta_1 & -\theta_2 & -\theta_3 & 0 \end{array} \right] \left[\begin{array}{c} p_1 \\ p_2 \\ p_3 \\ p_4 \end{array} \right]$
gilmore-lie-groups_ E08816	255	■1.000	$G_i=\left[\begin{array}{cc} 0 & M_i \\ -M_i & 0 \end{array}\right]$	$G_i = \left[\begin{array}{cc} 0 & M_i \\ -M_i & 0 \end{array} \right]$	$G_i = \left[\begin{array}{cc} 0 & M_i \\ -M_i & 0 \end{array} \right]$
gilmore-lie-groups_ E08820 (14.56)	256	■1.000	$\begin{aligned} & z_1=\frac{1}{\sqrt{2}}(s_1+i s_2) \quad z_2=\frac{1}{\sqrt{2}}(r_1+i r_2) \\ & z_3=\frac{1}{\sqrt{2}}(s_3+i s_4) \quad z_4=\frac{1}{\sqrt{2}}(r_3+i r_4) \end{aligned}$	$z_1=\frac{1}{\sqrt{2}}(s_1+i s_2) \quad z_2=\frac{1}{\sqrt{2}}(r_1+i r_2)$ $z_3=\frac{1}{\sqrt{2}}(s_3+i s_4) \quad z_4=\frac{1}{\sqrt{2}}(r_3+i r_4)$ $z_1^* z_1-z_2^* z_2+z_3^* z_3-z_4^* z_4=\zeta$	$z_1=\frac{1}{\sqrt{2}}(s_1+i s_2) \quad z_2=\frac{1}{\sqrt{2}}(r_1+i r_2)$ $z_3=\frac{1}{\sqrt{2}}(s_3+i s_4) \quad z_4=\frac{1}{\sqrt{2}}(r_3+i r_4)$ $z_1^* z_1-z_2^* z_2+z_3^* z_3-z_4^* z_4=\zeta$
gilmore-lie-groups_ E08821 (14.57)	257	■1.000	$\begin{aligned} & \mathbf{J}=\mathbf{r} \times \mathbf{p} \quad \text { angular momentum } \\ & \mathbf{M}=\frac{1}{2 m}(\mathbf{p} \times \mathbf{L}-\mathbf{L} \times \mathbf{p})-K \frac{\mathbf{r}}{r} \quad \text { Laplace-Runge-Lenz vector } \\ & \mathbf{A}=\frac{1}{2 m}(\mathbf{p} \times \mathbf{L}-\mathbf{L} \times \mathbf{p})+K \frac{\mathbf{r}}{r} \quad \text { dual vector } \\ & A_4=\mathbf{r} \cdot \mathbf{p}+\frac{3}{2} \frac{\hbar}{i} \quad \text { dual scalar } \end{aligned}$	$\mathbf{J}=\mathbf{r} \times \mathbf{p} \quad \text { angular momentum }$ $\mathbf{M}=\frac{1}{2 m}(\mathbf{p} \times \mathbf{L}-\mathbf{L} \times \mathbf{p})-K \frac{\mathbf{r}}{r} \quad \text { Laplace-Runge-Lenz vector }$ $\mathbf{A}=\frac{1}{2 m}(\mathbf{p} \times \mathbf{L}-\mathbf{L} \times \mathbf{p})+K \frac{\mathbf{r}}{r} \quad \text { dual vector }$ $A_4=\mathbf{r} \cdot \mathbf{p}+\frac{3}{2} \frac{\hbar}{i} \quad \text { dual scalar }$	$\mathbf{J}=\mathbf{r} \times \mathbf{p} \quad \text { angular momentum }$ $\mathbf{M}=\frac{1}{2 m}(\mathbf{p} \times \mathbf{L}-\mathbf{L} \times \mathbf{p})-K \frac{\mathbf{r}}{r} \quad \text { Laplace-Runge-Lenz vector }$ $\mathbf{A}=\frac{1}{2 m}(\mathbf{p} \times \mathbf{L}-\mathbf{L} \times \mathbf{p})+K \frac{\mathbf{r}}{r} \quad \text { dual vector }$ $A_4=\mathbf{r} \cdot \mathbf{p}+\frac{3}{2} \frac{\hbar}{i} \quad \text { dual scalar }$
gilmore-lie-groups_ E08824 (14.59)	258	■1.000	$H_D=-e \phi(r)+\beta m c^2+\gamma \cdot(c \mathbf{p}+e \mathbf{A})$	$H_D=-e \phi(r)+\beta m c^2+\gamma \cdot(c \mathbf{p}+e \mathbf{A})$	$H_D=-e \phi(r)+\beta m c^2+\gamma \cdot(c \mathbf{p}+e \mathbf{A})$

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gilmore-lie-groups_ EQ0825 (14.60)	258	■ 1.000	$\backslash beta=\backslash left[\backslash begin{array}{cc} I_{\backslash 2} \& \theta \backslash \backslash \theta \& \backslash -I_{\backslash 2} \backslash end{array}\backslash right] \quad \backslash gamma_{\backslash i}=\backslash left[\backslash begin{array}{cc} \theta \& \backslash sigma_{\backslash i} \backslash \backslash \backslash sigma_{\backslash i} \& \theta \backslash end{array}\backslash right]$	$\beta = \left[\begin{array}{cc} I_2 & 0 \\ 0 & -I_2 \end{array} \right] \quad \gamma_i = \left[\begin{array}{cc} 0 & \sigma_i \\ \sigma_i & 0 \end{array} \right]$	$\beta = \left[\begin{array}{cc} I_2 & 0 \\ 0 & -I_2 \end{array} \right] \quad \gamma_i = \left[\begin{array}{cc} 0 & \sigma_i \\ \sigma_i & 0 \end{array} \right]$
gilmore-lie-groups_ EQ0827 (14.62)	259	■ 1.000	$\backslash begin{aligned} \backslash Theta \& =r\backslash left\backslash \backslash left(H_{\backslash D}+\backslash frac{\backslash alpha}{r}\backslash right)^{\backslash 2}-\backslash left(E+\backslash frac{\backslash alpha}{r}\backslash right)^{\backslash 2} \backslash right\backslash \backslash \& =A\backslash left(\backslash Gamma_{\backslash 5}+\backslash Gamma_{\backslash 4}\backslash right)+B \backslash left(\backslash Gamma_{\backslash 5}-\backslash Gamma_{\backslash 4}\backslash right)+C \backslash end{aligned}$	$\Theta = r \left\{ \left(H_D + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\} \\ = A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$	$\Theta = r \left\{ \left(H_D + \frac{\alpha}{r} \right)^2 - \left(E + \frac{\alpha}{r} \right)^2 \right\} \\ = A (\Gamma_5 + \Gamma_4) + B (\Gamma_5 - \Gamma_4) + C$
gilmore-lie-groups_ EQ0828 (14.63)	259	■ 1.000	$\backslash left(r \backslash D^{\backslash 2}+\backslash frac{A}{r}\backslash right) \backslash R(r)=\theta$	$\left(r D^2 + \frac{A}{r} + B + C r \right) R(r) = 0$	$\left(r D^2 + \frac{A}{r} + B + C r \right) R(r) = 0$
gilmore-lie-groups_ EQ0830	260	■ 1.000	$\backslash begin{aligned} \& \backslash \backslash left[r \backslash D, \backslash frac{1}{r}\backslash right]=-\backslash frac{1}{r} \backslash \backslash \& \backslash \backslash left[r \backslash D^{\backslash 2}, \backslash frac{1}{r}\backslash right]=\backslash frac{2}{r^2}-\backslash frac{1}{r} \backslash D \backslash end{aligned}$	$\left[r D, \frac{1}{r} \right] = -\frac{1}{r} \\ \left[r D^2, \frac{1}{r} \right] = \frac{2}{r^2} - \frac{1}{r} D$	$\left[r D, \frac{1}{r} \right] = -\frac{1}{r} \\ \left[r D^2, \frac{1}{r} \right] = \frac{2}{r^2} - \frac{1}{r} D$
gilmore-lie-groups_ EQ0831 (14.65)	260	■ 1.000	$\backslash begin{aligned} \backslash Gamma_{\backslash 5} \& =\backslash frac{1}{2}\backslash left(r \backslash D^{\backslash 2}+\backslash frac{a}{r}-r\backslash right) \backslash \backslash \backslash Gamma_{\backslash 4} \& =\backslash frac{1}{2}\backslash left(r \backslash D^{\backslash 2}+\backslash frac{a}{r}+r\backslash right) \backslash \backslash M_{\backslash 4} \& =r \backslash D \backslash end{aligned}$	$\Gamma_5 = \frac{1}{2} \left(r D^2 + \frac{a}{r} - r \right) \\ \Gamma_4 = \frac{1}{2} \left(r D^2 + \frac{a}{r} + r \right) \\ M_4 = r D$	$\Gamma_5 = \frac{1}{2} \left(r D^2 + \frac{a}{r} - r \right) \\ \Gamma_4 = \frac{1}{2} \left(r D^2 + \frac{a}{r} + r \right) \\ M_4 = r D$
gilmore-lie-groups_ EQ0832 (14.66)	260	■ 1.000	$\backslash left(\backslash left(\backslash Gamma_{\backslash 5}+\backslash Gamma_{\backslash 4}\backslash right)+B+C\backslash left(\backslash Gamma_{\backslash 4}-\backslash Gamma_{\backslash 5}\backslash right)\backslash right) \backslash R(r)=\theta$	$((\Gamma_5 + \Gamma_4) + B + C (\Gamma_4 - \Gamma_5)) R(r) = 0$	$((\Gamma_5 + \Gamma_4) + B + C (\Gamma_4 - \Gamma_5)) R(r) = 0$
gilmore-lie-groups_ EQ0834 (14.68)	260	■ 1.000	$\backslash left[\backslash left(e^{\backslash -\backslash theta}-C \backslash e^{\backslash \backslash theta}\backslash right) \backslash Gamma_{\backslash 5}+\backslash left(e^{\backslash -\backslash theta}+C \backslash e^{\backslash \backslash theta}\backslash right) \backslash Gamma_{\backslash 4}+B\backslash right] \backslash e^{\backslash \backslash theta} \backslash M_{\backslash 4} \backslash R(r)=\theta$	$[(e^{-\theta} - C e^{\theta}) \Gamma_5 + (e^{-\theta} + C e^{\theta}) \Gamma_4 + B] e^{\theta M_4} R(r) = 0$	$[(e^{-\theta} - C e^{\theta}) \Gamma_5 + (e^{-\theta} + C e^{\theta}) \Gamma_4 + B] e^{\theta M_4} R(r) = 0$
gilmore-lie-groups_ EQ0835 (14.69)	260	■ 1.000	$\backslash left(2 \backslash sqrt{-C} \backslash Gamma_{\backslash 5}+B\backslash right) \backslash u(r)=\theta$	$\left(2\sqrt{-C} \Gamma_5 + B \right) u(r) = 0$	$\left(2\sqrt{-C} \Gamma_5 + B \right) u(r) = 0$
gilmore-lie-groups_ EQ0836 (14.70)	261	■ 1.000	$\backslash left(2 \backslash sqrt{C} \backslash Gamma_{\backslash 4}+B\backslash right) \backslash u(r)=\theta$	$\left(2\sqrt{C} \Gamma_4 + B \right) u(r) = 0$	$\left(2\sqrt{C} \Gamma_4 + B \right) u(r) = 0$
gilmore-lie-groups_ EQ0837 (14.71)	261	■ 1.000	$R_{\backslash k,\backslash l}(r) \backslash \sim \backslash sqrt{\backslash frac{2}{\backslash pi}} \backslash sin \backslash left(k \backslash r-\backslash frac{\backslash pi}{2} \backslash j+\backslash frac{\backslash alpha}{k}(\log(2 \backslash k \backslash r)+\backslash \Delta(j)) \backslash right)$	$R_{k,l}(r) \sim \sqrt{\frac{2}{\pi}} \sin \left(k r - \frac{\pi}{2} j + \frac{\alpha}{k} (\log(2 k r) + \delta(j)) \right)$	$R_{k,l}(r) \sim \sqrt{\frac{2}{\pi}} \sin \left(k r - \frac{\pi}{2} j + \frac{\alpha}{k} (\log(2 k r) + \delta(j)) \right)$
gilmore-lie-groups_ EQ0838 (14.72)	262	■ 1.000	$A \backslash \rightarrow A^{\backslash \backslash prime}=A+\backslash mu_{\backslash l}$	$A \rightarrow A' = A + \mu_l$	$A \rightarrow A' = A + \mu_l$
gilmore-lie-groups_ EQ0839 (14.73)	262	■ 1.000	$E_{\backslash N=\backslash n+\backslash l+1}=-\backslash frac{m \backslash c^{\backslash 2} \backslash alpha^{\backslash 2}}{2 \backslash N^{\backslash 2}} \backslash \rightarrow -\backslash frac{m \backslash c^{\backslash 2} \backslash alpha^{\backslash 2}}{2(\backslash N+\backslash \Delta(j))^{\backslash 2}}$	$E_{N=n+l+1} = -\frac{m c^2 \alpha^2}{2 N^2} \rightarrow -\frac{m c^2 \alpha^2}{2 (N + \Delta j)^2}$	$E_{N=n+l+1} = -\frac{m c^2 \alpha^2}{2 N^2} \rightarrow -\frac{m c^2 \alpha^2}{2 (N + \Delta j)^2}$

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gilmore-lie-groups_ E08840 (14.74)	262	■ 1.000	$\Delta \phi = -\frac{\pi}{2} \Delta j + \frac{\alpha}{\pi} \arg(\Gamma[j+1+\Delta j-i(\alpha/k)] - \Gamma[j+1-i(\alpha/k)])$	$\Delta \phi = -\frac{\pi}{2} \Delta j + \frac{\alpha}{\pi} \arg(\Gamma[j+1+\Delta j-i(\alpha/k)] - \Gamma[j+1-i(\alpha/k)])$	$\Delta \phi = -\frac{\pi}{2} \Delta j + \frac{\alpha}{\pi} \arg(\Gamma[j+1+\Delta j-i(\alpha/k)] - \Gamma[j+1-i(\alpha/k)])$
gilmore-lie-groups_ E08841	264	■ 1.000	$\begin{aligned} z &= x_3 = \theta_3 \cos \theta_2 \\ y &= x_2 = \theta_3 \sin \theta_2 \cos \theta_1 \\ x &= x_1 = \theta_3 \sin \theta_2 \sin \theta_1 \end{aligned}$	$\begin{aligned} z &= x_3 = \theta_3 \cos \theta_2 \\ y &= x_2 = \theta_3 \sin \theta_2 \cos \theta_1 \\ x &= x_1 = \theta_3 \sin \theta_2 \sin \theta_1 \end{aligned}$	$\begin{aligned} z &= x_3 = \theta_3 \cos \theta_2 \\ y &= x_2 = \theta_3 \sin \theta_2 \cos \theta_1 \\ x &= x_1 = \theta_3 \sin \theta_2 \sin \theta_1 \end{aligned}$
gilmore-lie-groups_ E08842	264	■ 1.000	$\sin^2 \theta_2 \mathcal{L}^2(S^2) = \left(\sin \theta_2 \frac{\partial}{\partial \theta_2} \right)^2 + \mathcal{L}^2(S^1)$	$\sin^2 \theta_2 \mathcal{L}^2(S^2) = \left(\sin \theta_2 \frac{\partial}{\partial \theta_2} \right)^2 + \mathcal{L}^2(S^1)$	$\sin^2 \theta_2 \mathcal{L}^2(S^2) = \left(\sin \theta_2 \frac{\partial}{\partial \theta_2} \right)^2 + \mathcal{L}^2(S^1)$
gilmore-lie-groups_ E08843	265	■ 1.000	$\begin{aligned} r &\rightarrow 0 \quad \left(\frac{d^2}{dr^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) \rightarrow \left(\frac{d^2}{dr^2} + \frac{A}{r^2} \right) R(r) = 0 \\ R(r) &\simeq r^\gamma \quad \gamma(\gamma-1) + A = 0 \\ r &\rightarrow \infty \quad \left(\frac{d^2}{dr^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) \rightarrow \left(\frac{d^2}{dr^2} + C \right) R(r) = 0 \\ R(r) &\simeq e^{\lambda r} \quad \lambda^2 + C = 0 \end{aligned}$	$\begin{aligned} r &\rightarrow 0 \quad \left(\frac{d^2}{dr^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) \rightarrow \left(\frac{d^2}{dr^2} + \frac{A}{r^2} \right) R(r) = 0 \\ R(r) &\simeq r^\gamma \quad \gamma(\gamma-1) + A = 0 \\ r &\rightarrow \infty \quad \left(\frac{d^2}{dr^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) \rightarrow \left(\frac{d^2}{dr^2} + C \right) R(r) = 0 \\ R(r) &\simeq e^{\lambda r} \quad \lambda^2 + C = 0 \end{aligned}$	$\begin{aligned} r &\rightarrow 0 \quad \left(\frac{d^2}{dr^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) \rightarrow \left(\frac{d^2}{dr^2} + \frac{A}{r^2} \right) R(r) = 0 \\ R(r) &\simeq r^\gamma \quad \gamma(\gamma-1) + A = 0 \\ r &\rightarrow \infty \quad \left(\frac{d^2}{dr^2} + \frac{A}{r^2} + \frac{B}{r} + C \right) R(r) \rightarrow \left(\frac{d^2}{dr^2} + C \right) R(r) = 0 \\ R(r) &\simeq e^{\lambda r} \quad \lambda^2 + C = 0 \end{aligned}$
gilmore-lie-groups_ E08844	265	■ 1.000	$[(j+1)j+2\gamma(j+1)]f_{j+1} + (2\lambda\gamma + 2\lambda j + B)f_j = 0$	$[(j+1)j+2\gamma(j+1)]f_{j+1} + (2\lambda\gamma + 2\lambda j + B)f_j = 0$	$[(j+1)j+2\gamma(j+1)]f_{j+1} + (2\lambda\gamma + 2\lambda j + B)f_j = 0$
gilmore-lie-groups_ E08845	265	■ 1.000	$f(r) = \sum_{j=0}^{\infty} \frac{\Gamma(j+\gamma+(B/2\lambda))}{\Gamma(\gamma+(B/2\lambda))} \frac{\Gamma(2\gamma)}{\Gamma(j+2\gamma)} \frac{(-2\lambda r)^j}{j!}$	$f(r) = \sum_{j=0}^{\infty} \frac{\Gamma(j+\gamma+(B/2\lambda))}{\Gamma(\gamma+(B/2\lambda))} \frac{\Gamma(2\gamma)}{\Gamma(j+2\gamma)} \frac{(-2\lambda r)^j}{j!}$	$f(r) = \sum_{j=0}^{\infty} \frac{\Gamma(j+\gamma+(B/2\lambda))}{\Gamma(\gamma+(B/2\lambda))} \frac{\Gamma(2\gamma)}{\Gamma(j+2\gamma)} \frac{(-2\lambda r)^j}{j!}$
gilmore-lie-groups_ E08846	265	■ 1.000	$n + \frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A} = \frac{B}{2\sqrt{-C}}$	$n + \frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A} = \frac{B}{2\sqrt{-C}}$	$n + \frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A} = \frac{B}{2\sqrt{-C}}$
gilmore-lie-groups_ E08847	266	■ 1.000	$\begin{aligned} E(n, l) &= \frac{mc^2}{\sqrt{1 + \frac{\alpha^2}{\left(n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}\right)^2}}} \\ W(n, l) &= -\frac{1}{2} mc^2 \alpha^2 \frac{1}{(n+l+1)^2} \end{aligned}$	$\begin{aligned} E(n, l) &= \frac{mc^2}{\sqrt{1 + \frac{\alpha^2}{\left(n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}\right)^2}}} \\ W(n, l) &= -\frac{1}{2} mc^2 \alpha^2 \frac{1}{(n+l+1)^2} \end{aligned}$	$\begin{aligned} E(n, l) &= \frac{mc^2}{\sqrt{1 + \frac{\alpha^2}{\left(n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}\right)^2}}} \\ W(n, l) &= -\frac{1}{2} mc^2 \alpha^2 \frac{1}{(n+l+1)^2} \end{aligned}$
gilmore-lie-groups_ E08848	266	■ 1.000	$f(r) = \sum_{j=0}^n \frac{\Gamma(2\gamma)}{\Gamma(j+2\gamma)} \frac{n!}{(n-j)!j!} (2\lambda r)^j$	$f(r) = \sum_{j=0}^n \frac{\Gamma(2\gamma)}{\Gamma(j+2\gamma)} \frac{n!}{(n-j)!j!} (2\lambda r)^j$	$f(r) = \sum_{j=0}^n \frac{\Gamma(2\gamma)}{\Gamma(j+2\gamma)} \frac{n!}{(n-j)!j!} (2\lambda r)^j$

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gilmore-lie-groups_ E00849	266	■ 1.000	$\begin{aligned} E(n, l) = & \frac{m c^2}{\sqrt{1 + \left(\frac{\alpha}{N'}\right)^2}} \rightarrow m c^2 - m c^2 \frac{1}{2 N^2} \alpha^2 + m c^2 \left(\frac{3}{8 N^4} - \frac{1}{N^3 (2 l + 1)} \right) \alpha^4 \\ & + m c^2 \left(-\frac{5}{16 N^6} + \frac{3}{2 N^5 (2 l + 1)} - \frac{2 N + 3 (2 l + 1)}{2 N^4 (2 l + 1)^3} \right) \alpha^6 + m c^2 \\ & \times \left(\frac{35}{128 N^8} - \frac{15}{8 N^7 (2 l + 1)} + \frac{6 N + 9 (2 l + 1)}{4 N^6 (2 l + 1)^3} \right. \\ & \left. - \frac{2 N^2 + 3 N (2 l + 1) + 2 (2 l + 1)^2}{N^5 (2 l + 1)^5} \right) \alpha^8 + \mathcal{O}(\alpha^{10}) \end{aligned}$	$E(n, l) = \frac{m c^2}{\sqrt{1 + \left(\frac{\alpha}{N'}\right)^2}} \rightarrow m c^2 - m c^2 \frac{1}{2 N^2} \alpha^2 + m c^2 \left(\frac{3}{8 N^4} - \frac{1}{N^3 (2 l + 1)} \right) \alpha^4$ $+ m c^2 \left(-\frac{5}{16 N^6} + \frac{3}{2 N^5 (2 l + 1)} - \frac{2 N + 3 (2 l + 1)}{2 N^4 (2 l + 1)^3} \right) \alpha^6 + m c^2$ $\times \left(\frac{35}{128 N^8} - \frac{15}{8 N^7 (2 l + 1)} + \frac{6 N + 9 (2 l + 1)}{4 N^6 (2 l + 1)^3} \right.$ $\left. - \frac{2 N^2 + 3 N (2 l + 1) + 2 (2 l + 1)^2}{N^5 (2 l + 1)^5} \right) \alpha^8 + \mathcal{O}(\alpha^{10})$	$E(n, l) = \frac{m c^2}{\sqrt{1 + \left(\frac{\alpha}{N'}\right)^2}} \rightarrow m c^2 - m c^2 \frac{1}{2 N^2} \alpha^2 + m c^2 \left(\frac{3}{8 N^4} - \frac{1}{N^3 (2 l + 1)} \right) \alpha^4$ $+ m c^2 \left(-\frac{5}{16 N^6} + \frac{3}{2 N^5 (2 l + 1)} - \frac{2 N + 3 (2 l + 1)}{2 N^4 (2 l + 1)^3} \right) \alpha^6 + m c^2$ $\times \left(\frac{35}{128 N^8} - \frac{15}{8 N^7 (2 l + 1)} + \frac{6 N + 9 (2 l + 1)}{4 N^6 (2 l + 1)^3} \right.$ $\left. - \frac{2 N^2 + 3 N (2 l + 1) + 2 (2 l + 1)^2}{N^5 (2 l + 1)^5} \right) \alpha^8 + \mathcal{O}(\alpha^{10})$
gilmore-lie-groups_ E00850	266	■ 1.000	$\begin{array}{l} \text{\texttt{\textbackslash begin{array}{lll} \text{ relativistic } \& a(n, l) = \sqrt{\left(N^{\prime}\right)^2 + \alpha^2} a_B} \\ \text{\texttt{\textbackslash sqrt{\left(N^{\prime}\right)^2 + \alpha^2} a_B} \& N' = n + \frac{1}{2} + \sqrt{\left(l + \frac{1}{2}\right)^2 - \alpha^2}} \\ \& N^{\prime} = n + l + 1 \\ \text{\texttt{\textbackslash nonrelativistic } \& a(n, l) = N a_B} \\ \& N = n + l + 1 \\ \text{\texttt{\textbackslash end{array}}} \end{array}$	$\begin{array}{l} \text{relativistic} \quad a(n, l) = \sqrt{(N')^2 + \alpha^2} a_B \quad N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2} \\ \text{nonrelativistic} \quad a(n, l) = N a_B \quad N = n + l + 1 \end{array}$	$\begin{array}{l} \text{relativistic} \quad a(n, l) = \sqrt{(N')^2 + \alpha^2} a_B \quad N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2} \\ \text{nonrelativistic} \quad a(n, l) = N a_B \quad N = n + l + 1 \end{array}$
gilmore-lie-groups_ E00851	267	■ 1.000	$f = -\frac{K}{r^2} + \frac{C}{r^3}$	$f = -\frac{K}{r^2} + \frac{C}{r^3}$	$f = -\frac{K}{r^2} + \frac{C}{r^3}$
gilmore-lie-groups_ E00852	267	■ 1.000	$r = \frac{a \left(1 - \epsilon^2 \right)}{1 + \epsilon \cos(\alpha \theta)}$	$r = \frac{a \left(1 - \epsilon^2 \right)}{1 + \epsilon \cos(\alpha \theta)}$	$r = \frac{a(1 - \epsilon^2)}{1 + \epsilon \cos(\alpha \theta)}$
gilmore-lie-groups_ E00853	268	■ 1.000	$H = m c^2 + \frac{p^2}{2 m} - \frac{K'}{r} + \frac{C'}{r^2}$	$H = m c^2 + \frac{p^2}{2 m} - \frac{K'}{r} + \frac{C'}{r^2}$	$H = m c^2 + \frac{p^2}{2 m} - \frac{K'}{r} + \frac{C'}{r^2}$
gilmore-lie-groups_ E00854	268	■ 1.000	$\omega = \frac{\partial}{\partial L} \left(\frac{1}{T} \int_0^T p(r) dt \right) = \frac{\partial}{\partial L} \left(\frac{m}{L T} \oint r^2 p(r) d\theta \right)$	$\omega = \frac{\partial}{\partial L} \left(\frac{1}{T} \int_0^T p(r) dt \right) = \frac{\partial}{\partial L} \left(\frac{m}{L T} \oint r^2 p(r) d\theta \right)$	$\omega = \frac{\partial}{\partial L} \left(\frac{1}{T} \int_0^T p(r) dt \right) = \frac{\partial}{\partial L} \left(\frac{m}{L T} \oint r^2 p(r) d\theta \right)$
gilmore-lie-groups_ E00855	268	■ 1.000	$\begin{array}{l} \text{\texttt{\textbackslash begin{array}{ll} \text{ special relativity } C = \frac{K L^2}{2 m^2 c^2} \& \omega = \frac{\pi K^2}{T L^2 c^2}} \\ \text{\texttt{\textbackslash c^2} \& \omega = \frac{6 \pi K^2}{T L^2 c^2}} \\ \text{\texttt{\textbackslash general relativity } C = 6 \times \frac{K L^2}{2 m^2 c^2} \& \omega = \frac{6 \pi K^2}{T L^2 c^2}} \\ \text{\texttt{\textbackslash end{array}}} \end{array}$	$\begin{array}{l} \text{special relativity } C = \frac{K L^2}{2 m^2 c^2} \quad \omega = \frac{\pi K^2}{T L^2 c^2} \\ \text{general relativity } C = 6 \times \frac{K L^2}{2 m^2 c^2} \quad \omega = \frac{6 \pi K^2}{T L^2 c^2} \end{array}$	$\begin{array}{l} \text{special relativity } C = \frac{K L^2}{2 m^2 c^2} \quad \omega = \frac{\pi K^2}{T L^2 c^2} \\ \text{general relativity } C = 6 \times \frac{K L^2}{2 m^2 c^2} \quad \omega = \frac{6 \pi K^2}{T L^2 c^2} \end{array}$

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gilmore-lie-groups_ EQ0856	269	■ 1.000	$\left(-\frac{\partial}{\partial\theta}+\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\phi}\right)\frac{\partial}{\partial\theta}\left(\frac{\partial}{\partial\theta}+\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\phi}\right)P_{-l}^l(\theta)e^{-il\phi}=e^{-il\phi}\left(-\frac{\partial}{\partial\theta}+l\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\phi}\right)P_{-l}^l(\theta)$	$\left(-\frac{\partial}{\partial\theta}+i\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\phi}\right)P_{-l}^l(\theta)e^{-il\phi}=e^{-il\phi}\left(-\frac{\partial}{\partial\theta}+l\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\phi}\right)P_{-l}^l(\theta)$	$\left(-\frac{\partial}{\partial\theta}+i\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\phi}\right)P_{-l}^l(\theta)e^{-il\phi}=e^{-il\phi}\left(-\frac{\partial}{\partial\theta}+l\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\phi}\right)P_{-l}^l(\theta)$
gilmore-lie-groups_ EQ0857	269	■ 1.000	$\int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi \left N_l \sin^l\theta e^{-il\phi}\right ^2=1$	$\int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi \left N_l \sin^l\theta e^{-il\phi}\right ^2=1$	$\int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi N_l \sin^l\theta e^{-il\phi} ^2=1$
gilmore-lie-groups_ EQ0858	269	■ 1.000	$N_l=\sqrt{\frac{1}{4\pi}}\sqrt{\frac{(2l+1)!!}{(2l)!!}}$	$N_l=\sqrt{\frac{1}{4\pi}}\sqrt{\frac{(2l+1)!!}{(2l)!!}}$	$N_l=\sqrt{\frac{1}{4\pi}}\sqrt{\frac{(2l+1)!!}{(2l)!!}}$
gilmore-lie-groups_ EQ0859	269	■ 1.000	$N_l=\sqrt{\frac{2l+1}{2l}}N_{l-1}$	$N_l=\sqrt{\frac{2l+1}{2l}}N_{l-1}$	$N_l=\sqrt{\frac{2l+1}{2l}}N_{l-1}$
gilmore-lie-groups_ EQ0860	269	■ 1.000	$\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right)=\frac{\dot{\mathbf{r}}(\mathbf{r}\cdot\mathbf{r})-\mathbf{r}(\mathbf{r}\cdot\dot{\mathbf{r}})}{r^3}=-\frac{\mathbf{r}\times(\mathbf{r}\times\dot{\mathbf{r}})}{r^3}$	$\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right)=\frac{\dot{\mathbf{r}}(\mathbf{r}\cdot\mathbf{r})-\mathbf{r}(\mathbf{r}\cdot\dot{\mathbf{r}})}{r^3}=-\frac{\mathbf{r}\times(\mathbf{r}\times\dot{\mathbf{r}})}{r^3}$	$\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right)=\frac{\dot{\mathbf{r}}(\mathbf{r}\cdot\mathbf{r})-\mathbf{r}(\mathbf{r}\cdot\dot{\mathbf{r}})}{r^3}=-\frac{\mathbf{r}\times(\mathbf{r}\times\dot{\mathbf{r}})}{r^3}$
gilmore-lie-groups_ EQ0861	270	■ 1.000	$\frac{\mathbf{p}}{p_0}=\frac{\mathbf{u}}{1-\mathbf{u}\cdot\mathbf{w}}$	$\frac{\mathbf{p}}{p_0}=\frac{\mathbf{u}}{1-\mathbf{u}\cdot\mathbf{w}}$	$\frac{\mathbf{p}}{p_0}=\frac{\mathbf{u}}{1-\mathbf{u}\cdot\mathbf{w}}$
gilmore-lie-groups_ EQ0862	271	■ 1.000	$\frac{(a+b+1)!}{(a+\frac{1}{2})!(b+\frac{1}{2})!}=\frac{(a+b)!}{(a-\frac{1}{2})!(b+\frac{1}{2})!}+\frac{(a+b)!}{(a+\frac{1}{2})!(b-\frac{1}{2})!}$	$\frac{(a+b+1)!}{(a+\frac{1}{2})!(b+\frac{1}{2})!}=\frac{(a+b)!}{(a-\frac{1}{2})!(b+\frac{1}{2})!}+\frac{(a+b)!}{(a+\frac{1}{2})!(b-\frac{1}{2})!}$	$\frac{(a+b+1)!}{(a+\frac{1}{2})!(b+\frac{1}{2})!}=\frac{(a+b)!}{(a-\frac{1}{2})!(b+\frac{1}{2})!}+\frac{(a+b)!}{(a+\frac{1}{2})!(b-\frac{1}{2})!}$
gilmore-lie-groups_ EQ0863	271	■ 1.000	$\dim \mathcal{Y}^l(S^n)=\dim \mathcal{Y}^{l-1}(S^n)+\dim \mathcal{Y}^l(S^{n-1})$	$\dim \mathcal{Y}^l(S^n)=\dim \mathcal{Y}^{l-1}(S^n)+\dim \mathcal{Y}^l(S^{n-1})$	$\dim \mathcal{Y}^l(S^n)=\dim \mathcal{Y}^{l-1}(S^n)+\dim \mathcal{Y}^l(S^{n-1})$
gilmore-lie-groups_ EQ0864	271	■ 1.000	$\nabla^2=\left(\frac{1}{r^{D-1/2}}\frac{\partial}{\partial r}r^{D-1/2}\right)^2+\frac{\mathcal{L}^2}{r^2}$	$\nabla^2=\left(\frac{1}{r^{D-1/2}}\frac{\partial}{\partial r}r^{D-1/2}\right)^2+\frac{\mathcal{L}^2}{r^2}$	$\nabla^2=\left(\frac{1}{r^{D-1/2}}\frac{\partial}{\partial r}r^{D-1/2}\right)^2+\frac{\mathcal{L}^2}{r^2}$
gilmore-lie-groups_ EQ0865	271	■ 1.000	$\left(l+\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\rightarrow\left(l+\frac{D-2}{2}\right)^2-\left(\frac{D-2}{2}\right)^2$	$\left(l+\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\rightarrow\left(l+\frac{D-2}{2}\right)^2-\left(\frac{D-2}{2}\right)^2$	$\left(l+\frac{1}{2}\right)^2-\left(\frac{1}{2}\right)^2\rightarrow\left(l+\frac{D-2}{2}\right)^2-\left(\frac{D-2}{2}\right)^2$
gilmore-lie-groups_ EQ0866	271	■ 1.000	$\begin{array}{ll} \text{relativistic} & N'\rightarrow n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2+l(D-3)-\alpha^2} \\ \text{nonrelativistic} & N\rightarrow n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2+l(D-3)} \end{array}$	$\begin{array}{ll} \text{relativistic} & N'\rightarrow n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2+l(D-3)-\alpha^2} \\ \text{nonrelativistic} & N\rightarrow n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2+l(D-3)} \end{array}$	$\begin{array}{ll} \text{relativistic} & N'\rightarrow n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2+l(D-3)-\alpha^2} \\ \text{nonrelativistic} & N\rightarrow n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2+l(D-3)} \end{array}$
gilmore-lie-groups_ EQ0867	272	■ 1.000	$H=\sum_{i=1}^n\hbar\omega\left(a_i^\dagger a_i+\frac{1}{2}\right)$	$H=\sum_{i=1}^n\hbar\omega\left(a_i^\dagger a_i+\frac{1}{2}\right)$	$H=\sum_{i=1}^n\hbar\omega\left(a_i^\dagger a_i+\frac{1}{2}\right)$

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gilmore-lie-groups_ E00868 (15.1)	275	■ 1.000	$x^2+y^2+z^2-(c\,t)^2=x^{\prime 2}+y^{\prime 2}+z^{\prime 2}-\left(c\,t^{\prime }\right)^2=0$	$x^2+y^2+z^2-(ct)^2=x'^2+y'^2+z'^2-(ct')^2=0$	$x^2+y^2+z^2-(ct)^2=x'^2+y'^2+z'^2-(ct')^2=0$
gilmore-lie-groups_ E00870 (15.3)	275	■ 1.000	$\Lambda \rightarrowtail I_4+\epsilon \left[\begin{array}{cccc} 0 & +\theta _3 & -\theta _2 & ib_1 \\ -\theta _3 & 0 & +\theta _1 & ib_2 \\ +\theta _2 & -\theta _1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right]=I_4+\epsilon (\theta \cdot \mathbf{J}+\mathbf{b}\cdot \mathbf{K})$	$\Lambda \rightarrow I_4+\epsilon \left[\begin{array}{cccc} 0 & +\theta _3 & -\theta _2 & ib_1 \\ -\theta _3 & 0 & +\theta _1 & ib_2 \\ +\theta _2 & -\theta _1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right]=I_4+\epsilon (\theta \cdot \mathbf{J}+\mathbf{b}\cdot \mathbf{K})$	$\Lambda \rightarrow I_4+\epsilon \left[\begin{array}{cccc} 0 & +\theta _3 & -\theta _2 & ib_1 \\ -\theta _3 & 0 & +\theta _1 & ib_2 \\ +\theta _2 & -\theta _1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right]=I_4+\epsilon (\theta \cdot \mathbf{J}+\mathbf{b}\cdot \mathbf{K})$
gilmore-lie-groups_ E00871 (15.4)	275	■ 1.000	$\begin{aligned} \left\{J_i,J_j\right\}&=-\epsilon _{ijk}J_k \\ \left\{J_i,K_j\right\}&=-\epsilon _{ijk}K_k \\ \left\{K_i,K_j\right\}&=+\epsilon _{ijk}J_k \end{aligned}$	$\begin{aligned} \left[J_i,J_j\right]&=-\epsilon _{ijk}J_k \\ \left[J_i,K_j\right]&=-\epsilon _{ijk}K_k \\ \left[K_i,K_j\right]&=+\epsilon _{ijk}J_k \end{aligned}$	$\begin{aligned} \left[J_i,J_j\right]&=-\epsilon _{ijk}J_k \\ \left[J_i,K_j\right]&=-\epsilon _{ijk}K_k \\ \left[K_i,K_j\right]&=+\epsilon _{ijk}J_k \end{aligned}$
gilmore-lie-groups_ E00872 (15.5)	275	■ 1.000	$\left(x_2-x_1\right)^2+\left(y_2-y_1\right)^2+\left(z_2-z_1\right)^2-\left(c\,t_2-c\,t_1\right)^2=\text {invariant}$	$(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2-(ct_2-ct_1)^2=\text {invariant}$	$(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2-(ct_2-ct_1)^2=\text {invariant}$
gilmore-lie-groups_ E00874 (15.7)	276	■ 1.000	$\begin{gathered} \left\{\Lambda _2,a_2\right\}\left\{\Lambda _1,a_1\right\}=\left\{\Lambda _2\Lambda _1,a_2+\Lambda _2a_1\right\} \\ \left\{I,a\right\}\left\{\Lambda ,0\right\}=\left\{\Lambda ,a\right\}=\left\{\Lambda ,0\right\}\left\{I,\Lambda ^{-1}a\right\} \end{gathered}$	$\begin{aligned} \left\{\Lambda _2,a_2\right\}\left\{\Lambda _1,a_1\right\}&=\left\{\Lambda _2\Lambda _1,a_2+\Lambda _2a_1\right\} \\ \left\{I,a\right\}\left\{\Lambda ,0\right\}&=\left\{\Lambda ,a\right\}=\left\{\Lambda ,0\right\}\left\{I,\Lambda ^{-1}a\right\} \end{aligned}$	$\begin{aligned} \left\{\Lambda _2,a_2\right\}\left\{\Lambda _1,a_1\right\}&=\left\{\Lambda _2\Lambda _1,a_2+\Lambda _2a_1\right\} \\ \left\{I,a\right\}\left\{\Lambda ,0\right\}&=\left\{\Lambda ,a\right\}=\left\{\Lambda ,0\right\}\left\{I,\Lambda ^{-1}a\right\} \end{aligned}$
gilmore-lie-groups_ E00875 (15.8)	276	■ 1.000	$\Gamma ^k(\{I,a\})=e^{ik\cdot a}$	$\Gamma ^k(\{I,a\})=e^{ik\cdot a}$	$\Gamma ^k(\{I,a\})=e^{ik\cdot a}$
gilmore-lie-groups_ E00876 (15.9)	276	■ 1.000	$\{I,a\} k\rangle = k'\rangle \langle k' \{I,a\} k\rangle = k'\rangle \delta (k'-k)\Gamma ^k(\{I,a\})= k\rangle e^{ik\cdot a}$	$\{I,a\} k\rangle = k'\rangle \langle k' \{I,a\} k\rangle = k'\rangle \delta (k'-k)\Gamma ^k(\{I,a\})= k\rangle e^{ik\cdot a}$	$\{I,a\} k\rangle = k'\rangle \langle k' \{I,a\} k\rangle = k'\rangle \delta (k'-k)\Gamma ^k(\{I,a\})= k\rangle e^{ik\cdot a}$
gilmore-lie-groups_ E00877 (15.10)	277	■ 1.000	$\begin{aligned} \mathbf{J}^{\{1\}}&=\frac{1}{2}(\mathbf{J}-i\mathbf{K}) \\ \mathbf{J}^{\{2\}}&=\frac{1}{2}(\mathbf{J}+i\mathbf{K}) \end{aligned}$	$\begin{aligned} \mathbf{J}^{\{1\}}&=\frac{1}{2}(\mathbf{J}-i\mathbf{K}) \\ \mathbf{J}^{\{2\}}&=\frac{1}{2}(\mathbf{J}+i\mathbf{K}) \end{aligned}$	$\begin{aligned} \mathbf{J}^{\{1\}}&=\frac{1}{2}(\mathbf{J}-i\mathbf{K}) \\ \mathbf{J}^{\{2\}}&=\frac{1}{2}(\mathbf{J}+i\mathbf{K}) \end{aligned}$
gilmore-lie-groups_ E00879 (15.12)	277	■ 1.000	$\begin{aligned} \operatorname{EXP}(\theta \cdot \mathbf{J}+\mathbf{b}\cdot \mathbf{K})&=\operatorname{EXP}\left[(\theta +i\mathbf{b})\cdot \mathbf{J}^{\{1\}}+(\theta -i\mathbf{b})\cdot \mathbf{J}^{\{2\}}\right] \\ &=D^j\left[(\theta +i\mathbf{b})\cdot \mathbf{J}^{\{1\}}\right]D^{j'}\left[(\theta -i\mathbf{b})\cdot \mathbf{J}^{\{2\}}\right] \end{aligned}$	$\begin{aligned} \operatorname{EXP}(\theta \cdot \mathbf{J}+\mathbf{b}\cdot \mathbf{K})&=\operatorname{EXP}\left[(\theta +i\mathbf{b})\cdot \mathbf{J}^{\{1\}}+(\theta -i\mathbf{b})\cdot \mathbf{J}^{\{2\}}\right] \\ &=D^j\left[(\theta +i\mathbf{b})\cdot \mathbf{J}^{\{1\}}\right]D^{j'}\left[(\theta -i\mathbf{b})\cdot \mathbf{J}^{\{2\}}\right] \end{aligned}$	$\begin{aligned} \operatorname{EXP}(\theta \cdot \mathbf{J}+\mathbf{b}\cdot \mathbf{K})&=\operatorname{EXP}\left[(\theta +i\mathbf{b})\cdot \mathbf{J}^{\{1\}}+(\theta -i\mathbf{b})\cdot \mathbf{J}^{\{2\}}\right] \\ &=D^j\left[(\theta +i\mathbf{b})\cdot \mathbf{J}^{\{1\}}\right]D^{j'}\left[(\theta -i\mathbf{b})\cdot \mathbf{J}^{\{2\}}\right] \end{aligned}$

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gilmore-lie-groups_ EQ0880 (15.13)	277	■ 1.000	$\begin{array}{ll} \mathbf{J}=\frac{i}{2}\sigma & \mathbf{J}=\frac{i}{2}\sigma \\ \mathbf{K}=-\frac{1}{2}\sigma & \mathbf{K}=-\frac{1}{2}\sigma \end{array}$	$\mathbf{J}=\frac{i}{2}\sigma \quad \mathbf{J}=\frac{i}{2}\sigma$ $\mathbf{K}=-\frac{1}{2}\sigma \quad \mathbf{K}=-\frac{1}{2}\sigma$	$\mathbf{J}=\frac{i}{2}\sigma \quad \mathbf{J}=\frac{i}{2}\sigma$ $\mathbf{K}=-\frac{1}{2}\sigma \quad \mathbf{K}=-\frac{1}{2}\sigma$
gilmore-lie-groups_ EQ0881 (15.14)	277	■ 1.000	$\begin{array}{cc} D^{\{j\}0} & D^{0j} \\ \mathbf{K}^{(j)}=i\mathbf{J}^{(j)} & \mathbf{K}^{(j)}=-i\mathbf{J}^{(j)} \end{array}$	$D^{j0} \quad D^{0j}$ $\mathbf{K}^{(j)}=i\mathbf{J}^{(j)} \quad \mathbf{K}^{(j)}=-i\mathbf{J}^{(j)}$	$D^{j0} \quad D^{0j}$ $\mathbf{K}^{(j)}=i\mathbf{J}^{(j)} \quad \mathbf{K}^{(j)}=-i\mathbf{J}^{(j)}$
gilmore-lie-groups_ EQ0883 (15.16)	278	■ 1.000	$D^{\{j\}j'}(\theta\cdot\mathbf{J}+\mathbf{b}\cdot\mathbf{K})=\text{EXP}\left[(\theta+i\mathbf{b})\cdot\mathbf{J}^{(j)}\right]\text{EXP}\left[(\theta-i\mathbf{b})\cdot\mathbf{J}^{(j')}\right]=D^{jj'}(\Lambda)$	$D^{jj'}(\theta\cdot\mathbf{J}+\mathbf{b}\cdot\mathbf{K})=\text{EXP}\left[(\theta+i\mathbf{b})\cdot\mathbf{J}^{(j)}\right]\text{EXP}\left[(\theta-i\mathbf{b})\cdot\mathbf{J}^{(j')}\right]=D^{jj'}(\Lambda)$	$D^{jj'}(\theta\cdot\mathbf{J}+\mathbf{b}\cdot\mathbf{K})=\text{EXP}\left[(\theta+i\mathbf{b})\cdot\mathbf{J}^{(j)}\right]\text{EXP}\left[(\theta-i\mathbf{b})\cdot\mathbf{J}^{(j')}\right]=D^{jj'}(\Lambda)$
gilmore-lie-groups_ EQ0885 (15.18)	278	■ 1.000	$\begin{gathered} D^{\{j\}j'}(\Lambda)\xrightarrow{\Lambda\downarrow SO(3)}D^j[SO(3)]\times D^{j'}[SO(3)]=\sum_{j''}D^{jj''}[SO(3)] \\ j-j' \leq j''\leq j+j' \end{gathered}$	$D^{jj'}(\Lambda)\xrightarrow{\Lambda\downarrow SO(3)}D^j[SO(3)]\times D^{j'}[SO(3)]=\sum_{j''}D^{jj''}[SO(3)]$ $ j-j' \leq j''\leq j+j'$	$D^{jj'}(\Lambda)\xrightarrow{\Lambda\downarrow SO(3)}D^j[SO(3)]\times D^{j'}[SO(3)]=\sum_{j''}D^{jj''}[SO(3)]$ $ j-j' \leq j''\leq j+j'$
gilmore-lie-groups_ EQ0889 (15.22)	279	■ 1.000	$\begin{aligned} \{I,a\}[\{\Lambda,0\} k\rangle]&=\{\Lambda,0\}\{I,\Lambda^{-1}a\} k\rangle \\ &=[\{\Lambda,0\} k\rangle]e^{ik\cdot\Lambda^{-1}a} \\ &=[\{\Lambda,0\} k\rangle]e^{i\Lambda k\cdot a}= \Lambda k\rangle e^{i\Lambda k\cdot a} \end{aligned}$	$\{I,a\}[\{\Lambda,0\} k\rangle]=\{\Lambda,0\}\{I,\Lambda^{-1}a\} k\rangle$ $=[\{\Lambda,0\} k\rangle]e^{ik\cdot\Lambda^{-1}a}$ $=[\{\Lambda,0\} k\rangle]e^{i\Lambda k\cdot a}= \Lambda k\rangle e^{i\Lambda k\cdot a}$	$\{I,a\}[\{\Lambda,0\} k\rangle]=\{\Lambda,0\}\{I,\Lambda^{-1}a\} k\rangle$ $=[\{\Lambda,0\} k\rangle]e^{ik\cdot\Lambda^{-1}a}$ $=[\{\Lambda,0\} k\rangle]e^{i\Lambda k\cdot a}= \Lambda k\rangle e^{i\Lambda k\cdot a}$
gilmore-lie-groups_ EQ0893 (15.27)	280	■ 1.000	$\{I,a\} k;\xi\rangle= k;\xi\rangle e^{ik\cdot a}$	$\{I,a\} k;\xi\rangle= k;\xi\rangle e^{ik\cdot a}$	$\{I,a\} k;\xi\rangle= k;\xi\rangle e^{ik\cdot a}$
gilmore-lie-groups_ EQ0894 (15.28)	280	■ 1.000	$\begin{aligned} \{I,a\}[\{\Lambda,0\} k;\xi\rangle]&=\{\Lambda,0\}\{I,\Lambda^{-1}a\} k;\xi\rangle \\ &=[\{\Lambda,0\} k;\xi\rangle]e^{ik\cdot\Lambda^{-1}a} \\ &=[\{\Lambda,0\} k;\xi\rangle]e^{i\Lambda k\cdot a} \end{aligned}$	$\{I,a\}[\{\Lambda,0\} k;\xi\rangle]=\{\Lambda,0\}\{I,\Lambda^{-1}a\} k;\xi\rangle$ $=[\{\Lambda,0\} k;\xi\rangle]e^{ik\cdot\Lambda^{-1}a}$ $=[\{\Lambda,0\} k;\xi\rangle]e^{i\Lambda k\cdot a}$	$\{I,a\}[\{\Lambda,0\} k;\xi\rangle]=\{\Lambda,0\}\{I,\Lambda^{-1}a\} k;\xi\rangle$ $=[\{\Lambda,0\} k;\xi\rangle]e^{ik\cdot\Lambda^{-1}a}$ $=[\{\Lambda,0\} k;\xi\rangle]e^{i\Lambda k\cdot a}$
gilmore-lie-groups_ EQ0895 (15.29)	280	■ 1.000	$\{\Lambda,0\} k;\xi\rangle= \Lambda k;\xi'\rangle M_{\xi'\xi}(\Lambda)$	$\{\Lambda,0\} k;\xi\rangle= \Lambda k;\xi'\rangle M_{\xi'\xi}(\Lambda)$	$\{\Lambda,0\} k;\xi\rangle= \Lambda k;\xi'\rangle M_{\xi'\xi}(\Lambda)$
gilmore-lie-groups_ EQ0896 (15.30)	280	■ 1.000	$k^{\prime}=\Lambda k$	$k'=\Lambda k$	$k'=\Lambda k$
gilmore-lie-groups_ EQ0899 (15.33)	281	■ 1.000	$\begin{aligned} H_k k^0 &= k^0 \\ C_k k^0 &= k \end{aligned}$	$H_k k^0 = k^0$ $C_k k^0 = k$	$H_k k^0 = k^0$ $C_k k^0 = k$
gilmore-lie-groups_ EQ0900 (15.34)	281	■ 1.000	$C_k k^0 = k = \Lambda k^0$	$C_k k^0 = k = \Lambda k^0$	$C_k k^0 = k = \Lambda k^0$

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gilmore-lie-groups_ E09901 (15.35)	281	■ 1.000	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & ib_1 \\ & -\theta_3 & 0 & +\theta_1 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ 0 \end{array}\right] = \left[\begin{array}{c} -\theta_2 \\ +\theta_1 \\ 0 \\ -ib_3 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & ib_1 \\ & -\theta_3 & 0 & +\theta_1 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ 0 \end{array}\right] = \left[\begin{array}{c} -\theta_2 \\ +\theta_1 \\ 0 \\ -ib_3 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & ib_1 \\ -\theta_3 & 0 & +\theta_1 & ib_2 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ 0 \end{array}\right] = \left[\begin{array}{c} -\theta_2 \\ +\theta_1 \\ 0 \\ -ib_3 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$
gilmore-lie-groups_ E09904 (15.38)	282	■ 1.000	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & -i\theta_2 \\ & -i\theta_2 & -\theta_3 & 0 \\ +\theta_2 & -\theta_1 & 0 & 0 \\ i\theta_2 & -i\theta_1 & 0 & 0 \end{array}\right] = \sum_i \theta_i Y_i \quad \begin{array}{l} Y_1 = J_1 + K_2 \\ Y_2 = J_2 - K_1 \\ Y_3 = J_3 \end{array}$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & -i\theta_2 \\ -\theta_3 & 0 & +\theta_1 & i\theta_1 \\ +\theta_2 & -\theta_1 & 0 & 0 \\ i\theta_2 & -i\theta_1 & 0 & 0 \end{array}\right] = \sum_i \theta_i Y_i \quad \begin{array}{l} Y_1 = J_1 + K_2 \\ Y_2 = J_2 - K_1 \\ Y_3 = J_3 \end{array}$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & -i\theta_2 \\ -\theta_3 & 0 & +\theta_1 & i\theta_1 \\ +\theta_2 & -\theta_1 & 0 & 0 \\ i\theta_2 & -i\theta_1 & 0 & 0 \end{array}\right] = \sum_i \theta_i Y_i \quad \begin{array}{l} Y_1 = J_1 + K_2 \\ Y_2 = J_2 - K_1 \\ Y_3 = J_3 \end{array}$
gilmore-lie-groups_ E09907 (15.41)	283	■ 1.000	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & ib_1 \\ & -\theta_3 & 0 & +\theta_1 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & ib_1 \\ & -\theta_3 & 0 & +\theta_1 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$	$\left[\begin{array}{cccc} 0 & +\theta_3 & -\theta_2 & ib_1 \\ -\theta_3 & 0 & +\theta_1 & ib_2 \\ +\theta_2 & -\theta_1 & 0 & ib_3 \\ -ib_1 & -ib_2 & -ib_3 & 0 \end{array}\right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right]$
gilmore-lie-groups_ E09908 (15.42)	283	■ 1.000	$\begin{aligned} & H_{k^0} k^0; \xi\rangle = H_{k^0} k^0; \xi'\rangle D_{\xi'\xi}(H_{k^0}) \\ & = k^0; \xi'\rangle D_{\xi'\xi}(H_{k^0}) \end{aligned}$	$\begin{aligned} H_{k^0} k^0; \xi\rangle &= H_{k^0} k^0; \xi'\rangle D_{\xi'\xi}(H_{k^0}) \\ &= k^0; \xi'\rangle D_{\xi'\xi}(H_{k^0}) \end{aligned}$	$\begin{aligned} H_{k^0} k^0; \xi\rangle &= H_{k^0} k^0; \xi'\rangle D_{\xi'\xi}(H_{k^0}) \\ &= k^0; \xi'\rangle D_{\xi'\xi}(H_{k^0}) \end{aligned}$
gilmore-lie-groups_ E09909 (15.44)	284	■ 1.000	$\operatorname{EXP}(\theta_3 Y_3 + \theta_1 Y_1 + \theta_2 Y_2) = e^{i\xi\theta_3}$	$\operatorname{EXP}(\theta_3 Y_3 + \theta_1 Y_1 + \theta_2 Y_2) = e^{i\xi\theta_3}$	$\operatorname{EXP}(\theta_3 Y_3 + \theta_1 Y_1 + \theta_2 Y_2) = e^{i\xi\theta_3}$
gilmore-lie-groups_ E09910 (15.45)	284	■ 1.000	$C_k k^0; \xi\rangle = k; \xi\rangle$	$C_k k^0; \xi\rangle = k; \xi\rangle$	$C_k k^0; \xi\rangle = k; \xi\rangle$

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gilmore-lie-groups_ E08943	291	1.000	$\mathbf{J} \cdot \nabla = \left[\begin{array}{ccccc} +2\partial_0 & \sqrt{4}\partial_+ & 0 & 0 & 0 \\ \sqrt{4}\partial_- & +1\partial_0 & \sqrt{6}\partial_+ & 0 & 0 \\ 0 & \sqrt{6}\partial_- & 0\partial_0 & \sqrt{6}\partial_+ & 0 \\ 0 & 0 & \sqrt{6}\partial_- & -1\partial_0 & \sqrt{4}\partial_+ \\ 0 & 0 & 0 & \sqrt{6}\partial_- & -2\partial_0 \end{array} \right]$	$\mathbf{J} \cdot \nabla = \left[\begin{array}{ccccc} +2\partial_0 & \sqrt{4}\partial_+ & 0 & 0 & 0 \\ \sqrt{4}\partial_- & +1\partial_0 & \sqrt{6}\partial_+ & 0 & 0 \\ 0 & \sqrt{6}\partial_- & 0\partial_0 & \sqrt{6}\partial_+ & 0 \\ 0 & 0 & \sqrt{6}\partial_- & -1\partial_0 & \sqrt{4}\partial_+ \\ 0 & 0 & 0 & \sqrt{6}\partial_- & -2\partial_0 \end{array} \right]$	
gilmore-lie-groups_ E08945	292	1.000	$F_{ij} = \left(\begin{array}{ccc} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{array} \right) = \left(\begin{array}{ccc} F_4 - \frac{1}{\sqrt{3}}F_5 & F_1 & F_3 \\ F_1 & -F_4 - \frac{1}{\sqrt{3}}F_5 & F_2 \\ F_3 & F_2 & +\frac{2}{\sqrt{3}}F_5 \end{array} \right)$	$F_{ij} = \left(\begin{array}{ccc} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{array} \right) = \left(\begin{array}{ccc} F_4 - \frac{1}{\sqrt{3}}F_5 & F_1 & F_3 \\ F_1 & -F_4 - \frac{1}{\sqrt{3}}F_5 & F_2 \\ F_3 & F_2 & +\frac{2}{\sqrt{3}}F_5 \end{array} \right)$	$F_{ij} = \left(\begin{array}{ccc} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{array} \right) = \left(\begin{array}{ccc} F_4 - \frac{1}{\sqrt{3}}F_5 & F_1 & F_3 \\ F_1 & -F_4 - \frac{1}{\sqrt{3}}F_5 & F_2 \\ F_3 & F_2 & +\frac{2}{\sqrt{3}}F_5 \end{array} \right)$
gilmore-lie-groups_ E08947	294	1.000	$\oint \mathbf{E} \cdot d\mathbf{S} = \int 4\pi \rho dV$	$\oint \mathbf{E} \cdot d\mathbf{S} = \int 4\pi \rho dV$	$\oint \mathbf{E} \cdot d\mathbf{S} = \int 4\pi \rho dV$
gilmore-lie-groups_ E08948	294	1.000	$4\pi a^2 \mathbf{E}(a) = 4\pi q$	$4\pi a^2 \mathbf{E}(a) = 4\pi q$	$4\pi a^2 \mathbf{E}(a) = 4\pi q$
gilmore-lie-groups_ E08949	294	1.000	$\mathbf{E}(a) = \frac{q}{a^2} \frac{\mathbf{a}}{ \mathbf{a} }$	$\mathbf{E}(a) = \frac{q}{a^2} \frac{\mathbf{a}}{ \mathbf{a} }$	$\mathbf{E}(a) = \frac{q}{a^2} \frac{\mathbf{a}}{ \mathbf{a} }$
gilmore-lie-groups_ E08950	295	1.000	$ \mathbf{E} = \frac{q}{R \sin \theta} = \frac{q}{R \sin(a/R)}$	$ \mathbf{E} = \frac{q}{R \sin \theta} = \frac{q}{R \sin(a/R)}$	$ \mathbf{E} = \frac{q}{R \sin \theta} = \frac{q}{R \sin(a/R)}$
gilmore-lie-groups_ E08951	295	1.000	$ \mathbf{E} = \frac{q}{R \sin(a/R)} = \frac{q(a)}{a} \quad q(a) = q \left(\frac{a/R}{\sin(a/R)} \right)$	$ \mathbf{E} = \frac{q}{R \sin(a/R)} = \frac{q(a)}{a} \quad q(a) = q \left(\frac{a/R}{\sin(a/R)} \right)$	$ \mathbf{E} = \frac{q}{R \sin(a/R)} = \frac{q(a)}{a} \quad q(a) = q \left(\frac{a/R}{\sin(a/R)} \right)$
gilmore-lie-groups_ E08952	295	1.000	$ds^2 = \frac{dr^2}{1 - kr^2} + r^2 \sum_{j=2}^n (\sin \theta_2 \sin \theta_3 \cdots \sin \theta_{j-1} d\theta_j)^2$	$ds^2 = \frac{dr^2}{1 - kr^2} + r^2 \sum_{j=2}^n (\sin \theta_2 \sin \theta_3 \cdots \sin \theta_{j-1} d\theta_j)^2$	$ds^2 = \frac{dr^2}{1 - kr^2} + r^2 \sum_{j=2}^n (\sin \theta_2 \sin \theta_3 \cdots \sin \theta_{j-1} d\theta_j)^2$
gilmore-lie-groups_ E08954	295	1.000	$\oint \mathbf{E} \cdot d\mathbf{S} = \int \Omega \rho(x) dV$	$\oint \mathbf{E} \cdot d\mathbf{S} = \int \Omega \rho(x) dV$	$\oint \mathbf{E} \cdot d\mathbf{S} = \int \Omega \rho(x) dV$
gilmore-lie-groups_ E08955	295	1.000	$ \mathbf{E} a^{n-1} = q$	$ \mathbf{E} a^{n-1} = q$	$ \mathbf{E} a^{n-1} = q$
gilmore-lie-groups_ E08956	296	1.000	$d(a) = \int_0^a \frac{dr}{\sqrt{1 - kr^2}} \rightarrow \begin{cases} \sinh^{-1} a & k = -1 \\ a & k = 0 \\ \sin^{-1} a & k = +1 \end{cases}$	$d(a) = \int_0^a \frac{dr}{\sqrt{1 - kr^2}} \rightarrow \begin{cases} \sinh^{-1} a & k = -1 \\ a & k = 0 \\ \sin^{-1} a & k = +1 \end{cases}$	$d(a) = \int_0^a \frac{dr}{\sqrt{1 - kr^2}} \rightarrow \begin{cases} \sinh^{-1} a & k = -1 \\ a & k = 0 \\ \sin^{-1} a & k = +1 \end{cases}$
gilmore-lie-groups_ E08958 (16.1)	299	1.000	$\frac{dy}{dx} = g(x)$	$\frac{dy}{dx} = g(x)$	$\frac{dy}{dx} = g(x)$

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gilmore-lie-groups_ EQ0959 (16.2)	299	■ 1.000	$y=G(x)=\int g(x) \, d x \quad(+ \, \text{additive constant})=G(x)+c$	$y = G(x) = \int g(x) dx \quad (+ \, \text{additive constant}) = G(x) + c$	$y = G(x) = \int g(x) \, dx \quad (+ \, \text{additive constant}) = G(x) + c$
gilmore-lie-groups_ EQ0960 (16.3)	299	■ 1.000	$e^{c \, \partial/\partial y} [y - G(x) = 0] = y + c - G(x) = 0$	$e^{c \partial/\partial y} [y - G(x) = 0] = y + c - G(x) = 0$	$e^{c \partial/\partial y} [y - G(x) = 0] = y + c - G(x) = 0$
gilmore-lie-groups_ EQ0963 (16.6)	301	■ 1.000	$\begin{array}{ll} x \rightarrow \bar{x}(\epsilon) = x + \epsilon \xi(x, y) + \mathcal{O}(\epsilon^2) & \bar{x}(\epsilon = 0) = x \\ y \rightarrow \bar{y}(\epsilon) = y + \epsilon \eta(x, y) + \mathcal{O}(\epsilon^2) & \bar{y}(\epsilon = 0) = y \\ p \rightarrow \bar{p}(\epsilon) = p + \epsilon \zeta(x, y, p) + \mathcal{O}(\epsilon^2) & \bar{p}(\epsilon = 0) = p \end{array}$	$\begin{array}{ll} x \rightarrow \bar{x}(\epsilon) = x + \epsilon \xi(x, y) + \mathcal{O}(\epsilon^2) & \bar{x}(\epsilon = 0) = x \\ y \rightarrow \bar{y}(\epsilon) = y + \epsilon \eta(x, y) + \mathcal{O}(\epsilon^2) & \bar{y}(\epsilon = 0) = y \\ p \rightarrow \bar{p}(\epsilon) = p + \epsilon \zeta(x, y, p) + \mathcal{O}(\epsilon^2) & \bar{p}(\epsilon = 0) = p \end{array}$	$\begin{array}{ll} x \rightarrow \bar{x}(\epsilon) = x + \epsilon \xi(x, y) + \mathcal{O}(\epsilon^2) & \bar{x}(\epsilon = 0) = x \\ y \rightarrow \bar{y}(\epsilon) = y + \epsilon \eta(x, y) + \mathcal{O}(\epsilon^2) & \bar{y}(\epsilon = 0) = y \\ p \rightarrow \bar{p}(\epsilon) = p + \epsilon \zeta(x, y, p) + \mathcal{O}(\epsilon^2) & \bar{p}(\epsilon = 0) = p \end{array}$
gilmore-lie-groups_ EQ0964 (16.7)	301	■ 1.000	$\bar{p} = \frac{d\bar{y}}{d\bar{x}} = \frac{d\bar{y}/dx}{d\bar{x}/dx} = \frac{p + \epsilon(\eta_x + \eta_y p)}{1 + \epsilon(\xi_x + \xi_y p)} \longrightarrow p + \epsilon[\eta_x + (\eta_y - \xi_x)p - \xi_y p^2]$	$\bar{p} = \frac{d\bar{y}}{d\bar{x}} = \frac{d\bar{y}/dx}{d\bar{x}/dx} = \frac{p + \epsilon(\eta_x + \eta_y p)}{1 + \epsilon(\xi_x + \xi_y p)} \longrightarrow p + \epsilon[\eta_x + (\eta_y - \xi_x)p - \xi_y p^2]$	$\bar{p} = \frac{d\bar{y}}{d\bar{x}} = \frac{d\bar{y}/dx}{d\bar{x}/dx} = \frac{p + \epsilon(\eta_x + \eta_y p)}{1 + \epsilon(\xi_x + \xi_y p)} \longrightarrow p + \epsilon[\eta_x + (\eta_y - \xi_x)p - \xi_y p^2]$
gilmore-lie-groups_ EQ0965 (16.8)	301	■ 1.000	$\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)}) = \eta_x + (\eta_y - \xi_x)p - \xi_y p^2$	$\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)}) = \eta_x + (\eta_y - \xi_x)p - \xi_y p^2$	$\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)}) = \eta_x + (\eta_y - \xi_x)p - \xi_y p^2$
gilmore-lie-groups_ EQ0967 (16.10)	301	■ 1.000	$F(\bar{x}(\epsilon), \bar{y}(\epsilon), \bar{p}(\epsilon)) = e^{\epsilon X} F(x, y, p) = 0$	$F(\bar{x}(\epsilon), \bar{y}(\epsilon), \bar{p}(\epsilon)) = e^{\epsilon X} F(x, y, p) = 0$	$F(\bar{x}(\epsilon), \bar{y}(\epsilon), \bar{p}(\epsilon)) = e^{\epsilon X} F(x, y, p) = 0$
gilmore-lie-groups_ EQ0968 (16.11)	302	■ 1.000	$X = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} + \zeta \frac{\partial}{\partial p}$	$X = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} + \zeta \frac{\partial}{\partial p}$	$X = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} + \zeta \frac{\partial}{\partial p}$
gilmore-lie-groups_ EQ0969 (16.12)	302	■ 1.000	$F(x, y, p) = 0 \quad \text{and} \quad XF(x, y, p) = 0$	$F(x, y, p) = 0 \quad \text{and} \quad XF(x, y, p) = 0$	$F(x, y, p) = 0 \quad \text{and} \quad XF(x, y, p) = 0$
gilmore-lie-groups_ EQ0971 (16.14)	303	■ 1.000	$XF = 0 \quad X \begin{bmatrix} R(x, y) \\ S(x, y) \\ T(x, y, p) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	$XF = 0 \quad X \begin{bmatrix} R(x, y) \\ S(x, y) \\ T(x, y, p) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	$XF = 0 \quad X \begin{bmatrix} R(x, y) \\ S(x, y) \\ T(x, y, p) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
gilmore-lie-groups_ EQ0972 (16.15)	303	■ 1.000	$\left(\xi(x, y) \frac{\partial}{\partial x} + \eta(x, y) \frac{\partial}{\partial y} \right) S(x, y) = 1$	$\left(\xi(x, y) \frac{\partial}{\partial x} + \eta(x, y) \frac{\partial}{\partial y} \right) S(x, y) = 1$	$\left(\xi(x, y) \frac{\partial}{\partial x} + \eta(x, y) \frac{\partial}{\partial y} \right) S(x, y) = 1$
gilmore-lie-groups_ EQ0974 (16.17)	303	■ 1.000	$\frac{dx}{\xi(x, y)} = \frac{dy}{\eta(x, y)} = \frac{dp}{\zeta(x, y, p)}$	$\frac{dx}{\xi(x, y)} = \frac{dy}{\eta(x, y)} = \frac{dp}{\zeta(x, y, p)}$	$\frac{dx}{\xi(x, y)} = \frac{dy}{\eta(x, y)} = \frac{dp}{\zeta(x, y, p)}$
gilmore-lie-groups_ EQ0975 (16.18)	304	■ 1.000	$\frac{dS}{dR} = \frac{dS(x, y)}{dR(x, y)} = \frac{dS/dx}{dR/dx} = \frac{S_x + S_y p}{R_x + R_y p}$	$\frac{dS}{dR} = \frac{dS(x, y)}{dR(x, y)} = \frac{dS/dx}{dR/dx} = \frac{S_x + S_y p}{R_x + R_y p}$	$\frac{dS}{dR} = \frac{dS(x, y)}{dR(x, y)} = \frac{dS/dx}{dR/dx} = \frac{S_x + S_y p}{R_x + R_y p}$
gilmore-lie-groups_ EQ0977 (16.20)	304	■ 1.000	$S = \int f(R, -, T(R)) \, dR + c$	$S = \int f(R, -, T(R)) dR + c$	$S = \int f(R, -, T(R)) \, dR + c$
gilmore-lie-groups_ EQ0978 (16.21)	305	■ 1.000	$F(x, y, p) = xp + y - xy^2 = 0$	$F(x, y, p) = xp + y - xy^2 = 0$	$F(x, y, p) = xp + y - xy^2 = 0$

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gilmore-lie-groups_ EQ0979 (16.22)	305	■ 1.000	$\xi\left(p-y^2\right)+\eta(1-2xy)+\left[\eta_x+\left(\eta_y-\xi_x\right)p-\xi_y p^2\right] x=0$	$\xi\left(p-y^2\right)+\eta(1-2xy)+\left[\eta_x+\left(\eta_y-\xi_x\right)p-\xi_y p^2\right] x=0$	$\xi(p-y^2)+\eta(1-2xy)+[\eta_x+(\eta_y-\xi_x)p-\xi_y p^2]x=0$
gilmore-lie-groups_ EQ0980 (16.23)	306	■ 1.000	$\xi\left(-\frac{y}{x}\right)+\eta(1-2xy)+\eta_x x+\left(\eta_y-\xi_x\right)\left(x y^2-y\right)-\xi_y\left(x y^4-2 y^3+\frac{y^2}{x}\right)=0$	$\xi\left(-\frac{y}{x}\right)+\eta(1-2xy)+\eta_x x+\left(\eta_y-\xi_x\right)\left(x y^2-y\right)-\xi_y\left(x y^4-2 y^3+\frac{y^2}{x}\right)=0$	$\xi\left(-\frac{y}{x}\right)+\eta(1-2xy)+\eta_x x+\left(\eta_y-\xi_x\right)\left(x y^2-y\right)-\xi_y\left(x y^4-2 y^3+\frac{y^2}{x}\right)=0$
gilmore-lie-groups_ EQ0982 (16.25)	306	■ 1.000	$\begin{aligned} & \xi\left(-\frac{y}{x}\right)+\eta(1-2xy)+\eta_x x+\left(\eta_y-\xi_x\right)\left(x y^2-y\right)-\xi_y\left(x y^4-2 y^3+\frac{y^2}{x}\right)=0 \\ & \xi\left(-\frac{y}{x}\right)+\eta(1-2xy)+\eta_x x+\left(\eta_y-\xi_x\right)\left(x y^2-y\right)-\xi_y\left(x y^4-2 y^3+\frac{y^2}{x}\right)=0 \end{aligned}$	$\left(\xi_{00}+\xi_{10} x+\xi_{01} y\right)\left(-\frac{y}{x}\right)+\left(\eta_{00}+\eta_{10} x+\eta_{01} y\right)(1-2xy)+\eta_{10} x$ $+\left(\eta_{01}-\xi_{10}\right)\left(x y^2-y\right)-\xi_{01}\left(x y^4-2 y^3+\frac{y^2}{x}\right)=0$	$\left(\xi_{00}+\xi_{10} x+\xi_{01} y\right)\left(-\frac{y}{x}\right)+\left(\eta_{00}+\eta_{10} x+\eta_{01} y\right)(1-2xy)+\eta_{10} x$ $+\left(\eta_{01}-\xi_{10}\right)\left(x y^2-y\right)-\xi_{01}\left(x y^4-2 y^3+\frac{y^2}{x}\right)=0$
gilmore-lie-groups_ EQ0984 (16.27)	307	■ 1.000	$X=x \frac{\partial}{\partial x}-y \frac{\partial}{\partial y}-2 p \frac{\partial}{\partial p}$	$X=x \frac{\partial}{\partial x}-y \frac{\partial}{\partial y}-2 p \frac{\partial}{\partial p}$	$X=x \frac{\partial}{\partial x}-y \frac{\partial}{\partial y}-2 p \frac{\partial}{\partial p}$
gilmore-lie-groups_ EQ0985 (16.28)	307	■ 1.000	$\left(x \frac{\partial}{\partial x}-y \frac{\partial}{\partial y}\right) S(x, y)=1$	$\left(x \frac{\partial}{\partial x}-y \frac{\partial}{\partial y}\right) S(x, y)=1$	$\left(x \frac{\partial}{\partial x}-y \frac{\partial}{\partial y}\right) S(x, y)=1$
gilmore-lie-groups_ EQ0986 (16.29)	307	■ 1.000	$\frac{d x}{x}=\frac{d y}{-y}=\frac{d p}{-2 p}$	$\frac{d x}{x}=\frac{d y}{-y}=\frac{d p}{-2 p}$	$\frac{d x}{x}=\frac{d y}{-y}=\frac{d p}{-2 p}$
gilmore-lie-groups_ EQ0988 (16.31)	307	■ 1.000	$F(x, y, p)=x p+y-x y^2=0 \longrightarrow e^S\left[\frac{T}{R}+1-R\right]=0$	$F(x, y, p)=x p+y-x y^2=0 \longrightarrow e^S\left[\frac{T}{R}+1-R\right]=0$	$F(x, y, p)=x p+y-x y^2=0 \longrightarrow e^S\left[\frac{T}{R}+1-R\right]=0$
gilmore-lie-groups_ EQ0989 (16.32)	308	■ 1.000	$\frac{d S}{d R}=\frac{d(\ln y)}{d(x y)}=\frac{p / y}{y+x p}=\frac{T e^S / R^2}{e^S+\left(R e^{-S}\right)\left(T e^{2 S} / R^2\right)}$	$\frac{d S}{d R}=\frac{d(\ln y)}{d(x y)}=\frac{p / y}{y+x p}=\frac{T e^S / R^2}{e^S+\left(R e^{-S}\right)\left(T e^{2 S} / R^2\right)}$	$\frac{d S}{d R}=\frac{d(\ln y)}{d(x y)}=\frac{p / y}{y+x p}=\frac{T e^S / R^2}{e^S+\left(R e^{-S}\right)\left(T e^{2 S} / R^2\right)}$
gilmore-lie-groups_ EQ0991 (16.34)	308	■ 1.000	$\frac{d S}{d R}=\frac{1}{R}-\frac{1}{R^2} \Longrightarrow S=\ln (R)+\frac{1}{R}+c$	$\frac{d S}{d R}=\frac{1}{R}-\frac{1}{R^2} \Longrightarrow S=\ln (R)+\frac{1}{R}+c$	$\frac{d S}{d R}=\frac{1}{R}-\frac{1}{R^2} \Longrightarrow S=\ln (R)+\frac{1}{R}+c$
gilmore-lie-groups_ EQ0992 (16.35)	309	■ 1.000	$y=\frac{-1}{x(c+\ln x)}$	$y=\frac{-1}{x(c+\ln x)}$	$y=\frac{-1}{x(c+\ln x)}$
gilmore-lie-groups_ EQ0994 (16.37)	310	■ 1.000	$X=\xi \frac{\partial}{\partial x}+\eta \frac{\partial}{\partial y}+\eta^{(1)} \frac{\partial}{\partial y^{(1)}}+\eta^{(2)} \frac{\partial}{\partial y^{(2)}}$	$X=\xi \frac{\partial}{\partial x}+\eta \frac{\partial}{\partial y}+\eta^{(1)} \frac{\partial}{\partial y^{(1)}}+\eta^{(2)} \frac{\partial}{\partial y^{(2)}}$	$X=\xi \frac{\partial}{\partial x}+\eta \frac{\partial}{\partial y}+\eta^{(1)} \frac{\partial}{\partial y^{(1)}}+\eta^{(2)} \frac{\partial}{\partial y^{(2)}}$

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gilmore-lie-groups_ EQ0995 (16.38)	310	■ 1.000	$\begin{aligned} \frac{d^2 \bar{y}}{d\bar{x}^2} &= \frac{d}{d\bar{x}} \left(\frac{d\bar{y}}{d\bar{x}} \right) = \frac{d}{d\bar{x}} \left(p + \epsilon \eta^{(1)} \right) = \frac{D^{(1)}(p + \epsilon \eta^{(1)})}{D^{(0)}(x + \epsilon \xi)} = \frac{y^{(2)} + \epsilon D^{(1)} \eta^{(1)}}{1 + \epsilon D^{(0)} \xi} \\ &= y^{(2)} + \epsilon \left(D^{(1)} \eta^{(1)} - y^{(2)} D^{(0)} \xi \right) \end{aligned} \quad (16.)$	$\frac{d^2 \bar{y}}{d\bar{x}^2} = \frac{d}{d\bar{x}} \left(\frac{d\bar{y}}{d\bar{x}} \right) = \frac{d}{d\bar{x}} \left(p + \epsilon \eta^{(1)} \right) = \frac{D^{(1)}(p + \epsilon \eta^{(1)})}{D^{(0)}(x + \epsilon \xi)} = \frac{y^{(2)} + \epsilon D^{(1)} \eta^{(1)}}{1 + \epsilon D^{(0)} \xi}$	$\frac{d^2 \bar{y}}{d\bar{x}^2} = \frac{d}{d\bar{x}} \left(\frac{d\bar{y}}{d\bar{x}} \right) = \frac{d}{d\bar{x}} (p + \epsilon \eta^{(1)}) = \frac{D^{(1)}(p + \epsilon \eta^{(1)})}{D^{(0)}(x + \epsilon \xi)} = \frac{y^{(2)} + \epsilon D^{(1)} \eta^{(1)}}{1 + \epsilon D^{(0)} \xi}$
gilmore-lie-groups_ EQ0996 (16.39)	310	■ 1.000	$\eta^{(2)}(x, y, y^{(1)}, y^{(2)}) = D^{(1)} \eta^{(1)} - y^{(2)} D^{(0)} \xi$	$\eta^{(2)}(x, y, y^{(1)}, y^{(2)}) = D^{(1)} \eta^{(1)} - y^{(2)} D^{(0)} \xi$	$\eta^{(2)}(x, y, y^{(1)}, y^{(2)}) = D^{(1)} \eta^{(1)} - y^{(2)} D^{(0)} \xi$
gilmore-lie-groups_ EQ0997 (16.40)	311	■ 1.000	$D^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \cdots + y^{(n+1)} \frac{\partial}{\partial y^{(n)}}$	$D^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \cdots + y^{(n+1)} \frac{\partial}{\partial y^{(n)}}$	$D^{(n)} = \frac{\partial}{\partial x} + \frac{dy}{dx} \frac{\partial}{\partial y} + \frac{dy^{(1)}}{dx} \frac{\partial}{\partial y^{(1)}} + \cdots + y^{(n+1)} \frac{\partial}{\partial y^{(n)}}$
gilmore-lie-groups_ EQ0998 (16.41)	311	■ 1.000	$\eta^{(2)} = \eta_{xx} + (2\eta_{xy} - \xi_{xx}) y' + (\eta_{yy} - 2\xi_{xy}) y'^2 - \xi_{yy} y'^3 + (\eta_y - 2\xi_x - 3\xi_{yy'}) y''$	$\eta^{(2)} = \eta_{xx} + (2\eta_{xy} - \xi_{xx}) y' + (\eta_{yy} - 2\xi_{xy}) y'^2 - \xi_{yy} y'^3 + (\eta_y - 2\xi_x - 3\xi_{yy'}) y''$	$\eta^{(2)} = \eta_{xx} + (2\eta_{xy} - \xi_{xx}) y' + (\eta_{yy} - 2\xi_{xy}) y'^2 - \xi_{yy} y'^3 + (\eta_y - 2\xi_x - 3\xi_{yy'}) y''$
gilmore-lie-groups_ EQ0999 (16.42)	311	■ 1.000	$F \left(x, y, y^{(1)}, y^{(2)} \right) = 0 \quad X \left(x, y, y^{(1)}, y^{(2)} \right) F \left(x, y, y^{(1)}, y^{(2)} \right) = 0$	$F \left(x, y, y^{(1)}, y^{(2)} \right) = 0 \quad X \left(x, y, y^{(1)}, y^{(2)} \right) F \left(x, y, y^{(1)}, y^{(2)} \right) = 0$	$F(x, y, y^{(1)}, y^{(2)}) = 0 \quad X(x, y, y^{(1)}, y^{(2)}) F(x, y, y^{(1)}, y^{(2)}) = 0$
gilmore-lie-groups_ EQ1000 (16.43)	312	■ 1.000	$\frac{dT}{dR} = \frac{dT/dx}{dR/dx} = \frac{T_x + T_y y^{(1)} + T_{y^{(1)}} y^{(2)}}{R_x + R_y y^{(1)}}$	$\frac{dT}{dR} = \frac{dT/dx}{dR/dx} = \frac{T_x + T_y y^{(1)} + T_{y^{(1)}} y^{(2)}}{R_x + R_y y^{(1)}}$	$\frac{dT}{dR} = \frac{dT/dx}{dR/dx} = \frac{T_x + T_y y^{(1)} + T_{y^{(1)}} y^{(2)}}{R_x + R_y y^{(1)}}$
gilmore-lie-groups_ EQ1001 (16.44)	312	■ 1.000	$\frac{dT}{dR} = \frac{2xy' + x^2 y''}{y + xy'}$	$\frac{dT}{dR} = \frac{2xy' + x^2 y''}{y + xy'}$	$\frac{dT}{dR} = \frac{2xy' + x^2 y''}{y + xy'}$
gilmore-lie-groups_ EQ1002 (16.45)	312	■ 1.000	$G \left(R, -, T, \frac{dT}{dR} \right) = 0$	$G \left(R, -, T, \frac{dT}{dR} \right) = 0$	$G \left(R, -, T, \frac{dT}{dR} \right) = 0$
gilmore-lie-groups_ EQ1003 (16.46)	312	■ 1.000	$\frac{dT}{dR} = \frac{x^2 y'' + 2xy'}{y + xy'} = g(xy, x^2 y') = g(R, T)$	$\frac{dT}{dR} = \frac{x^2 y'' + 2xy'}{y + xy'} = g(xy, x^2 y') = g(R, T)$	$\frac{dT}{dR} = \frac{x^2 y'' + 2xy'}{y + xy'} = g(xy, x^2 y') = g(R, T)$
gilmore-lie-groups_ EQ1004 (16.47)	313	■ 1.000	$X = \xi \frac{\partial}{\partial x} + \eta^{(0)} \frac{\partial}{\partial y^{(0)}} + \eta^{(1)} \frac{\partial}{\partial y^{(1)}} + \cdots + \eta^{(n)} \frac{\partial}{\partial y^{(n)}} = \xi \frac{\partial}{\partial x} + \sum_{j=0}^n \eta^{(j)} \frac{\partial}{\partial y^{(j)}}$	$X = \xi \frac{\partial}{\partial x} + \eta^{(0)} \frac{\partial}{\partial y^{(0)}} + \eta^{(1)} \frac{\partial}{\partial y^{(1)}} + \cdots + \eta^{(n)} \frac{\partial}{\partial y^{(n)}} = \xi \frac{\partial}{\partial x} + \sum_{j=0}^n \eta^{(j)} \frac{\partial}{\partial y^{(j)}}$	$X = \xi \frac{\partial}{\partial x} + \eta^{(0)} \frac{\partial}{\partial y^{(0)}} + \eta^{(1)} \frac{\partial}{\partial y^{(1)}} + \cdots + \eta^{(n)} \frac{\partial}{\partial y^{(n)}} = \xi \frac{\partial}{\partial x} + \sum_{j=0}^n \eta^{(j)} \frac{\partial}{\partial y^{(j)}}$

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gilmore-lie-groups_ EQ1023	319	■ 1.000	$\left[\begin{array}{c} i_x \\ i_y \\ i_t \end{array}\right] = \left[\begin{array}{cccc} x^2 - y^2 + t^2 & 2xy & 2xt & -xu \\ 2yx & -x^2 + y^2 + t^2 & 2yt & -yu \\ 2tx & 2ty & x^2 + y^2 + t^2 & -tu \end{array}\right] \left[\begin{array}{c} \partial_x \\ \partial_y \\ \partial_t \end{array}\right]$		
gilmore-lie-groups_ EQ1025	320	■ 1.000	$S = t^2 \frac{\partial}{\partial t} + t \mathbf{x} \cdot \nabla - \frac{1}{4} (\mathbf{x} \cdot \mathbf{x} + 2nt) u \frac{\partial}{\partial u}$		
gilmore-lie-groups_ EQ1026	320	■ 1.000	$I = \int_{t_1}^{t_2} L(t, x, \dot{x}) dt \quad \delta I = 0$		
gilmore-lie-groups_ EQ1027	320	■ 1.000	$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) - \frac{\partial L}{\partial x_i} = 0$		
gilmore-lie-groups_ EQ1028	320	■ 1.000	$I = \int_{t'_1}^{t'_2} L(t', x', \dot{x}') dt' = \int_{t_1}^{t_2} L(t', x', \dot{x}') \frac{dt'}{dt} dt$		
gilmore-lie-groups_ EQ1029	320	■ 1.000	$\int_{t_1}^{t_2} \left(\xi \frac{\partial L}{\partial t} + \eta_i \frac{\partial L}{\partial x_i} + \eta_i^{(1)} \frac{\partial L}{\partial \dot{x}_i} + \frac{d\xi}{dt} L \right) dt = 0$		
gilmore-lie-groups_ EQ1030	320	■ 1.000	$\frac{d}{dt} [\xi L + (\eta_i - \xi \dot{x}_i) L_{\dot{x}_i}] = 0$		
gilmore-lie-groups_ EQ1031	321	■ 1.000	$\mathbf{v} = \sum_{i=1}^p \xi^i(x, u) \frac{\partial}{\partial x^i} + \sum_{\alpha=1}^q \phi^\alpha(x, u) \frac{\partial}{\partial u^\alpha}$		
gilmore-lie-groups_ EQ1032	321	■ 1.000	$P^i = \xi^i \mathcal{L} + \sum_{\alpha=1}^q \phi^\alpha(x, u) \frac{\partial \mathcal{L}}{\partial u_i^\alpha} - \sum_{\alpha=1}^q \sum_{j=1}^p \xi^j u_j^\alpha \frac{\partial \mathcal{L}}{\partial u_i^\alpha}$		
gilmore-lie-groups_ EQ1033	321	■ 1.000	$\nabla P = \operatorname{div} P = \frac{\partial P^i}{\partial x^i} = 0$		

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gilmore-lie-groups_ EQ1035	321	■ 1.000	$\left\langle g \left \begin{array}{c} \lambda \\ \mu \nu \end{array} \right. \right\rangle = \sqrt{\frac{\dim \lambda}{\text{Vol}(G)}} \Gamma_{\mu \nu}^{\lambda}(g) \quad \left\langle \begin{array}{c} \lambda \\ \mu \nu \end{array} \left g \right. \right\rangle = \sqrt{\frac{\dim \lambda}{\text{Vol}(G)}} \Gamma_{\mu \nu}^{\lambda*}(g)$	$\left\langle g \left \begin{array}{c} \lambda \\ \mu \nu \end{array} \right. \right\rangle = \sqrt{\frac{\dim \lambda}{\text{Vol}(G)}} \Gamma_{\mu \nu}^{\lambda}(g) \quad \left\langle \begin{array}{c} \lambda \\ \mu \nu \end{array} \left g \right. \right\rangle = \sqrt{\frac{\dim \lambda}{\text{Vol}(G)}} \Gamma_{\mu \nu}^{\lambda*}(g)$	